

SUPPORTING INFORMATION: AUTHOR MANUSCRIPT v8 (CORRECTED DRAFT)

Recording context and data splitting can inflate acoustic re-identification accuracy across species and studies. Graham Wallington and Jackie Lighten.

Appendix S1. Methods detail

S1.1 Reading BirdNET

BirdNET v2.4 was read at the layer before its species predictions, which gives 1,024 numbers for each 3 s of sound at 48 kHz. Sound was read as whole numbers and scaled, mixed to one channel and cut into 3-s segments at the recording's own sample rate. Each segment was then resampled to 48 kHz by the Fourier method, and the last segment was padded with silence. The embedding of a clip is the mean of its segment embeddings. This order matters: the Fourier method gives a different result if the whole recording is resampled before it is cut. Our reader reproduced the embeddings of birdnet==1.1.0 to within 3×10^{-6} on eight great tit clips. BirdNET was given the sound at its recorded level, except that BirdPark clips carry a common endpoint gain (0.95 divided by the endpoint peak), preserving relative but not absolute levels.

Seven layers inside BirdNET can be read: the outputs of four processing stages of falling time resolution, two layers after the last stage and the final embedding. Each was averaged over time, and each was also summarised by its mean and spread over time, giving 14 candidate embeddings. These were used only in the analysis of BirdNET's internal layers (S1.13).

S1.2 Neural networks trained on animal or general sound

The 19 networks (Table S12) were loaded through the Bacpipe 1.3.5 library (Kather et al. 2026), with the weights it supplies, except the two AVEX networks, which were loaded through the Earth Species Project's avex library. A network entered the comparison only after it had loaded on our computer and produced an embedding for a real clip. Each clip was resampled to the network's own sample rate with an anti-aliasing filter and cut into windows of the network's own input length (for AVEX, its 5-s training window); the last

window, and any clip shorter than one window, was padded with silence. Each window passed through the network's published preprocessing, and the clip's embedding was the mean over its windows. The sound was given to the network at its recorded level, apart from the common BirdPark endpoint gain described in S1.1 and the network's own preprocessing.

For networks built from a stack of repeated transformer blocks (AvesEcho, Bird-MAE, ProtoCLR, BirdAVES, AVES, NatureBEATs, BioLingual, BEATs, AudioMAE and AVEX sl-BEATs-all), the output of every block, averaged over its time steps and then over windows, was a candidate embedding alongside the network's own embedding, which is the output read under the own-final-output rule (S1.5). The convolutional networks (Perch 2.0, Perch, BirdNET v3, SurfPerch, AVEX EfficientNet-B0, AudioProtoPNet, ConvNeXt, RCL_FS_BSED and VGGish) offer their own embedding only; Perch 2.0 and BirdNET v3 are distributed in a format whose internal layers cannot be read. For each network, the sample rate, window length and the named blocks that were read are listed in Table S12, from a record made by reloading each network and checking that it produced the same candidate names as the measurements. Every weight file was checked against its recorded checksum before that reload.

S1.3 Speech neural networks

Seventeen speech network checkpoints were measured: sixteen in an earlier run and again for this study, and XEUS. One of the sixteen duplicates another (below), so 16 speech networks are compared. The input to the first transformer block and the output of every block were candidates, at three playback speeds: the recording's own speed, one half and one third. Slowing was applied by dividing the recording's sample rate, so a 250 kHz bat clip was treated as 250, 125 or 83.3 kHz, and the highest original frequency reaching the 8 kHz range of these networks was 24 kHz. Sound was brought to 16 kHz by polyphase resampling with an anti-aliasing filter. Unlike BirdNET and the networks in S1.2, the speech networks were given each clip with its mean removed and scaled to a peak of one, so they could not use a clip's absolute level; the loudness control could. A clip shorter than a network accepts was

padding with silence to the shortest length it accepts. On the chiffchaff and tree pipit datasets, the WavLM combinations that exceeded the 16 GB graphics card were read in 60-s windows, each layer averaged over every frame of every window.

The speaker-verification version of WavLM Base+ is loaded through the same interface as the other networks, which reads the transformer and not the speaker-recognition layers; its weights are identical to those of WavLM Base+, so it gives the same values at every layer and speed (it was run at the card's reduced precision on the across-year tree pipit, which changed Fisher ratios in the sixth significant figure and no choice or accuracy). It was measured because it was specified in advance, but it is left out of every count, rank and correction.

The 16 networks of the earlier run were run again on every dataset, keeping every candidate embedding, so that both layer rules could be applied to the same embeddings. Each candidate's re-ID accuracy and its right or wrong answer on every test clip were compared with the earlier run. Of 8,297 candidates on the 12 datasets of the earlier run, 8,295 were identical. The other two were layers 16 and 20 of the wav2vec 2.0 conformer at one-third speed on the across-year chiffchaff (0.120 against 0.125, and 0.150 against 0.155), where the graphics card's matrix arithmetic differed between the runs. Neither was chosen by either rule, and every value reported from that run was identical on every dataset (Table S19). The graphics card's matrix arithmetic had not been recorded in the earlier run. For 28 combinations of network and dataset (24 at full 32-bit precision and 4 at the card's reduced TF32 precision, on the great tit, the within-year and across-year chiffchaff and the across-year tree pipit), the rerun reproduced the earlier run only with the arithmetic that run had used; each of these networks was rerun whole on that dataset with that setting, which is recorded in its result file.

S1.4 Classifier

Embeddings were scaled to unit length and centred on the mean of the enrolment embeddings. Kernel ridge regression used a linear kernel and a fixed penalty of 1.0, and

predicted a one-hot code for each individual; a test clip was assigned to the individual with the highest predicted value. Embeddings of three or fewer numbers (the controls) were standardised on the enrolment clips instead of scaled to unit length.

S1.5 The layer rules

Fisher rule. For each network, the candidate (layer and, for speech networks, playback speed) with the highest Fisher ratio on the enrolment clips was chosen: the ratio of variation between individuals to variation within individuals, averaged over the embedding's dimensions. The Fisher ratio was computed on raw candidate embeddings, using unweighted means over individuals; the classifier instead used the feature preparation in S1.4. The test clips were not used.

Held-out-session rule. Each individual's enrolment sessions were dealt into five groups by a fixed hash. For each group in turn, the classifier was fitted on the enrolment clips of the other four and scored on that group, and the candidate with the highest accuracy on the held-out sessions was chosen; ties went to the Fisher ratio. Individuals with one enrolment session, or whose sessions are not recorded, were used to fit the classifier but were never scored. The rule could be applied to seven datasets, with sessions as defined in Table 1: the fruit bat, the zebra finch recorded singly, both BirdPark groups, both great tit datasets and the rook. In the Stowell et al. datasets the enrolment clips carry no session within a bird, all little penguin enrolment clips come from one night, and the cockatoo release records no session; on these datasets the rule can only choose what the Fisher rule chose, so it is summarised on the six primary datasets where it applies.

Own final output. Specified after the fourth round of review and exploratory: each network was read at the output its authors provide, the network's own embedding for networks trained on animal or general sound, the last transformer layer at the recording's own speed for speech networks (layer 12 of base and layer 24 of large networks, layer 18 of XEUS), and the output at the recording's own speed for the speaker networks. Networks with one

embedding, BirdNET and the controls are unchanged. The accuracy and right or wrong answer of every candidate were already stored, so no network was run again.

S1.6 Datasets

BirdPark. From the BirdPark release (Rüttimann et al. 2024) we kept vocal segments that had been attributed to one bird by its transmitter, did not overlap another bird's syllable, and were recorded on the shared wall microphone. Each group was divided by recording file, with the three files of lowest hash as the test side.

Rook. Each rook recording stores several microphones as channels, and the channel index is not the same physical microphone on every day. We used channel 0 throughout, so that each call was counted once. Birds with at least 20 calls in each year were kept (11 of 15), recordings from 2020 were used for enrolment and from 2021 for testing, and enrolment was capped at 90 clips per bird, drawn with a fixed seed. For Diagnostic 3, the 2020 recordings were also divided by day, and the same calls were enrolled on channel 0 and tested on channel 1 (Table S4).

Great tit with 50 birds. From the release of Merino Recalde et al. (2024), a bird was included when its ring was recorded as the father of a nest attempt and also as a male great tit in the morphometric records, and when songs from his nest attempts existed in two or more years. Ten songs per bird and year were drawn by a hash of the file name; songs from the first year were used for enrolment and from the next for testing. As in the 16-bird dataset, a song is attributed to a bird through the nest attempt, not through an observation of the singer.

S1.7 Beecher's information statistic

On every embedding wider than three numbers, embeddings were scaled to unit length and reduced to principal components fitted on all call clips of the dataset, with at most one fewer component than individuals. For each component, a one-way analysis of variance gave F , and the component contributed $\log_2 \sqrt{((F + n_0 - 1)/n_0)}$ bits, where n_0 is the effective group size of the unbalanced design; components with F below one contributed nothing (Beecher 1989).

S1.8 Paired-clip test

All pairs of test clips were formed, or 800,000 drawn at random with a fixed seed where more exist, and scored by the cosine similarity of their unit-length embeddings. Each pair was classed by whether the two clips came from the same individual and from the same recording. The reported difference is the mean similarity of animal pairs (same individual, different recordings) minus that of recording pairs (different individuals, same recording). The standardised difference divides it by the square root of the mean of the two population variances. The equal error rate approximates the crossing of false-accept and false-reject rates. A preprint audit found that the original implementation could split tied scores according to their labels. The corrected implementation groups equal scores and includes rejection of all scores, then reports the mean of the two error rates at the attainable threshold closest to equality. The historical paired-clip error rates in Tables S3 and S4 await recomputation from archived scores and are not validated by this draft.

S1.9 Within-recording permutation test

Enrolment labels were shuffled 9,999 times, each time only among clips from the same recording, and the classifier was refitted. A recording holding one animal keeps its label, so the test can only shuffle clips from recordings with two or more animals. In the fruit bat, the unit is the recording file. Only 8 of 6,188 bat enrolment recordings held two or more bats, so 16 of 6,252 enrolment clips could be shuffled. In the zebra finch recorded singly, the unit is the day, and 6 of the 15 enrolment days held more than one bird.

S1.10 Background tests

Four pairings. The classifier was fitted on calls or on background recordings and tested on calls or on backgrounds, with the same individuals, splits and seed as the call-only test. Each pairing had its own label-permutation test (9,999 shuffles) and interval over individuals (10,000 draws).

Added background. Each test clip was mixed with one background from its own individual and one from another individual (the donor), at 10 dB above, equal to and 10 dB below the clip's own level. Each individual's donor was the next individual in sorted order, and a hash

of the test clip's name chose which of the own and donor backgrounds was used, from the test side of the split, and the classifier fitted on the original enrolment clips was applied. We report the difference in balanced accuracy between the two mixtures, with a paired interval over individuals, and the proportion of answers that the donor's background changed. Each mixture was peak-scaled to 0.95, whereas the original arm was unscaled. Own-versus-other mixtures share that scaling rule; comparisons with the original arm, including the donor-naming baseline, also differ in level preprocessing. The mixed clips were written once per dataset and read by every network.

Donor naming. When BirdNET named the wrong individual for a clip mixed with a donor's background, we counted how often the wrong answer was the donor, and compared that share with the same share among BirdNET's errors on the unmixed test clips, which measures how often it already confused each individual with its fixed donor, and with 9,999 random reassignments of donors that kept each clip's own individual excluded. This analysis was specified after results were seen and is exploratory.

Planted-signal test. Calls from the ten within-year chiffchaffs with the most enrolment calls were mixed into the across-year chiffchaff background recordings, each bird into the territory of one across-year bird, enrolment calls into enrolment-year backgrounds and test calls into test-year backgrounds, at 10 dB above, equal to and 10 dB below the background. Each planted bird therefore had its own site, as in the real data, while the call it carried was known to differ between birds. The three datasets were scored with BirdNET as in Table 2.

Right whale. The four pairings were also run with BirdNET on 234 upcalls and 234 paired recordings of tag noise from 11 North Atlantic right whales carrying suction-cup tags, released with Tolkova et al. (2026); these data are not among the 13 datasets. Each whale carried one tag deployment, so no division by session exists. Within each whale, a call and its paired noise went to the test side when the first byte of a SHA-256 hash of a fixed salt followed by the clip stem was below 77 and to enrolment otherwise, putting about three in ten clips on the test side and keeping each call with its own noise. Individuals with no test

clip after this split were omitted; the archived table reports 11 retained whales, which remains to be rechecked from the original manifest. Before BirdNET read each clip, the 50 to 500 Hz band was shifted upward by 0 to 10,000 Hz in steps of 1,000 Hz, as in Tolkova et al. (2026), identically for calls and noise. The shift for the result was specified in advance as the one with the highest Fisher ratio on the enrolment calls, which was 10,000 Hz.

Equivalence test on the little owl. We asked whether background-only accuracy lay below chance plus one tenth of the call's excess over chance (0.044 above chance), passing if the 95th percentile of the interval over individuals fell below that bound. The margin was set after the background estimate was seen but derived from the call result alone.

S1.11 Remedies after recording

Background augmentation (Stowell et al. 2019) mixed every enrolment call with a background from each other individual, at the clips' own levels, and added the mixtures to the enrolment clips; test clips were not changed. It was applied with BirdNET to the within-year chiffchaff, the little owl and the within-year tree pipit.

Spectral subtraction. For each individual within each side of the split, the mean short-time magnitude spectrum (Hann window of 1,024 samples, step 256) of its background recordings was computed. That spectrum, times one or times two, was subtracted from every call and background clip of the individual, keeping at least 5% of each clip's own magnitude in every frequency bin and keeping the phase. Because it uses each individual's own backgrounds, it asks whether the site can be removed when the site of every test clip is known.

Adaptive score normalisation (Matějka et al. 2017). Scores were the class-mean classifier's cosine similarities on unit-scaled, enrolment-centred embeddings. Duration and loudness controls are excluded because cosine unit normalisation removes their magnitude. The comparison set was every background recording on the enrolment side. For each enrolled individual and each test clip, the mean and spread of its 50 highest scores against that set were computed, and each score was replaced by the average of its two standardised values.

Session compensation. Four methods from human speaker recognition were applied to the fruit bat's BirdNET embeddings: subtracting each session's mean computed over all clips of that day on the same side of the split, every bat included, subtracting a running session mean that uses only the clips already heard (using the global enrolment mean until five clips have been heard), within-class covariance normalisation (Hatch et al. 2006) at eight levels of shrinkage, and removing 1 to 32 directions of variation between sessions. How well the recording day could still be predicted was measured by a classifier fitted on all but one bat and tested on the one left out, for each bat in turn. Because a day's mean includes every bat, subtracting it places each bat's clips opposite the other bats' clips of that day, so this classifier predicts that bat's days systematically wrongly after mean subtraction. Within-class covariance normalisation was also applied to the little owl.

S1.12 Open-set allocation

Within each of four groups of individuals, individuals were ordered by a hash of a fixed salt, the group and the individual, dealt round the four roles (known and strangers, each for calibration and for testing), and rotated so that every individual held every role once per group, giving 16 allocations. Each role held four individuals, two or three on the rook (11 birds) and 12 or 13 on the 50-bird great tit. The test gallery holds only the test-known individuals, so naming the correct individual is a choice among four (chance 0.25). The 36 neural-network embeddings were scaled to unit length, without per-dimension standardisation or enrolment centring, and scored by cosine similarity. Duration and loudness controls are excluded from this analysis because unit normalisation removes their magnitude. The separately reported acquisition-context control retains its own calibration-only standardisation. For each stranger budget, the calibrated threshold was the lowest score at which no more than that proportion of the calibration strangers' test clips was accepted; thresholds were set before the test individuals were scored. The test-derived threshold is the best of all thresholds that accept at most the budgeted proportion of the test strangers, found with the test strangers themselves. A calibrated threshold may accept more test strangers than the budget, and the proportion it accepted is reported. Balanced

accuracy on known individuals is the mean over known individuals of the proportion of their clips accepted and correctly named; balanced accuracy on strangers is the mean over strangers of each stranger's proportion of clips rejected; their geometric mean is the AnimalCLEF 2025 score (Adam et al. 2025).

S1.13 BirdNET's internal layers, learned combination and sequence pooling

Of the 14 candidates in S1.1, the one with the highest enrolment Fisher ratio was chosen, and its permutation p was multiplied by the number of distinct candidates and capped at one. Two of the 14 repeat others (the post-convolution mean is the final embedding, and the final embedding's spread over time is zero), so the multiplier was 12. The learned combination took each of the seven layers averaged over time, scaled and centred it, projected it onto at most 16 directions that best separate individuals relative to variation within them, and joined the projections, all fitted on enrolment clips only; its permutation test refitted the projection for every shuffle. Learned sequence pooling weighted each time step of one of BirdNET's five time-resolved layers by a learned attention vector and mapped the weighted mean and spread to 64 numbers, trained on about two fifths of the individuals and scored on the others, which it never saw; with 3 to 7 individuals scored per dataset, its values are not comparable in scale with Table 2.

S1.14 Provenance and archive

Each result file records its code version, clip lists and their checksums, splits, random seeds and computing environment. Every run was made on one computer under a resource lease that records the job. The statistics added in this version (Tables S13, S14, S21 and S22 and the counts in the text) were computed from the same result files on a second computer, after every table and figure of the previous version had been reproduced there byte for byte. Every result file is archived in cloud storage under the SHA-256 checksum of its contents, and an archived copy is accepted only when its size and MD5 match the local file. Before the original tables were written, an evidence gate checked the presence, provenance and role-specific contracts of all 224 result files. It checked the stated resampling counts for most roles, but did not enforce them for corrected-sweep, speech, selection-refresh and

speech-replay results. Those roles therefore require separate count verification. Every table and figure was then produced from those files by scripts that record the file and checksum behind every value.

Tables

Table S1. How closely identity is tied to each recording variable stored with the clips, over all clips of each dataset.

Dataset	Recording variable	Levels	NMI	Cramér's V	Levels holding two or more individuals	Individuals on two or more levels
Zebra finch, group of four	microphone	1	0.000	0.000	1 of 1	0 of 4
Zebra finch, group of four	session	7	0.473	0.644	7 of 7	4 of 4
Zebra finch, group of eight	microphone	1	0.000	0.000	1 of 1	0 of 8
Zebra finch, group of eight	session	7	0.407	0.494	6 of 7	8 of 8
Rook, across year	microphone channel	1	0.000	0.000	1 of 1	0 of 11
Rook, across year	calling event	327	0.515	0.629	50 of 327	11 of 11
Rook, across year	recording time	23	0.383	0.496	23 of 23	11 of 11
Rook, across year	recording	81	0.549	0.649	67 of 81	11 of 11
Rook, across year	channels stored	3	0.120	0.271	3 of 3	11 of 11
Rook, across year	year	2	0.185	0.476	2 of 2	11 of 11
Zebra finch, one bird per recording	date	24	0.827	0.863	7 of 24	9 of 9
Zebra finch, one bird per recording	day	24	0.827	0.863	7 of 24	9 of 9
Chiffchaff, within year	session span	1	0.000	0.000	1 of 1	0 of 13
Chiffchaff, across year	session span	1	0.000	0.000	1 of 1	0 of 10
Little owl, across year	session span	1	0.000	0.000	1 of 1	0 of 16
Tree pipit, within year	session span	1	0.000	0.000	1 of 1	0 of 10
Tree pipit, across year	session span	1	0.000	0.000	1 of 1	0 of 10
Great tit, across year	cohort	3	1.000	1.000	3 of 3	0 of 16
Great tit, across year	nest box	32	1.000	1.000	0 of 32	16 of 16
Great tit, across year	recording	215	1.000	1.000	0 of 215	16 of 16
Great tit, across year	year	3	0.358	0.500	3 of 3	16 of 16
Little penguin, across night	call type	1	0.000	0.000	1 of 1	0 of 15
Little penguin, across night	night	2	0.027	0.138	2 of 2	13 of 15
Little penguin, across night	recording time	3	0.493	0.576	2 of 3	5 of 13
Little penguin, across night	day	3	0.493	0.576	2 of 3	5 of 13
Egyptian fruit bat, across year	microphone channel	4	0.398	0.498	4 of 4	6 of 6
Egyptian fruit bat, across year	folder	6	0.057	0.186	6 of 6	6 of 6
Egyptian fruit bat, across year	recording time	7099	0.998	0.998	18 of 7099	6 of 6
Egyptian fruit bat, across year	day	76	0.136	0.285	75 of 76	6 of 6
Egyptian fruit bat, across year	recording	7109	0.998	0.998	12 of 7109	6 of 6
Egyptian fruit bat, across year	treatment	5	0.440	0.489	5 of 5	6 of 6

Notes: NMI, mutual information divided by the smaller of the two entropies: 0, the variable is unrelated to identity; 1, either determines the other. It is 1 both when every level holds one individual (each recording one animal) and when every individual sits on one level (each bird in one year cohort); the last two columns show which. A variable with one level carries no information. The Stowell et al. datasets store no recording variable within a split.

Table S2. Great tit place control: for each bird of the 16-bird dataset, whether any nest box or recording contributed to both its enrolment and test clips, whether the years differ, and the distance between its enrolment and test nest boxes.

Bird	Same nest box on both sides	Same recording on both sides	Years differ	Distance between nest boxes (m)
GT-01	no	no	yes	59.4
GT-02	no	no	yes	77.8
GT-03	no	no	yes	65.0
GT-04	no	no	yes	54.5
GT-05	no	no	yes	79.8
GT-06	no	no	yes	115.4
GT-07	no	no	yes	57.1
GT-08	no	no	yes	228.5
GT-09	no	no	yes	194.3
GT-10	no	no	yes	151.6
GT-11	no	no	yes	145.6
GT-12	no	no	yes	52.0
GT-13	no	no	yes	174.5
GT-14	no	no	yes	63.9
GT-15	no	no	yes	113.3
GT-16	no	no	yes	90.0

Notes: Bird codes are arbitrary; ring numbers are not published. Distances ranged from 52.0 to 228.5 m (median 84.9 m). In the 50-bird dataset, no recording contributed to both sides, but 21 of 50 birds were enrolled and tested at the same nest box; distances ranged from 0.0 to 733.4 m (median 29.2 m).

Table S3. Closed-set re-ID with BirdNET and the kernel ridge classifier on every dataset, all quantities.

Dataset	n	Enrolment / test clips	Re-ID accuracy (95% interval)	Three-stage interval	Unifor m chance	Majorit y class	Permutatio n 95th percentile	p	Within-recording 95th percentile (p)	Paired-clip difference	Standardised difference	Equal error rate
Zebra finch, group of four	4	199 / 218	0.807 (0.669 to 0.963)	0.613 to 0.963	0.250	0.445	0.491	0.0001	0.587 (0.0001)	0.161	1.675	0.173
Zebra finch, group of eight	8	260 / 229	0.498 (0.176 to 0.725)	0.124 to 0.754	0.125	0.358	0.293	0.0002	0.362 (0.0002)	-0.003	-0.033	0.524
Rook, across year	11	990 / 4,417	0.375 (0.260 to 0.510)	0.250 to 0.529	0.091	0.241	0.127	0.0001	0.224 (0.0001)	0.030	0.260	0.449
Zebra finch, one bird per recording	9	90 / 90	0.967 (0.933 to 1.000)	0.906 to 1.000	0.111	0.111	0.244	0.0001	0.856 (0.0002)	n/a	n/a	n/a
Chiffchaff, within year	13	5,107 / 1,131	0.765 (0.681 to 0.846)	n/a	0.077	0.393	0.154	0.0001	n/a	n/a	n/a	n/a
Chiffchaff, across year	10	324 / 200	0.160 (0.035 to 0.333)	n/a	0.100	0.160	0.190	0.1327	n/a	n/a	n/a	n/a
Little owl, across year	16	545 / 407	0.501 (0.371 to 0.634)	n/a	0.062	0.084	0.123	0.0001	n/a	n/a	n/a	n/a
Tree pipit, within year	10	409 / 303	0.376 (0.193 to 0.566)	n/a	0.100	0.125	0.162	0.0001	n/a	n/a	n/a	n/a
Tree pipit, across year	10	409 / 313	0.214 (0.116 to 0.334)	n/a	0.100	0.160	0.157	0.0022	n/a	n/a	n/a	n/a
Great tit, across year	16	160 / 160	0.444 (0.319 to 0.575)	0.297 to 0.594	0.062	0.062	0.106	0.0001	n/a	n/a	n/a	n/a
Little penguin, across night	15	2,429 / 792	0.693 (0.604 to 0.772)	0.594 to 0.782	0.067	0.136	0.115	0.0001	n/a	n/a	n/a	n/a
Red-tailed black cockatoo, random split	16	1,428 / 357	0.961 (0.929 to 0.980)	n/a	0.062	0.216	0.182	0.0001	n/a	n/a	n/a	n/a
Egyptian fruit bat, across year	6	6,252 / 928	0.381 (0.189 to 0.569)	0.188 to 0.565	0.167	0.297	0.239	0.0001	n/a	n/a	n/a	n/a
Great tit, 50 birds, across year	50	500 / 500	0.270 (0.202 to 0.342)	0.188 to 0.357	0.020	0.020	0.036	0.0001	n/a	n/a	n/a	n/a

Notes: As Table 2. Paired-clip test: mean cosine similarity of animal pairs (one individual, different recordings) minus that of recording pairs (different individuals, one recording), its standardised value and the equal error rate for telling the two kinds of pair apart from a single pair. n/a: the dataset has no recording holding two individuals, or does not identify recordings. The fruit bat has only five pairs of test clips from two bats in one recording and 16 enrolment clips that the within-recording test can shuffle, so neither test is reported. The little penguin's test clips record only a calendar date, across nests, so a pair from one date is not a pair from one recording. The equal error rates are historical values awaiting verification with corrected tie handling; see S1.8.

Table S4. The rook under five divisions, with BirdNET.

Division	n	Enrolment / test clips	Re-ID accuracy (95% interval)	Three-stage interval	Permutation 95th percentile	p	Within-recording p	Standardised paired-clip difference	Equal error rate
Enrolled in 2020, tested in 2021	11	8,226 / 4,417	0.422 (0.251 to 0.561)	0.247 to 0.578	0.137	0.0001	0.0001	0.260	0.449
The same, enrolment capped at 90 clips per bird	11	990 / 4,417	0.375 (0.260 to 0.510)	0.250 to 0.529	0.127	0.0001	0.0001	0.260	0.449
The same clips, divided at random within each bird	11	8,226 / 4,417	0.547 (0.351 to 0.683)	0.343 to 0.694	0.127	0.0001	0.0001	0.102	0.482
Enrolled and tested on different 2020 days	14	5,656 / 4,143	0.395 (0.248 to 0.512)	0.233 to 0.599	0.148	0.0001	0.0001	0.036	0.496
The same calls, enrolled on channel 0, tested on channel 1	14	5,656 / 4,143	0.405 (0.259 to 0.515)	0.238 to 0.607	0.156	0.0001	0.0001	-0.045	0.510

Notes: Channel 0 throughout unless stated. The channel index is not the same physical microphone on every recording day. The equal error rates are historical values awaiting verification with corrected tie handling; see S1.8.

Table S5. Re-ID from calls and from background recordings for all 38 embeddings and controls, each at its layer chosen by the Fisher rule.

Dataset	Embedding	C→C	B→B	C→B	B→C
Chiffchaff, within year	BirdNET v2.4	0.765 (0.681 to 0.846); 0.0001	0.737 (0.599 to 0.813); 0.0001	0.456 (0.344 to 0.625); 0.0001	0.508 (0.281 to 0.651); 0.0001
Chiffchaff, within year	Perch 2.0	0.831 (0.756 to 0.885); 0.0001	0.787 (0.674 to 0.861); 0.0001	0.536 (0.457 to 0.649); 0.0001	0.599 (0.368 to 0.724); 0.0001
Chiffchaff, within year	Perch	0.778 (0.656 to 0.858); 0.0001	0.795 (0.663 to 0.865); 0.0001	0.553 (0.478 to 0.670); 0.0001	0.462 (0.334 to 0.583); 0.0001
Chiffchaff, within year	SurfPerch	0.729 (0.633 to 0.812); 0.0001	0.615 (0.501 to 0.695); 0.0001	0.239 (0.141 to 0.398); 0.0042	0.147 (0.086 to 0.226); 0.1438
Chiffchaff, within year	BirdNET v3 (preview)	0.624 (0.372 to 0.767); 0.0001	0.620 (0.416 to 0.731); 0.0001	0.545 (0.294 to 0.681); 0.0001	0.332 (0.233 to 0.504); 0.0001
Chiffchaff, within year	AVEX sl-BEATs	0.737 (0.588 to 0.818); 0.0001	0.685 (0.526 to 0.765); 0.0001	0.494 (0.341 to 0.609); 0.0001	0.378 (0.286 to 0.496); 0.0003
Chiffchaff, within year	AVEX EfficientNet-B0	0.691 (0.576 to 0.761); 0.0001	0.608 (0.446 to 0.698); 0.0001	0.262 (0.136 to 0.382); 0.0012	0.207 (0.132 to 0.385); 0.0532
Chiffchaff, within year	AvesEcho	0.691 (0.522 to 0.774); 0.0001	0.655 (0.486 to 0.733); 0.0001	0.402 (0.247 to 0.506); 0.0001	0.589 (0.362 to 0.709); 0.0001
Chiffchaff, within year	AudioProtoPNet	0.806 (0.649 to 0.889); 0.0001	0.815 (0.687 to 0.880); 0.0001	0.675 (0.516 to 0.746); 0.0001	0.574 (0.377 to 0.686); 0.0001
Chiffchaff, within year	ConvNeXt (BirdSet)	0.805 (0.688 to 0.866); 0.0001	0.835 (0.693 to 0.902); 0.0001	0.671 (0.498 to 0.751); 0.0001	0.559 (0.379 to 0.665); 0.0001
Chiffchaff, within year	Bird-MAE	0.319 (0.110 to 0.446); 0.0001	0.270 (0.150 to 0.428); 0.0002	0.363 (0.095 to 0.519); 0.0001	0.126 (0.025 to 0.327); 0.2143
Chiffchaff, within year	ProtoCLR	0.325 (0.211 to 0.477); 0.0001	0.285 (0.190 to 0.461); 0.0009	0.185 (0.103 to 0.321); 0.0832	0.111 (0.023 to 0.293); 0.2838
Chiffchaff, within year	RCL FS BSSED	0.256 (0.119 to 0.496); 0.0154	0.283 (0.171 to 0.485); 0.0060	0.234 (0.107 to 0.468); 0.0660	0.134 (0.018 to 0.382); 0.1790
Chiffchaff, within year	BirdAVES	0.420 (0.329 to 0.589); 0.0001	0.363 (0.249 to 0.553); 0.0001	0.311 (0.188 to 0.487); 0.0029	0.172 (0.057 to 0.359); 0.1106
Chiffchaff, within year	AVES	0.397 (0.310 to 0.555); 0.0001	0.345 (0.236 to 0.518); 0.0001	0.357 (0.173 to 0.497); 0.0005	0.159 (0.046 to 0.357); 0.1242
Chiffchaff, within year	NatureBEATs	0.741 (0.579 to 0.816); 0.0001	0.682 (0.530 to 0.756); 0.0001	0.511 (0.330 to 0.612); 0.0001	0.553 (0.399 to 0.629); 0.0001
Chiffchaff, within year	BioLingual	0.637 (0.451 to 0.747); 0.0001	0.583 (0.398 to 0.695); 0.0001	0.200 (0.078 to 0.453); 0.0766	0.254 (0.130 to 0.485); 0.0597
Chiffchaff, within year	BEATs	0.558 (0.343 to 0.672); 0.0001	0.617 (0.435 to 0.713); 0.0001	0.494 (0.243 to 0.635); 0.0001	0.263 (0.134 to 0.493); 0.0072
Chiffchaff, within year	AudioMAE	0.184 (0.124 to 0.280); 0.0882	0.185 (0.053 to 0.471); 0.0912	0.142 (0.047 to 0.267); 0.1817	0.077 (0.021 to 0.186); 0.5387
Chiffchaff, within year	VGGish	0.458 (0.256 to 0.560); 0.0001	0.417 (0.230 to 0.511); 0.0001	0.209 (0.117 to 0.374); 0.0498	0.399 (0.125 to 0.556); 0.0001
Chiffchaff, within year	ECAPA-TDNN (speaker)	0.416 (0.303 to 0.547); 0.0001	0.498 (0.347 to 0.587); 0.0001	0.198 (0.101 to 0.386); 0.0078	0.309 (0.189 to 0.383); 0.0002
Chiffchaff, within year	ResNet (speaker)	0.523 (0.326 to 0.633); 0.0001	0.455 (0.310 to 0.550); 0.0001	0.228 (0.140 to 0.416); 0.0009	0.414 (0.158 to 0.539); 0.0001
Chiffchaff, within year	x-vector (speaker)	0.517 (0.325 to 0.634); 0.0001	0.536 (0.333 to 0.639); 0.0001	0.177 (0.076 to 0.397); 0.0779	0.294 (0.148 to 0.366); 0.0079
Chiffchaff, within year	wav2vec 2.0 Base	0.331 (0.234 to 0.528); 0.0001	0.323 (0.217 to 0.516); 0.0001	0.271 (0.142 to 0.392); 0.0063	0.185 (0.074 to 0.381); 0.0831
Chiffchaff, within year	wav2vec 2.0 Large (robust)	0.305 (0.208 to 0.444); 0.0004	0.289 (0.187 to 0.445); 0.0008	0.285 (0.140 to 0.442); 0.0030	0.141 (0.032 to 0.340); 0.1609
Chiffchaff, within year	XLS-R 300M	0.321 (0.214 to 0.444); 0.0001	0.285 (0.189 to 0.425); 0.0012	0.301 (0.137 to 0.445); 0.0010	0.160 (0.060 to 0.332); 0.1286
Chiffchaff, within year	wav2vec 2.0 Conformer Large	0.414 (0.306 to 0.577); 0.0001	0.425 (0.299 to 0.561); 0.0001	0.268 (0.157 to 0.492); 0.0046	0.221 (0.137 to 0.362); 0.0619
Chiffchaff, within year	MMS 300M	0.324 (0.225 to 0.473); 0.0001	0.278 (0.184 to 0.428); 0.0005	0.300 (0.148 to 0.439); 0.0004	0.161 (0.055 to 0.344); 0.1289
Chiffchaff, within year	HuBERT Base	0.314 (0.215 to 0.504); 0.0002	0.322 (0.225 to 0.497); 0.0002	0.251 (0.122 to 0.364); 0.0141	0.155 (0.040 to 0.358); 0.1306
Chiffchaff, within year	HuBERT Large	0.316 (0.214 to 0.444); 0.0003	0.283 (0.188 to 0.435); 0.0032	0.311 (0.144 to 0.474); 0.0023	0.135 (0.031 to 0.330); 0.1674
Chiffchaff, within year	data2vec Base (100 h)	0.382 (0.278 to 0.505); 0.0001	0.375 (0.227 to 0.471); 0.0001	0.229 (0.117 to 0.443); 0.0250	0.123 (0.031 to 0.297); 0.2099
Chiffchaff, within year	data2vec Base (960 h)	0.378 (0.270 to 0.494); 0.0001	0.330 (0.208 to 0.440); 0.0001	0.270 (0.168 to 0.461); 0.0058	0.161 (0.060 to 0.333); 0.1227
Chiffchaff, within year	WavLM Base+	0.327 (0.232 to 0.518); 0.0001	0.319 (0.226 to 0.499); 0.0002	0.284 (0.172 to 0.415); 0.0031	0.164 (0.063 to 0.340); 0.1141
Chiffchaff, within year	WavLM Large	0.347 (0.226 to 0.472); 0.0002	0.299 (0.198 to 0.434); 0.0014	0.309 (0.142 to 0.444); 0.0019	0.141 (0.040 to 0.330); 0.1615
Chiffchaff, within year	UniSpeech-SAT Base+	0.317 (0.219 to 0.504); 0.0001	0.304 (0.210 to 0.477); 0.0004	0.265 (0.155 to 0.401); 0.0092	0.193 (0.081 to 0.394); 0.0869
Chiffchaff, within year	XEUS	0.341 (0.236 to 0.467); 0.0001	0.334 (0.216 to 0.473); 0.0002	0.313 (0.181 to 0.459); 0.0018	0.154 (0.056 to 0.327); 0.1395
Chiffchaff, within year	Loudness control	0.094 (0.012 to 0.269); 0.4053	0.118 (0.016 to 0.307); 0.2596	0.073 (0.002 to 0.255); 0.5224	0.116 (0.029 to 0.281); 0.2341
Chiffchaff, within year	Duration control	0.040 (0.000 to 0.146); 0.8848	0.088 (0.000 to 0.304); 0.4116	0.050 (0.000 to 0.191); 0.6576	0.047 (0.000 to 0.161); 0.8036
Little owl, across year	BirdNET v2.4	0.501 (0.371 to 0.634); 0.0001	0.105 (0.027 to 0.203); 0.0860	0.083 (0.009 to 0.198); 0.2263	0.076 (0.004 to 0.205); 0.3198
Little owl, across year	Perch 2.0	0.651 (0.489 to 0.785); 0.0001	0.029 (0.008 to 0.054); 0.9597	0.108 (0.017 to 0.245); 0.0474	0.052 (0.000 to 0.127); 0.6445
Little owl, across year	Perch	0.646 (0.499 to 0.776); 0.0001	0.064 (0.024 to 0.114); 0.4603	0.115 (0.037 to 0.215); 0.0127	0.052 (0.000 to 0.147); 0.6578
Little owl, across year	SurfPerch	0.442 (0.290 to 0.591); 0.0001	0.147 (0.068 to 0.238); 0.0005	0.120 (0.028 to 0.232); 0.0133	0.049 (0.003 to 0.126); 0.6800
Little owl, across year	BirdNET v3 (preview)	0.373 (0.233 to 0.530); 0.0001	0.086 (0.012 to 0.194); 0.1735	0.156 (0.018 to 0.332); 0.0004	0.005 (0.000 to 0.016); 0.9977
Little owl, across year	AVEX sl-BEATs	0.194 (0.081 to 0.329); 0.0005	0.149 (0.031 to 0.307); 0.0122	0.068 (0.000 to 0.210); 0.4289	0.054 (0.000 to 0.159); 0.6706
Little owl, across year	AVEX EfficientNet-B0	0.373 (0.234 to 0.522); 0.0001	0.098 (0.025 to 0.191); 0.0626	0.098 (0.017 to 0.200); 0.0727	0.084 (0.012 to 0.189); 0.2328

Dataset	Embedding	C→C	B→B	C→B	B→C
Little owl, across year	AvesEcho	0.138 (0.028 to 0.264); 0.0439	0.086 (0.023 to 0.173); 0.2243	0.066 (0.000 to 0.202); 0.4412	0.091 (0.000 to 0.224); 0.1555
Little owl, across year	AudioProtoPNet	0.631 (0.469 to 0.781); 0.0001	0.068 (0.027 to 0.126); 0.3611	0.088 (0.011 to 0.188); 0.1376	0.047 (0.000 to 0.147); 0.7284
Little owl, across year	ConvNeXt (BirdSet)	0.624 (0.477 to 0.751); 0.0001	0.061 (0.023 to 0.106); 0.4979	0.068 (0.013 to 0.148); 0.4036	0.074 (0.005 to 0.174); 0.3434
Little owl, across year	Bird-MAE	0.251 (0.102 to 0.407); 0.0003	0.098 (0.010 to 0.221); 0.1489	0.066 (0.000 to 0.207); 0.4540	0.044 (0.000 to 0.145); 0.8725
Little owl, across year	ProtoCLR	0.172 (0.065 to 0.305); 0.0077	0.054 (0.000 to 0.147); 0.5654	0.110 (0.000 to 0.272); 0.0287	0.044 (0.000 to 0.145); 0.8910
Little owl, across year	RCL FS BSED	0.189 (0.068 to 0.328); 0.0015	0.064 (0.004 to 0.169); 0.4495	0.103 (0.000 to 0.254); 0.1423	0.059 (0.000 to 0.149); 0.5083
Little owl, across year	BirdAVES	0.285 (0.158 to 0.430); 0.0001	0.076 (0.012 to 0.184); 0.2973	0.095 (0.000 to 0.248); 0.0941	0.101 (0.000 to 0.244); 0.1078
Little owl, across year	AVES	0.285 (0.162 to 0.410); 0.0001	0.108 (0.021 to 0.214); 0.0989	0.076 (0.000 to 0.203); 0.3371	0.143 (0.021 to 0.299); 0.0110
Little owl, across year	ResNet (speaker)	0.241 (0.134 to 0.357); 0.0002	0.132 (0.025 to 0.272); 0.0295	0.066 (0.000 to 0.207); 0.4569	0.066 (0.000 to 0.151); 0.4489
Little owl, across year	BioLingual	0.133 (0.023 to 0.254); 0.0578	0.064 (0.000 to 0.186); 0.4593	0.064 (0.000 to 0.199); 0.5268	0.044 (0.000 to 0.145); 0.9168
Little owl, across year	BEATs	0.216 (0.088 to 0.369); 0.0001	0.149 (0.034 to 0.307); 0.0177	0.066 (0.000 to 0.207); 0.4734	0.054 (0.000 to 0.174); 0.7038
Little owl, across year	AudioMAE	0.170 (0.066 to 0.288); 0.0087	0.081 (0.000 to 0.214); 0.3146	0.066 (0.000 to 0.207); 0.4807	0.044 (0.000 to 0.145); 0.9251
Little owl, across year	VGGish	0.231 (0.098 to 0.379); 0.0001	0.037 (0.011 to 0.074); 0.8080	0.034 (0.000 to 0.089); 0.8451	0.091 (0.014 to 0.214); 0.1602
Little owl, across year	ECAPA-TDNN (speaker)	0.437 (0.278 to 0.598); 0.0001	0.071 (0.015 to 0.163); 0.3360	0.122 (0.000 to 0.261); 0.0103	0.076 (0.009 to 0.173); 0.3076
Little owl, across year	ResNet (speaker)	0.437 (0.280 to 0.591); 0.0001	0.098 (0.037 to 0.165); 0.0658	0.081 (0.000 to 0.216); 0.2028	0.071 (0.000 to 0.203); 0.3688
Little owl, across year	x-vector (speaker)	0.393 (0.252 to 0.537); 0.0001	0.105 (0.027 to 0.204); 0.0683	0.120 (0.000 to 0.299); 0.0062	0.167 (0.005 to 0.352); 0.0021
Little owl, across year	wav2vec 2.0 Base	0.263 (0.105 to 0.444); 0.0002	0.132 (0.035 to 0.246); 0.0265	0.103 (0.002 to 0.250); 0.1012	0.150 (0.027 to 0.302); 0.0135
Little owl, across year	wav2vec 2.0 Large (robust)	0.270 (0.132 to 0.419); 0.0002	0.090 (0.007 to 0.188); 0.2107	0.110 (0.004 to 0.250); 0.0883	0.098 (0.013 to 0.235); 0.1600
Little owl, across year	XLS-R 300M	0.292 (0.138 to 0.458); 0.0001	0.081 (0.004 to 0.176); 0.2789	0.115 (0.004 to 0.265); 0.0671	0.103 (0.010 to 0.233); 0.1391
Little owl, across year	wav2vec 2.0 Conformer Large	0.349 (0.198 to 0.507); 0.0001	0.117 (0.034 to 0.220); 0.0748	0.115 (0.005 to 0.264); 0.0699	0.081 (0.000 to 0.223); 0.2878
Little owl, across year	MMS 300M	0.290 (0.137 to 0.459); 0.0001	0.090 (0.014 to 0.188); 0.1986	0.120 (0.004 to 0.272); 0.0501	0.115 (0.012 to 0.262); 0.0833
Little owl, across year	HuBERT Base	0.290 (0.151 to 0.440); 0.0001	0.152 (0.041 to 0.287); 0.0133	0.086 (0.005 to 0.222); 0.2233	0.108 (0.010 to 0.253); 0.1171
Little owl, across year	HuBERT Large	0.187 (0.089 to 0.309); 0.0073	0.110 (0.021 to 0.223); 0.1099	0.088 (0.000 to 0.234); 0.1473	0.059 (0.000 to 0.167); 0.5474
Little owl, across year	data2vec Base (100 h)	0.361 (0.214 to 0.521); 0.0001	0.108 (0.031 to 0.201); 0.1049	0.117 (0.005 to 0.266); 0.0587	0.091 (0.000 to 0.231); 0.2006
Little owl, across year	data2vec Base (960 h)	0.378 (0.222 to 0.551); 0.0001	0.098 (0.025 to 0.189); 0.1535	0.117 (0.005 to 0.271); 0.0579	0.079 (0.000 to 0.216); 0.2981
Little owl, across year	WavLM Base+	0.305 (0.162 to 0.455); 0.0001	0.149 (0.044 to 0.278); 0.0137	0.117 (0.000 to 0.280); 0.0558	0.064 (0.000 to 0.201); 0.4642
Little owl, across year	WavLM Large	0.268 (0.131 to 0.414); 0.0001	0.086 (0.013 to 0.176); 0.2450	0.110 (0.000 to 0.264); 0.0891	0.101 (0.005 to 0.247); 0.1644
Little owl, across year	UniSpeech-SAT Base+	0.270 (0.130 to 0.427); 0.0001	0.161 (0.050 to 0.295); 0.0062	0.110 (0.005 to 0.267); 0.0801	0.098 (0.005 to 0.241); 0.1630
Little owl, across year	XEUS	0.297 (0.155 to 0.454); 0.0001	0.112 (0.024 to 0.234); 0.0891	0.117 (0.000 to 0.274); 0.0642	0.088 (0.003 to 0.230); 0.2368
Little owl, across year	Loudness control	0.069 (0.014 to 0.136); 0.4105	0.020 (0.000 to 0.048); 0.9465	0.020 (0.003 to 0.041); 0.9470	0.059 (0.000 to 0.151); 0.5832
Little owl, across year	Duration control	0.115 (0.000 to 0.294); 0.0945	0.054 (0.000 to 0.172); 0.5617	0.068 (0.000 to 0.214); 0.3975	0.044 (0.000 to 0.145); 0.9470
Tree pipit, within year	BirdNET v2.4	0.376 (0.193 to 0.566); 0.0001	0.177 (0.106 to 0.246); 0.0408	0.147 (0.046 to 0.257); 0.1293	0.106 (0.023 to 0.219); 0.4515
Tree pipit, within year	Perch 2.0	0.548 (0.357 to 0.736); 0.0001	0.256 (0.109 to 0.426); 0.0004	0.150 (0.016 to 0.379); 0.0928	0.083 (0.000 to 0.262); 0.7466
Tree pipit, within year	Perch	0.426 (0.225 to 0.628); 0.0001	0.242 (0.111 to 0.393); 0.0004	0.225 (0.053 to 0.436); 0.0008	0.129 (0.027 to 0.284); 0.1858
Tree pipit, within year	SurfPerch	0.455 (0.261 to 0.639); 0.0001	0.280 (0.140 to 0.436); 0.0001	0.188 (0.040 to 0.415); 0.0042	0.165 (0.050 to 0.295); 0.0263
Tree pipit, within year	BirdNET v3 (preview)	0.257 (0.127 to 0.414); 0.0001	0.259 (0.147 to 0.378); 0.0004	0.215 (0.076 to 0.399); 0.0024	0.086 (0.000 to 0.266); 0.7436
Tree pipit, within year	AVEX sl-BEATs	0.162 (0.054 to 0.285); 0.0515	0.119 (0.040 to 0.213); 0.3006	0.099 (0.004 to 0.275); 0.5300	0.145 (0.043 to 0.269); 0.0987
Tree pipit, within year	AVEX EfficientNet-B0	0.281 (0.108 to 0.473); 0.0001	0.198 (0.077 to 0.329); 0.0023	0.154 (0.027 to 0.354); 0.0615	0.191 (0.061 to 0.341); 0.0047
Tree pipit, within year	AvesEcho	0.135 (0.015 to 0.327); 0.1992	0.089 (0.007 to 0.212); 0.5810	0.116 (0.000 to 0.334); 0.3580	0.102 (0.013 to 0.240); 0.4698
Tree pipit, within year	AudioProtoPNet	0.442 (0.234 to 0.655); 0.0001	0.253 (0.129 to 0.409); 0.0005	0.246 (0.089 to 0.441); 0.0004	0.089 (0.000 to 0.269); 0.6622
Tree pipit, within year	ConvNeXt (BirdSet)	0.502 (0.290 to 0.705); 0.0001	0.246 (0.122 to 0.404); 0.0004	0.225 (0.081 to 0.410); 0.0008	0.102 (0.016 to 0.245); 0.5015
Tree pipit, within year	Bird-MAE	0.069 (0.019 to 0.131); 0.8566	0.109 (0.023 to 0.233); 0.4521	0.085 (0.000 to 0.207); 0.6953	0.063 (0.007 to 0.130); 0.9126
Tree pipit, within year	ProtoCLR	0.152 (0.026 to 0.309); 0.0891	0.137 (0.033 to 0.271); 0.1454	0.143 (0.013 to 0.365); 0.0735	0.102 (0.000 to 0.272); 0.4982
Tree pipit, within year	RCL FS BSED	0.172 (0.053 to 0.329); 0.1120	0.092 (0.013 to 0.216); 0.5313	0.212 (0.019 to 0.420); 0.0282	0.096 (0.021 to 0.194); 0.5178
Tree pipit, within year	BirdAVES	0.300 (0.104 to 0.503); 0.0003	0.218 (0.076 to 0.409); 0.0173	0.218 (0.033 to 0.446); 0.0095	0.228 (0.048 to 0.445); 0.0043
Tree pipit, within year	AVES	0.228 (0.064 to 0.424); 0.0065	0.212 (0.083 to 0.385); 0.0210	0.188 (0.017 to 0.426); 0.0401	0.277 (0.069 to 0.503); 0.0006

Dataset	Embedding	C→C	B→B	C→B	B→C
Tree pipit, within year	NatureBEATs	0.182 (0.083 to 0.283); 0.0242	0.113 (0.039 to 0.198); 0.3616	0.109 (0.011 to 0.255); 0.4074	0.158 (0.050 to 0.280); 0.0571
Tree pipit, within year	BioLingual	0.112 (0.014 to 0.227); 0.3898	0.109 (0.037 to 0.199); 0.4210	0.099 (0.007 to 0.257); 0.5148	0.086 (0.010 to 0.210); 0.6694
Tree pipit, within year	BEATs	0.211 (0.095 to 0.329); 0.0052	0.140 (0.060 to 0.229); 0.1677	0.126 (0.048 to 0.224); 0.2568	0.129 (0.021 to 0.259); 0.2320
Tree pipit, within year	AudioMAE	0.125 (0.024 to 0.240); 0.2976	0.130 (0.000 to 0.292); 0.2772	0.167 (0.000 to 0.396); 0.0485	0.112 (0.000 to 0.288); 0.3141
Tree pipit, within year	VGGish	0.116 (0.028 to 0.218); 0.3378	0.164 (0.060 to 0.292); 0.0800	0.068 (0.007 to 0.152); 0.8217	0.033 (0.000 to 0.072); 0.9801
Tree pipit, within year	ECAPA-TDNN (speaker)	0.356 (0.166 to 0.565); 0.0001	0.222 (0.103 to 0.350); 0.0010	0.171 (0.018 to 0.400); 0.0285	0.195 (0.050 to 0.375); 0.0080
Tree pipit, within year	ResNet (speaker)	0.264 (0.127 to 0.426); 0.0001	0.208 (0.110 to 0.320); 0.0011	0.160 (0.014 to 0.368); 0.0310	0.162 (0.048 to 0.306); 0.0360
Tree pipit, within year	x-vector (speaker)	0.304 (0.117 to 0.512); 0.0001	0.184 (0.058 to 0.334); 0.0348	0.130 (0.007 to 0.338); 0.2151	0.155 (0.034 to 0.301); 0.0796
Tree pipit, within year	wav2vec 2.0 Base	0.231 (0.060 to 0.419); 0.0053	0.195 (0.062 to 0.380); 0.0290	0.082 (0.024 to 0.150); 0.6838	0.155 (0.022 to 0.343); 0.1023
Tree pipit, within year	wav2vec 2.0 Large (robust)	0.168 (0.004 to 0.388); 0.1049	0.137 (0.023 to 0.328); 0.2251	0.109 (0.015 to 0.222); 0.4228	0.205 (0.010 to 0.444); 0.0252
Tree pipit, within year	XLS-R 300M	0.135 (0.014 to 0.327); 0.2424	0.150 (0.022 to 0.343); 0.1641	0.167 (0.004 to 0.411); 0.0754	0.201 (0.025 to 0.424); 0.0307
Tree pipit, within year	wav2vec 2.0 Conformer Large	0.149 (0.020 to 0.342); 0.1782	0.174 (0.051 to 0.366); 0.0837	0.191 (0.010 to 0.418); 0.0270	0.231 (0.022 to 0.461); 0.0099
Tree pipit, within year	MMS 300M	0.125 (0.016 to 0.316); 0.2906	0.143 (0.020 to 0.336); 0.1920	0.150 (0.007 to 0.362); 0.1362	0.201 (0.029 to 0.414); 0.0292
Tree pipit, within year	HuBERT Base	0.191 (0.035 to 0.384); 0.0429	0.150 (0.040 to 0.321); 0.1522	0.143 (0.033 to 0.280); 0.1515	0.198 (0.017 to 0.413); 0.0304
Tree pipit, within year	HuBERT Large	0.188 (0.016 to 0.419); 0.0624	0.150 (0.032 to 0.342); 0.1625	0.078 (0.011 to 0.178); 0.7111	0.221 (0.019 to 0.480); 0.0113
Tree pipit, within year	data2vec Base (100 h)	0.182 (0.038 to 0.374); 0.0644	0.191 (0.059 to 0.386); 0.0463	0.116 (0.028 to 0.218); 0.3642	0.224 (0.039 to 0.440); 0.0102
Tree pipit, within year	data2vec Base (960 h)	0.172 (0.041 to 0.356); 0.0870	0.177 (0.049 to 0.372); 0.0690	0.102 (0.028 to 0.188); 0.4934	0.224 (0.047 to 0.439); 0.0094
Tree pipit, within year	WavLM Base+	0.158 (0.025 to 0.350); 0.1245	0.184 (0.053 to 0.360); 0.0512	0.167 (0.031 to 0.351); 0.0637	0.198 (0.010 to 0.426); 0.0310
Tree pipit, within year	WavLM Large	0.165 (0.009 to 0.374); 0.1156	0.164 (0.044 to 0.353); 0.1106	0.085 (0.017 to 0.177); 0.6563	0.221 (0.010 to 0.486); 0.0113
Tree pipit, within year	UniSpeech-SAT Base+	0.145 (0.018 to 0.325); 0.1816	0.167 (0.054 to 0.322); 0.0885	0.113 (0.027 to 0.233); 0.3851	0.162 (0.013 to 0.372); 0.1059
Tree pipit, within year	XEUS	0.139 (0.010 to 0.329); 0.2213	0.177 (0.048 to 0.372); 0.0718	0.160 (0.023 to 0.332); 0.0948	0.218 (0.020 to 0.449); 0.0157
Tree pipit, within year	Loudness control	0.228 (0.032 to 0.451); 0.0160	0.246 (0.052 to 0.475); 0.0063	0.215 (0.026 to 0.443); 0.0190	0.185 (0.045 to 0.378); 0.0603
Tree pipit, within year	Duration control	0.145 (0.000 to 0.336); 0.1041	0.126 (0.000 to 0.310); 0.2911	0.126 (0.000 to 0.355); 0.2220	0.116 (0.000 to 0.328); 0.2804
Tree pipit, across year	BirdNET v2.4	0.214 (0.116 to 0.334); 0.0022	0.065 (0.029 to 0.102); 0.8142	0.075 (0.015 to 0.145); 0.7089	0.089 (0.020 to 0.184); 0.6125
Tree pipit, across year	Perch 2.0	0.447 (0.293 to 0.624); 0.0001	0.163 (0.085 to 0.247); 0.0389	0.105 (0.020 to 0.275); 0.4218	0.093 (0.000 to 0.302); 0.5698
Tree pipit, across year	Perch	0.351 (0.219 to 0.518); 0.0001	0.078 (0.022 to 0.142); 0.7554	0.075 (0.033 to 0.128); 0.7816	0.086 (0.004 to 0.224); 0.6723
Tree pipit, across year	SurfPerch	0.243 (0.151 to 0.358); 0.0001	0.072 (0.039 to 0.105); 0.8485	0.072 (0.004 to 0.171); 0.8377	0.083 (0.031 to 0.139); 0.6997
Tree pipit, across year	BirdNET v3 (preview)	0.160 (0.095 to 0.227); 0.0388	0.150 (0.066 to 0.235); 0.0915	0.160 (0.036 to 0.294); 0.0677	0.093 (0.004 to 0.278); 0.5612
Tree pipit, across year	AVEX sl-BEATs	0.096 (0.060 to 0.130); 0.5067	0.039 (0.009 to 0.081); 0.9857	0.049 (0.007 to 0.115); 0.9577	0.042 (0.006 to 0.094); 0.9651
Tree pipit, across year	AVEX EfficientNet-B0	0.211 (0.112 to 0.323); 0.0003	0.131 (0.067 to 0.207); 0.1525	0.101 (0.026 to 0.218); 0.4591	0.093 (0.029 to 0.172); 0.5766
Tree pipit, across year	AvesEcho	0.019 (0.000 to 0.039); 0.9988	0.033 (0.004 to 0.076); 0.9804	0.078 (0.000 to 0.244); 0.7513	0.099 (0.000 to 0.279); 0.4976
Tree pipit, across year	AudioProtoPNet	0.201 (0.103 to 0.315); 0.0023	0.095 (0.039 to 0.159); 0.5396	0.098 (0.019 to 0.210); 0.4980	0.086 (0.000 to 0.274); 0.6571
Tree pipit, across year	ConvNeXt (BirdSet)	0.288 (0.141 to 0.464); 0.0001	0.098 (0.037 to 0.155); 0.4963	0.072 (0.007 to 0.147); 0.7903	0.102 (0.000 to 0.275); 0.4558
Tree pipit, across year	Bird-MAE	0.032 (0.000 to 0.096); 0.9861	0.059 (0.000 to 0.169); 0.9225	0.065 (0.000 to 0.195); 0.8727	0.010 (0.000 to 0.020); 1.0000
Tree pipit, across year	ProtoCLR	0.083 (0.029 to 0.159); 0.6308	0.075 (0.025 to 0.139); 0.7971	0.105 (0.018 to 0.256); 0.4003	0.089 (0.008 to 0.246); 0.5785
Tree pipit, across year	RCL FS BSIED	0.045 (0.016 to 0.072); 0.8746	0.013 (0.000 to 0.038); 0.9792	0.003 (0.000 to 0.011); 0.9956	0.099 (0.000 to 0.269); 0.4667
Tree pipit, across year	BirdAVES	0.083 (0.041 to 0.116); 0.6309	0.029 (0.000 to 0.078); 0.9714	0.023 (0.000 to 0.060); 0.9834	0.115 (0.020 to 0.198); 0.3404
Tree pipit, across year	AVES	0.038 (0.016 to 0.060); 0.9660	0.013 (0.000 to 0.035); 0.9982	0.042 (0.000 to 0.139); 0.9059	0.102 (0.014 to 0.192); 0.4460
Tree pipit, across year	NatureBEATs	0.070 (0.036 to 0.100); 0.7896	0.029 (0.005 to 0.059); 0.9965	0.049 (0.007 to 0.114); 0.9513	0.038 (0.006 to 0.081); 0.9745
Tree pipit, across year	BioLingual	0.058 (0.000 to 0.135); 0.8599	0.042 (0.013 to 0.088); 0.9608	0.078 (0.004 to 0.230); 0.7020	0.070 (0.000 to 0.224); 0.7916
Tree pipit, across year	BEATs	0.086 (0.045 to 0.130); 0.5970	0.072 (0.012 to 0.152); 0.7487	0.046 (0.016 to 0.083); 0.9355	0.054 (0.000 to 0.123); 0.8877
Tree pipit, across year	AudioMAE	0.045 (0.000 to 0.140); 0.9118	0.039 (0.000 to 0.098); 0.9699	0.052 (0.000 to 0.141); 0.9422	0.067 (0.000 to 0.219); 0.8696
Tree pipit, across year	VGGish	0.070 (0.034 to 0.110); 0.7720	0.114 (0.040 to 0.225); 0.3392	0.039 (0.010 to 0.077); 0.9427	0.112 (0.036 to 0.201); 0.3590
Tree pipit, across year	ECAPA-TDNN (speaker)	0.214 (0.114 to 0.344); 0.0006	0.118 (0.072 to 0.176); 0.2628	0.101 (0.008 to 0.262); 0.4581	0.147 (0.060 to 0.235); 0.0635
Tree pipit, across year	ResNet (speaker)	0.137 (0.066 to 0.231); 0.0940	0.072 (0.030 to 0.118); 0.8537	0.085 (0.010 to 0.220); 0.6758	0.153 (0.054 to 0.241); 0.0397
Tree pipit, across year	x-vector (speaker)	0.195 (0.080 to 0.354); 0.0064	0.069 (0.031 to 0.124); 0.8089	0.105 (0.013 to 0.275); 0.4229	0.089 (0.016 to 0.193); 0.6015

Dataset	Embedding	C→C	B→B	C→B	B→C
Tree pipit, across year	wav2vec 2.0 Base	0.064 (0.000 to 0.170); 0.7734	0.020 (0.004 to 0.037); 0.9893	0.069 (0.016 to 0.143); 0.7453	0.147 (0.000 to 0.389); 0.1706
Tree pipit, across year	wav2vec 2.0 Large (robust)	0.032 (0.006 to 0.062); 0.9397	0.026 (0.000 to 0.054); 0.9487	0.092 (0.000 to 0.216); 0.5296	0.099 (0.000 to 0.208); 0.4728
Tree pipit, across year	XLS-R 300M	0.022 (0.000 to 0.053); 0.9742	0.020 (0.000 to 0.038); 0.9720	0.082 (0.000 to 0.193); 0.6110	0.086 (0.000 to 0.167); 0.5659
Tree pipit, across year	wav2vec 2.0 Conformer Large	0.026 (0.000 to 0.056); 0.9657	0.029 (0.000 to 0.071); 0.9337	0.033 (0.000 to 0.098); 0.9308	0.115 (0.000 to 0.260); 0.3624
Tree pipit, across year	MMS 300M	0.019 (0.000 to 0.044); 0.9828	0.023 (0.000 to 0.046); 0.9656	0.065 (0.000 to 0.159); 0.7542	0.089 (0.000 to 0.173); 0.5453
Tree pipit, across year	HuBERT Base	0.013 (0.000 to 0.027); 0.9964	0.029 (0.000 to 0.061); 0.9587	0.000 (0.000 to 0.000); 1.0000	0.134 (0.000 to 0.355); 0.2332
Tree pipit, across year	HuBERT Large	0.032 (0.000 to 0.069); 0.9326	0.033 (0.007 to 0.056); 0.9168	0.078 (0.000 to 0.207); 0.6415	0.102 (0.000 to 0.213); 0.4550
Tree pipit, across year	data2vec Base (100 h)	0.029 (0.004 to 0.058); 0.9654	0.010 (0.000 to 0.027); 0.9967	0.029 (0.000 to 0.095); 0.9532	0.093 (0.000 to 0.205); 0.5158
Tree pipit, across year	data2vec Base (960 h)	0.029 (0.009 to 0.051); 0.9674	0.010 (0.000 to 0.024); 0.9965	0.033 (0.000 to 0.105); 0.9414	0.073 (0.000 to 0.150); 0.6765
Tree pipit, across year	WavLM Base+	0.000 (0.000 to 0.000); 1.0000	0.026 (0.004 to 0.055); 0.9676	0.000 (0.000 to 0.000); 1.0000	0.118 (0.000 to 0.318); 0.3259
Tree pipit, across year	WavLM Large	0.026 (0.000 to 0.053); 0.9647	0.020 (0.004 to 0.036); 0.9693	0.059 (0.000 to 0.166); 0.7961	0.099 (0.011 to 0.208); 0.4818
Tree pipit, across year	UniSpeech-SAT Base+	0.016 (0.003 to 0.031); 0.9935	0.026 (0.004 to 0.055); 0.9682	0.007 (0.000 to 0.021); 0.9989	0.131 (0.000 to 0.354); 0.2512
Tree pipit, across year	XEUS	0.010 (0.000 to 0.019); 0.9981	0.016 (0.000 to 0.037); 0.9837	0.007 (0.000 to 0.021); 0.9964	0.115 (0.009 to 0.254); 0.3542
Tree pipit, across year	Loudness control	0.073 (0.019 to 0.139); 0.7154	0.036 (0.000 to 0.086); 0.9528	0.075 (0.014 to 0.147); 0.6927	0.121 (0.000 to 0.338); 0.2985
Tree pipit, across year	Duration control	0.083 (0.000 to 0.242); 0.6719	0.072 (0.000 to 0.188); 0.8394	0.092 (0.000 to 0.253); 0.5670	0.073 (0.000 to 0.247); 0.9383
Chiffchaff, across year	BirdNET v2.4	0.160 (0.035 to 0.333); 0.1327	0.132 (0.036 to 0.277); 0.2439	0.112 (0.005 to 0.318); 0.4019	0.110 (0.019 to 0.240); 0.4114
Chiffchaff, across year	Perch 2.0	0.160 (0.043 to 0.293); 0.0916	0.127 (0.036 to 0.262); 0.2586	0.147 (0.017 to 0.337); 0.1542	0.125 (0.000 to 0.262); 0.2596
Chiffchaff, across year	Perch	0.150 (0.046 to 0.279); 0.1178	0.137 (0.043 to 0.281); 0.1783	0.173 (0.036 to 0.362); 0.0552	0.075 (0.011 to 0.156); 0.7837
Chiffchaff, across year	SurfPerch	0.255 (0.104 to 0.428); 0.0009	0.193 (0.050 to 0.387); 0.0152	0.188 (0.036 to 0.380); 0.0163	0.190 (0.006 to 0.445); 0.0122
Chiffchaff, across year	BirdNET v3 (preview)	0.085 (0.033 to 0.150); 0.6650	0.071 (0.026 to 0.132); 0.7682	0.096 (0.000 to 0.214); 0.5231	0.005 (0.000 to 0.016); 1.0000
Chiffchaff, across year	AVEX sl-BEATs	0.210 (0.073 to 0.364); 0.0133	0.173 (0.040 to 0.342); 0.0602	0.173 (0.000 to 0.360); 0.0580	0.130 (0.017 to 0.291); 0.2144
Chiffchaff, across year	AVEX EfficientNet-B0	0.120 (0.035 to 0.236); 0.3143	0.147 (0.032 to 0.303); 0.1095	0.137 (0.031 to 0.292); 0.1738	0.085 (0.006 to 0.231); 0.6945
Chiffchaff, across year	AvesEcho	0.180 (0.015 to 0.383); 0.0780	0.086 (0.000 to 0.274); 0.5860	0.188 (0.000 to 0.391); 0.0755	0.120 (0.000 to 0.347); 0.3371
Chiffchaff, across year	AudioProtoPNet	0.115 (0.047 to 0.200); 0.3617	0.107 (0.024 to 0.236); 0.4211	0.142 (0.027 to 0.337); 0.1693	0.045 (0.000 to 0.114); 0.9732
Chiffchaff, across year	ConvNeXt (BirdSet)	0.135 (0.035 to 0.266); 0.2006	0.137 (0.027 to 0.296); 0.1908	0.137 (0.019 to 0.348); 0.2069	0.105 (0.009 to 0.236); 0.4655
Chiffchaff, across year	Bird-MAE	0.080 (0.014 to 0.178); 0.7886	0.081 (0.029 to 0.144); 0.6905	0.091 (0.025 to 0.180); 0.6050	0.120 (0.000 to 0.267); 0.2755
Chiffchaff, across year	ProtoCLR	0.195 (0.041 to 0.385); 0.0121	0.127 (0.021 to 0.332); 0.2535	0.117 (0.025 to 0.242); 0.3413	0.105 (0.034 to 0.190); 0.4577
Chiffchaff, across year	RCL FS BSIED	0.150 (0.016 to 0.318); 0.2351	0.107 (0.000 to 0.274); 0.4498	0.142 (0.031 to 0.268); 0.2606	0.170 (0.000 to 0.416); 0.1482
Chiffchaff, across year	BirdAVES	0.190 (0.021 to 0.399); 0.0691	0.107 (0.000 to 0.313); 0.4261	0.188 (0.034 to 0.369); 0.0671	0.105 (0.000 to 0.330); 0.4518
Chiffchaff, across year	AVES	0.205 (0.000 to 0.414); 0.0508	0.102 (0.000 to 0.312); 0.4544	0.203 (0.023 to 0.402); 0.0426	0.155 (0.000 to 0.397); 0.1568
Chiffchaff, across year	NatureBEATs	0.160 (0.032 to 0.304); 0.1046	0.107 (0.015 to 0.271); 0.4273	0.188 (0.000 to 0.384); 0.0373	0.105 (0.010 to 0.269); 0.4491
Chiffchaff, across year	BioLingual	0.160 (0.045 to 0.290); 0.0988	0.173 (0.064 to 0.316); 0.0415	0.137 (0.029 to 0.314); 0.1900	0.120 (0.000 to 0.302); 0.2969
Chiffchaff, across year	BEATs	0.155 (0.042 to 0.289); 0.1273	0.117 (0.019 to 0.271); 0.3498	0.147 (0.023 to 0.274); 0.1735	0.150 (0.000 to 0.368); 0.1404
Chiffchaff, across year	AudioMAE	0.150 (0.045 to 0.281); 0.1418	0.173 (0.021 to 0.374); 0.0523	0.157 (0.000 to 0.360); 0.0914	0.155 (0.000 to 0.379); 0.0916
Chiffchaff, across year	VGGish	0.095 (0.000 to 0.278); 0.5181	0.066 (0.000 to 0.210); 0.7368	0.046 (0.000 to 0.117); 0.8757	0.155 (0.000 to 0.369); 0.1480
Chiffchaff, across year	ECAPA-TDNN (speaker)	0.150 (0.056 to 0.280); 0.1265	0.147 (0.012 to 0.306); 0.1281	0.147 (0.027 to 0.354); 0.1550	0.170 (0.012 to 0.384); 0.0415
Chiffchaff, across year	ResNet (speaker)	0.145 (0.046 to 0.290); 0.1644	0.112 (0.022 to 0.226); 0.3711	0.086 (0.022 to 0.168); 0.6370	0.130 (0.022 to 0.273); 0.2330
Chiffchaff, across year	x-vector (speaker)	0.140 (0.032 to 0.318); 0.2328	0.168 (0.000 to 0.389); 0.1158	0.122 (0.011 to 0.319); 0.3304	0.110 (0.016 to 0.229); 0.4192
Chiffchaff, across year	wav2vec 2.0 Base	0.165 (0.000 to 0.389); 0.1430	0.132 (0.016 to 0.322); 0.2649	0.223 (0.000 to 0.478); 0.0144	0.175 (0.000 to 0.395); 0.0945
Chiffchaff, across year	wav2vec 2.0 Large (robust)	0.210 (0.016 to 0.433); 0.0486	0.102 (0.000 to 0.283); 0.4563	0.178 (0.039 to 0.322); 0.0880	0.170 (0.000 to 0.418); 0.1285
Chiffchaff, across year	XLS-R 300M	0.235 (0.056 to 0.441); 0.0213	0.117 (0.000 to 0.321); 0.3510	0.193 (0.033 to 0.359); 0.0548	0.175 (0.000 to 0.429); 0.1096
Chiffchaff, across year	wav2vec 2.0 Conformer Large	0.205 (0.000 to 0.389); 0.0497	0.107 (0.000 to 0.286); 0.4178	0.183 (0.023 to 0.353); 0.0690	0.160 (0.000 to 0.394); 0.1511
Chiffchaff, across year	MMS 300M	0.220 (0.032 to 0.436); 0.0347	0.107 (0.000 to 0.295); 0.4213	0.198 (0.029 to 0.374); 0.0448	0.165 (0.000 to 0.410); 0.1395
Chiffchaff, across year	HuBERT Base	0.125 (0.000 to 0.348); 0.3242	0.112 (0.000 to 0.313); 0.3894	0.193 (0.017 to 0.440); 0.0392	0.155 (0.000 to 0.393); 0.1620
Chiffchaff, across year	HuBERT Large	0.215 (0.000 to 0.445); 0.0422	0.127 (0.000 to 0.339); 0.2895	0.208 (0.043 to 0.380); 0.0379	0.180 (0.000 to 0.438); 0.0956

Dataset	Embedding	C→C	B→B	C→B	B→C
Chiffchaff, across year	data2vec Base (100 h)	0.180 (0.000 to 0.398); 0.0995	0.107 (0.000 to 0.308); 0.4238	0.152 (0.000 to 0.321); 0.1637	0.175 (0.000 to 0.414); 0.1020
Chiffchaff, across year	data2vec Base (960 h)	0.180 (0.000 to 0.394); 0.1000	0.096 (0.000 to 0.286); 0.4923	0.152 (0.000 to 0.330); 0.1596	0.155 (0.000 to 0.379); 0.1647
Chiffchaff, across year	WavLM Base+	0.125 (0.000 to 0.348); 0.3240	0.102 (0.000 to 0.303); 0.4601	0.218 (0.034 to 0.451); 0.0193	0.150 (0.016 to 0.367); 0.1906
Chiffchaff, across year	WavLM Large	0.200 (0.024 to 0.386); 0.0611	0.107 (0.000 to 0.291); 0.4201	0.168 (0.030 to 0.315); 0.1099	0.185 (0.000 to 0.451); 0.0810
Chiffchaff, across year	UniSpeech-SAT Base+	0.130 (0.000 to 0.349); 0.2931	0.102 (0.000 to 0.293); 0.4630	0.234 (0.052 to 0.467); 0.0093	0.150 (0.000 to 0.389); 0.1823
Chiffchaff, across year	XEUS	0.205 (0.016 to 0.407); 0.0551	0.107 (0.006 to 0.292); 0.4219	0.218 (0.016 to 0.412); 0.0258	0.135 (0.000 to 0.339); 0.2592
Chiffchaff, across year	Loudness control	0.025 (0.000 to 0.058); 0.9672	0.036 (0.000 to 0.077); 0.9682	0.086 (0.000 to 0.238); 0.6503	0.030 (0.000 to 0.094); 0.9676
Chiffchaff, across year	Duration control	0.105 (0.000 to 0.297); 0.4988	0.117 (0.000 to 0.324); 0.3154	0.041 (0.000 to 0.138); 0.9426	0.085 (0.000 to 0.240); 0.6863

Notes: C→B: classifier fitted on calls and tested on backgrounds; the other columns likewise. Each cell: re-ID accuracy (95% interval over individuals, 10,000 draws); permutation p (9,999 shuffles), unadjusted. Entries whose B→B p is below 0.05 after Holm's correction over the 38 entries of a dataset: Chiffchaff, within year 35; Little owl, across year 1; Tree pipit, within year 8; Tree pipit, across year 0; Chiffchaff, across year 0 (unadjusted: 35; 8; 16; 1; 2; about 2 of 38 expected by chance).

Table S6. Effect of adding a background recording to test clips, for all 38 embeddings and controls: own-donor minus other-donor balanced accuracy.

Dataset	Embedding	Background +10 dB	Background 0 dB	Background -10 dB
Chiffchaff, within year	BirdNET v2.4	0.325 (0.194 to 0.465)	0.219 (0.089 to 0.362)	0.095 (0.035 to 0.165)
Chiffchaff, within year	Perch 2.0	0.450 (0.300 to 0.591)	0.337 (0.221 to 0.461)	0.124 (0.054 to 0.204)
Chiffchaff, within year	Perch	0.458 (0.305 to 0.613)	0.329 (0.183 to 0.492)	0.151 (0.060 to 0.268)
Chiffchaff, within year	SurfPerch	0.215 (0.103 to 0.343)	0.164 (0.078 to 0.269)	0.067 (0.028 to 0.106)
Chiffchaff, within year	BirdNET v3 (preview)	0.311 (0.188 to 0.457)	0.250 (0.144 to 0.378)	0.107 (0.039 to 0.176)
Chiffchaff, within year	AVEX sl-BEATs	0.215 (0.090 to 0.351)	0.184 (0.080 to 0.303)	0.112 (0.066 to 0.161)
Chiffchaff, within year	AVEX EfficientNet-B0	0.243 (0.124 to 0.371)	0.225 (0.129 to 0.330)	0.098 (0.049 to 0.148)
Chiffchaff, within year	AvesEcho	0.105 (0.031 to 0.185)	0.053 (-0.014 to 0.123)	0.072 (0.019 to 0.127)
Chiffchaff, within year	AudioProtoPNet	0.429 (0.278 to 0.582)	0.348 (0.208 to 0.498)	0.215 (0.104 to 0.340)
Chiffchaff, within year	ConvNeXt (BirdSet)	0.417 (0.250 to 0.585)	0.314 (0.181 to 0.451)	0.200 (0.120 to 0.291)
Chiffchaff, within year	Bird-MAE	0.082 (0.0003 to 0.192)	0.073 (-0.001 to 0.172)	0.031 (-0.002 to 0.073)
Chiffchaff, within year	ProtoCLR	0.118 (0.053 to 0.192)	0.130 (0.041 to 0.250)	0.080 (0.028 to 0.147)
Chiffchaff, within year	RCL FS BSSED	0.018 (-0.014 to 0.064)	0.022 (-0.002 to 0.067)	0.029 (0.000 to 0.084)
Chiffchaff, within year	BirdAVES	0.105 (0.003 to 0.221)	0.080 (0.005 to 0.161)	0.075 (0.027 to 0.138)
Chiffchaff, within year	AVES	0.058 (-0.016 to 0.142)	0.050 (0.009 to 0.108)	0.038 (-0.009 to 0.089)
Chiffchaff, within year	NatureBEATs	0.113 (0.033 to 0.214)	0.109 (0.029 to 0.217)	0.107 (0.040 to 0.188)
Chiffchaff, within year	BioLingual	0.250 (0.124 to 0.398)	0.157 (0.044 to 0.290)	0.074 (0.025 to 0.118)
Chiffchaff, within year	BEATs	0.229 (0.095 to 0.371)	0.210 (0.075 to 0.346)	0.155 (0.073 to 0.246)
Chiffchaff, within year	AudioMAE	-0.013 (-0.081 to 0.031)	0.018 (-0.003 to 0.048)	0.001 (-0.007 to 0.009)
Chiffchaff, within year	VGGish	0.066 (-0.028 to 0.166)	0.071 (0.012 to 0.139)	0.046 (-0.020 to 0.106)
Chiffchaff, within year	ECAPA-TDNN (speaker)	0.178 (0.074 to 0.283)	0.172 (0.073 to 0.289)	0.085 (0.011 to 0.172)
Chiffchaff, within year	ResNet (speaker)	0.222 (0.117 to 0.341)	0.187 (0.093 to 0.294)	0.110 (0.047 to 0.179)
Chiffchaff, within year	x-vector (speaker)	0.198 (0.081 to 0.331)	0.195 (0.083 to 0.317)	0.092 (0.041 to 0.148)
Chiffchaff, within year	wav2vec 2.0 Base	0.140 (0.025 to 0.265)	0.144 (0.041 to 0.259)	0.046 (-0.018 to 0.122)
Chiffchaff, within year	wav2vec 2.0 Large (robust)	0.088 (-0.008 to 0.196)	0.061 (0.002 to 0.145)	0.044 (-0.009 to 0.106)
Chiffchaff, within year	XLS-R 300M	0.049 (-0.071 to 0.166)	0.087 (0.020 to 0.176)	0.052 (0.004 to 0.108)
Chiffchaff, within year	wav2vec 2.0 Conformer Large	0.189 (0.032 to 0.352)	0.224 (0.100 to 0.363)	0.096 (0.025 to 0.169)
Chiffchaff, within year	MMS 300M	0.066 (-0.014 to 0.159)	0.085 (0.019 to 0.176)	0.042 (-0.008 to 0.103)
Chiffchaff, within year	HuBERT Base	0.098 (-0.020 to 0.224)	0.133 (0.023 to 0.262)	0.053 (-0.019 to 0.149)
Chiffchaff, within year	HuBERT Large	0.086 (-0.021 to 0.221)	0.093 (-0.002 to 0.212)	0.036 (-0.019 to 0.111)
Chiffchaff, within year	data2vec Base (100 h)	0.125 (0.037 to 0.218)	0.113 (0.015 to 0.213)	0.059 (-0.014 to 0.134)
Chiffchaff, within year	data2vec Base (960 h)	0.150 (0.049 to 0.266)	0.140 (0.063 to 0.231)	0.057 (-0.017 to 0.136)
Chiffchaff, within year	WavLM Base+	0.142 (0.012 to 0.274)	0.155 (0.025 to 0.304)	0.091 (0.010 to 0.194)
Chiffchaff, within year	WavLM Large	0.078 (-0.040 to 0.208)	0.090 (0.004 to 0.197)	0.049 (-0.007 to 0.118)
Chiffchaff, within year	UniSpeech-SAT Base+	0.141 (0.026 to 0.253)	0.115 (0.007 to 0.244)	0.059 (-0.008 to 0.146)
Chiffchaff, within year	XEUS	0.129 (0.039 to 0.228)	0.122 (0.036 to 0.217)	0.033 (-0.021 to 0.093)
Chiffchaff, within year	Loudness control	0.004 (0.000 to 0.013)	-0.002 (-0.007 to 0.000)	0.000 (0.000 to 0.000)
Chiffchaff, within year	Duration control	0.000 (0.000 to 0.000)	0.000 (0.000 to 0.000)	0.000 (0.000 to 0.000)
Little owl, across year	BirdNET v2.4	-0.100 (-0.268 to 0.051)	-0.071 (-0.179 to 0.026)	0.000 (-0.033 to 0.035)
Little owl, across year	Perch 2.0	-0.069 (-0.197 to 0.047)	-0.033 (-0.076 to 0.001)	0.003 (-0.034 to 0.036)
Little owl, across year	Perch	-0.052 (-0.133 to 0.020)	-0.038 (-0.085 to 0.010)	-0.007 (-0.035 to 0.022)
Little owl, across year	SurfPerch	-0.018 (-0.095 to 0.055)	-0.002 (-0.084 to 0.079)	-0.016 (-0.071 to 0.037)
Little owl, across year	BirdNET v3 (preview)	0.021 (-0.073 to 0.116)	-0.040 (-0.107 to 0.024)	0.022 (-0.015 to 0.064)
Little owl, across year	AVEX sl-BEATs	0.026 (-0.006 to 0.083)	0.021 (0.000 to 0.062)	0.012 (-0.004 to 0.032)
Little owl, across year	AVEX EfficientNet-B0	-0.074 (-0.167 to 0.017)	-0.025 (-0.084 to 0.035)	-0.023 (-0.047 to 0.001)

Dataset	Embedding	Background +10 dB	Background 0 dB	Background -10 dB
Little owl, across year	AvesEcho	-0.002 (-0.013 to 0.007)	-0.002 (-0.013 to 0.007)	-0.006 (-0.019 to 0.000)
Little owl, across year	AudioProtoPNet	-0.103 (-0.211 to -0.003)	-0.078 (-0.154 to -0.011)	-0.047 (-0.092 to -0.011)
Little owl, across year	ConvNeXt (BirdSet)	-0.041 (-0.119 to 0.044)	-0.044 (-0.112 to 0.021)	-0.019 (-0.062 to 0.018)
Little owl, across year	Bird-MAE	-0.047 (-0.119 to 0.008)	-0.037 (-0.086 to 0.007)	-0.010 (-0.027 to 0.006)
Little owl, across year	ProtoCLR	0.049 (-0.015 to 0.158)	0.014 (-0.039 to 0.092)	0.013 (-0.023 to 0.056)
Little owl, across year	RCL FS BSED	-0.016 (-0.047 to 0.000)	-0.025 (-0.074 to 0.000)	-0.007 (-0.020 to 0.000)
Little owl, across year	BirdAVES	-0.039 (-0.128 to 0.034)	-0.033 (-0.094 to 0.015)	-0.020 (-0.063 to 0.018)
Little owl, across year	AVES	-0.007 (-0.076 to 0.054)	-0.028 (-0.089 to 0.017)	-0.024 (-0.067 to 0.009)
Little owl, across year	NatureBEATs	0.007 (-0.022 to 0.042)	0.003 (0.000 to 0.010)	0.000 (0.000 to 0.000)
Little owl, across year	BioLingual	0.000 (0.000 to 0.000)	0.000 (0.000 to 0.000)	0.000 (0.000 to 0.000)
Little owl, across year	BEATs	0.000 (0.000 to 0.000)	0.000 (0.000 to 0.000)	-0.002 (-0.006 to 0.000)
Little owl, across year	AudioMAE	0.026 (0.004 to 0.052)	-0.009 (-0.033 to 0.013)	0.000 (0.000 to 0.000)
Little owl, across year	VGGish	-0.009 (-0.082 to 0.083)	0.027 (-0.021 to 0.106)	0.016 (-0.006 to 0.043)
Little owl, across year	ECAPA-TDNN (speaker)	0.008 (-0.113 to 0.115)	-0.003 (-0.058 to 0.044)	-0.005 (-0.044 to 0.036)
Little owl, across year	ResNet (speaker)	-0.045 (-0.143 to 0.038)	-0.058 (-0.153 to 0.012)	-0.031 (-0.063 to -0.001)
Little owl, across year	x-vector (speaker)	0.010 (-0.068 to 0.080)	-0.029 (-0.093 to 0.028)	-0.033 (-0.054 to -0.013)
Little owl, across year	wav2vec 2.0 Base	0.001 (-0.037 to 0.041)	-0.010 (-0.065 to 0.035)	-0.035 (-0.061 to -0.009)
Little owl, across year	wav2vec 2.0 Large (robust)	-0.041 (-0.128 to 0.032)	-0.077 (-0.181 to 0.024)	-0.004 (-0.041 to 0.036)
Little owl, across year	XLS-R 300M	-0.080 (-0.158 to -0.008)	-0.079 (-0.187 to 0.028)	-0.018 (-0.061 to 0.025)
Little owl, across year	wav2vec 2.0 Conformer Large	-0.084 (-0.191 to 0.009)	-0.059 (-0.138 to 0.016)	-0.021 (-0.066 to 0.032)
Little owl, across year	MMS 300M	-0.096 (-0.170 to -0.023)	-0.122 (-0.242 to -0.013)	-0.029 (-0.074 to 0.013)
Little owl, across year	HuBERT Base	-0.006 (-0.127 to 0.100)	-0.038 (-0.120 to 0.037)	-0.022 (-0.066 to 0.015)
Little owl, across year	HuBERT Large	-0.102 (-0.194 to -0.022)	-0.074 (-0.157 to 0.011)	-0.026 (-0.059 to 0.003)
Little owl, across year	data2vec Base (100 h)	-0.024 (-0.142 to 0.079)	-0.090 (-0.153 to -0.033)	-0.045 (-0.082 to -0.007)
Little owl, across year	data2vec Base (960 h)	-0.045 (-0.152 to 0.049)	-0.109 (-0.175 to -0.051)	-0.071 (-0.154 to -0.009)
Little owl, across year	WavLM Base+	-0.026 (-0.122 to 0.068)	-0.076 (-0.155 to -0.009)	-0.017 (-0.058 to 0.022)
Little owl, across year	WavLM Large	-0.101 (-0.183 to -0.032)	-0.106 (-0.209 to -0.010)	-0.050 (-0.109 to -0.001)
Little owl, across year	UniSpeech-SAT Base+	0.004 (-0.107 to 0.121)	-0.047 (-0.146 to 0.044)	-0.030 (-0.078 to 0.014)
Little owl, across year	XEUS	-0.023 (-0.135 to 0.086)	-0.061 (-0.144 to 0.030)	-0.020 (-0.066 to 0.028)
Little owl, across year	Loudness control	0.012 (0.000 to 0.033)	0.020 (0.000 to 0.054)	0.005 (-0.011 to 0.026)
Little owl, across year	Duration control	0.000 (0.000 to 0.000)	0.000 (0.000 to 0.000)	0.000 (0.000 to 0.000)
Tree pipit, within year	BirdNET v2.4	0.043 (-0.036 to 0.131)	-0.003 (-0.068 to 0.057)	-0.027 (-0.091 to 0.031)
Tree pipit, within year	Perch 2.0	-0.012 (-0.172 to 0.134)	-0.001 (-0.117 to 0.110)	-0.004 (-0.102 to 0.072)
Tree pipit, within year	Perch	0.058 (-0.101 to 0.207)	0.031 (-0.083 to 0.141)	-0.025 (-0.112 to 0.035)
Tree pipit, within year	SurfPerch	0.026 (-0.110 to 0.162)	-0.027 (-0.158 to 0.096)	-0.075 (-0.151 to -0.007)
Tree pipit, within year	BirdNET v3 (preview)	0.066 (0.007 to 0.152)	0.063 (0.003 to 0.142)	0.009 (-0.030 to 0.065)
Tree pipit, within year	AVEX sl-BEATs	0.025 (-0.025 to 0.079)	0.005 (-0.054 to 0.064)	-0.026 (-0.076 to 0.017)
Tree pipit, within year	AVEX EfficientNet-B0	0.010 (-0.062 to 0.073)	0.045 (-0.010 to 0.100)	0.019 (0.006 to 0.033)
Tree pipit, within year	AvesEcho	0.015 (-0.020 to 0.064)	0.012 (-0.017 to 0.043)	-0.022 (-0.061 to 0.004)
Tree pipit, within year	AudioProtoPNet	0.125 (-0.043 to 0.264)	0.072 (-0.092 to 0.195)	-0.026 (-0.139 to 0.052)
Tree pipit, within year	ConvNeXt (BirdSet)	0.137 (0.008 to 0.271)	0.090 (-0.012 to 0.196)	0.012 (-0.091 to 0.099)
Tree pipit, within year	Bird-MAE	0.005 (-0.013 to 0.027)	0.003 (0.000 to 0.009)	0.006 (-0.011 to 0.027)
Tree pipit, within year	ProtoCLR	-0.024 (-0.057 to 0.006)	-0.021 (-0.053 to 0.008)	0.006 (-0.008 to 0.026)
Tree pipit, within year	RCL FS BSED	0.105 (0.000 to 0.292)	0.070 (-0.014 to 0.225)	0.029 (-0.034 to 0.121)
Tree pipit, within year	BirdAVES	0.024 (-0.124 to 0.190)	0.011 (-0.140 to 0.157)	-0.014 (-0.081 to 0.053)
Tree pipit, within year	AVES	0.087 (-0.039 to 0.274)	0.040 (-0.090 to 0.214)	-0.038 (-0.089 to 0.003)
Tree pipit, within year	NatureBEATs	0.018 (-0.014 to 0.055)	-0.003 (-0.063 to 0.058)	0.005 (-0.033 to 0.045)

Dataset	Embedding	Background +10 dB	Background 0 dB	Background -10 dB
Tree pipit, within year	BioLingual	0.003 (-0.013 to 0.024)	-0.012 (-0.024 to 0.000)	-0.000 (-0.021 to 0.016)
Tree pipit, within year	BEATs	0.087 (0.001 to 0.206)	0.070 (-0.001 to 0.152)	0.012 (-0.030 to 0.050)
Tree pipit, within year	AudioMAE	0.028 (-0.032 to 0.114)	-0.004 (-0.050 to 0.037)	-0.004 (-0.018 to 0.011)
Tree pipit, within year	VGGish	-0.012 (-0.103 to 0.104)	-0.055 (-0.109 to -0.005)	-0.024 (-0.070 to 0.017)
Tree pipit, within year	ECAPA-TDNN (speaker)	0.001 (-0.072 to 0.065)	-0.047 (-0.134 to 0.031)	-0.041 (-0.102 to 0.015)
Tree pipit, within year	ResNet (speaker)	0.039 (-0.026 to 0.111)	-0.015 (-0.057 to 0.023)	-0.051 (-0.119 to 0.010)
Tree pipit, within year	x-vector (speaker)	-0.037 (-0.124 to 0.027)	-0.031 (-0.092 to 0.018)	-0.005 (-0.038 to 0.028)
Tree pipit, within year	wav2vec 2.0 Base	0.015 (-0.089 to 0.135)	0.016 (-0.164 to 0.212)	0.010 (-0.061 to 0.108)
Tree pipit, within year	wav2vec 2.0 Large (robust)	0.028 (-0.161 to 0.279)	-0.007 (-0.200 to 0.244)	0.005 (-0.115 to 0.167)
Tree pipit, within year	XLS-R 300M	0.045 (-0.156 to 0.292)	-0.004 (-0.197 to 0.244)	-0.020 (-0.107 to 0.067)
Tree pipit, within year	wav2vec 2.0 Conformer Large	0.042 (-0.144 to 0.286)	-0.029 (-0.238 to 0.230)	-0.021 (-0.131 to 0.104)
Tree pipit, within year	MMS 300M	0.043 (-0.157 to 0.289)	0.001 (-0.191 to 0.240)	-0.021 (-0.115 to 0.077)
Tree pipit, within year	HuBERT Base	-0.020 (-0.209 to 0.216)	-0.011 (-0.206 to 0.226)	0.047 (-0.087 to 0.199)
Tree pipit, within year	HuBERT Large	0.035 (-0.118 to 0.225)	0.031 (-0.145 to 0.263)	0.021 (-0.100 to 0.169)
Tree pipit, within year	data2vec Base (100 h)	0.014 (-0.169 to 0.258)	0.011 (-0.170 to 0.247)	0.041 (-0.071 to 0.207)
Tree pipit, within year	data2vec Base (960 h)	0.009 (-0.189 to 0.252)	-0.005 (-0.187 to 0.227)	0.017 (-0.095 to 0.167)
Tree pipit, within year	WavLM Base+	0.000 (-0.214 to 0.257)	-0.007 (-0.245 to 0.252)	0.029 (-0.080 to 0.155)
Tree pipit, within year	WavLM Large	0.034 (-0.106 to 0.231)	0.004 (-0.185 to 0.239)	-0.012 (-0.130 to 0.133)
Tree pipit, within year	UniSpeech-SAT Base+	0.037 (-0.171 to 0.282)	0.001 (-0.209 to 0.236)	0.031 (-0.078 to 0.151)
Tree pipit, within year	XEUS	0.002 (-0.245 to 0.279)	-0.027 (-0.274 to 0.240)	-0.003 (-0.140 to 0.164)
Tree pipit, within year	Loudness control	0.078 (0.000 to 0.172)	0.033 (-0.003 to 0.079)	0.003 (0.000 to 0.010)
Tree pipit, within year	Duration control	0.000 (0.000 to 0.000)	0.000 (0.000 to 0.000)	0.000 (0.000 to 0.000)
Tree pipit, across year	BirdNET v2.4	-0.104 (-0.284 to 0.045)	-0.090 (-0.206 to 0.022)	-0.028 (-0.083 to 0.030)
Tree pipit, across year	Perch 2.0	-0.003 (-0.140 to 0.118)	0.024 (-0.039 to 0.086)	-0.013 (-0.070 to 0.047)
Tree pipit, across year	Perch	-0.095 (-0.239 to 0.049)	-0.052 (-0.179 to 0.079)	-0.047 (-0.154 to 0.052)
Tree pipit, across year	SurfPerch	-0.103 (-0.251 to 0.012)	-0.038 (-0.148 to 0.045)	0.009 (-0.040 to 0.064)
Tree pipit, across year	BirdNET v3 (preview)	0.002 (-0.074 to 0.072)	-0.011 (-0.068 to 0.035)	-0.004 (-0.032 to 0.021)
Tree pipit, across year	AVEX sl-BEATs	-0.137 (-0.320 to -0.013)	-0.096 (-0.250 to 0.004)	-0.055 (-0.172 to 0.022)
Tree pipit, across year	AVEX EfficientNet-B0	-0.056 (-0.189 to 0.049)	-0.050 (-0.111 to 0.007)	-0.009 (-0.054 to 0.049)
Tree pipit, across year	AvesEcho	-0.055 (-0.126 to 0.004)	-0.049 (-0.120 to 0.001)	-0.011 (-0.040 to 0.015)
Tree pipit, across year	AudioProtoPNet	-0.070 (-0.234 to 0.078)	-0.052 (-0.201 to 0.081)	-0.054 (-0.152 to 0.029)
Tree pipit, across year	ConvNeXt (BirdSet)	-0.096 (-0.271 to 0.044)	-0.110 (-0.259 to 0.016)	-0.061 (-0.143 to 0.019)
Tree pipit, across year	Bird-MAE	-0.033 (-0.080 to 0.000)	-0.023 (-0.063 to 0.000)	-0.016 (-0.037 to 0.000)
Tree pipit, across year	ProtoCLR	-0.015 (-0.040 to 0.000)	-0.004 (-0.025 to 0.012)	-0.004 (-0.012 to 0.004)
Tree pipit, across year	RCL_FS_BSED	-0.143 (-0.337 to 0.000)	-0.080 (-0.213 to 0.000)	-0.053 (-0.160 to 0.000)
Tree pipit, across year	BirdAVES	-0.181 (-0.409 to -0.010)	-0.161 (-0.368 to 0.000)	-0.107 (-0.256 to 0.002)
Tree pipit, across year	AVES	-0.194 (-0.434 to -0.013)	-0.167 (-0.387 to 0.000)	-0.084 (-0.197 to 0.000)
Tree pipit, across year	NatureBEATs	-0.108 (-0.293 to 0.013)	-0.099 (-0.271 to 0.003)	-0.067 (-0.227 to 0.023)
Tree pipit, across year	BioLingual	-0.047 (-0.137 to 0.007)	-0.023 (-0.080 to 0.010)	-0.017 (-0.043 to 0.000)
Tree pipit, across year	BEATs	-0.129 (-0.292 to -0.008)	-0.076 (-0.213 to 0.016)	-0.043 (-0.117 to 0.000)
Tree pipit, across year	AudioMAE	-0.010 (-0.030 to 0.000)	-0.007 (-0.020 to 0.000)	-0.003 (-0.010 to 0.000)
Tree pipit, across year	VGGish	-0.070 (-0.197 to 0.000)	-0.006 (-0.031 to 0.024)	-0.012 (-0.047 to 0.017)
Tree pipit, across year	ECAPA-TDNN (speaker)	-0.053 (-0.099 to -0.010)	0.021 (-0.051 to 0.099)	-0.033 (-0.078 to 0.014)
Tree pipit, across year	ResNet (speaker)	-0.000 (-0.072 to 0.086)	-0.010 (-0.089 to 0.068)	0.003 (-0.027 to 0.034)
Tree pipit, across year	x-vector (speaker)	-0.030 (-0.057 to -0.007)	0.024 (-0.024 to 0.075)	-0.043 (-0.082 to -0.004)
Tree pipit, across year	wav2vec 2.0 Base	-0.107 (-0.285 to 0.018)	-0.062 (-0.140 to 0.005)	-0.034 (-0.063 to -0.011)
Tree pipit, across year	wav2vec 2.0 Large (robust)	-0.075 (-0.274 to 0.045)	-0.062 (-0.199 to 0.013)	0.001 (-0.021 to 0.022)

Dataset	Embedding	Background +10 dB	Background 0 dB	Background -10 dB
Tree pipit, across year	XLS-R 300M	-0.074 (-0.274 to 0.043)	-0.054 (-0.177 to 0.013)	0.010 (-0.013 to 0.033)
Tree pipit, across year	wav2vec 2.0 Conformer Large	-0.098 (-0.279 to 0.000)	-0.086 (-0.251 to 0.007)	-0.006 (-0.052 to 0.034)
Tree pipit, across year	MMS 300M	-0.076 (-0.276 to 0.041)	-0.059 (-0.190 to 0.016)	0.007 (-0.013 to 0.027)
Tree pipit, across year	HuBERT Base	-0.224 (-0.454 to -0.028)	-0.111 (-0.226 to -0.020)	-0.022 (-0.046 to -0.001)
Tree pipit, across year	HuBERT Large	-0.068 (-0.248 to 0.040)	-0.067 (-0.210 to 0.010)	-0.019 (-0.041 to 0.000)
Tree pipit, across year	data2vec Base (100 h)	-0.082 (-0.274 to 0.035)	-0.067 (-0.187 to 0.011)	-0.006 (-0.042 to 0.024)
Tree pipit, across year	data2vec Base (960 h)	-0.092 (-0.276 to 0.023)	-0.064 (-0.191 to 0.016)	-0.007 (-0.042 to 0.022)
Tree pipit, across year	WavLM Base+	-0.235 (-0.482 to -0.039)	-0.120 (-0.253 to -0.018)	-0.012 (-0.034 to 0.004)
Tree pipit, across year	WavLM Large	-0.080 (-0.274 to 0.040)	-0.064 (-0.194 to 0.007)	-0.004 (-0.020 to 0.007)
Tree pipit, across year	UniSpeech-SAT Base+	-0.175 (-0.383 to -0.021)	-0.060 (-0.150 to -0.004)	-0.009 (-0.034 to 0.010)
Tree pipit, across year	XEUS	-0.082 (-0.268 to 0.019)	-0.075 (-0.251 to 0.025)	-0.019 (-0.070 to 0.013)
Tree pipit, across year	Loudness control	-0.018 (-0.100 to 0.048)	-0.015 (-0.084 to 0.043)	-0.009 (-0.028 to 0.000)
Tree pipit, across year	Duration control	0.000 (0.000 to 0.000)	0.000 (0.000 to 0.000)	0.000 (0.000 to 0.000)
Chiffchaff, across year	BirdNET v2.4	-0.034 (-0.182 to 0.131)	-0.067 (-0.174 to 0.046)	-0.067 (-0.159 to 0.010)
Chiffchaff, across year	Perch 2.0	-0.083 (-0.222 to 0.064)	-0.048 (-0.132 to 0.021)	-0.003 (-0.051 to 0.046)
Chiffchaff, across year	Perch	-0.051 (-0.261 to 0.147)	-0.066 (-0.216 to 0.066)	-0.018 (-0.068 to 0.039)
Chiffchaff, across year	SurfPerch	0.048 (-0.022 to 0.136)	0.058 (-0.007 to 0.137)	0.008 (-0.023 to 0.044)
Chiffchaff, across year	BirdNET v3 (preview)	-0.073 (-0.211 to 0.065)	-0.037 (-0.105 to 0.036)	-0.046 (-0.091 to -0.001)
Chiffchaff, across year	AVEX sl-BEATs	0.005 (-0.085 to 0.085)	0.029 (-0.047 to 0.140)	0.041 (0.003 to 0.083)
Chiffchaff, across year	AVEX EfficientNet-B0	0.000 (-0.101 to 0.094)	-0.032 (-0.103 to 0.048)	0.006 (-0.033 to 0.049)
Chiffchaff, across year	AvesEcho	-0.032 (-0.272 to 0.149)	0.041 (-0.027 to 0.116)	0.014 (-0.009 to 0.039)
Chiffchaff, across year	AudioProtoPNet	-0.075 (-0.214 to 0.074)	-0.065 (-0.161 to 0.033)	0.002 (-0.035 to 0.042)
Chiffchaff, across year	ConvNeXt (BirdSet)	-0.021 (-0.166 to 0.157)	-0.010 (-0.088 to 0.093)	0.016 (-0.017 to 0.053)
Chiffchaff, across year	Bird-MAE	0.006 (-0.037 to 0.061)	-0.010 (-0.065 to 0.052)	-0.015 (-0.048 to 0.010)
Chiffchaff, across year	ProtoCLR	-0.030 (-0.124 to 0.069)	-0.042 (-0.145 to 0.066)	-0.054 (-0.131 to 0.011)
Chiffchaff, across year	RCL FS BSSED	0.045 (-0.068 to 0.188)	0.005 (-0.082 to 0.106)	-0.023 (-0.056 to 0.000)
Chiffchaff, across year	BirdAVES	0.036 (-0.071 to 0.207)	-0.012 (-0.086 to 0.055)	-0.003 (-0.058 to 0.041)
Chiffchaff, across year	AVES	0.040 (-0.110 to 0.237)	0.025 (-0.029 to 0.086)	-0.007 (-0.079 to 0.063)
Chiffchaff, across year	NatureBEATs	-0.040 (-0.115 to 0.030)	0.005 (-0.009 to 0.021)	-0.001 (-0.025 to 0.024)
Chiffchaff, across year	BioLingual	-0.051 (-0.092 to -0.017)	-0.015 (-0.063 to 0.046)	-0.027 (-0.067 to 0.013)
Chiffchaff, across year	BEATs	0.042 (-0.097 to 0.168)	0.037 (-0.013 to 0.101)	0.014 (-0.047 to 0.066)
Chiffchaff, across year	AudioMAE	-0.023 (-0.050 to 0.003)	0.003 (-0.021 to 0.028)	0.032 (0.005 to 0.066)
Chiffchaff, across year	VGish	0.081 (-0.025 to 0.263)	0.021 (-0.026 to 0.080)	0.009 (0.000 to 0.027)
Chiffchaff, across year	ECAPA-TDNN (speaker)	0.053 (-0.062 to 0.181)	0.068 (0.009 to 0.157)	0.025 (-0.017 to 0.072)
Chiffchaff, across year	ResNet (speaker)	-0.021 (-0.167 to 0.133)	-0.056 (-0.138 to 0.012)	-0.007 (-0.058 to 0.038)
Chiffchaff, across year	x-vector (speaker)	0.009 (-0.079 to 0.129)	-0.023 (-0.094 to 0.060)	-0.054 (-0.101 to -0.014)
Chiffchaff, across year	wav2vec 2.0 Base	0.041 (-0.094 to 0.232)	0.003 (-0.084 to 0.094)	-0.004 (-0.025 to 0.016)
Chiffchaff, across year	wav2vec 2.0 Large (robust)	-0.050 (-0.245 to 0.158)	-0.034 (-0.151 to 0.077)	-0.024 (-0.050 to -0.001)
Chiffchaff, across year	XLS-R 300M	-0.044 (-0.281 to 0.191)	-0.062 (-0.207 to 0.064)	-0.033 (-0.084 to 0.013)
Chiffchaff, across year	wav2vec 2.0 Conformer Large	-0.082 (-0.310 to 0.141)	-0.038 (-0.164 to 0.108)	-0.019 (-0.072 to 0.037)
Chiffchaff, across year	MMS 300M	-0.006 (-0.226 to 0.213)	-0.030 (-0.175 to 0.101)	-0.005 (-0.043 to 0.029)
Chiffchaff, across year	HuBERT Base	-0.094 (-0.258 to 0.055)	-0.041 (-0.113 to 0.018)	-0.005 (-0.016 to 0.000)
Chiffchaff, across year	HuBERT Large	-0.011 (-0.242 to 0.237)	-0.033 (-0.127 to 0.060)	0.002 (-0.032 to 0.037)
Chiffchaff, across year	data2vec Base (100 h)	0.011 (-0.164 to 0.203)	-0.005 (-0.156 to 0.145)	0.002 (-0.033 to 0.035)
Chiffchaff, across year	data2vec Base (960 h)	-0.010 (-0.180 to 0.174)	-0.032 (-0.177 to 0.108)	-0.004 (-0.047 to 0.034)
Chiffchaff, across year	WavLM Base+	-0.096 (-0.257 to 0.051)	-0.039 (-0.109 to 0.015)	0.000 (0.000 to 0.000)
Chiffchaff, across year	WavLM Large	-0.066 (-0.329 to 0.175)	-0.084 (-0.230 to 0.052)	-0.046 (-0.105 to 0.002)

Dataset	Embedding	Background +10 dB	Background 0 dB	Background -10 dB
Chiffchaff, across year	UniSpeech-SAT Base+	-0.078 (-0.268 to 0.104)	-0.043 (-0.120 to 0.023)	-0.010 (-0.025 to 0.000)
Chiffchaff, across year	XEUS	-0.020 (-0.321 to 0.273)	-0.016 (-0.210 to 0.168)	-0.033 (-0.122 to 0.050)
Chiffchaff, across year	Loudness control	-0.026 (-0.069 to 0.000)	-0.007 (-0.021 to 0.000)	0.000 (0.000 to 0.000)
Chiffchaff, across year	Duration control	0.000 (0.000 to 0.000)	0.000 (0.000 to 0.000)	0.000 (0.000 to 0.000)

Notes: Background level relative to the test clip. Positive values mean the classifier's answers follow the background; negative values mean a clip mixed with its own individual's test-year background was named less accurately than one mixed with another individual's. Interval over individuals, 10,000 draws.

Table S7. Exploratory analysis, specified after the results were seen: when BirdNET named the wrong individual for a clip mixed with another individual's background, the share of those errors that named the donor.

Dataset	Added background relative to test clip	Wrong answers	Named the donor, no background added	Named the donor, background added	Under reassigned donors	p
Chiffchaff, within year	+10 dB	886	1.9%	24.9%	7.7%	0.0001
Chiffchaff, within year	0 dB	679	1.9%	20.2%	8.0%	0.0001
Chiffchaff, within year	-10 dB	410	1.9%	11.7%	6.7%	0.0001
Little owl, across year	+10 dB	202	2.0%	1.0%	6.3%	1.0000
Little owl, across year	0 dB	183	2.0%	1.1%	6.8%	1.0000
Little owl, across year	-10 dB	208	2.0%	1.4%	5.6%	0.9996
Tree pipit, within year	+10 dB	246	12.2%	13.0%	9.5%	0.0402
Tree pipit, within year	0 dB	215	12.2%	15.8%	9.4%	0.0011
Tree pipit, within year	-10 dB	200	12.2%	12.5%	9.8%	0.1146
Tree pipit, across year	+10 dB	245	14.2%	8.2%	11.0%	0.9487
Tree pipit, across year	0 dB	239	14.2%	15.5%	10.7%	0.0114
Tree pipit, across year	-10 dB	243	14.2%	16.0%	11.0%	0.0083
Chiffchaff, across year	+10 dB	168	3.6%	12.5%	10.5%	0.2116
Chiffchaff, across year	0 dB	158	3.6%	10.1%	11.2%	0.7212
Chiffchaff, across year	-10 dB	159	3.6%	2.5%	10.4%	1.0000

Notes: Each individual's donor is the next individual in sorted order, so the donor is fixed for all of an individual's clips; a hash chooses only which of the donor's background recordings is used. No background added: the same share among the errors on the unmixed test clips, which shows how often BirdNET already confused each individual with its donor.

Reassigned donors: 9,999 random reassignments of donors among the wrong answers, never giving a clip its own individual; p is the share at or above the observed value. This comparison does not control for the fixed pairing, so the column without background is the one to compare with.

Table S8. Planted-signal test: calls of ten within-year chiffchaffs mixed into the across-year chiffchaff backgrounds, one bird per territory, scored with BirdNET.

Mixture	Re-ID accuracy (95% interval)	p	Uniform chance	Majority class
Call 10 dB above background	0.641 (0.562 to 0.825)	0.0001	0.100	0.436
Equal level	0.362 (0.231 to 0.646)	0.0003	0.100	0.436
Call 10 dB below background	0.171 (0.055 to 0.448)	0.1208	0.100	0.436

Notes: Interval over individuals, 10,000 draws; p from 9,999 label shuffles.

Table S9. Background augmentation and spectral subtraction with BirdNET, scored through the four pairings.

Dataset	Remedy	Arm	C→C	B→B	C→B	B→C
Chiffchaff, within year	Background augmentation	Original	0.765 (0.683 to 0.846); 0.0001	0.737 (0.597 to 0.810); 0.0001	0.456 (0.343 to 0.627); 0.0001	0.508 (0.278 to 0.654); 0.0001
Chiffchaff, within year	Background augmentation	Augmented	0.814 (0.770 to 0.890); 0.0001	0.737 (0.597 to 0.810); 0.0001	0.260 (0.148 to 0.420); 0.0001	0.508 (0.278 to 0.654); 0.0001
Little owl, across year	Background augmentation	Original	0.501 (0.376 to 0.630); 0.0001	0.105 (0.028 to 0.202); 0.0832	0.083 (0.009 to 0.199); 0.2334	0.076 (0.005 to 0.204); 0.3273
Little owl, across year	Background augmentation	Augmented	0.560 (0.406 to 0.713); 0.0001	0.105 (0.028 to 0.202); 0.0832	0.059 (0.000 to 0.174); 0.5312	0.076 (0.005 to 0.204); 0.3273
Tree pipit, within year	Background augmentation	Original	0.376 (0.192 to 0.561); 0.0001	0.177 (0.106 to 0.245); 0.0407	0.147 (0.044 to 0.259); 0.1326	0.106 (0.023 to 0.221); 0.4514
Tree pipit, within year	Background augmentation	Augmented	0.587 (0.397 to 0.750); 0.0001	0.177 (0.106 to 0.245); 0.0407	0.085 (0.004 to 0.253); 0.6807	0.106 (0.023 to 0.221); 0.4514
Chiffchaff, within year	Spectral subtraction	Original	0.765 (0.681 to 0.846); 0.0001	0.737 (0.599 to 0.813); 0.0001	0.456 (0.344 to 0.625); 0.0001	0.508 (0.281 to 0.651); 0.0001
Chiffchaff, within year	Spectral subtraction	Subtracted once	0.721 (0.657 to 0.838); 0.0001	0.681 (0.606 to 0.753); 0.0001	0.438 (0.342 to 0.595); 0.0001	0.267 (0.184 to 0.344); 0.0194
Chiffchaff, within year	Spectral subtraction	Subtracted twice	0.736 (0.658 to 0.829); 0.0001	0.708 (0.599 to 0.780); 0.0001	0.470 (0.362 to 0.646); 0.0001	0.271 (0.151 to 0.345); 0.0197
Little owl, across year	Spectral subtraction	Original	0.501 (0.371 to 0.634); 0.0001	0.105 (0.027 to 0.203); 0.0860	0.083 (0.009 to 0.198); 0.2263	0.076 (0.004 to 0.205); 0.3198
Little owl, across year	Spectral subtraction	Subtracted once	0.545 (0.415 to 0.680); 0.0001	0.083 (0.021 to 0.179); 0.1875	0.022 (0.004 to 0.044); 0.9820	0.052 (0.000 to 0.154); 0.6380
Little owl, across year	Spectral subtraction	Subtracted twice	0.523 (0.390 to 0.664); 0.0001	0.061 (0.016 to 0.121); 0.4950	0.100 (0.017 to 0.206); 0.0596	0.106 (0.005 to 0.277); 0.0893
Tree pipit, within year	Spectral subtraction	Original	0.376 (0.193 to 0.566); 0.0001	0.177 (0.106 to 0.246); 0.0408	0.147 (0.046 to 0.257); 0.1293	0.106 (0.023 to 0.219); 0.4515
Tree pipit, within year	Spectral subtraction	Subtracted once	0.485 (0.308 to 0.670); 0.0001	0.225 (0.084 to 0.432); 0.0027	0.167 (0.007 to 0.406); 0.0401	0.096 (0.003 to 0.277); 0.5803
Tree pipit, within year	Spectral subtraction	Subtracted twice	0.502 (0.316 to 0.692); 0.0001	0.181 (0.074 to 0.313); 0.0200	0.154 (0.007 to 0.383); 0.0660	0.092 (0.006 to 0.267); 0.6352
Tree pipit, across year	Spectral subtraction	Original	0.214 (0.116 to 0.334); 0.0022	0.065 (0.029 to 0.102); 0.8142	0.075 (0.015 to 0.145); 0.7089	0.089 (0.020 to 0.184); 0.6125
Tree pipit, across year	Spectral subtraction	Subtracted once	0.358 (0.219 to 0.520); 0.0001	0.098 (0.054 to 0.151); 0.4966	0.108 (0.000 to 0.278); 0.3893	0.099 (0.004 to 0.298); 0.5082
Tree pipit, across year	Spectral subtraction	Subtracted twice	0.323 (0.194 to 0.473); 0.0001	0.075 (0.039 to 0.110); 0.7739	0.101 (0.000 to 0.264); 0.4536	0.102 (0.000 to 0.298); 0.4620
Chiffchaff, across year	Spectral subtraction	Original	0.160 (0.035 to 0.333); 0.1327	0.132 (0.036 to 0.277); 0.2439	0.112 (0.005 to 0.318); 0.4019	0.110 (0.019 to 0.240); 0.4114
Chiffchaff, across year	Spectral subtraction	Subtracted once	0.205 (0.050 to 0.395); 0.0202	0.168 (0.070 to 0.296); 0.0727	0.203 (0.018 to 0.448); 0.0168	0.115 (0.010 to 0.268); 0.3568
Chiffchaff, across year	Spectral subtraction	Subtracted twice	0.215 (0.055 to 0.422); 0.0074	0.137 (0.033 to 0.289); 0.1817	0.168 (0.000 to 0.403); 0.0607	0.120 (0.000 to 0.323); 0.2935

Notes: Each cell: re-ID accuracy (95% interval over individuals, 10,000 draws); permutation p (9,999 shuffles). Background augmentation changes only the enrolment calls, so the two pairings fitted on backgrounds are unchanged. Spectral subtraction removes each individual's mean background spectrum, once or twice, from all its clips. The original arm was scored with its own draws of individuals, so its interval bounds can differ slightly from those in Table 4.

Table S10. Adaptive score normalisation with the class-mean classifier: re-ID accuracy before and after, for all 36 neural networks.

Dataset	Embedding	Calls tested	Backgrounds tested
Chiffchaff, within year	BirdNET v2.4	0.549 to 0.386	0.265 to 0.192
Chiffchaff, within year	Perch 2.0	0.542 to 0.463	0.269 to 0.311
Chiffchaff, within year	Perch	0.448 to 0.324	0.237 to 0.315
Chiffchaff, within year	SurfPerch	0.321 to 0.398	0.106 to 0.125
Chiffchaff, within year	BirdNET v3 (preview)	0.407 to 0.445	0.304 to 0.395
Chiffchaff, within year	AVEX sl-BEATs	0.375 to 0.317	0.162 to 0.181
Chiffchaff, within year	AVEX EfficientNet-B0	0.383 to 0.378	0.108 to 0.222
Chiffchaff, within year	AvesEcho	0.374 to 0.299	0.176 to 0.167
Chiffchaff, within year	AudioProtoPNet	0.489 to 0.274	0.244 to 0.339
Chiffchaff, within year	ConvNeXt (BirdSet)	0.494 to 0.270	0.258 to 0.240
Chiffchaff, within year	Bird-MAE	0.148 to 0.133	0.109 to 0.134
Chiffchaff, within year	ProtoCLR	0.094 to 0.148	0.068 to 0.089
Chiffchaff, within year	RCL FS BSED	0.131 to 0.139	0.166 to 0.161
Chiffchaff, within year	BirdAVES	0.199 to 0.164	0.152 to 0.181
Chiffchaff, within year	AVES	0.224 to 0.163	0.198 to 0.220
Chiffchaff, within year	NatureBEATs	0.408 to 0.341	0.236 to 0.268
Chiffchaff, within year	BioLingual	0.256 to 0.378	0.080 to 0.121
Chiffchaff, within year	BEATs	0.182 to 0.133	0.281 to 0.177
Chiffchaff, within year	AudioMAE	0.065 to 0.135	0.068 to 0.099
Chiffchaff, within year	VGGish	0.202 to 0.322	0.070 to 0.106
Chiffchaff, within year	ECAPA-TDNN (speaker)	0.270 to 0.224	0.119 to 0.136
Chiffchaff, within year	ResNet (speaker)	0.385 to 0.307	0.217 to 0.205
Chiffchaff, within year	x-vector (speaker)	0.292 to 0.342	0.116 to 0.186
Chiffchaff, within year	wav2vec 2.0 Base	0.134 to 0.144	0.082 to 0.123
Chiffchaff, within year	wav2vec 2.0 Large (robust)	0.163 to 0.148	0.100 to 0.112
Chiffchaff, within year	XLS-R 300M	0.156 to 0.138	0.096 to 0.115
Chiffchaff, within year	wav2vec 2.0 Conformer Large	0.220 to 0.210	0.145 to 0.175
Chiffchaff, within year	MMS 300M	0.178 to 0.141	0.115 to 0.125
Chiffchaff, within year	HuBERT Base	0.151 to 0.153	0.085 to 0.135
Chiffchaff, within year	HuBERT Large	0.180 to 0.159	0.110 to 0.146
Chiffchaff, within year	data2vec Base (100 h)	0.149 to 0.164	0.113 to 0.203
Chiffchaff, within year	data2vec Base (960 h)	0.183 to 0.160	0.195 to 0.183
Chiffchaff, within year	WavLM Base+	0.159 to 0.134	0.095 to 0.151
Chiffchaff, within year	WavLM Large	0.194 to 0.169	0.112 to 0.151
Chiffchaff, within year	UniSpeech-SAT Base+	0.161 to 0.139	0.091 to 0.157
Chiffchaff, within year	XEUS	0.210 to 0.166	0.186 to 0.211
Little owl, across year	BirdNET v2.4	0.393 to 0.386	0.071 to 0.066
Little owl, across year	Perch 2.0	0.526 to 0.423	0.073 to 0.105
Little owl, across year	Perch	0.577 to 0.501	0.125 to 0.115
Little owl, across year	SurfPerch	0.359 to 0.337	0.120 to 0.115
Little owl, across year	BirdNET v3 (preview)	0.140 to 0.143	0.078 to 0.049
Little owl, across year	AVEX sl-BEATs	0.182 to 0.189	0.012 to 0.020
Little owl, across year	AVEX EfficientNet-B0	0.263 to 0.248	0.117 to 0.112
Little owl, across year	AvesEcho	0.143 to 0.177	0.090 to 0.088
Little owl, across year	AudioProtoPNet	0.437 to 0.297	0.088 to 0.081

Dataset	Embedding	Calls tested	Backgrounds tested
Little owl, across year	ConvNeXt (BirdSet)	0.479 to 0.396	0.100 to 0.100
Little owl, across year	Bird-MAE	0.256 to 0.312	0.068 to 0.044
Little owl, across year	ProtoCLR	0.263 to 0.297	0.093 to 0.105
Little owl, across year	RCL FS BSSED	0.172 to 0.123	0.088 to 0.088
Little owl, across year	BirdAVES	0.219 to 0.214	0.046 to 0.046
Little owl, across year	AVES	0.241 to 0.270	0.051 to 0.054
Little owl, across year	NatureBEATs	0.243 to 0.265	0.049 to 0.032
Little owl, across year	BioLingual	0.123 to 0.219	0.066 to 0.059
Little owl, across year	BEATs	0.204 to 0.162	0.064 to 0.034
Little owl, across year	AudioMAE	0.209 to 0.258	0.066 to 0.061
Little owl, across year	VGGish	0.197 to 0.145	0.083 to 0.061
Little owl, across year	ECAPA-TDNN (speaker)	0.287 to 0.283	0.117 to 0.117
Little owl, across year	ResNet (speaker)	0.354 to 0.280	0.051 to 0.054
Little owl, across year	x-vector (speaker)	0.243 to 0.248	0.054 to 0.064
Little owl, across year	wav2vec 2.0 Base	0.160 to 0.120	0.051 to 0.044
Little owl, across year	wav2vec 2.0 Large (robust)	0.133 to 0.155	0.056 to 0.064
Little owl, across year	XLS-R 300M	0.147 to 0.231	0.059 to 0.066
Little owl, across year	wav2vec 2.0 Conformer Large	0.199 to 0.233	0.051 to 0.054
Little owl, across year	MMS 300M	0.160 to 0.243	0.054 to 0.071
Little owl, across year	HuBERT Base	0.165 to 0.155	0.051 to 0.051
Little owl, across year	HuBERT Large	0.214 to 0.194	0.076 to 0.081
Little owl, across year	data2vec Base (100 h)	0.236 to 0.251	0.051 to 0.078
Little owl, across year	data2vec Base (960 h)	0.265 to 0.263	0.051 to 0.071
Little owl, across year	WavLM Base+	0.147 to 0.138	0.051 to 0.051
Little owl, across year	WavLM Large	0.167 to 0.201	0.051 to 0.061
Little owl, across year	UniSpeech-SAT Base+	0.162 to 0.162	0.051 to 0.054
Little owl, across year	XEUS	0.204 to 0.197	0.051 to 0.051
Tree pipit, within year	BirdNET v2.4	0.327 to 0.317	0.184 to 0.137
Tree pipit, within year	Perch 2.0	0.469 to 0.436	0.116 to 0.181
Tree pipit, within year	Perch	0.300 to 0.294	0.184 to 0.208
Tree pipit, within year	SurfPerch	0.330 to 0.287	0.133 to 0.181
Tree pipit, within year	BirdNET v3 (preview)	0.218 to 0.132	0.218 to 0.287
Tree pipit, within year	AVEX sl-BEATs	0.125 to 0.125	0.075 to 0.065
Tree pipit, within year	AVEX EfficientNet-B0	0.244 to 0.254	0.123 to 0.143
Tree pipit, within year	AvesEcho	0.089 to 0.089	0.075 to 0.075
Tree pipit, within year	AudioProtoPNet	0.337 to 0.310	0.225 to 0.300
Tree pipit, within year	ConvNeXt (BirdSet)	0.360 to 0.376	0.177 to 0.229
Tree pipit, within year	Bird-MAE	0.116 to 0.132	0.082 to 0.102
Tree pipit, within year	ProtoCLR	0.135 to 0.139	0.119 to 0.102
Tree pipit, within year	RCL FS BSSED	0.106 to 0.116	0.140 to 0.140
Tree pipit, within year	BirdAVES	0.221 to 0.241	0.184 to 0.184
Tree pipit, within year	AVES	0.248 to 0.185	0.218 to 0.167
Tree pipit, within year	NatureBEATs	0.122 to 0.129	0.065 to 0.072
Tree pipit, within year	BioLingual	0.069 to 0.066	0.126 to 0.130
Tree pipit, within year	BEATs	0.149 to 0.185	0.092 to 0.082
Tree pipit, within year	AudioMAE	0.099 to 0.083	0.137 to 0.130
Tree pipit, within year	VGGish	0.089 to 0.122	0.061 to 0.051

Dataset	Embedding	Calls tested	Backgrounds tested
Tree pipit, within year	ECAPA-TDNN (speaker)	0.300 to 0.307	0.191 to 0.208
Tree pipit, within year	ResNet (speaker)	0.201 to 0.201	0.143 to 0.133
Tree pipit, within year	x-vector (speaker)	0.281 to 0.267	0.137 to 0.157
Tree pipit, within year	wav2vec 2.0 Base	0.191 to 0.158	0.130 to 0.137
Tree pipit, within year	wav2vec 2.0 Large (robust)	0.158 to 0.178	0.113 to 0.171
Tree pipit, within year	XLS-R 300M	0.155 to 0.158	0.109 to 0.171
Tree pipit, within year	wav2vec 2.0 Conformer Large	0.155 to 0.149	0.137 to 0.205
Tree pipit, within year	MMS 300M	0.142 to 0.132	0.099 to 0.147
Tree pipit, within year	HuBERT Base	0.195 to 0.165	0.171 to 0.147
Tree pipit, within year	HuBERT Large	0.162 to 0.155	0.113 to 0.171
Tree pipit, within year	data2vec Base (100 h)	0.158 to 0.165	0.102 to 0.143
Tree pipit, within year	data2vec Base (960 h)	0.155 to 0.165	0.102 to 0.137
Tree pipit, within year	WavLM Base+	0.208 to 0.152	0.164 to 0.140
Tree pipit, within year	WavLM Large	0.152 to 0.165	0.116 to 0.184
Tree pipit, within year	UniSpeech-SAT Base+	0.172 to 0.149	0.150 to 0.137
Tree pipit, within year	XEUS	0.175 to 0.155	0.106 to 0.171
Tree pipit, across year	BirdNET v2.4	0.173 to 0.214	0.056 to 0.078
Tree pipit, across year	Perch 2.0	0.345 to 0.281	0.075 to 0.124
Tree pipit, across year	Perch	0.195 to 0.243	0.062 to 0.114
Tree pipit, across year	SurfPerch	0.166 to 0.134	0.065 to 0.042
Tree pipit, across year	BirdNET v3 (preview)	0.141 to 0.112	0.176 to 0.199
Tree pipit, across year	AVEX sl-BEATs	0.121 to 0.086	0.114 to 0.121
Tree pipit, across year	AVEX EfficientNet-B0	0.153 to 0.163	0.111 to 0.114
Tree pipit, across year	AvesEcho	0.035 to 0.070	0.098 to 0.085
Tree pipit, across year	AudioProtoPNet	0.137 to 0.125	0.078 to 0.147
Tree pipit, across year	ConvNeXt (BirdSet)	0.214 to 0.224	0.029 to 0.088
Tree pipit, across year	Bird-MAE	0.058 to 0.080	0.098 to 0.085
Tree pipit, across year	ProtoCLR	0.093 to 0.112	0.111 to 0.092
Tree pipit, across year	RCL FS BSSED	0.061 to 0.093	0.016 to 0.007
Tree pipit, across year	BirdAVES	0.054 to 0.125	0.039 to 0.013
Tree pipit, across year	AVES	0.038 to 0.112	0.013 to 0.029
Tree pipit, across year	NatureBEATs	0.125 to 0.099	0.121 to 0.127
Tree pipit, across year	BioLingual	0.054 to 0.077	0.101 to 0.105
Tree pipit, across year	BEATs	0.128 to 0.083	0.137 to 0.095
Tree pipit, across year	AudioMAE	0.051 to 0.058	0.078 to 0.075
Tree pipit, across year	VGGish	0.102 to 0.099	0.023 to 0.056
Tree pipit, across year	ECAPA-TDNN (speaker)	0.195 to 0.208	0.108 to 0.114
Tree pipit, across year	ResNet (speaker)	0.157 to 0.141	0.105 to 0.118
Tree pipit, across year	x-vector (speaker)	0.163 to 0.166	0.118 to 0.124
Tree pipit, across year	wav2vec 2.0 Base	0.070 to 0.064	0.134 to 0.114
Tree pipit, across year	wav2vec 2.0 Large (robust)	0.073 to 0.064	0.121 to 0.114
Tree pipit, across year	XLS-R 300M	0.058 to 0.026	0.124 to 0.108
Tree pipit, across year	wav2vec 2.0 Conformer Large	0.048 to 0.029	0.108 to 0.085
Tree pipit, across year	MMS 300M	0.061 to 0.038	0.121 to 0.114
Tree pipit, across year	HuBERT Base	0.045 to 0.045	0.059 to 0.033
Tree pipit, across year	HuBERT Large	0.070 to 0.048	0.127 to 0.108
Tree pipit, across year	data2vec Base (100 h)	0.077 to 0.054	0.065 to 0.056

Dataset	Embedding	Calls tested	Backgrounds tested
Tree pipit, across year	data2vec Base (960 h)	0.070 to 0.045	0.069 to 0.059
Tree pipit, across year	WavLM Base+	0.035 to 0.045	0.042 to 0.013
Tree pipit, across year	WavLM Large	0.067 to 0.038	0.124 to 0.098
Tree pipit, across year	UniSpeech-SAT Base+	0.032 to 0.045	0.056 to 0.026
Tree pipit, across year	XEUS	0.048 to 0.054	0.059 to 0.042
Chiffchaff, across year	BirdNET v2.4	0.125 to 0.150	0.096 to 0.142
Chiffchaff, across year	Perch 2.0	0.115 to 0.105	0.132 to 0.117
Chiffchaff, across year	Perch	0.105 to 0.115	0.152 to 0.152
Chiffchaff, across year	SurfPerch	0.120 to 0.165	0.127 to 0.168
Chiffchaff, across year	BirdNET v3 (preview)	0.065 to 0.065	0.091 to 0.107
Chiffchaff, across year	AVEX sl-BEATs	0.150 to 0.120	0.137 to 0.132
Chiffchaff, across year	AVEX EfficientNet-B0	0.075 to 0.065	0.142 to 0.162
Chiffchaff, across year	AvesEcho	0.135 to 0.150	0.162 to 0.203
Chiffchaff, across year	AudioProtoPNet	0.085 to 0.095	0.157 to 0.152
Chiffchaff, across year	ConvNeXt (BirdSet)	0.105 to 0.120	0.117 to 0.137
Chiffchaff, across year	Bird-MAE	0.115 to 0.135	0.081 to 0.112
Chiffchaff, across year	ProtoCLR	0.185 to 0.140	0.112 to 0.112
Chiffchaff, across year	RCL FS BSED	0.135 to 0.060	0.081 to 0.112
Chiffchaff, across year	BirdAVES	0.150 to 0.105	0.183 to 0.178
Chiffchaff, across year	AVES	0.175 to 0.165	0.193 to 0.203
Chiffchaff, across year	NatureBEATs	0.120 to 0.140	0.102 to 0.132
Chiffchaff, across year	BioLingual	0.090 to 0.125	0.117 to 0.117
Chiffchaff, across year	BEATs	0.155 to 0.170	0.096 to 0.117
Chiffchaff, across year	AudioMAE	0.095 to 0.110	0.102 to 0.096
Chiffchaff, across year	VGGish	0.120 to 0.080	0.091 to 0.086
Chiffchaff, across year	ECAPA-TDNN (speaker)	0.105 to 0.085	0.152 to 0.132
Chiffchaff, across year	ResNet (speaker)	0.140 to 0.175	0.066 to 0.102
Chiffchaff, across year	x-vector (speaker)	0.100 to 0.095	0.112 to 0.102
Chiffchaff, across year	wav2vec 2.0 Base	0.180 to 0.105	0.168 to 0.168
Chiffchaff, across year	wav2vec 2.0 Large (robust)	0.160 to 0.175	0.137 to 0.122
Chiffchaff, across year	XLS-R 300M	0.155 to 0.170	0.132 to 0.122
Chiffchaff, across year	wav2vec 2.0 Conformer Large	0.165 to 0.155	0.127 to 0.122
Chiffchaff, across year	MMS 300M	0.165 to 0.170	0.137 to 0.117
Chiffchaff, across year	HuBERT Base	0.150 to 0.140	0.198 to 0.188
Chiffchaff, across year	HuBERT Large	0.170 to 0.170	0.137 to 0.132
Chiffchaff, across year	data2vec Base (100 h)	0.185 to 0.160	0.137 to 0.152
Chiffchaff, across year	data2vec Base (960 h)	0.185 to 0.155	0.112 to 0.147
Chiffchaff, across year	WavLM Base+	0.140 to 0.130	0.193 to 0.188
Chiffchaff, across year	WavLM Large	0.150 to 0.170	0.076 to 0.132
Chiffchaff, across year	UniSpeech-SAT Base+	0.140 to 0.130	0.198 to 0.193
Chiffchaff, across year	XEUS	0.160 to 0.180	0.122 to 0.132

Notes: Backgrounds tested: classifier fitted on backgrounds and tested on backgrounds. Intervals are in the numerical ledger. Duration and loudness controls are omitted because their unit-scaled values do not measure the intended magnitude controls; their kernel-ridge results elsewhere are retained.

Table S11. Session compensation on the fruit bat with BirdNET: re-ID accuracy and how well the recording day could still be predicted, for every setting specified in advance.

Method	Setting	Re-ID accuracy	Day predicted	Day, majority rate
directions of between-session variation removed	0	0.381	0.258	0.166
directions of between-session variation removed	1	0.370	0.261	0.166
directions of between-session variation removed	2	0.362	0.264	0.166
directions of between-session variation removed	4	0.369	0.261	0.166
directions of between-session variation removed	8	0.383	0.269	0.166
directions of between-session variation removed	16	0.331	0.261	0.166
directions of between-session variation removed	32	0.306	0.250	0.166
none	n/a	0.381	0.258	0.166
session mean subtracted, whole session	n/a	0.284	0.016	0.166
session mean subtracted, running	n/a	0.279	0.131	0.166
within-class covariance normalisation	0.0	0.281	0.265	0.166
within-class covariance normalisation	0.01	0.300	0.256	0.166
within-class covariance normalisation	0.03	0.312	0.252	0.166
within-class covariance normalisation	0.1	0.333	0.263	0.166
within-class covariance normalisation	0.3	0.352	0.269	0.166
within-class covariance normalisation	0.6	0.374	0.273	0.166
within-class covariance normalisation	0.9	0.378	0.278	0.166
within-class covariance normalisation	1.0	0.381	0.258	0.166

Method	Setting	Classifier	Re-ID accuracy (95% interval)
none	n/a	kernel ridge	0.501 (0.374 to 0.631)
none	n/a	class mean	0.393 (0.244 to 0.553)
none	n/a	nearest clip	0.548 (0.407 to 0.696)
within-class covariance normalisation	0.0	kernel ridge	0.484 (0.360 to 0.609)
within-class covariance normalisation	0.0	class mean	0.501 (0.378 to 0.632)
within-class covariance normalisation	0.0	nearest clip	0.501 (0.381 to 0.629)
within-class covariance normalisation	0.01	kernel ridge	0.516 (0.394 to 0.643)
within-class covariance normalisation	0.01	class mean	0.526 (0.411 to 0.650)
within-class covariance normalisation	0.01	nearest clip	0.514 (0.390 to 0.638)
within-class covariance normalisation	0.03	kernel ridge	0.526 (0.409 to 0.656)
within-class covariance normalisation	0.03	class mean	0.528 (0.406 to 0.648)
within-class covariance normalisation	0.03	nearest clip	0.533 (0.414 to 0.654)
within-class covariance normalisation	0.1	kernel ridge	0.541 (0.405 to 0.680)
within-class covariance normalisation	0.1	class mean	0.533 (0.405 to 0.657)
within-class covariance normalisation	0.1	nearest clip	0.555 (0.426 to 0.688)
within-class covariance normalisation	0.3	kernel ridge	0.533 (0.389 to 0.672)
within-class covariance normalisation	0.3	class mean	0.543 (0.410 to 0.672)
within-class covariance normalisation	0.3	nearest clip	0.560 (0.426 to 0.685)
within-class covariance normalisation	0.6	kernel ridge	0.526 (0.394 to 0.661)
within-class covariance normalisation	0.6	class mean	0.521 (0.391 to 0.651)
within-class covariance normalisation	0.6	nearest clip	0.565 (0.426 to 0.693)
within-class covariance normalisation	0.9	kernel ridge	0.523 (0.379 to 0.658)
within-class covariance normalisation	0.9	class mean	0.457 (0.324 to 0.590)

Method	Setting	Classifier	Re-ID accuracy (95% interval)
within-class covariance normalisation	0.9	nearest clip	0.572 (0.436 to 0.707)
within-class covariance normalisation	1.0	kernel ridge	0.501 (0.377 to 0.631)
within-class covariance normalisation	1.0	class mean	0.393 (0.250 to 0.562)
within-class covariance normalisation	1.0	nearest clip	0.548 (0.394 to 0.689)

Notes: Upper: fruit bat. Day predicted: accuracy of a classifier fitted on all but one bat and tested on the one left out, for each bat in turn. Lower: within-class covariance normalisation on the little owl with three classifiers. Removing 0 directions and shrinkage 1.0 leave the embedding unchanged and repeat the row without compensation. Spearman correlation between re-ID accuracy and day prediction over the 16 distinct bat settings: 0.616 ($p = 0.0111$); without the two session-mean rows, 0.477 ($p = 0.0843$). Subtracting each day's mean over all bats places a bat's clips opposite the other bats' clips of the same day, so a day classifier fitted on the other bats is systematically wrong; the day prediction of 0.016 is a result of that design.

Table S12. The 35 neural networks compared with BirdNET v2.4.

Neural network	Group	Source	Input sample rate (Hz)	Highest frequency analysed (kHz)	Window (s)	Candidate embeddings per dataset	Playback slowing
Perch 2.0	animal or general sound	van Merriënboer et al. (2025)	32,000	16	5	1	1
Perch	animal or general sound	Ghani et al. (2023)	32,000	16	5	1	1
SurfPerch	animal or general sound	Williams et al. (2025)	32,000	16	5	1	1
BirdNET v3 (preview)	animal or general sound	developer preview, as supplied by Bacpipe 1.3.5 (Kather et al. 2026)	32,000	16	3	1	1
AVEX sl-BEATs	animal or general sound	Miron et al. (2026)	16,000	8	5	13	1
AVEX EfficientNet-B0	animal or general sound	Miron et al. (2026)	16,000	8	5	1	1
AvesEcho	animal or general sound	Ghani et al. (2025)	32,000	16	3	13	1
AudioProtoPNet	animal or general sound	Heinrich et al. (2025)	32,000	16	5	1	1
ConvNeXt (BirdSet)	animal or general sound	Rauch et al. (2025b)	32,000	16	5	1	1
Bird-MAE	animal or general sound	Rauch et al. (2025a)	32,000	16	5	33	1
ProtoCLR	animal or general sound	Moummad et al. (2026)	16,000	8	6	14	1
RCL_FS_BSED	animal or general sound	Moummad et al. (2024)	22,050	11.025	0.2	1	1
BirdAVES	animal or general sound	Hagiwara (2023)	16,000	8	1	25	1
AVES	animal or general sound	Hagiwara (2023)	16,000	8	1	13	1
NatureBEATs	animal or general sound	Robinson et al. (2025)	16,000	8	5	13	1
BioLingual	animal or general sound	Robinson et al. (2024)	48,000	24	10	13	1
BEATs	animal or general sound	Chen et al. (2023)	16,000	8	5	13	1
AudioMAE	animal or general sound	Huang et al. (2022)	16,000	8	10	13	1
VGGish	animal or general sound	Hershey et al. (2017)	16,000	8	1	1	1
ECAPA-TDNN (speaker)	human speech	Desplanques et al. (2020); Ravanelli et al. (2021); speechbrain/spkrec-ecapa-voxceleb	16,000	8; 16 and 24 at slowed playback	whole clip	3	1, 2, 3
ResNet (speaker)	human speech	Villalba et al. (2020); Ravanelli et al. (2021); speechbrain/spkrec-resnet-voxceleb	16,000	8; 16 and 24 at slowed playback	whole clip	3	1, 2, 3

Neural network x-vector (speaker)	Group human speech	Source Snyder et al. (2018); Ravanelli et al. (2021); speechbrain/spkrec-xvect-voxceleb	Input sample rate (Hz) 16,000	Highest frequency analysed (kHz) 8; 16 and 24 at slowed playback	Window (s) whole clip	Candidate embeddings per dataset 3	Playback slowing 1, 2, 3
wav2vec 2.0 Base	human speech	Baevski et al. (2020); facebook/wav2vec2-base	16,000	8; 16 and 24 at slowed playback	whole clip	39	1, 2, 3
wav2vec 2.0 Large (robust)	human speech	Hsu et al. (2021b); facebook/wav2vec2-large-robust	16,000	8; 16 and 24 at slowed playback	whole clip	75	1, 2, 3
XLS-R 300M	human speech	Babu et al. (2022); facebook/wav2vec2-xls-r-300m	16,000	8; 16 and 24 at slowed playback	whole clip	75	1, 2, 3
wav2vec 2.0 Conformer Large	human speech	Gulati et al. (2020); Wang et al. (2020); facebook/wav2vec2-conformer-ropelarge	16,000	8; 16 and 24 at slowed playback	whole clip	75	1, 2, 3
MMS 300M	human speech	Pratap et al. (2024); facebook/mms-300m	16,000	8; 16 and 24 at slowed playback	whole clip	75	1, 2, 3
HuBERT Base	human speech	Hsu et al. (2021a); facebook/hubert-base-ls960	16,000	8; 16 and 24 at slowed playback	whole clip	39	1, 2, 3
HuBERT Large	human speech	Hsu et al. (2021a); facebook/hubert-large-ll60k	16,000	8; 16 and 24 at slowed playback	whole clip	75	1, 2, 3
data2vec Base (100 h)	human speech	Baevski et al. (2022); facebook/data2vec-audio-base-100h	16,000	8; 16 and 24 at slowed playback	whole clip	39	1, 2, 3
data2vec Base (960 h)	human speech	Baevski et al. (2022); facebook/data2vec-audio-base-960h	16,000	8; 16 and 24 at slowed playback	whole clip	39	1, 2, 3
WavLM Base+	human speech	Chen et al. (2022a); microsoft/wavlm-base-plus	16,000	8; 16 and 24 at slowed playback	whole clip	39	1, 2, 3
WavLM Large	human speech	Chen et al. (2022a); microsoft/wavlm-large	16,000	8; 16 and 24 at slowed playback	whole clip	75	1, 2, 3
UniSpeech-SAT Base+	human speech	Chen et al. (2022b); microsoft/unispeech-sat-base-plus	16,000	8; 16 and 24 at slowed playback	whole clip	39	1, 2, 3
XEUS	human speech	Chen et al. (2024); espnet/xeus	16,000	8; 16 and 24 at slowed playback	whole clip	57	1, 2, 3

Notes: Speech networks are identified by their method papers and published checkpoint names. Window: the input length each clip was cut into; speech networks read whole clips, or 60-s windows where a clip exceeded the graphics card's memory. Candidate embeddings: layers or blocks read, times playback speeds, where computed on that dataset. Playback slowing: 1, the recording's own speed; 2 and 3, half and one third of it. Highest frequency analysed: half the input sample rate, the most a network can represent; for speech networks, 8 kHz at the recording's own speed and 16 and 24 kHz at half and one third of it. BirdNET v2.4 reads 48 kHz audio. Sample rates and windows come from a record made by reloading each network. The speaker-verification checkpoint of WavLM Base+ (microsoft/wavlm-base-plus-sv) was also measured; the layers read have the weights of WavLM Base+ and every value was identical, so it is not counted separately. BirdNET v3 (preview) is the ONNX file Bacpipe 1.3.5 supplies (SHA-256 6f58d7ffa4c33bf49c8c67ac27bc5265a940a139cb67254477997bad41efc16d).

Table S13. Each neural network and control against BirdNET v2.4 on every dataset, under the two layer rules specified in advance.

Zebra finch, group of four: BirdNET v2.4 0.807 (0.669 to 0.963).

Neural network	Fisher rule: layer, re-ID accuracy	Difference from BirdNET	Holm p	Held-out rule: layer, re-ID accuracy	Difference from BirdNET	Holm p
Perch 2.0	embedding: 0.830 (0.679 to 0.923)	0.023 (−0.037 to 0.073)	1.0000	embedding: 0.830 (0.679 to 0.923)	0.023 (−0.037 to 0.073)	1.0000
Perch	embedding: 0.803 (0.717 to 0.838)	−0.005 (−0.125 to 0.079)	1.0000	embedding: 0.803 (0.717 to 0.838)	−0.005 (−0.125 to 0.079)	1.0000
SurfPerch	embedding: 0.780 (0.682 to 0.860)	−0.028 (−0.092 to 0.057)	1.0000	embedding: 0.780 (0.682 to 0.860)	−0.028 (−0.092 to 0.057)	1.0000
BirdNET v3 (preview)	embedding: 0.794 (0.696 to 0.900)	−0.014 (−0.062 to 0.027)	1.0000	embedding: 0.794 (0.696 to 0.900)	−0.014 (−0.062 to 0.027)	1.0000
AVEX sl-BEATs	block 0: 0.417 (0.058 to 0.675)	−0.390 (−0.904 to −0.035)	0.8274	block 5: 0.683 (0.604 to 0.776)	−0.124 (−0.183 to −0.027)	0.8556
AVEX EfficientNet-B0	embedding: 0.798 (0.660 to 0.867)	−0.009 (−0.094 to 0.055)	1.0000	embedding: 0.798 (0.660 to 0.867)	−0.009 (−0.094 to 0.055)	1.0000
AvesEcho	block 3: 0.541 (0.236 to 0.921)	−0.266 (−0.421 to −0.042)	0.2184	embedding: 0.798 (0.702 to 0.942)	−0.009 (−0.036 to 0.075)	1.0000
AudioProtoPNet	embedding: 0.835 (0.679 to 0.950)	0.028 (−0.013 to 0.064)	1.0000	embedding: 0.835 (0.679 to 0.950)	0.028 (−0.013 to 0.064)	1.0000
ConvNeXt (BirdSet)	embedding: 0.839 (0.698 to 0.950)	0.032 (−0.010 to 0.070)	1.0000	embedding: 0.839 (0.698 to 0.950)	0.032 (−0.010 to 0.070)	1.0000
Bird-MAE	block 12: 0.367 (0.123 to 0.638)	−0.440 (−0.588 to −0.264)	< 0.0074	block 18: 0.404 (0.202 to 0.625)	−0.404 (−0.509 to −0.264)	< 0.0074
ProtoCLR	block 0: 0.372 (0.086 to 0.598)	−0.436 (−0.878 to −0.047)	0.2184	block 12: 0.610 (0.541 to 0.688)	−0.197 (−0.375 to −0.019)	0.2494
RCL FS_BSED	embedding: 0.661 (0.491 to 0.835)	−0.147 (−0.273 to −0.027)	0.8880	embedding: 0.661 (0.491 to 0.835)	−0.147 (−0.273 to −0.027)	0.9768
BirdAVES	block 23: 0.638 (0.396 to 0.774)	−0.170 (−0.319 to −0.123)	< 0.0074	block 9: 0.560 (0.270 to 0.801)	−0.248 (−0.360 to −0.125)	< 0.0074
AVES	block 6: 0.445 (0.228 to 0.683)	−0.362 (−0.482 to −0.182)	< 0.0074	block 5: 0.454 (0.211 to 0.721)	−0.353 (−0.500 to −0.160)	< 0.0074
NatureBEATs	block 6: 0.734 (0.585 to 0.892)	−0.073 (−0.108 to −0.047)	0.2184	block 10: 0.784 (0.682 to 0.900)	−0.023 (−0.062 to 0.019)	1.0000
BioLingual	block 9: 0.417 (0.053 to 0.896)	−0.390 (−0.658 to −0.068)	< 0.0074	block 4: 0.679 (0.482 to 0.954)	−0.128 (−0.245 to 0.000)	1.0000
BEATs	block 4: 0.495 (0.059 to 0.859)	−0.312 (−0.892 to 0.116)	1.0000	block 10: 0.734 (0.574 to 0.938)	−0.073 (−0.110 to −0.024)	< 0.0074
AudioMAE	block 6: 0.509 (0.263 to 0.851)	−0.298 (−0.447 to −0.113)	< 0.0074	block 11: 0.683 (0.482 to 0.958)	−0.124 (−0.251 to 0.023)	1.0000
VGGish	embedding: 0.702 (0.486 to 0.873)	−0.106 (−0.182 to −0.058)	< 0.0074	embedding: 0.702 (0.486 to 0.873)	−0.106 (−0.182 to −0.058)	< 0.0074
ECAPA-TDNN (speaker)	embedding, 1/3 speed: 0.711 (0.509 to 0.796)	−0.096 (−0.218 to −0.016)	0.2184	embedding, 1/3 speed: 0.711 (0.509 to 0.796)	−0.096 (−0.218 to −0.016)	0.2494
ResNet (speaker)	embedding, 1/2 speed: 0.771 (0.647 to 0.816)	−0.037 (−0.188 to 0.073)	1.0000	embedding: 0.794 (0.669 to 0.896)	−0.014 (−0.059 to 0.038)	1.0000
x-vector (speaker)	embedding, 1/3 speed: 0.789 (0.623 to 0.941)	−0.018 (−0.059 to −0.003)	0.2184	embedding, 1/3 speed: 0.789 (0.623 to 0.941)	−0.018 (−0.059 to −0.003)	0.2494
wav2vec 2.0 Base	layer 1, 1/2 speed: 0.826 (0.660 to 0.879)	0.018 (−0.118 to 0.120)	1.0000	layer 4, 1/3 speed: 0.697 (0.585 to 0.733)	−0.110 (−0.254 to −0.009)	0.2494
wav2vec 2.0 Large (robust)	layer 21, 1/2 speed: 0.555 (0.272 to 0.865)	−0.252 (−0.439 to −0.017)	0.2184	layer 22, 1/2 speed: 0.394 (0.224 to 0.610)	−0.413 (−0.682 to −0.113)	0.2494
XLS-R 300M	layer 1, 1/3 speed: 0.789 (0.608 to 0.891)	−0.018 (−0.126 to 0.030)	1.0000	layer 2, full speed: 0.839 (0.736 to 0.958)	0.032 (−0.004 to 0.075)	1.0000
wav2vec 2.0 Conformer Large	layer 1, 1/3 speed: 0.780 (0.601 to 0.875)	−0.028 (−0.129 to 0.040)	1.0000	layer 4, full speed: 0.780 (0.689 to 0.827)	−0.028 (−0.141 to 0.059)	1.0000
MMS 300M	layer 1, 1/3 speed: 0.798 (0.622 to 0.923)	−0.009 (−0.092 to 0.023)	1.0000	layer 3, 1/2 speed: 0.784 (0.615 to 0.959)	−0.023 (−0.053 to 0.016)	1.0000
HuBERT Base	layer 2, full speed: 0.638 (0.471 to 0.760)	−0.170 (−0.475 to 0.026)	1.0000	layer 1, full speed: 0.656 (0.506 to 0.766)	−0.151 (−0.425 to 0.028)	1.0000
HuBERT Large	layer 3, 1/3 speed: 0.716 (0.412 to 0.923)	−0.092 (−0.257 to 0.000)	1.0000	layer 11, 1/3 speed: 0.679 (0.351 to 0.923)	−0.128 (−0.299 to 0.010)	1.0000
data2vec Base (100 h)	layer 1, 1/2 speed: 0.743 (0.622 to 0.892)	−0.064 (−0.073 to −0.038)	0.2184	layer 1, full speed: 0.739 (0.628 to 0.808)	−0.069 (−0.153 to 0.000)	1.0000
data2vec Base (960 h)	layer 1, 1/2 speed: 0.743 (0.581 to 0.882)	−0.064 (−0.118 to −0.045)	< 0.0074	layer 1, full speed: 0.725 (0.574 to 0.846)	−0.083 (−0.118 to −0.038)	0.2494

Neural network WavLM Base+	Fisher rule: layer, re-ID accuracy layer 0, 1/3 speed: 0.817 (0.681 to 0.852)	Difference from BirdNET 0.009 (−0.141 to 0.105)	Holm p 1.0000	Held-out rule: layer, re-ID accuracy layer 5, 1/2 speed: 0.716 (0.534 to 0.864)	Difference from BirdNET −0.092 (−0.151 to −0.068)	Holm p < 0.0074
WavLM Large	layer 23, 1/3 speed: 0.743 (0.472 to 0.921)	−0.064 (−0.208 to −0.018)	< 0.0074	layer 19, full speed: 0.711 (0.585 to 0.829)	−0.096 (−0.135 to −0.057)	0.2340
UniSpeech-SAT Base+	layer 0, 1/3 speed: 0.794 (0.630 to 0.837)	−0.014 (−0.165 to 0.083)	1.0000	layer 1, full speed: 0.720 (0.565 to 0.838)	−0.087 (−0.367 to 0.098)	1.0000
XEUS	layer 0, 1/2 speed: 0.817 (0.660 to 0.905)	0.009 (−0.059 to 0.064)	1.0000	layer 11, 1/2 speed: 0.812 (0.642 to 0.955)	0.005 (−0.050 to 0.025)	1.0000
Loudness control	summary: 0.381 (0.018 to 0.846)	−0.427 (−0.693 to −0.118)	< 0.0074	summary: 0.381 (0.018 to 0.846)	−0.427 (−0.693 to −0.118)	< 0.0074
Duration control	summary: 0.468 (0.000 to 0.850)	−0.339 (−0.906 to 0.107)	1.0000	summary: 0.468 (0.000 to 0.850)	−0.339 (−0.906 to 0.107)	1.0000

Zebra finch, one bird per recording: BirdNET v2.4 0.967 (0.933 to 0.989).

Neural network Perch 2.0	Fisher rule: layer, re-ID accuracy embedding: 0.956 (0.911 to 1.000)	Difference from BirdNET −0.011 (−0.044 to 0.022)	Holm p 1.0000	Held-out rule: layer, re-ID accuracy embedding: 0.956 (0.911 to 1.000)	Difference from BirdNET −0.011 (−0.044 to 0.022)	Holm p 1.0000
Perch	embedding: 0.878 (0.722 to 0.978)	−0.089 (−0.222 to 0.000)	0.4256	embedding: 0.878 (0.722 to 0.978)	−0.089 (−0.222 to 0.000)	0.4820
SurfPerch	embedding: 0.900 (0.833 to 0.967)	−0.067 (−0.111 to −0.022)	0.0182	embedding: 0.900 (0.833 to 0.967)	−0.067 (−0.111 to −0.022)	0.0252
BirdNET v3 (preview)	embedding: 0.844 (0.744 to 0.933)	−0.122 (−0.200 to −0.044)	0.0240	embedding: 0.844 (0.744 to 0.933)	−0.122 (−0.200 to −0.044)	0.0336
AVEX sl-BEATs	block 6: 0.733 (0.533 to 0.878)	−0.233 (−0.444 to −0.078)	< 0.0074	block 11: 0.889 (0.811 to 0.967)	−0.078 (−0.156 to −0.022)	0.4820
AVEX EfficientNet-B0	embedding: 0.878 (0.767 to 0.978)	−0.089 (−0.189 to 0.000)	0.4416	embedding: 0.878 (0.767 to 0.978)	−0.089 (−0.189 to 0.000)	0.4820
AvesEcho	block 4: 0.756 (0.567 to 0.911)	−0.211 (−0.389 to −0.056)	0.0306	embedding: 0.889 (0.756 to 0.978)	−0.078 (−0.189 to 0.000)	0.4820
AudioProtoPNet	embedding: 0.933 (0.844 to 1.000)	−0.033 (−0.111 to 0.022)	1.0000	embedding: 0.933 (0.844 to 1.000)	−0.033 (−0.111 to 0.022)	1.0000
ConvNeXt (BirdSet)	embedding: 0.978 (0.933 to 1.000)	0.011 (−0.022 to 0.044)	1.0000	embedding: 0.978 (0.933 to 1.000)	0.011 (−0.022 to 0.044)	1.0000
Bird-MAE	block 16: 0.567 (0.356 to 0.744)	−0.400 (−0.600 to −0.222)	< 0.0074	block 24: 0.656 (0.444 to 0.833)	−0.311 (−0.511 to −0.133)	< 0.0074
ProtoCLR	block 11: 0.378 (0.167 to 0.589)	−0.589 (−0.789 to −0.389)	< 0.0074	block 1: 0.344 (0.144 to 0.578)	−0.622 (−0.811 to −0.411)	< 0.0074
RCL FS_BSED	embedding: 0.456 (0.189 to 0.722)	−0.511 (−0.756 to −0.267)	< 0.0074	embedding: 0.456 (0.189 to 0.722)	−0.511 (−0.756 to −0.267)	< 0.0074
BirdAVES	block 23: 0.511 (0.333 to 0.678)	−0.456 (−0.611 to −0.300)	< 0.0074	block 22: 0.556 (0.367 to 0.744)	−0.411 (−0.578 to −0.244)	< 0.0074
AVES	block 10: 0.556 (0.389 to 0.711)	−0.411 (−0.556 to −0.267)	< 0.0074	block 10: 0.556 (0.389 to 0.711)	−0.411 (−0.556 to −0.267)	< 0.0074
NatureBEATs	block 2: 0.600 (0.400 to 0.789)	−0.367 (−0.567 to −0.189)	< 0.0074	block 7: 0.700 (0.578 to 0.811)	−0.267 (−0.389 to −0.167)	< 0.0074
BioLingual	embedding: 0.956 (0.911 to 0.989)	−0.011 (−0.044 to 0.022)	1.0000	embedding: 0.956 (0.911 to 0.989)	−0.011 (−0.044 to 0.022)	1.0000
BEATs	block 4: 0.500 (0.278 to 0.722)	−0.467 (−0.689 to −0.244)	< 0.0074	block 9: 0.767 (0.544 to 0.933)	−0.200 (−0.400 to −0.056)	0.0050
AudioMAE	block 10: 0.733 (0.578 to 0.867)	−0.233 (−0.400 to −0.089)	0.0120	block 9: 0.667 (0.511 to 0.800)	−0.300 (−0.456 to −0.167)	< 0.0074
VGGish	embedding: 0.533 (0.333 to 0.733)	−0.433 (−0.622 to −0.244)	< 0.0074	embedding: 0.533 (0.333 to 0.733)	−0.433 (−0.622 to −0.244)	< 0.0074
ECAPA-TDNN (speaker)	embedding, 1/2 speed: 0.844 (0.744 to 0.933)	−0.122 (−0.222 to −0.033)	0.0168	embedding, 1/2 speed: 0.844 (0.744 to 0.933)	−0.122 (−0.222 to −0.033)	0.0228
ResNet (speaker)	embedding: 0.878 (0.756 to 0.967)	−0.089 (−0.200 to 0.000)	0.4416	embedding, 1/3 speed: 0.889 (0.822 to 0.944)	−0.078 (−0.133 to −0.033)	0.0252
x-vector (speaker)	embedding, 1/2 speed: 0.767 (0.611 to 0.911)	−0.200 (−0.333 to −0.078)	< 0.0074	embedding, 1/2 speed: 0.767 (0.611 to 0.911)	−0.200 (−0.333 to −0.078)	< 0.0074
wav2vec 2.0 Base	layer 1, 1/3 speed: 0.622 (0.389 to 0.844)	−0.344 (−0.544 to −0.144)	0.0040	layer 0, 1/3 speed: 0.722 (0.556 to 0.889)	−0.244 (−0.389 to −0.100)	0.0050
wav2vec 2.0 Large (robust)	layer 24, 1/2 speed: 0.256 (0.056 to 0.500)	−0.711 (−0.900 to −0.478)	< 0.0074	layer 2, full speed: 0.700 (0.467 to 0.900)	−0.267 (−0.478 to −0.089)	< 0.0074
XLS-R 300M	layer 0, 1/3 speed: 0.756 (0.533 to 0.911)	−0.211 (−0.411 to −0.078)	< 0.0074	layer 6, 1/3 speed: 0.767 (0.478 to 0.978)	−0.200 (−0.456 to −0.011)	0.4820
wav2vec 2.0 Conformer Large	layer 1, full speed: 0.700 (0.467 to 0.911)	−0.267 (−0.489 to −0.078)	0.0072	layer 5, full speed: 0.689 (0.456 to 0.889)	−0.278 (−0.478 to −0.100)	0.0050

Neural network MMS 300M	Fisher rule: layer, re-ID accuracy layer 0, full speed: 0.722 (0.489 to 0.911)	Difference from BirdNET -0.244 (-0.456 to -0.067)	Holm p 0.0072	Held-out rule: layer, re-ID accuracy layer 16, 1/3 speed: 0.767 (0.556 to 0.933)	Difference from BirdNET -0.200 (-0.378 to -0.044)	Holm p 0.1274
HuBERT Base	layer 0, full speed: 0.744 (0.511 to 0.933)	-0.222 (-0.433 to -0.056)	0.0240	layer 0, full speed: 0.744 (0.511 to 0.933)	-0.222 (-0.433 to -0.056)	0.0320
HuBERT Large	layer 2, full speed: 0.656 (0.411 to 0.867)	-0.311 (-0.533 to -0.122)	< 0.0074	layer 0, full speed: 0.733 (0.500 to 0.922)	-0.233 (-0.444 to -0.067)	0.0080
data2vec Base (100 h)	layer 5, full speed: 0.478 (0.267 to 0.689)	-0.489 (-0.667 to -0.300)	< 0.0074	layer 0, full speed: 0.744 (0.522 to 0.911)	-0.222 (-0.422 to -0.067)	0.0050
data2vec Base (960 h)	layer 10, 1/3 speed: 0.422 (0.233 to 0.611)	-0.544 (-0.722 to -0.367)	< 0.0074	layer 0, full speed: 0.744 (0.522 to 0.911)	-0.222 (-0.422 to -0.067)	0.0050
WavLM Base+	layer 0, full speed: 0.756 (0.533 to 0.933)	-0.211 (-0.411 to -0.056)	0.0240	layer 0, full speed: 0.756 (0.533 to 0.933)	-0.211 (-0.411 to -0.056)	0.0320
WavLM Large	layer 0, full speed: 0.711 (0.489 to 0.900)	-0.256 (-0.456 to -0.089)	0.0072	layer 5, full speed: 0.744 (0.467 to 0.956)	-0.222 (-0.467 to -0.022)	0.1274
UniSpeech-SAT Base+	layer 2, 1/3 speed: 0.789 (0.689 to 0.900)	-0.178 (-0.256 to -0.089)	0.0040	layer 1, full speed: 0.767 (0.522 to 0.956)	-0.200 (-0.411 to -0.033)	0.1274
XEUS	layer 0, 1/3 speed: 0.867 (0.744 to 0.978)	-0.100 (-0.200 to 0.000)	0.4256	layer 6, full speed: 0.778 (0.511 to 0.978)	-0.189 (-0.422 to -0.011)	0.4820
Loudness control	summary: 0.233 (0.100 to 0.389)	-0.733 (-0.878 to -0.578)	< 0.0074	summary: 0.233 (0.100 to 0.389)	-0.733 (-0.878 to -0.578)	< 0.0074
Duration control	summary: 0.189 (0.000 to 0.456)	-0.778 (-0.967 to -0.533)	< 0.0074	summary: 0.189 (0.000 to 0.456)	-0.778 (-0.967 to -0.533)	< 0.0074

Zebra finch, group of eight: BirdNET v2.4 0.498 (0.181 to 0.731).

Neural network Perch 2.0	Fisher rule: layer, re-ID accuracy embedding: 0.559 (0.374 to 0.719)	Difference from BirdNET 0.061 (-0.034 to 0.209)	Holm p 1.0000	Held-out rule: layer, re-ID accuracy embedding: 0.559 (0.374 to 0.719)	Difference from BirdNET 0.061 (-0.034 to 0.209)	Holm p 1.0000
Perch	embedding: 0.528 (0.365 to 0.725)	0.031 (-0.093 to 0.199)	1.0000	embedding: 0.528 (0.365 to 0.725)	0.031 (-0.093 to 0.199)	1.0000
SurfPerch	embedding: 0.397 (0.171 to 0.689)	-0.100 (-0.213 to 0.016)	1.0000	embedding: 0.397 (0.171 to 0.689)	-0.100 (-0.213 to 0.016)	1.0000
BirdNET v3 (preview)	embedding: 0.515 (0.347 to 0.692)	0.017 (-0.099 to 0.179)	1.0000	embedding: 0.515 (0.347 to 0.692)	0.017 (-0.099 to 0.179)	1.0000
AVEX sl-BEATs	block 0: 0.201 (0.014 to 0.593)	-0.297 (-0.544 to -0.019)	0.5890	block 11: 0.437 (0.232 to 0.664)	-0.061 (-0.163 to 0.085)	1.0000
AVEX EfficientNet-B0	embedding: 0.485 (0.234 to 0.705)	-0.013 (-0.070 to 0.067)	1.0000	embedding: 0.485 (0.234 to 0.705)	-0.013 (-0.070 to 0.067)	1.0000
AvesEcho	block 7: 0.297 (0.094 to 0.667)	-0.201 (-0.391 to -0.009)	0.8064	block 9: 0.301 (0.105 to 0.630)	-0.197 (-0.360 to -0.036)	< 0.0074
AudioProtoPNet	embedding: 0.576 (0.397 to 0.748)	0.079 (-0.037 to 0.228)	1.0000	embedding: 0.576 (0.397 to 0.748)	0.079 (-0.037 to 0.228)	1.0000
ConvNeXt (BirdSet)	embedding: 0.502 (0.284 to 0.743)	0.004 (-0.095 to 0.128)	1.0000	embedding: 0.502 (0.284 to 0.743)	0.004 (-0.095 to 0.128)	1.0000
Bird-MAE	block 5: 0.188 (0.012 to 0.567)	-0.310 (-0.550 to -0.035)	0.3740	block 0: 0.188 (0.025 to 0.551)	-0.310 (-0.552 to -0.024)	0.5016
ProtoCLR	embedding: 0.118 (0.000 to 0.450)	-0.380 (-0.640 to -0.047)	0.5056	block 2: 0.183 (0.030 to 0.515)	-0.314 (-0.534 to -0.031)	0.8384
RCL FS BSED	embedding: 0.210 (0.013 to 0.627)	-0.288 (-0.539 to -0.004)	1.0000	embedding: 0.210 (0.013 to 0.627)	-0.288 (-0.539 to -0.004)	1.0000
BirdAVES	block 23: 0.245 (0.091 to 0.481)	-0.253 (-0.409 to -0.059)	< 0.0074	block 13: 0.197 (0.092 to 0.381)	-0.301 (-0.494 to -0.043)	0.8580
AVES	block 11: 0.175 (0.061 to 0.406)	-0.323 (-0.550 to -0.038)	0.4422	block 4: 0.157 (0.052 to 0.361)	-0.341 (-0.566 to -0.044)	0.3710
NatureBEATs	block 0: 0.201 (0.014 to 0.593)	-0.297 (-0.544 to -0.019)	0.5890	block 8: 0.397 (0.241 to 0.654)	-0.100 (-0.313 to 0.168)	1.0000
BioLingual	block 5: 0.192 (0.022 to 0.554)	-0.306 (-0.540 to -0.036)	0.1610	block 11: 0.354 (0.130 to 0.680)	-0.144 (-0.314 to 0.107)	1.0000
BEATs	block 5: 0.197 (0.020 to 0.576)	-0.301 (-0.537 to -0.019)	0.5890	block 9: 0.485 (0.250 to 0.690)	-0.013 (-0.149 to 0.104)	1.0000
AudioMAE	block 7: 0.275 (0.077 to 0.681)	-0.223 (-0.470 to 0.068)	1.0000	block 5: 0.223 (0.045 to 0.580)	-0.275 (-0.518 to -0.017)	0.4692
VGGish	embedding: 0.445 (0.114 to 0.681)	-0.052 (-0.182 to 0.010)	1.0000	embedding: 0.445 (0.114 to 0.681)	-0.052 (-0.182 to 0.010)	1.0000
ECAPA-TDNN (speaker)	embedding, 1/2 speed: 0.454 (0.323 to 0.620)	-0.044 (-0.184 to 0.157)	1.0000	embedding: 0.358 (0.222 to 0.535)	-0.140 (-0.291 to 0.076)	1.0000
ResNet (speaker)	embedding: 0.467 (0.314 to 0.720)	-0.031 (-0.180 to 0.155)	1.0000	embedding: 0.467 (0.314 to 0.720)	-0.031 (-0.180 to 0.155)	1.0000
x-vector (speaker)	embedding, 1/2 speed: 0.454 (0.193 to 0.644)	-0.044 (-0.227 to 0.083)	1.0000	embedding, 1/2 speed: 0.454 (0.193 to 0.644)	-0.044 (-0.227 to 0.083)	1.0000

Neural network wav2vec 2.0 Base	Fisher rule: layer, re-ID accuracy layer 0, 1/2 speed: 0.380 (0.230 to 0.669)	Difference from BirdNET −0.118 (−0.326 to 0.116)	Holm p 1.0000	Held-out rule: layer, re-ID accuracy layer 12, 1/3 speed: 0.262 (0.126 to 0.504)	Difference from BirdNET −0.236 (−0.423 to 0.007)	Holm p 1.0000
wav2vec 2.0 Large (robust)	layer 0, 1/2 speed: 0.310 (0.104 to 0.711)	−0.188 (−0.475 to 0.124)	1.0000	layer 21, 1/2 speed: 0.240 (0.059 to 0.586)	−0.258 (−0.535 to 0.048)	1.0000
XLS-R 300M	layer 0, 1/2 speed: 0.367 (0.159 to 0.763)	−0.131 (−0.427 to 0.210)	1.0000	layer 21, 1/3 speed: 0.380 (0.220 to 0.671)	−0.118 (−0.330 to 0.110)	1.0000
wav2vec 2.0 Conformer Large	layer 0, 1/3 speed: 0.406 (0.214 to 0.748)	−0.092 (−0.290 to 0.183)	1.0000	layer 0, 1/3 speed: 0.406 (0.214 to 0.748)	−0.092 (−0.290 to 0.183)	1.0000
MMS 300M	layer 0, 1/2 speed: 0.402 (0.205 to 0.780)	−0.096 (−0.373 to 0.222)	1.0000	layer 23, 1/2 speed: 0.223 (0.054 to 0.543)	−0.275 (−0.551 to 0.040)	1.0000
HuBERT Base	layer 0, 1/3 speed: 0.376 (0.202 to 0.698)	−0.122 (−0.318 to 0.124)	1.0000	layer 7, 1/2 speed: 0.271 (0.121 to 0.554)	−0.227 (−0.454 to 0.070)	1.0000
HuBERT Large	layer 18, 1/3 speed: 0.218 (0.038 to 0.562)	−0.279 (−0.552 to 0.054)	1.0000	layer 13, full speed: 0.306 (0.110 to 0.548)	−0.192 (−0.383 to 0.034)	1.0000
data2vec Base (100 h)	layer 3, full speed: 0.480 (0.215 to 0.658)	−0.017 (−0.245 to 0.164)	1.0000	layer 3, 1/2 speed: 0.432 (0.197 to 0.646)	−0.066 (−0.213 to 0.064)	1.0000
data2vec Base (960 h)	layer 4, 1/2 speed: 0.371 (0.068 to 0.628)	−0.127 (−0.260 to −0.045)	< 0.0074	layer 12, 1/3 speed: 0.170 (0.034 to 0.474)	−0.328 (−0.549 to −0.040)	0.8384
WavLM Base+	layer 0, 1/3 speed: 0.397 (0.237 to 0.683)	−0.100 (−0.285 to 0.117)	1.0000	layer 5, full speed: 0.384 (0.237 to 0.591)	−0.114 (−0.333 to 0.196)	1.0000
WavLM Large	layer 0, 1/3 speed: 0.314 (0.166 to 0.622)	−0.183 (−0.420 to 0.098)	1.0000	layer 13, full speed: 0.476 (0.237 to 0.683)	−0.022 (−0.198 to 0.153)	1.0000
UniSpeech-SAT Base+	layer 0, 1/3 speed: 0.410 (0.247 to 0.702)	−0.087 (−0.271 to 0.124)	1.0000	layer 5, full speed: 0.476 (0.189 to 0.698)	−0.022 (−0.211 to 0.106)	1.0000
XEUS	layer 0, full speed: 0.445 (0.317 to 0.660)	−0.052 (−0.230 to 0.174)	1.0000	layer 18, 1/2 speed: 0.284 (0.091 to 0.549)	−0.214 (−0.341 to −0.065)	< 0.0074
Loudness control	summary: 0.210 (0.014 to 0.608)	−0.288 (−0.565 to 0.079)	1.0000	summary: 0.210 (0.014 to 0.608)	−0.288 (−0.565 to 0.079)	1.0000
Duration control	summary: 0.284 (0.000 to 0.572)	−0.214 (−0.661 to 0.000)	1.0000	summary: 0.284 (0.000 to 0.572)	−0.214 (−0.661 to 0.000)	1.0000

Great tit, across year. BirdNET v2.4 0.444 (0.319 to 0.569).

Neural network Perch 2.0	Fisher rule: layer, re-ID accuracy embedding: 0.450 (0.312 to 0.588)	Difference from BirdNET 0.006 (−0.094 to 0.100)	Holm p 0.9264	Held-out rule: layer, re-ID accuracy embedding: 0.450 (0.312 to 0.588)	Difference from BirdNET 0.006 (−0.094 to 0.100)	Holm p 1.0000
Perch	embedding: 0.294 (0.194 to 0.400)	−0.150 (−0.256 to −0.056)	0.0112	embedding: 0.294 (0.194 to 0.400)	−0.150 (−0.256 to −0.056)	0.0116
SurfPerch	embedding: 0.375 (0.275 to 0.475)	−0.069 (−0.131 to −0.012)	0.0968	embedding: 0.375 (0.275 to 0.475)	−0.069 (−0.131 to −0.012)	0.1210
BirdNET v3 (preview)	embedding: 0.156 (0.106 to 0.206)	−0.287 (−0.437 to −0.144)	< 0.0074	embedding: 0.156 (0.106 to 0.206)	−0.287 (−0.437 to −0.144)	< 0.0074
AVEX sl-BEATS	block 6: 0.306 (0.200 to 0.412)	−0.137 (−0.244 to −0.037)	0.0768	block 9: 0.331 (0.219 to 0.450)	−0.112 (−0.206 to −0.019)	0.1164
AVEX EfficientNet-B0	embedding: 0.338 (0.225 to 0.456)	−0.106 (−0.231 to 0.000)	0.1188	embedding: 0.338 (0.225 to 0.456)	−0.106 (−0.231 to 0.000)	0.1782
AvesEcho	block 5: 0.119 (0.050 to 0.206)	−0.325 (−0.450 to −0.181)	< 0.0074	embedding: 0.438 (0.294 to 0.581)	−0.006 (−0.069 to 0.056)	1.0000
AudioProtoPNet	embedding: 0.294 (0.194 to 0.394)	−0.150 (−0.269 to −0.044)	0.0616	embedding: 0.294 (0.194 to 0.394)	−0.150 (−0.269 to −0.044)	0.0616
ConvNeXt (BirdSet)	embedding: 0.325 (0.200 to 0.463)	−0.119 (−0.225 to −0.025)	0.0768	embedding: 0.325 (0.200 to 0.463)	−0.119 (−0.225 to −0.025)	0.0884
Bird-MAE	block 12: 0.138 (0.044 to 0.250)	−0.306 (−0.469 to −0.131)	0.0192	block 9: 0.125 (0.037 to 0.231)	−0.319 (−0.481 to −0.144)	0.0116
ProtoCLR	embedding: 0.113 (0.031 to 0.212)	−0.331 (−0.500 to −0.150)	0.0192	embedding: 0.113 (0.031 to 0.212)	−0.331 (−0.500 to −0.150)	0.0192
RCL FS BSIED	embedding: 0.219 (0.094 to 0.362)	−0.225 (−0.338 to −0.106)	0.0112	embedding: 0.219 (0.094 to 0.362)	−0.225 (−0.338 to −0.106)	0.0116
BirdAVES	block 0: 0.287 (0.169 to 0.419)	−0.156 (−0.269 to −0.050)	0.0540	block 0: 0.287 (0.169 to 0.419)	−0.156 (−0.269 to −0.050)	0.0540
AVES	block 0: 0.225 (0.113 to 0.344)	−0.219 (−0.331 to −0.106)	0.0060	block 0: 0.225 (0.113 to 0.344)	−0.219 (−0.331 to −0.106)	0.0062
NatureBEATS	block 1: 0.200 (0.106 to 0.306)	−0.244 (−0.394 to −0.088)	0.0294	block 9: 0.244 (0.106 to 0.394)	−0.200 (−0.319 to −0.081)	0.0342
BioLingual	block 4: 0.188 (0.094 to 0.294)	−0.256 (−0.400 to −0.119)	0.0060	embedding: 0.319 (0.206 to 0.444)	−0.125 (−0.206 to −0.050)	0.0220

Neural network	Fisher rule: layer, re-ID accuracy	Difference from BirdNET	Holm p	Held-out rule: layer, re-ID accuracy	Difference from BirdNET	Holm p
BEATs	block 9: 0.300 (0.200 to 0.400)	-0.144 (-0.250 to -0.044)	0.0768	block 8: 0.231 (0.125 to 0.344)	-0.212 (-0.344 to -0.081)	0.0192
AudioMAE	block 3: 0.100 (0.025 to 0.194)	-0.344 (-0.500 to -0.175)	< 0.0074	block 11: 0.212 (0.131 to 0.294)	-0.231 (-0.350 to -0.125)	< 0.0074
VGGish	embedding: 0.169 (0.087 to 0.250)	-0.275 (-0.394 to -0.150)	< 0.0074	embedding: 0.169 (0.087 to 0.250)	-0.275 (-0.394 to -0.150)	< 0.0074
ECAPA-TDNN (speaker)	embedding, 1/3 speed: 0.256 (0.150 to 0.375)	-0.188 (-0.331 to -0.069)	0.0112	embedding, 1/3 speed: 0.256 (0.150 to 0.375)	-0.188 (-0.331 to -0.069)	0.0116
ResNet (speaker)	embedding, 1/3 speed: 0.256 (0.175 to 0.344)	-0.188 (-0.312 to -0.069)	0.0396	embedding, 1/3 speed: 0.256 (0.175 to 0.344)	-0.188 (-0.312 to -0.069)	0.0396
x-vector (speaker)	embedding, 1/3 speed: 0.206 (0.106 to 0.319)	-0.237 (-0.344 to -0.131)	< 0.0074	embedding, 1/2 speed: 0.194 (0.113 to 0.300)	-0.250 (-0.356 to -0.150)	< 0.0074
wav2vec 2.0 Base	layer 0, full speed: 0.231 (0.113 to 0.369)	-0.212 (-0.344 to -0.081)	0.0342	layer 2, full speed: 0.231 (0.119 to 0.356)	-0.212 (-0.325 to -0.100)	0.0062
wav2vec 2.0 Large (robust)	layer 0, full speed: 0.219 (0.113 to 0.331)	-0.225 (-0.338 to -0.100)	0.0294	layer 0, full speed: 0.219 (0.113 to 0.331)	-0.225 (-0.338 to -0.100)	0.0294
XLS-R 300M	layer 0, 1/3 speed: 0.181 (0.087 to 0.287)	-0.262 (-0.394 to -0.119)	0.0192	layer 1, full speed: 0.231 (0.125 to 0.338)	-0.212 (-0.331 to -0.094)	0.0150
wav2vec 2.0 Conformer Large	layer 1, full speed: 0.237 (0.138 to 0.350)	-0.206 (-0.325 to -0.075)	0.0448	layer 1, full speed: 0.237 (0.138 to 0.350)	-0.206 (-0.325 to -0.075)	0.0448
MMS 300M	layer 0, 1/3 speed: 0.244 (0.144 to 0.350)	-0.200 (-0.319 to -0.062)	0.0650	layer 0, full speed: 0.269 (0.169 to 0.381)	-0.175 (-0.294 to -0.050)	0.0884
HuBERT Base	layer 0, full speed: 0.269 (0.144 to 0.406)	-0.175 (-0.306 to -0.044)	0.0768	layer 0, full speed: 0.269 (0.144 to 0.406)	-0.175 (-0.306 to -0.044)	0.0884
HuBERT Large	layer 0, full speed: 0.269 (0.163 to 0.381)	-0.175 (-0.294 to -0.044)	0.0768	layer 0, full speed: 0.269 (0.163 to 0.381)	-0.175 (-0.294 to -0.044)	0.0884
data2vec Base (100 h)	layer 0, full speed: 0.263 (0.156 to 0.381)	-0.181 (-0.306 to -0.050)	0.0768	layer 0, full speed: 0.263 (0.156 to 0.381)	-0.181 (-0.306 to -0.050)	0.0884
data2vec Base (960 h)	layer 0, full speed: 0.281 (0.175 to 0.400)	-0.162 (-0.300 to -0.019)	0.0968	layer 0, full speed: 0.281 (0.175 to 0.400)	-0.162 (-0.300 to -0.019)	0.1210
WavLM Base+	layer 0, full speed: 0.250 (0.125 to 0.388)	-0.194 (-0.306 to -0.075)	0.0396	layer 0, full speed: 0.250 (0.125 to 0.388)	-0.194 (-0.306 to -0.075)	0.0396
WavLM Large	layer 0, full speed: 0.287 (0.181 to 0.394)	-0.156 (-0.269 to -0.037)	0.0768	layer 2, full speed: 0.250 (0.131 to 0.381)	-0.194 (-0.306 to -0.081)	0.0294
UniSpeech-SAT Base+	layer 0, full speed: 0.281 (0.156 to 0.412)	-0.162 (-0.281 to -0.038)	0.0768	layer 0, full speed: 0.281 (0.156 to 0.412)	-0.162 (-0.281 to -0.038)	0.0884
XEUS	layer 0, full speed: 0.231 (0.119 to 0.350)	-0.212 (-0.325 to -0.100)	0.0112	layer 2, full speed: 0.275 (0.150 to 0.412)	-0.169 (-0.287 to -0.044)	0.0884
Loudness control	summary: 0.113 (0.031 to 0.206)	-0.331 (-0.494 to -0.169)	< 0.0074	summary: 0.113 (0.031 to 0.206)	-0.331 (-0.494 to -0.169)	< 0.0074
Duration control	summary: 0.069 (0.000 to 0.175)	-0.375 (-0.537 to -0.200)	< 0.0074	summary: 0.069 (0.000 to 0.175)	-0.375 (-0.537 to -0.200)	< 0.0074

Tree pipit, across year: BirdNET v2.4 0.214 (0.117 to 0.336).

Neural network	Fisher rule: layer, re-ID accuracy	Difference from BirdNET	Holm p	Held-out rule: layer, re-ID accuracy	Difference from BirdNET	Holm p
Perch 2.0	embedding: 0.447 (0.294 to 0.625)	0.233 (0.149 to 0.310)	< 0.0074	n/a	n/a	n/a
Perch	embedding: 0.351 (0.219 to 0.517)	0.137 (0.064 to 0.207)	0.0204	n/a	n/a	n/a
SurfPerch	embedding: 0.243 (0.153 to 0.358)	0.029 (-0.015 to 0.083)	1.0000	n/a	n/a	n/a
BirdNET v3 (preview)	embedding: 0.160 (0.096 to 0.228)	-0.054 (-0.173 to 0.035)	1.0000	n/a	n/a	n/a
AVEX sl-BEATs	block 0: 0.096 (0.059 to 0.130)	-0.118 (-0.241 to -0.015)	0.3144	n/a	n/a	n/a
AVEX EfficientNet-B0	embedding: 0.211 (0.114 to 0.319)	-0.003 (-0.091 to 0.091)	1.0000	n/a	n/a	n/a

Neural network	Fisher rule: layer, re-ID accuracy	Difference from BirdNET	Holm p	Held-out rule: layer, re-ID accuracy	Difference from BirdNET	Holm p
AvesEcho	block 0: 0.019 (0.000 to 0.039)	-0.195 (-0.325 to -0.091)	< 0.0074	n/a	n/a	n/a
AudioProtoPNet	embedding: 0.201 (0.102 to 0.317)	-0.013 (-0.070 to 0.036)	1.0000	n/a	n/a	n/a
ConvNeXt (BirdSet)	embedding: 0.288 (0.138 to 0.464)	0.073 (-0.023 to 0.156)	1.0000	n/a	n/a	n/a
Bird-MAE	block 0: 0.032 (0.000 to 0.096)	-0.182 (-0.292 to -0.100)	< 0.0074	n/a	n/a	n/a
ProtoCLR	embedding: 0.083 (0.030 to 0.156)	-0.131 (-0.261 to 0.000)	0.4950	n/a	n/a	n/a
RCL FS BSED	embedding: 0.045 (0.017 to 0.071)	-0.169 (-0.283 to -0.080)	< 0.0074	n/a	n/a	n/a
BirdAVES	block 0: 0.083 (0.042 to 0.117)	-0.131 (-0.252 to -0.030)	0.1230	n/a	n/a	n/a
AVES	block 0: 0.038 (0.016 to 0.060)	-0.176 (-0.296 to -0.084)	< 0.0074	n/a	n/a	n/a
NatureBEATs	block 0: 0.070 (0.037 to 0.099)	-0.144 (-0.272 to -0.038)	0.0864	n/a	n/a	n/a
BioLingual	block 0: 0.058 (0.000 to 0.133)	-0.157 (-0.278 to -0.028)	0.2782	n/a	n/a	n/a
BEATs	block 9: 0.086 (0.044 to 0.129)	-0.128 (-0.241 to -0.041)	< 0.0074	n/a	n/a	n/a
AudioMAE	block 0: 0.045 (0.000 to 0.137)	-0.169 (-0.318 to -0.015)	0.3916	n/a	n/a	n/a
VGGish	embedding: 0.070 (0.034 to 0.109)	-0.144 (-0.249 to -0.062)	< 0.0074	n/a	n/a	n/a
ECAPA-TDNN (speaker)	embedding, 1/3 speed: 0.214 (0.116 to 0.343)	0.000 (-0.100 to 0.098)	1.0000	n/a	n/a	n/a
ResNet (speaker)	embedding, 1/3 speed: 0.137 (0.065 to 0.228)	-0.077 (-0.177 to -0.003)	0.4200	n/a	n/a	n/a
x-vector (speaker)	embedding, 1/3 speed: 0.195 (0.081 to 0.351)	-0.019 (-0.127 to 0.097)	1.0000	n/a	n/a	n/a
wav2vec 2.0 Base	layer 1, 1/3 speed: 0.064 (0.000 to 0.168)	-0.150 (-0.239 to -0.081)	< 0.0074	n/a	n/a	n/a
wav2vec 2.0 Large (robust)	layer 0, 1/3 speed: 0.032 (0.006 to 0.062)	-0.182 (-0.306 to -0.089)	< 0.0074	n/a	n/a	n/a
XLS-R 300M	layer 0, 1/3 speed: 0.022 (0.000 to 0.054)	-0.192 (-0.322 to -0.096)	< 0.0074	n/a	n/a	n/a
wav2vec 2.0 Conformer Large	layer 0, 1/3 speed: 0.026 (0.000 to 0.057)	-0.188 (-0.318 to -0.092)	< 0.0074	n/a	n/a	n/a
MMS 300M	layer 0, 1/3 speed: 0.019 (0.000 to 0.045)	-0.195 (-0.321 to -0.100)	< 0.0074	n/a	n/a	n/a
HuBERT Base	layer 0, 1/3 speed: 0.013 (0.000 to 0.026)	-0.201 (-0.317 to -0.108)	< 0.0074	n/a	n/a	n/a
HuBERT Large	layer 0, 1/3 speed: 0.032 (0.000 to 0.070)	-0.182 (-0.314 to -0.084)	< 0.0074	n/a	n/a	n/a
data2vec Base (100 h)	layer 0, 1/3 speed: 0.029 (0.004 to 0.059)	-0.185 (-0.310 to -0.092)	< 0.0074	n/a	n/a	n/a
data2vec Base (960 h)	layer 0, 1/3 speed: 0.029 (0.009 to 0.051)	-0.185 (-0.307 to -0.091)	< 0.0074	n/a	n/a	n/a
WavLM Base+	layer 0, 1/3 speed: 0.000 (0.000 to 0.000)	-0.214 (-0.336 to -0.117)	< 0.0074	n/a	n/a	n/a
WavLM Large	layer 0, 1/3 speed: 0.026 (0.000 to 0.054)	-0.188 (-0.317 to -0.091)	< 0.0074	n/a	n/a	n/a
UniSpeech-SAT Base+	layer 0, 1/3 speed: 0.016 (0.003 to 0.030)	-0.198 (-0.314 to -0.105)	< 0.0074	n/a	n/a	n/a
XEUS	layer 0, 1/3 speed: 0.010 (0.000 to 0.019)	-0.204 (-0.323 to -0.109)	< 0.0074	n/a	n/a	n/a
Loudness control	summary: 0.073 (0.018 to 0.138)	-0.141 (-0.261 to -0.033)	0.1288	n/a	n/a	n/a
Duration control	summary: 0.083 (0.000 to 0.234)	-0.131 (-0.307 to 0.079)	1.0000	n/a	n/a	n/a

Chiffchaff, across year: BirdNET v2.4 0.160 (0.035 to 0.330).

Neural network	Fisher rule: layer, re-ID accuracy	Difference from BirdNET	Holm p	Held-out rule: layer, re-ID accuracy	Difference from BirdNET	Holm p
Perch 2.0	embedding: 0.160 (0.034 to 0.289)	0.000 (−0.091 to 0.087)	1.0000	n/a	n/a	n/a
Perch	embedding: 0.150 (0.044 to 0.277)	−0.010 (−0.078 to 0.068)	1.0000	n/a	n/a	n/a
SurfPerch	embedding: 0.255 (0.103 to 0.421)	0.095 (0.015 to 0.192)	0.6808	n/a	n/a	n/a
BirdNET v3 (preview)	embedding: 0.085 (0.033 to 0.151)	−0.075 (−0.245 to 0.077)	1.0000	n/a	n/a	n/a
AVEX sl-BEATs	block 9: 0.210 (0.074 to 0.361)	0.050 (−0.026 to 0.119)	1.0000	n/a	n/a	n/a
AVEX EfficientNet-B0	embedding: 0.120 (0.036 to 0.234)	−0.040 (−0.116 to 0.033)	1.0000	n/a	n/a	n/a
AvesEcho	block 6: 0.180 (0.012 to 0.382)	0.020 (−0.058 to 0.097)	1.0000	n/a	n/a	n/a
AudioProtoPNet	embedding: 0.115 (0.047 to 0.197)	−0.045 (−0.155 to 0.054)	1.0000	n/a	n/a	n/a
ConvNeXt (BirdSet)	embedding: 0.135 (0.034 to 0.262)	−0.025 (−0.098 to 0.051)	1.0000	n/a	n/a	n/a
Bird-MAE	block 11: 0.080 (0.014 to 0.173)	−0.080 (−0.177 to 0.019)	1.0000	n/a	n/a	n/a
ProtoCLR	embedding: 0.195 (0.041 to 0.379)	0.035 (−0.141 to 0.222)	1.0000	n/a	n/a	n/a
RCL FS BSED	embedding: 0.150 (0.016 to 0.312)	−0.010 (−0.083 to 0.064)	1.0000	n/a	n/a	n/a
BirdAVES	block 0: 0.190 (0.022 to 0.400)	0.030 (−0.056 to 0.111)	1.0000	n/a	n/a	n/a
AVES	block 0: 0.205 (0.000 to 0.414)	0.045 (−0.056 to 0.141)	1.0000	n/a	n/a	n/a
NatureBEATs	block 9: 0.160 (0.033 to 0.302)	0.000 (−0.065 to 0.065)	1.0000	n/a	n/a	n/a
BioLingual	embedding: 0.160 (0.031 to 0.289)	0.000 (−0.086 to 0.078)	1.0000	n/a	n/a	n/a
BEATs	block 9: 0.155 (0.043 to 0.287)	−0.005 (−0.070 to 0.053)	1.0000	n/a	n/a	n/a
AudioMAE	embedding: 0.150 (0.046 to 0.279)	−0.010 (−0.071 to 0.045)	1.0000	n/a	n/a	n/a
VGGish	embedding: 0.095 (0.000 to 0.275)	−0.065 (−0.126 to −0.005)	1.0000	n/a	n/a	n/a
ECAPA-TDNN (speaker)	embedding, 1/3 speed: 0.150 (0.054 to 0.275)	−0.010 (−0.090 to 0.075)	1.0000	n/a	n/a	n/a
ResNet (speaker)	embedding, 1/3 speed: 0.145 (0.046 to 0.289)	−0.015 (−0.166 to 0.140)	1.0000	n/a	n/a	n/a
x-vector (speaker)	embedding, 1/3 speed: 0.140 (0.033 to 0.313)	−0.020 (−0.084 to 0.040)	1.0000	n/a	n/a	n/a
wav2vec 2.0 Base	layer 1, full speed: 0.165 (0.000 to 0.380)	0.005 (−0.077 to 0.093)	1.0000	n/a	n/a	n/a
wav2vec 2.0 Large (robust)	layer 0, full speed: 0.210 (0.017 to 0.426)	0.050 (−0.045 to 0.168)	1.0000	n/a	n/a	n/a
XLS-R 300M	layer 0, full speed: 0.235 (0.057 to 0.439)	0.075 (−0.022 to 0.176)	1.0000	n/a	n/a	n/a
wav2vec 2.0 Conformer Large	layer 0, full speed: 0.205 (0.000 to 0.389)	0.045 (−0.067 to 0.161)	1.0000	n/a	n/a	n/a
MMS 300M	layer 0, full speed: 0.220 (0.034 to 0.433)	0.060 (−0.027 to 0.156)	1.0000	n/a	n/a	n/a
HuBERT Base	layer 0, full speed: 0.125 (0.000 to 0.344)	−0.035 (−0.104 to 0.036)	1.0000	n/a	n/a	n/a
HuBERT Large	layer 0, full speed: 0.215 (0.000 to 0.439)	0.055 (−0.052 to 0.176)	1.0000	n/a	n/a	n/a
data2vec Base (100 h)	layer 0, full speed: 0.180 (0.000 to 0.392)	0.020 (−0.067 to 0.123)	1.0000	n/a	n/a	n/a
data2vec Base (960 h)	layer 0, full speed: 0.180 (0.000 to 0.386)	0.020 (−0.065 to 0.112)	1.0000	n/a	n/a	n/a

Neural network WavLM Base+	Fisher rule: layer, re-ID accuracy layer 0, full speed: 0.125 (0.000 to 0.344)	Difference from BirdNET −0.035 (−0.104 to 0.036)	Holm p 1.0000	Held-out rule: layer, re-ID accuracy n/a	Difference from BirdNET n/a	Holm p n/a
WavLM Large	layer 0, 1/2 speed: 0.200 (0.029 to 0.384)	0.040 (−0.043 to 0.129)	1.0000	n/a	n/a	n/a
UniSpeech-SAT Base+	layer 0, full speed: 0.130 (0.000 to 0.346)	−0.030 (−0.096 to 0.038)	1.0000	n/a	n/a	n/a
XEUS	layer 0, 1/2 speed: 0.205 (0.017 to 0.407)	0.045 (−0.059 to 0.173)	1.0000	n/a	n/a	n/a
Loudness control	summary: 0.025 (0.000 to 0.058)	−0.135 (−0.308 to 0.000)	1.0000	n/a	n/a	n/a
Duration control	summary: 0.105 (0.000 to 0.298)	−0.055 (−0.271 to 0.156)	1.0000	n/a	n/a	n/a

Tree pipit, within year: BirdNET v2.4 0.376 (0.195 to 0.564).

Neural network Perch 2.0	Fisher rule: layer, re-ID accuracy embedding: 0.548 (0.361 to 0.738)	Difference from BirdNET 0.172 (0.103 to 0.251)	Holm p < 0.0074	Held-out rule: layer, re-ID accuracy n/a	Difference from BirdNET n/a	Holm p n/a
Perch	embedding: 0.426 (0.230 to 0.628)	0.050 (−0.054 to 0.171)	1.0000	n/a	n/a	n/a
SurfPerch	embedding: 0.455 (0.265 to 0.634)	0.079 (−0.010 to 0.171)	1.0000	n/a	n/a	n/a
BirdNET v3 (preview)	embedding: 0.257 (0.128 to 0.417)	−0.119 (−0.304 to 0.054)	1.0000	n/a	n/a	n/a
AVEX sl-BEATs	block 0: 0.162 (0.056 to 0.286)	−0.215 (−0.377 to −0.048)	0.3696	n/a	n/a	n/a
AVEX EfficientNet-B0	embedding: 0.281 (0.111 to 0.475)	−0.096 (−0.244 to 0.051)	1.0000	n/a	n/a	n/a
AvesEcho	block 0: 0.135 (0.015 to 0.329)	−0.241 (−0.469 to −0.025)	0.8004	n/a	n/a	n/a
AudioProtoPNet	embedding: 0.442 (0.239 to 0.657)	0.066 (−0.087 to 0.237)	1.0000	n/a	n/a	n/a
ConvNeXt (BirdSet)	embedding: 0.502 (0.298 to 0.704)	0.125 (0.012 to 0.265)	0.8456	n/a	n/a	n/a
Bird-MAE	block 0: 0.069 (0.020 to 0.134)	−0.307 (−0.511 to −0.108)	0.0748	n/a	n/a	n/a
ProtoCLR	embedding: 0.152 (0.025 to 0.308)	−0.224 (−0.448 to −0.012)	1.0000	n/a	n/a	n/a
RCL FS BSED	embedding: 0.172 (0.054 to 0.337)	−0.205 (−0.441 to 0.051)	1.0000	n/a	n/a	n/a
BirdAVES	block 0: 0.300 (0.105 to 0.515)	−0.076 (−0.269 to 0.157)	1.0000	n/a	n/a	n/a
AVES	block 0: 0.228 (0.065 to 0.429)	−0.149 (−0.343 to 0.100)	1.0000	n/a	n/a	n/a
NatureBEATs	block 0: 0.182 (0.085 to 0.281)	−0.195 (−0.350 to −0.028)	0.6120	n/a	n/a	n/a
BioLingual	block 0: 0.112 (0.014 to 0.228)	−0.264 (−0.427 to −0.116)	< 0.0074	n/a	n/a	n/a
BEATs	block 9: 0.211 (0.096 to 0.329)	−0.165 (−0.308 to −0.033)	0.4340	n/a	n/a	n/a
AudioMAE	block 0: 0.125 (0.023 to 0.241)	−0.251 (−0.412 to −0.099)	< 0.0074	n/a	n/a	n/a
VGGish	embedding: 0.116 (0.028 to 0.216)	−0.261 (−0.489 to −0.052)	0.3776	n/a	n/a	n/a
ECAPA-TDNN (speaker)	embedding, 1/3 speed: 0.356 (0.167 to 0.567)	−0.020 (−0.189 to 0.142)	1.0000	n/a	n/a	n/a
ResNet (speaker)	embedding, 1/3 speed: 0.264 (0.130 to 0.424)	−0.112 (−0.259 to 0.036)	1.0000	n/a	n/a	n/a
x-vector (speaker)	embedding, 1/3 speed: 0.304 (0.117 to 0.517)	−0.073 (−0.243 to 0.106)	1.0000	n/a	n/a	n/a
wav2vec 2.0 Base	layer 1, 1/3 speed: 0.231 (0.061 to 0.426)	−0.145 (−0.326 to 0.084)	1.0000	n/a	n/a	n/a
wav2vec 2.0 Large (robust)	layer 0, 1/3 speed: 0.168 (0.006 to 0.400)	−0.208 (−0.474 to 0.090)	1.0000	n/a	n/a	n/a
XLS-R 300M	layer 0, 1/3 speed: 0.135 (0.015 to 0.329)	−0.241 (−0.477 to 0.043)	1.0000	n/a	n/a	n/a

Neural network wav2vec 2.0 Conformer Large	Fisher rule: layer, re-ID accuracy layer 0, 1/3 speed: 0.149 (0.021 to 0.346)	Difference from BirdNET −0.228 (−0.466 to 0.058)	Holm p 1.0000	Held-out rule: layer, re-ID accuracy n/a	Difference from BirdNET n/a	Holm p n/a
MMS 300M	layer 0, 1/3 speed: 0.125 (0.016 to 0.316)	−0.251 (−0.486 to 0.036)	1.0000	n/a	n/a	n/a
HuBERT Base	layer 0, 1/3 speed: 0.191 (0.036 to 0.393)	−0.185 (−0.414 to 0.081)	1.0000	n/a	n/a	n/a
HuBERT Large	layer 0, 1/3 speed: 0.188 (0.016 to 0.429)	−0.188 (−0.457 to 0.099)	1.0000	n/a	n/a	n/a
data2vec Base (100 h)	layer 0, 1/3 speed: 0.182 (0.040 to 0.380)	−0.195 (−0.417 to 0.077)	1.0000	n/a	n/a	n/a
data2vec Base (960 h)	layer 0, 1/3 speed: 0.172 (0.041 to 0.360)	−0.205 (−0.426 to 0.062)	1.0000	n/a	n/a	n/a
WavLM Base+	layer 0, 1/3 speed: 0.158 (0.025 to 0.355)	−0.218 (−0.445 to 0.059)	1.0000	n/a	n/a	n/a
WavLM Large	layer 0, 1/3 speed: 0.165 (0.009 to 0.386)	−0.211 (−0.472 to 0.070)	1.0000	n/a	n/a	n/a
UniSpeech-SAT Base+	layer 0, 1/3 speed: 0.145 (0.018 to 0.329)	−0.231 (−0.452 to 0.033)	1.0000	n/a	n/a	n/a
XEUS	layer 0, 1/3 speed: 0.139 (0.011 to 0.333)	−0.238 (−0.481 to 0.047)	1.0000	n/a	n/a	n/a
Loudness control	summary: 0.228 (0.032 to 0.461)	−0.149 (−0.407 to 0.140)	1.0000	n/a	n/a	n/a
Duration control	summary: 0.145 (0.000 to 0.334)	−0.231 (−0.490 to 0.023)	1.0000	n/a	n/a	n/a

Little owl, across year: BirdNET v2.4 0.501 (0.372 to 0.631).

Neural network Perch 2.0	Fisher rule: layer, re-ID accuracy embedding: 0.651 (0.494 to 0.786)	Difference from BirdNET 0.150 (0.044 to 0.244)	Holm p 0.1932	Held-out rule: layer, re-ID accuracy n/a	Difference from BirdNET n/a	Holm p n/a
Perch	embedding: 0.646 (0.501 to 0.775)	0.145 (0.057 to 0.229)	0.0432	n/a	n/a	n/a
SurfPerch	embedding: 0.442 (0.291 to 0.593)	−0.059 (−0.218 to 0.097)	1.0000	n/a	n/a	n/a
BirdNET v3 (preview)	embedding: 0.373 (0.237 to 0.523)	−0.128 (−0.258 to 0.003)	0.4944	n/a	n/a	n/a
AVEX sl-BEATs	block 2: 0.194 (0.077 to 0.331)	−0.307 (−0.446 to −0.168)	< 0.0074	n/a	n/a	n/a
AVEX EfficientNet-B0	embedding: 0.373 (0.237 to 0.522)	−0.128 (−0.253 to −0.003)	0.4944	n/a	n/a	n/a
AvesEcho	block 0: 0.138 (0.029 to 0.266)	−0.364 (−0.532 to −0.181)	0.0168	n/a	n/a	n/a
AudioProtoPNet	embedding: 0.631 (0.472 to 0.782)	0.130 (0.033 to 0.246)	0.0920	n/a	n/a	n/a
ConvNeXt (BirdSet)	embedding: 0.624 (0.479 to 0.754)	0.123 (0.006 to 0.229)	0.4944	n/a	n/a	n/a
Bird-MAE	embedding: 0.251 (0.103 to 0.412)	−0.251 (−0.453 to −0.036)	0.3424	n/a	n/a	n/a
ProtoCLR	embedding: 0.172 (0.067 to 0.305)	−0.329 (−0.467 to −0.182)	< 0.0074	n/a	n/a	n/a
RCL_FS_BSED	embedding: 0.189 (0.071 to 0.331)	−0.312 (−0.495 to −0.111)	0.0864	n/a	n/a	n/a
BirdAVES	block 23: 0.285 (0.155 to 0.427)	−0.216 (−0.346 to −0.089)	0.0120	n/a	n/a	n/a
AVES	block 0: 0.285 (0.160 to 0.406)	−0.216 (−0.373 to −0.071)	0.0600	n/a	n/a	n/a
NatureBEATs	block 2: 0.241 (0.131 to 0.357)	−0.260 (−0.415 to −0.098)	0.0468	n/a	n/a	n/a
BioLingual	block 1: 0.133 (0.021 to 0.256)	−0.369 (−0.547 to −0.172)	0.0120	n/a	n/a	n/a
BEATs	block 2: 0.216 (0.087 to 0.369)	−0.285 (−0.447 to −0.091)	0.1364	n/a	n/a	n/a
AudioMAE	block 3: 0.170 (0.064 to 0.292)	−0.332 (−0.478 to −0.176)	< 0.0074	n/a	n/a	n/a
VGGish	embedding: 0.231 (0.098 to 0.379)	−0.270 (−0.398 to −0.157)	< 0.0074	n/a	n/a	n/a

Neural network ECAPA-TDNN (speaker)	Fisher rule: layer, re-ID accuracy embedding, 1/3 speed: 0.437 (0.279 to 0.595)	Difference from BirdNET −0.064 (−0.255 to 0.123)	Holm p 1.0000	Held-out rule: layer, re-ID accuracy n/a	Difference from BirdNET n/a	Holm p n/a
ResNet (speaker)	embedding, 1/3 speed: 0.437 (0.280 to 0.584)	−0.064 (−0.251 to 0.115)	1.0000	n/a	n/a	n/a
x-vector (speaker)	embedding, 1/3 speed: 0.393 (0.252 to 0.532)	−0.108 (−0.299 to 0.085)	1.0000	n/a	n/a	n/a
wav2vec 2.0 Base	layer 0, full speed: 0.263 (0.104 to 0.442)	−0.238 (−0.412 to −0.050)	0.2840	n/a	n/a	n/a
wav2vec 2.0 Large (robust)	layer 0, full speed: 0.270 (0.132 to 0.419)	−0.231 (−0.400 to −0.049)	0.2840	n/a	n/a	n/a
XLS-R 300M	layer 0, full speed: 0.292 (0.136 to 0.457)	−0.209 (−0.400 to −0.002)	0.4944	n/a	n/a	n/a
wav2vec 2.0 Conformer Large	layer 1, full speed: 0.349 (0.201 to 0.507)	−0.152 (−0.290 to 0.009)	0.4944	n/a	n/a	n/a
MMS 300M	layer 0, full speed: 0.290 (0.137 to 0.461)	−0.211 (−0.395 to −0.012)	0.4944	n/a	n/a	n/a
HuBERT Base	layer 0, full speed: 0.290 (0.155 to 0.439)	−0.211 (−0.356 to −0.045)	0.2840	n/a	n/a	n/a
HuBERT Large	layer 1, full speed: 0.187 (0.089 to 0.305)	−0.314 (−0.460 to −0.178)	< 0.0074	n/a	n/a	n/a
data2vec Base (100 h)	layer 0, full speed: 0.361 (0.217 to 0.518)	−0.140 (−0.281 to 0.018)	0.4944	n/a	n/a	n/a
data2vec Base (960 h)	layer 0, full speed: 0.378 (0.222 to 0.549)	−0.123 (−0.270 to 0.040)	0.6800	n/a	n/a	n/a
WavLM Base+	layer 0, full speed: 0.305 (0.164 to 0.455)	−0.197 (−0.332 to −0.034)	0.3424	n/a	n/a	n/a
WavLM Large	layer 0, full speed: 0.268 (0.129 to 0.419)	−0.233 (−0.413 to −0.030)	0.4116	n/a	n/a	n/a
UniSpeech-SAT Base+	layer 0, full speed: 0.270 (0.131 to 0.427)	−0.231 (−0.390 to −0.051)	0.2840	n/a	n/a	n/a
XEUS	layer 0, full speed: 0.297 (0.156 to 0.458)	−0.204 (−0.355 to −0.025)	0.4116	n/a	n/a	n/a
Loudness control	summary: 0.069 (0.014 to 0.134)	−0.432 (−0.570 to −0.305)	< 0.0074	n/a	n/a	n/a
Duration control	summary: 0.115 (0.000 to 0.293)	−0.386 (−0.526 to −0.243)	< 0.0074	n/a	n/a	n/a

Little penguin, across night: BirdNET v2.4 0.693 (0.607 to 0.773).

Neural network	Fisher rule: layer, re-ID accuracy	Difference from BirdNET	Holm p	Held-out rule: layer, re-ID accuracy	Difference from BirdNET	Holm p
Perch 2.0	embedding: 0.705 (0.599 to 0.791)	0.011 (−0.051 to 0.069)	1.0000	n/a	n/a	n/a
Perch	embedding: 0.635 (0.528 to 0.727)	−0.058 (−0.134 to 0.013)	1.0000	n/a	n/a	n/a
SurfPerch	embedding: 0.646 (0.551 to 0.728)	−0.047 (−0.107 to 0.011)	1.0000	n/a	n/a	n/a
BirdNET v3 (preview)	embedding: 0.501 (0.371 to 0.611)	−0.192 (−0.268 to −0.124)	< 0.0074	n/a	n/a	n/a
AVEX sl-BEATS	block 3: 0.518 (0.396 to 0.621)	−0.176 (−0.255 to −0.102)	< 0.0074	n/a	n/a	n/a
AVEX EfficientNet-B0	embedding: 0.569 (0.444 to 0.681)	−0.124 (−0.199 to −0.051)	0.0040	n/a	n/a	n/a
AvesEcho	block 1: 0.407 (0.271 to 0.520)	−0.287 (−0.390 to −0.188)	< 0.0074	n/a	n/a	n/a
AudioProtoPNet	embedding: 0.674 (0.569 to 0.760)	−0.019 (−0.076 to 0.036)	1.0000	n/a	n/a	n/a

Neural network	Fisher rule: layer, re-ID accuracy	Difference from BirdNET	Holm p	Held-out rule: layer, re-ID accuracy	Difference from BirdNET	Holm p
ConvNeXt (BirdSet)	embedding: 0.658 (0.538 to 0.755)	-0.035 (-0.130 to 0.052)	1.0000	n/a	n/a	n/a
Bird-MAE	block 18: 0.385 (0.222 to 0.523)	-0.308 (-0.434 to -0.196)	< 0.0074	n/a	n/a	n/a
ProtoCLR	embedding: 0.447 (0.310 to 0.560)	-0.246 (-0.366 to -0.132)	< 0.0074	n/a	n/a	n/a
RCL FS BSED	embedding: 0.622 (0.532 to 0.711)	-0.071 (-0.117 to -0.022)	0.0936	n/a	n/a	n/a
BirdAVES	block 0: 0.726 (0.631 to 0.818)	0.033 (-0.023 to 0.097)	1.0000	n/a	n/a	n/a
AVES	block 0: 0.761 (0.673 to 0.845)	0.068 (-0.002 to 0.149)	0.8736	n/a	n/a	n/a
NatureBEATs	block 1: 0.490 (0.373 to 0.593)	-0.203 (-0.290 to -0.120)	< 0.0074	n/a	n/a	n/a
BioLingual	block 2: 0.343 (0.184 to 0.489)	-0.350 (-0.490 to -0.225)	< 0.0074	n/a	n/a	n/a
BEATs	block 4: 0.564 (0.450 to 0.661)	-0.129 (-0.195 to -0.070)	< 0.0074	n/a	n/a	n/a
AudioMAE	block 3: 0.307 (0.168 to 0.431)	-0.386 (-0.512 to -0.266)	< 0.0074	n/a	n/a	n/a
VGGish	embedding: 0.463 (0.341 to 0.576)	-0.230 (-0.300 to -0.166)	< 0.0074	n/a	n/a	n/a
ECAPA-TDNN (speaker)	embedding, 1/3 speed: 0.604 (0.461 to 0.723)	-0.090 (-0.183 to -0.017)	0.2006	n/a	n/a	n/a
ResNet (speaker)	embedding, 1/3 speed: 0.558 (0.419 to 0.681)	-0.135 (-0.226 to -0.060)	< 0.0074	n/a	n/a	n/a
x-vector (speaker)	embedding, 1/3 speed: 0.518 (0.362 to 0.645)	-0.176 (-0.290 to -0.084)	0.0040	n/a	n/a	n/a
wav2vec 2.0 Base	layer 0, 1/3 speed: 0.740 (0.649 to 0.823)	0.047 (-0.003 to 0.094)	0.8736	n/a	n/a	n/a
wav2vec 2.0 Large (robust)	layer 0, 1/3 speed: 0.710 (0.591 to 0.819)	0.016 (-0.046 to 0.077)	1.0000	n/a	n/a	n/a
XLS-R 300M	layer 0, 1/3 speed: 0.722 (0.617 to 0.817)	0.029 (-0.039 to 0.102)	1.0000	n/a	n/a	n/a
wav2vec 2.0 Conformer Large	layer 0, 1/3 speed: 0.730 (0.625 to 0.825)	0.037 (-0.033 to 0.113)	1.0000	n/a	n/a	n/a
MMS 300M	layer 0, 1/3 speed: 0.737 (0.643 to 0.826)	0.044 (-0.013 to 0.107)	1.0000	n/a	n/a	n/a
HuBERT Base	layer 0, 1/3 speed: 0.760 (0.677 to 0.840)	0.067 (0.004 to 0.138)	0.5880	n/a	n/a	n/a
HuBERT Large	layer 0, 1/3 speed: 0.677 (0.550 to 0.792)	-0.016 (-0.090 to 0.054)	1.0000	n/a	n/a	n/a
data2vec Base (100 h)	layer 0, 1/3 speed: 0.794 (0.720 to 0.869)	0.101 (0.062 to 0.145)	< 0.0074	n/a	n/a	n/a
data2vec Base (960 h)	layer 0, 1/3 speed: 0.797 (0.716 to 0.875)	0.104 (0.055 to 0.161)	< 0.0074	n/a	n/a	n/a
WavLM Base+	layer 0, 1/3 speed: 0.797 (0.723 to 0.870)	0.104 (0.055 to 0.160)	< 0.0074	n/a	n/a	n/a
WavLM Large	layer 0, 1/3 speed: 0.686 (0.560 to 0.802)	-0.008 (-0.094 to 0.077)	1.0000	n/a	n/a	n/a
UniSpeech-SAT Base+	layer 0, 1/3 speed: 0.764 (0.670 to 0.853)	0.071 (0.008 to 0.136)	0.4288	n/a	n/a	n/a
XEUS	layer 0, 1/3 speed: 0.798 (0.719 to 0.875)	0.105 (0.051 to 0.173)	< 0.0074	n/a	n/a	n/a
Loudness control	summary: 0.159 (0.048 to 0.267)	-0.534 (-0.659 to -0.411)	< 0.0074	n/a	n/a	n/a
Duration control	summary: 0.167 (0.000 to 0.394)	-0.527 (-0.738 to -0.287)	< 0.0074	n/a	n/a	n/a

Red-tailed black cockatoo, random split: BirdNET v2.4 0.961 (0.930 to 0.980).

Neural network	Fisher rule: layer, re-ID accuracy	Difference from BirdNET	Holm p	Held-out rule: layer, re-ID accuracy	Difference from BirdNET	Holm p
Perch 2.0	embedding: 0.961 (0.933 to 0.979)	0.000 (−0.015 to 0.023)	1.0000	n/a	n/a	n/a
Perch	embedding: 0.947 (0.920 to 0.973)	−0.014 (−0.043 to 0.029)	1.0000	n/a	n/a	n/a
SurfPerch	embedding: 0.958 (0.920 to 0.981)	−0.003 (−0.019 to 0.017)	1.0000	n/a	n/a	n/a
BirdNET v3 (preview)	embedding: 0.854 (0.791 to 0.892)	−0.106 (−0.158 to −0.070)	< 0.0074	n/a	n/a	n/a
AVEX sl-BEATs	block 0: 0.577 (0.279 to 0.742)	−0.384 (−0.672 to −0.223)	< 0.0074	n/a	n/a	n/a
AVEX EfficientNet-B0	embedding: 0.891 (0.839 to 0.925)	−0.070 (−0.107 to −0.034)	0.0080	n/a	n/a	n/a
AvesEcho	block 6: 0.798 (0.570 to 0.909)	−0.162 (−0.371 to −0.062)	< 0.0074	n/a	n/a	n/a
AudioProtoPNet	embedding: 0.941 (0.910 to 0.967)	−0.020 (−0.050 to 0.023)	1.0000	n/a	n/a	n/a
ConvNeXt (BirdSet)	embedding: 0.969 (0.936 to 0.991)	0.008 (−0.019 to 0.045)	1.0000	n/a	n/a	n/a
Bird-MAE	block 15: 0.504 (0.133 to 0.731)	−0.457 (−0.810 to −0.236)	< 0.0074	n/a	n/a	n/a
ProtoCLR	embedding: 0.515 (0.219 to 0.713)	−0.445 (−0.726 to −0.259)	< 0.0074	n/a	n/a	n/a
RCL FS BSED	embedding: 0.709 (0.436 to 0.841)	−0.252 (−0.510 to −0.122)	< 0.0074	n/a	n/a	n/a
BirdAVES	block 0: 0.910 (0.838 to 0.956)	−0.050 (−0.106 to −0.014)	0.0408	n/a	n/a	n/a
AVES	block 0: 0.964 (0.917 to 0.986)	0.003 (−0.025 to 0.021)	1.0000	n/a	n/a	n/a
NatureBEATs	block 9: 0.947 (0.885 to 0.978)	−0.014 (−0.056 to 0.009)	1.0000	n/a	n/a	n/a
BioLingual	block 1: 0.468 (0.009 to 0.729)	−0.493 (−0.926 to −0.241)	< 0.0074	n/a	n/a	n/a
BEATs	block 4: 0.678 (0.413 to 0.829)	−0.283 (−0.535 to −0.142)	< 0.0074	n/a	n/a	n/a
AudioMAE	block 3: 0.476 (0.024 to 0.733)	−0.485 (−0.913 to −0.236)	< 0.0074	n/a	n/a	n/a
VGGish	embedding: 0.812 (0.680 to 0.902)	−0.148 (−0.258 to −0.071)	< 0.0074	n/a	n/a	n/a
ECAPA-TDNN (speaker)	embedding, 1/3 speed: 0.810 (0.705 to 0.868)	−0.151 (−0.241 to −0.095)	< 0.0074	n/a	n/a	n/a
ResNet (speaker)	embedding, 1/3 speed: 0.810 (0.709 to 0.864)	−0.151 (−0.231 to −0.102)	< 0.0074	n/a	n/a	n/a
x-vector (speaker)	embedding, 1/3 speed: 0.728 (0.548 to 0.829)	−0.232 (−0.393 to −0.135)	< 0.0074	n/a	n/a	n/a
wav2vec 2.0 Base	layer 1, full speed: 0.871 (0.787 to 0.915)	−0.090 (−0.154 to −0.053)	< 0.0074	n/a	n/a	n/a
wav2vec 2.0 Large (robust)	layer 20, 1/2 speed: 0.922 (0.822 to 0.969)	−0.039 (−0.122 to 0.004)	1.0000	n/a	n/a	n/a
XLS-R 300M	layer 0, 1/3 speed: 0.910 (0.825 to 0.950)	−0.050 (−0.121 to −0.014)	0.1066	n/a	n/a	n/a
wav2vec 2.0 Conformer Large	layer 0, 1/3 speed: 0.894 (0.801 to 0.938)	−0.067 (−0.146 to −0.021)	0.0420	n/a	n/a	n/a
MMS 300M	layer 0, 1/3 speed: 0.908 (0.814 to 0.951)	−0.053 (−0.131 to −0.016)	0.0504	n/a	n/a	n/a
HuBERT Base	layer 0, 1/3 speed: 0.916 (0.843 to 0.950)	−0.045 (−0.104 to −0.014)	0.0408	n/a	n/a	n/a
HuBERT Large	layer 0, 1/3 speed: 0.846 (0.717 to 0.917)	−0.115 (−0.231 to −0.046)	< 0.0074	n/a	n/a	n/a
data2vec Base (100 h)	layer 0, 1/3 speed: 0.936 (0.877 to 0.967)	−0.025 (−0.072 to 0.008)	1.0000	n/a	n/a	n/a
data2vec Base (960 h)	layer 0, 1/3 speed: 0.938 (0.878 to 0.968)	−0.022 (−0.073 to 0.012)	1.0000	n/a	n/a	n/a
WavLM Base+	layer 0, 1/3 speed: 0.924 (0.856 to 0.954)	−0.036 (−0.092 to −0.007)	0.1536	n/a	n/a	n/a
WavLM Large	layer 0, 1/3 speed: 0.846 (0.730 to 0.906)	−0.115 (−0.219 to −0.054)	0.0080	n/a	n/a	n/a

Neural network UniSpeech-SAT Base+	Fisher rule: layer, re-ID accuracy layer 0, 1/3 speed: 0.916 (0.843 to 0.947)	Difference from BirdNET -0.045 (-0.103 to -0.016)	Holm p 0.0180	Held-out rule: layer, re-ID accuracy n/a	Difference from BirdNET n/a	Holm p n/a
XEUS	layer 0, 1/3 speed: 0.899 (0.822 to 0.943)	-0.062 (-0.133 to -0.010)	0.1826	n/a	n/a	n/a
Loudness control	summary: 0.406 (0.153 to 0.587)	-0.555 (-0.793 to -0.375)	< 0.0074	n/a	n/a	n/a
Duration control	summary: 0.336 (0.000 to 0.666)	-0.625 (-0.958 to -0.304)	< 0.0074	n/a	n/a	n/a

Egyptian fruit bat, across year: BirdNET v2.4 0.381 (0.189 to 0.561).

Neural network Perch 2.0	Fisher rule: layer, re-ID accuracy embedding: 0.350 (0.161 to 0.532)	Difference from BirdNET -0.031 (-0.115 to 0.104)	Holm p 1.0000	Held-out rule: layer, re-ID accuracy embedding: 0.350 (0.161 to 0.532)	Difference from BirdNET -0.031 (-0.115 to 0.104)	Holm p 1.0000
Perch	embedding: 0.268 (0.131 to 0.537)	-0.113 (-0.353 to 0.180)	1.0000	embedding: 0.268 (0.131 to 0.537)	-0.113 (-0.353 to 0.180)	1.0000
SurfPerch	embedding: 0.353 (0.202 to 0.496)	-0.028 (-0.136 to 0.104)	1.0000	embedding: 0.353 (0.202 to 0.496)	-0.028 (-0.136 to 0.104)	1.0000
BirdNET v3 (preview)	embedding: 0.293 (0.194 to 0.475)	-0.088 (-0.272 to 0.103)	1.0000	embedding: 0.293 (0.194 to 0.475)	-0.088 (-0.272 to 0.103)	1.0000
AVEX sl-BEATs	block 0: 0.114 (0.037 to 0.238)	-0.267 (-0.495 to -0.054)	0.0072	block 10: 0.378 (0.207 to 0.512)	-0.003 (-0.082 to 0.085)	1.0000
AVEX EfficientNet-B0	embedding: 0.307 (0.258 to 0.359)	-0.074 (-0.237 to 0.088)	1.0000	embedding: 0.307 (0.258 to 0.359)	-0.074 (-0.237 to 0.088)	1.0000
AvesEcho	block 5: 0.151 (0.028 to 0.381)	-0.231 (-0.472 to 0.028)	1.0000	embedding: 0.223 (0.160 to 0.292)	-0.158 (-0.299 to -0.027)	0.0518
AudioProtoPNet	embedding: 0.305 (0.131 to 0.512)	-0.077 (-0.189 to 0.100)	1.0000	embedding: 0.305 (0.131 to 0.512)	-0.077 (-0.189 to 0.100)	1.0000
ConvNeXt (BirdSet)	embedding: 0.333 (0.127 to 0.530)	-0.048 (-0.121 to 0.061)	1.0000	embedding: 0.333 (0.127 to 0.530)	-0.048 (-0.121 to 0.061)	1.0000
Bird-MAE	block 8: 0.114 (0.046 to 0.200)	-0.267 (-0.490 to -0.041)	0.1768	block 25: 0.123 (0.048 to 0.220)	-0.259 (-0.479 to -0.031)	0.4968
ProtoCLR	block 10: 0.184 (0.019 to 0.384)	-0.197 (-0.518 to 0.172)	1.0000	embedding: 0.133 (0.052 to 0.233)	-0.249 (-0.494 to -0.001)	1.0000
RCL FS BSED	embedding: 0.191 (0.077 to 0.327)	-0.191 (-0.467 to 0.056)	1.0000	embedding: 0.191 (0.077 to 0.327)	-0.191 (-0.467 to 0.056)	1.0000
BirdAVES	block 0: 0.246 (0.138 to 0.329)	-0.136 (-0.336 to 0.100)	1.0000	block 21: 0.348 (0.123 to 0.569)	-0.033 (-0.163 to 0.041)	1.0000
AVES	block 0: 0.283 (0.024 to 0.473)	-0.098 (-0.311 to 0.057)	1.0000	block 2: 0.280 (0.041 to 0.473)	-0.101 (-0.294 to 0.030)	1.0000
NatureBEATs	block 3: 0.131 (0.051 to 0.258)	-0.250 (-0.481 to -0.019)	1.0000	block 9: 0.350 (0.114 to 0.584)	-0.031 (-0.111 to 0.023)	1.0000
BioLingual	block 2: 0.119 (0.034 to 0.231)	-0.263 (-0.513 to -0.010)	1.0000	embedding: 0.209 (0.122 to 0.319)	-0.172 (-0.417 to 0.054)	1.0000
BEATs	block 7: 0.223 (0.156 to 0.298)	-0.158 (-0.382 to 0.062)	1.0000	block 9: 0.259 (0.180 to 0.395)	-0.123 (-0.352 to 0.050)	1.0000
AudioMAE	block 1: 0.190 (0.005 to 0.415)	-0.192 (-0.534 to 0.204)	1.0000	block 10: 0.184 (0.104 to 0.328)	-0.197 (-0.423 to 0.037)	1.0000
VGGish	embedding: 0.166 (0.081 to 0.216)	-0.216 (-0.370 to -0.033)	0.4950	embedding: 0.166 (0.081 to 0.216)	-0.216 (-0.370 to -0.033)	0.5250
ECAPA-TDNN (speaker)	embedding, 1/2 speed: 0.297 (0.187 to 0.411)	-0.084 (-0.199 to 0.048)	1.0000	embedding, 1/3 speed: 0.272 (0.147 to 0.406)	-0.110 (-0.232 to 0.034)	1.0000
ResNet (speaker)	embedding: 0.319 (0.204 to 0.412)	-0.062 (-0.244 to 0.144)	1.0000	embedding, 1/3 speed: 0.265 (0.182 to 0.375)	-0.116 (-0.287 to 0.053)	1.0000
x-vector (speaker)	embedding, 1/3 speed: 0.305 (0.144 to 0.422)	-0.077 (-0.186 to 0.057)	1.0000	embedding, 1/3 speed: 0.305 (0.144 to 0.422)	-0.077 (-0.186 to 0.057)	1.0000
wav2vec 2.0 Base	layer 0, full speed: 0.233 (0.063 to 0.364)	-0.149 (-0.384 to 0.125)	1.0000	layer 3, 1/3 speed: 0.365 (0.139 to 0.561)	-0.016 (-0.120 to 0.053)	1.0000
wav2vec 2.0 Large (robust)	layer 0, full speed: 0.231 (0.045 to 0.349)	-0.151 (-0.320 to 0.104)	1.0000	layer 3, 1/2 speed: 0.394 (0.159 to 0.597)	0.013 (-0.103 to 0.133)	1.0000
XLS-R 300M	layer 1, full speed: 0.258 (0.146 to 0.349)	-0.124 (-0.234 to -0.007)	1.0000	layer 5, 1/3 speed: 0.265 (0.134 to 0.456)	-0.116 (-0.361 to 0.140)	1.0000
wav2vec 2.0 Conformer Large	layer 2, full speed: 0.283 (0.091 to 0.429)	-0.098 (-0.288 to 0.171)	1.0000	layer 5, 1/3 speed: 0.310 (0.153 to 0.486)	-0.071 (-0.233 to 0.125)	1.0000
MMS 300M	layer 0, full speed: 0.218 (0.060 to 0.320)	-0.164 (-0.269 to -0.001)	1.0000	layer 4, 1/3 speed: 0.367 (0.166 to 0.530)	-0.014 (-0.147 to 0.155)	1.0000
HuBERT Base	layer 0, full speed: 0.249 (0.026 to 0.385)	-0.133 (-0.337 to 0.116)	1.0000	layer 2, 1/3 speed: 0.345 (0.103 to 0.578)	-0.037 (-0.216 to 0.068)	1.0000

Neural network HuBERT Large	Fisher rule: layer, re-ID accuracy layer 0, full speed: 0.216 (0.049 to 0.358)	Difference from BirdNET -0.166 (-0.242 to -0.060)	Holm p 0.0700	Held-out rule: layer, re-ID accuracy layer 5, 1/3 speed: 0.319 (0.159 to 0.475)	Difference from BirdNET -0.062 (-0.217 to 0.118)	Holm p 1.0000
data2vec Base (100 h)	layer 0, full speed: 0.255 (0.027 to 0.413)	-0.126 (-0.335 to 0.169)	1.0000	layer 1, 1/3 speed: 0.337 (0.196 to 0.483)	-0.044 (-0.215 to 0.154)	1.0000
data2vec Base (960 h)	layer 0, full speed: 0.258 (0.034 to 0.419)	-0.124 (-0.327 to 0.170)	1.0000	layer 1, 1/3 speed: 0.341 (0.243 to 0.412)	-0.041 (-0.189 to 0.118)	1.0000
WavLM Base+	layer 0, full speed: 0.250 (0.040 to 0.389)	-0.131 (-0.328 to 0.128)	1.0000	layer 1, 1/3 speed: 0.369 (0.136 to 0.586)	-0.013 (-0.187 to 0.088)	1.0000
WavLM Large	layer 0, full speed: 0.238 (0.018 to 0.458)	-0.143 (-0.214 to -0.093)	< 0.0074	layer 6, 1/3 speed: 0.414 (0.157 to 0.636)	0.032 (-0.086 to 0.096)	1.0000
UniSpeech-SAT Base+	layer 0, full speed: 0.235 (0.023 to 0.420)	-0.147 (-0.377 to 0.180)	1.0000	layer 2, 1/3 speed: 0.334 (0.132 to 0.518)	-0.047 (-0.153 to 0.019)	1.0000
XEUS	layer 0, 1/2 speed: 0.306 (0.047 to 0.547)	-0.075 (-0.262 to 0.034)	1.0000	layer 4, 1/3 speed: 0.364 (0.075 to 0.626)	-0.017 (-0.213 to 0.103)	1.0000
Loudness control	summary: 0.248 (0.013 to 0.620)	-0.134 (-0.531 to 0.395)	1.0000	summary: 0.248 (0.013 to 0.620)	-0.134 (-0.531 to 0.395)	1.0000
Duration control	summary: 0.208 (0.000 to 0.462)	-0.173 (-0.537 to 0.249)	1.0000	summary: 0.208 (0.000 to 0.462)	-0.173 (-0.537 to 0.249)	1.0000

Chiffchaff, within year: BirdNET v2.4 0.765 (0.680 to 0.841).

Neural network Perch 2.0	Fisher rule: layer, re-ID accuracy embedding: 0.831 (0.756 to 0.884)	Difference from BirdNET 0.066 (-0.008 to 0.122)	Holm p 0.5782	Held-out rule: layer, re-ID accuracy n/a	Difference from BirdNET n/a	Holm p n/a
Perch	embedding: 0.778 (0.656 to 0.858)	0.013 (-0.105 to 0.094)	1.0000	n/a	n/a	n/a
SurfPerch	embedding: 0.729 (0.632 to 0.808)	-0.036 (-0.115 to 0.013)	0.8892	n/a	n/a	n/a
BirdNET v3 (preview)	embedding: 0.624 (0.370 to 0.770)	-0.141 (-0.381 to -0.003)	0.3504	n/a	n/a	n/a
AVEX sl-BEATs	block 9: 0.737 (0.585 to 0.817)	-0.027 (-0.127 to 0.023)	1.0000	n/a	n/a	n/a
AVEX EfficientNet-B0	embedding: 0.691 (0.577 to 0.760)	-0.074 (-0.173 to -0.021)	0.0400	n/a	n/a	n/a
AvesEcho	embedding: 0.691 (0.518 to 0.772)	-0.073 (-0.217 to -0.008)	0.2160	n/a	n/a	n/a
AudioProtoPNet	embedding: 0.806 (0.646 to 0.890)	0.042 (-0.096 to 0.114)	1.0000	n/a	n/a	n/a
ConvNeXt (BirdSet)	embedding: 0.805 (0.687 to 0.866)	0.041 (-0.066 to 0.089)	1.0000	n/a	n/a	n/a
Bird-MAE	block 12: 0.319 (0.114 to 0.447)	-0.446 (-0.633 to -0.339)	< 0.0074	n/a	n/a	n/a
ProtoCLR	embedding: 0.325 (0.211 to 0.477)	-0.439 (-0.548 to -0.305)	< 0.0074	n/a	n/a	n/a
RCL FS BSED	embedding: 0.256 (0.121 to 0.494)	-0.508 (-0.643 to -0.289)	< 0.0074	n/a	n/a	n/a
BirdAVES	block 0: 0.420 (0.331 to 0.588)	-0.345 (-0.430 to -0.205)	< 0.0074	n/a	n/a	n/a
AVES	block 0: 0.397 (0.309 to 0.554)	-0.368 (-0.438 to -0.251)	< 0.0074	n/a	n/a	n/a
NatureBEATs	block 9: 0.741 (0.577 to 0.816)	-0.024 (-0.156 to 0.035)	1.0000	n/a	n/a	n/a
BioLingual	block 10: 0.637 (0.449 to 0.743)	-0.127 (-0.281 to -0.052)	< 0.0074	n/a	n/a	n/a
BEATs	block 9: 0.558 (0.340 to 0.673)	-0.207 (-0.401 to -0.118)	< 0.0074	n/a	n/a	n/a
AudioMAE	block 2: 0.184 (0.122 to 0.279)	-0.581 (-0.658 to -0.468)	< 0.0074	n/a	n/a	n/a
VGGish	embedding: 0.458 (0.256 to 0.561)	-0.307 (-0.501 to -0.223)	< 0.0074	n/a	n/a	n/a
ECAPA-TDNN (speaker)	embedding, 1/3 speed: 0.416 (0.303 to 0.545)	-0.348 (-0.438 to -0.258)	< 0.0074	n/a	n/a	n/a
ResNet (speaker)	embedding, 1/3 speed: 0.523 (0.327 to 0.633)	-0.242 (-0.411 to -0.153)	< 0.0074	n/a	n/a	n/a
x-vector (speaker)	embedding, 1/2 speed: 0.517 (0.326 to 0.634)	-0.248 (-0.404 to -0.170)	< 0.0074	n/a	n/a	n/a
wav2vec 2.0 Base	layer 0, full speed: 0.331 (0.234 to 0.522)	-0.434 (-0.519 to -0.274)	< 0.0074	n/a	n/a	n/a

Neural network wav2vec 2.0 Large (robust)	Fisher rule: layer, re-ID accuracy layer 0, 1/2 speed: 0.305 (0.208 to 0.444)	Difference from BirdNET -0.460 (-0.555 to -0.343)	Holm p < 0.0074	Held-out rule: layer, re-ID accuracy n/a	Difference from BirdNET n/a	Holm p n/a
XLS-R 300M	layer 0, 1/2 speed: 0.321 (0.213 to 0.440)	-0.444 (-0.548 to -0.345)	< 0.0074	n/a	n/a	n/a
wav2vec 2.0 Conformer Large	layer 2, full speed: 0.414 (0.304 to 0.575)	-0.351 (-0.425 to -0.236)	< 0.0074	n/a	n/a	n/a
MMS 300M	layer 0, 1/2 speed: 0.324 (0.224 to 0.473)	-0.441 (-0.531 to -0.319)	< 0.0074	n/a	n/a	n/a
HuBERT Base	layer 0, full speed: 0.314 (0.218 to 0.496)	-0.451 (-0.536 to -0.298)	< 0.0074	n/a	n/a	n/a
HuBERT Large	layer 0, 1/2 speed: 0.316 (0.215 to 0.442)	-0.449 (-0.540 to -0.351)	< 0.0074	n/a	n/a	n/a
data2vec Base (100 h)	layer 1, 1/3 speed: 0.382 (0.279 to 0.503)	-0.383 (-0.489 to -0.276)	< 0.0074	n/a	n/a	n/a
data2vec Base (960 h)	layer 0, 1/3 speed: 0.378 (0.268 to 0.491)	-0.386 (-0.487 to -0.298)	< 0.0074	n/a	n/a	n/a
WavLM Base+	layer 0, full speed: 0.327 (0.233 to 0.512)	-0.438 (-0.519 to -0.283)	< 0.0074	n/a	n/a	n/a
WavLM Large	layer 0, 1/2 speed: 0.347 (0.225 to 0.470)	-0.417 (-0.524 to -0.325)	< 0.0074	n/a	n/a	n/a
UniSpeech-SAT Base+	layer 0, full speed: 0.317 (0.222 to 0.497)	-0.447 (-0.534 to -0.296)	< 0.0074	n/a	n/a	n/a
XEUS	layer 0, 1/2 speed: 0.341 (0.236 to 0.465)	-0.424 (-0.518 to -0.324)	< 0.0074	n/a	n/a	n/a
Loudness control	summary: 0.094 (0.010 to 0.267)	-0.671 (-0.747 to -0.522)	< 0.0074	n/a	n/a	n/a
Duration control	summary: 0.040 (0.000 to 0.147)	-0.725 (-0.794 to -0.607)	< 0.0074	n/a	n/a	n/a

Rook, across year: BirdNET v2.4 0.375 (0.261 to 0.506).

Neural network Perch 2.0	Fisher rule: layer, re-ID accuracy embedding: 0.418 (0.306 to 0.535)	Difference from BirdNET 0.043 (0.017 to 0.065)	Holm p 0.0384	Held-out rule: layer, re-ID accuracy embedding: 0.418 (0.306 to 0.535)	Difference from BirdNET 0.043 (0.017 to 0.065)	Holm p 0.0336
Perch	embedding: 0.393 (0.282 to 0.516)	0.018 (-0.027 to 0.062)	0.9816	embedding: 0.393 (0.282 to 0.516)	0.018 (-0.027 to 0.062)	1.0000
SurfPerch	embedding: 0.306 (0.200 to 0.425)	-0.069 (-0.106 to -0.038)	0.0050	embedding: 0.306 (0.200 to 0.425)	-0.069 (-0.106 to -0.038)	0.0046
BirdNET v3 (preview)	embedding: 0.328 (0.212 to 0.439)	-0.048 (-0.098 to -0.008)	0.1632	embedding: 0.328 (0.212 to 0.439)	-0.048 (-0.098 to -0.008)	0.1632
AVEX sl-BEATs	block 5: 0.284 (0.153 to 0.442)	-0.092 (-0.177 to -0.020)	0.1152	block 10: 0.342 (0.202 to 0.457)	-0.033 (-0.102 to 0.019)	1.0000
AVEX EfficientNet-B0	embedding: 0.345 (0.247 to 0.442)	-0.031 (-0.073 to -0.001)	0.2580	embedding: 0.345 (0.247 to 0.442)	-0.031 (-0.073 to -0.001)	0.3010
AvesEcho	embedding: 0.343 (0.223 to 0.437)	-0.033 (-0.125 to 0.024)	0.9816	embedding: 0.343 (0.223 to 0.437)	-0.033 (-0.125 to 0.024)	1.0000
AudioProtoPNet	embedding: 0.386 (0.269 to 0.506)	0.011 (-0.008 to 0.024)	0.9248	embedding: 0.386 (0.269 to 0.506)	0.011 (-0.008 to 0.024)	1.0000
ConvNeXt (BirdSet)	embedding: 0.409 (0.260 to 0.536)	0.034 (-0.044 to 0.083)	0.9816	embedding: 0.409 (0.260 to 0.536)	0.034 (-0.044 to 0.083)	1.0000
Bird-MAE	block 1: 0.125 (0.055 to 0.225)	-0.250 (-0.341 to -0.145)	< 0.0074	block 0: 0.095 (0.038 to 0.200)	-0.281 (-0.375 to -0.166)	< 0.0074
ProtoCLR	block 1: 0.101 (0.039 to 0.211)	-0.274 (-0.352 to -0.164)	< 0.0074	block 6: 0.079 (0.015 to 0.213)	-0.296 (-0.383 to -0.167)	< 0.0074
RCL FS BSIED	embedding: 0.105 (0.032 to 0.215)	-0.270 (-0.350 to -0.178)	< 0.0074	embedding: 0.105 (0.032 to 0.215)	-0.270 (-0.350 to -0.178)	< 0.0074
BirdAVES	block 11: 0.151 (0.032 to 0.366)	-0.225 (-0.322 to -0.095)	0.0050	block 23: 0.174 (0.053 to 0.384)	-0.202 (-0.305 to -0.073)	0.0144
AVES	block 10: 0.179 (0.055 to 0.375)	-0.197 (-0.283 to -0.091)	0.0050	block 9: 0.173 (0.047 to 0.374)	-0.202 (-0.293 to -0.093)	0.0046
NatureBEATs	block 3: 0.166 (0.044 to 0.394)	-0.209 (-0.320 to -0.064)	0.0660	block 7: 0.277 (0.109 to 0.455)	-0.099 (-0.206 to -0.014)	0.1458
BioLingual	block 11: 0.281 (0.084 to 0.479)	-0.095 (-0.233 to 0.018)	0.5330	embedding: 0.303 (0.186 to 0.404)	-0.073 (-0.139 to -0.025)	0.0170
BEATs	block 4: 0.207 (0.086 to 0.410)	-0.168 (-0.272 to -0.044)	0.0728	block 10: 0.341 (0.208 to 0.456)	-0.034 (-0.092 to 0.005)	0.5436

Neural network	Fisher rule: layer, re-ID accuracy	Difference from BirdNET	Holm p	Held-out rule: layer, re-ID accuracy	Difference from BirdNET	Holm p
AudioMAE	block 7: 0.185 (0.038 to 0.418)	-0.191 (-0.310 to -0.038)	0.1520	block 11: 0.234 (0.115 to 0.410)	-0.141 (-0.226 to -0.048)	0.0270
VGGish	embedding: 0.180 (0.086 to 0.329)	-0.195 (-0.266 to -0.117)	< 0.0074	embedding: 0.180 (0.086 to 0.329)	-0.195 (-0.266 to -0.117)	< 0.0074
ECAPA-TDNN (speaker)	embedding, 1/3 speed: 0.271 (0.180 to 0.365)	-0.105 (-0.200 to -0.023)	0.1520	embedding, 1/2 speed: 0.281 (0.169 to 0.400)	-0.094 (-0.160 to -0.035)	0.0552
ResNet (speaker)	embedding, 1/2 speed: 0.297 (0.152 to 0.451)	-0.078 (-0.165 to -0.016)	0.1092	embedding: 0.239 (0.125 to 0.435)	-0.137 (-0.221 to -0.035)	0.0552
x-vector (speaker)	embedding, 1/2 speed: 0.296 (0.161 to 0.447)	-0.079 (-0.174 to -0.004)	0.2534	embedding: 0.262 (0.136 to 0.417)	-0.114 (-0.163 to -0.064)	< 0.0074
wav2vec 2.0 Base	layer 0, full speed: 0.153 (0.056 to 0.344)	-0.222 (-0.327 to -0.091)	< 0.0074	layer 3, full speed: 0.202 (0.096 to 0.369)	-0.174 (-0.258 to -0.082)	< 0.0074
wav2vec 2.0 Large (robust)	layer 24, full speed: 0.125 (0.012 to 0.267)	-0.250 (-0.409 to -0.089)	0.0136	layer 23, full speed: 0.125 (0.012 to 0.264)	-0.251 (-0.408 to -0.091)	0.0120
XLS-R 300M	layer 22, full speed: 0.187 (0.067 to 0.384)	-0.188 (-0.278 to -0.083)	0.0080	layer 14, full speed: 0.226 (0.088 to 0.370)	-0.150 (-0.224 to -0.101)	< 0.0074
wav2vec 2.0 Conformer Large	layer 11, 1/2 speed: 0.256 (0.138 to 0.395)	-0.120 (-0.227 to -0.024)	0.1152	layer 8, full speed: 0.281 (0.134 to 0.418)	-0.094 (-0.164 to -0.049)	< 0.0074
MMS 300M	layer 24, 1/2 speed: 0.163 (0.048 to 0.344)	-0.212 (-0.316 to -0.112)	< 0.0074	layer 8, full speed: 0.224 (0.090 to 0.397)	-0.151 (-0.234 to -0.075)	< 0.0074
HuBERT Base	layer 12, 1/2 speed: 0.235 (0.089 to 0.400)	-0.140 (-0.226 to -0.071)	< 0.0074	layer 8, full speed: 0.250 (0.112 to 0.420)	-0.125 (-0.212 to -0.051)	0.0224
HuBERT Large	layer 1, 1/3 speed: 0.197 (0.058 to 0.319)	-0.178 (-0.295 to -0.113)	< 0.0074	layer 17, full speed: 0.170 (0.070 to 0.317)	-0.206 (-0.299 to -0.122)	< 0.0074
data2vec Base (100 h)	layer 11, full speed: 0.169 (0.060 to 0.319)	-0.207 (-0.293 to -0.137)	< 0.0074	layer 5, full speed: 0.213 (0.078 to 0.377)	-0.162 (-0.240 to -0.094)	< 0.0074
data2vec Base (960 h)	layer 2, full speed: 0.236 (0.095 to 0.392)	-0.139 (-0.223 to -0.077)	< 0.0074	layer 9, full speed: 0.163 (0.050 to 0.349)	-0.213 (-0.306 to -0.114)	< 0.0074
WavLM Base+	layer 12, 1/2 speed: 0.105 (0.022 to 0.287)	-0.270 (-0.367 to -0.151)	< 0.0074	layer 0, full speed: 0.162 (0.061 to 0.354)	-0.213 (-0.310 to -0.098)	< 0.0074
WavLM Large	layer 1, 1/3 speed: 0.166 (0.049 to 0.365)	-0.210 (-0.312 to -0.094)	0.0050	layer 3, 1/2 speed: 0.210 (0.091 to 0.371)	-0.165 (-0.289 to -0.057)	0.0120
UniSpeech-SAT Base+	layer 1, full speed: 0.181 (0.059 to 0.375)	-0.195 (-0.293 to -0.084)	0.0080	layer 9, 1/2 speed: 0.205 (0.076 to 0.392)	-0.170 (-0.287 to -0.055)	0.0336
XEUS	layer 1, 1/2 speed: 0.247 (0.111 to 0.408)	-0.128 (-0.236 to -0.043)	0.0108	layer 18, 1/2 speed: 0.254 (0.077 to 0.430)	-0.122 (-0.260 to -0.021)	0.0880
Loudness control	summary: 0.126 (0.019 to 0.342)	-0.249 (-0.363 to -0.108)	< 0.0074	summary: 0.126 (0.019 to 0.342)	-0.249 (-0.363 to -0.108)	< 0.0074
Duration control	summary: 0.030 (0.000 to 0.113)	-0.345 (-0.467 to -0.210)	0.0050	summary: 0.030 (0.000 to 0.113)	-0.345 (-0.467 to -0.210)	0.0046

Notes: Re-ID accuracy with a 95% interval over individuals; difference from BirdNET with a paired interval over the same 10,000 draws of individuals; two-sided bootstrap $p = \min(1, 2 \times \min(\text{proportion of differences} \leq 0, \text{proportion} \geq 0))$, with zero differences counted on both sides, then adjusted by Holm's method over the 37 comparisons on that dataset and rule. When no draw crossed zero the empirical estimate is zero; < 0.0074 denotes the display resolution $37 \times 2 / 10,000$, not a confidence bound on the underlying p value. BirdNET's interval comes from the same paired draws, so its bounds can differ slightly from those in Table 2. n/a: the held-out rule does not apply, because the dataset has fewer than two enrolment sessions per individual, and on such datasets it would choose what the Fisher rule chose.

Table S14. Consistency of the ranking of embeddings across datasets under the three layer rules, and each entry's average rank.

Layer rule	Entries ranked	Friedman χ^2	p	Kendall's W	Nemenyi critical difference
Fisher, 13 datasets	38	224.7	< 0.0001	0.47	16.8
Held-out, 6 datasets	38	138.3	< 0.0001	0.62	24.8
Own final output, 13 datasets	38	307.7	< 0.0001	0.64	16.8

Embedding	Average rank, Fisher rule	Average rank, held-out rule	Average rank, own final output
Perch 2.0	4.12	3.67	3.38
ConvNeXt (BirdSet)	6.00	5.58	5.77
BirdNET v2.4	6.27	4.00	5.62
SurfPerch	7.69	10.50	6.96
AudioProtoPNet	8.08	7.33	7.81
Perch	8.35	9.83	7.92
AVEX EfficientNet-B0	11.92	9.58	12.12
ResNet (speaker)	14.50	15.92	18.81
ECAPA-TDNN (speaker)	15.19	18.92	19.73
XEUS	16.04	13.92	22.46
x-vector (speaker)	16.08	18.42	18.62
BirdNET v3 (preview)	16.19	15.67	15.81
data2vec Base (960 h)	17.42	22.67	30.54
data2vec Base (100 h)	17.54	18.42	30.38
wav2vec 2.0 Conformer Large	17.73	19.58	19.31
BirdAVES	19.00	25.08	14.08
wav2vec 2.0 Base	19.42	23.00	27.04
AVES	19.58	31.08	19.46
XLS-R 300M	19.85	18.83	26.77
WavLM Base+	20.08	19.75	24.04
UniSpeech-SAT Base+	20.19	16.67	21.00
MMS 300M	20.46	17.50	25.00
HuBERT Base	20.88	21.25	21.00
WavLM Large	21.19	16.83	18.00
AVEX sl-BEATs	21.62	11.08	9.15
HuBERT Large	22.31	23.42	22.73
NatureBEATs	22.73	17.83	9.96
BEATs	22.77	17.75	18.77
wav2vec 2.0 Large (robust)	23.69	26.50	33.31
AvesEcho	25.38	13.58	9.81
VGGish	26.38	27.83	25.88
BioLingual	26.58	17.67	12.58
RCL FS BSED	28.04	32.67	28.58
AudioMAE	29.85	28.83	19.46
ProtoCLR	30.19	35.75	30.54
Loudness control	31.58	34.50	33.88
Duration control	33.04	34.42	34.69
Bird-MAE	33.08	35.17	30.04

Notes: Rank 1 is the highest re-ID accuracy on a dataset. Held-out rule: the six datasets where it applies. Own final output: specified after the review, so exploratory. Comparisons with BirdNET after Holm's correction: Fisher rule 9 of 481 above and 183 below; held-out rule 1 of 222 above and 79 below; own final output 5 of 481 above and 200 below. On the six datasets, the held-out rule changed the chosen layer in 130 of 222 comparisons, raising re-ID accuracy in 82 and lowering it in 45.

Table S15. Re-ID with strangers present for all 36 neural networks: medians over 16 allocations, with ranges.

Dataset	Embedding	Names correctly	Separation	Accepted and named, calibrated	Strangers accepted, calibrated	Balanced accuracy, known	Balanced accuracy, strangers	Geometric mean
Great tit, across year	BirdNET v2.4	0.600 (0.375 to 0.700)	0.617 (0.466 to 0.733)	0.175 (0.050 to 0.350)	0.087 (0.000 to 0.350)	0.175 (0.050 to 0.350)	0.913 (0.650 to 1.000)	0.394 (0.221 to 0.494)
Great tit, across year	Perch 2.0	0.625 (0.400 to 0.700)	0.616 (0.497 to 0.667)	0.225 (0.025 to 0.350)	0.087 (0.000 to 0.275)	0.225 (0.025 to 0.350)	0.913 (0.725 to 1.000)	0.419 (0.158 to 0.556)
Great tit, across year	Perch	0.425 (0.275 to 0.625)	0.534 (0.324 to 0.676)	0.062 (0.000 to 0.225)	0.087 (0.025 to 0.300)	0.062 (0.000 to 0.225)	0.913 (0.700 to 0.975)	0.239 (0.000 to 0.400)
Great tit, across year	SurfPerch	0.562 (0.350 to 0.750)	0.584 (0.443 to 0.675)	0.150 (0.025 to 0.250)	0.087 (0.000 to 0.350)	0.150 (0.025 to 0.250)	0.912 (0.650 to 1.000)	0.359 (0.156 to 0.468)
Great tit, across year	BirdNET v3 (preview)	0.325 (0.200 to 0.500)	0.540 (0.349 to 0.610)	0.038 (0.000 to 0.125)	0.087 (0.000 to 0.350)	0.038 (0.000 to 0.125)	0.913 (0.650 to 1.000)	0.176 (0.000 to 0.285)
Great tit, across year	AVEX sl-BEATs	0.412 (0.200 to 0.625)	0.530 (0.440 to 0.666)	0.062 (0.000 to 0.225)	0.100 (0.000 to 0.375)	0.062 (0.000 to 0.225)	0.900 (0.625 to 1.000)	0.238 (0.000 to 0.404)
Great tit, across year	AVEX EfficientNet-B0	0.500 (0.375 to 0.600)	0.603 (0.437 to 0.728)	0.138 (0.025 to 0.200)	0.087 (0.000 to 0.250)	0.138 (0.025 to 0.200)	0.913 (0.750 to 1.000)	0.349 (0.141 to 0.402)
Great tit, across year	AvesEcho	0.388 (0.225 to 0.550)	0.559 (0.351 to 0.700)	0.062 (0.000 to 0.325)	0.100 (0.000 to 0.650)	0.062 (0.000 to 0.325)	0.900 (0.350 to 1.000)	0.223 (0.000 to 0.406)
Great tit, across year	AudioProtoPNet	0.463 (0.350 to 0.600)	0.535 (0.454 to 0.614)	0.075 (0.025 to 0.175)	0.087 (0.025 to 0.350)	0.075 (0.025 to 0.175)	0.913 (0.650 to 0.975)	0.267 (0.135 to 0.391)
Great tit, across year	ConvNeXt (BirdSet)	0.450 (0.275 to 0.650)	0.523 (0.408 to 0.654)	0.062 (0.000 to 0.150)	0.087 (0.000 to 0.375)	0.062 (0.000 to 0.150)	0.913 (0.625 to 1.000)	0.240 (0.000 to 0.387)
Great tit, across year	Bird-MAE	0.375 (0.200 to 0.575)	0.554 (0.342 to 0.667)	0.100 (0.000 to 0.200)	0.075 (0.000 to 0.325)	0.100 (0.000 to 0.200)	0.925 (0.675 to 1.000)	0.293 (0.000 to 0.424)
Great tit, across year	ProtoCLR	0.438 (0.275 to 0.625)	0.526 (0.367 to 0.649)	0.050 (0.000 to 0.225)	0.087 (0.000 to 0.350)	0.050 (0.000 to 0.225)	0.913 (0.650 to 1.000)	0.211 (0.000 to 0.382)
Great tit, across year	RCL_FS_BSED	0.450 (0.200 to 0.825)	0.615 (0.410 to 0.862)	0.087 (0.000 to 0.375)	0.087 (0.000 to 0.650)	0.088 (0.000 to 0.375)	0.913 (0.350 to 1.000)	0.266 (0.000 to 0.466)
Great tit, across year	BirdAVES	0.487 (0.350 to 0.625)	0.600 (0.506 to 0.801)	0.225 (0.000 to 0.375)	0.087 (0.000 to 0.425)	0.225 (0.000 to 0.375)	0.913 (0.575 to 1.000)	0.439 (0.000 to 0.556)
Great tit, across year	AVES	0.500 (0.250 to 0.650)	0.597 (0.445 to 0.767)	0.175 (0.050 to 0.300)	0.087 (0.000 to 0.300)	0.175 (0.050 to 0.300)	0.913 (0.700 to 1.000)	0.380 (0.224 to 0.512)
Great tit, across year	NatureBEATs	0.375 (0.150 to 0.550)	0.524 (0.389 to 0.664)	0.050 (0.000 to 0.225)	0.075 (0.000 to 0.375)	0.050 (0.000 to 0.225)	0.925 (0.625 to 1.000)	0.217 (0.000 to 0.375)
Great tit, across year	BioLingual	0.450 (0.300 to 0.625)	0.542 (0.431 to 0.649)	0.050 (0.000 to 0.200)	0.087 (0.000 to 0.275)	0.050 (0.000 to 0.200)	0.912 (0.725 to 1.000)	0.221 (0.000 to 0.394)
Great tit, across year	BEATs	0.438 (0.250 to 0.625)	0.593 (0.341 to 0.749)	0.075 (0.000 to 0.450)	0.087 (0.000 to 0.400)	0.075 (0.000 to 0.450)	0.913 (0.600 to 1.000)	0.254 (0.000 to 0.636)
Great tit, across year	AudioMAE	0.375 (0.200 to 0.550)	0.536 (0.296 to 0.664)	0.075 (0.000 to 0.175)	0.087 (0.000 to 0.325)	0.075 (0.000 to 0.175)	0.913 (0.675 to 1.000)	0.263 (0.000 to 0.391)
Great tit, across year	VGGish	0.362 (0.200 to 0.750)	0.542 (0.370 to 0.793)	0.075 (0.025 to 0.200)	0.087 (0.000 to 0.250)	0.075 (0.025 to 0.200)	0.913 (0.750 to 1.000)	0.260 (0.139 to 0.424)
Great tit, across year	ECAPA-TDNN (speaker)	0.438 (0.175 to 0.575)	0.528 (0.415 to 0.680)	0.100 (0.000 to 0.225)	0.087 (0.025 to 0.200)	0.100 (0.000 to 0.225)	0.913 (0.800 to 0.975)	0.292 (0.000 to 0.431)
Great tit, across year	ResNet (speaker)	0.450 (0.200 to 0.625)	0.537 (0.412 to 0.671)	0.025 (0.000 to 0.350)	0.087 (0.000 to 0.400)	0.025 (0.000 to 0.350)	0.913 (0.600 to 1.000)	0.155 (0.000 to 0.458)
Great tit, across year	x-vector (speaker)	0.388 (0.175 to 0.550)	0.562 (0.394 to 0.731)	0.050 (0.000 to 0.125)	0.087 (0.000 to 0.300)	0.050 (0.000 to 0.125)	0.913 (0.700 to 1.000)	0.213 (0.000 to 0.304)

Dataset	Embedding	Names correctly	Separation	Accepted and named, calibrated	Strangers accepted, calibrated	Balanced accuracy, known	Balanced accuracy, strangers	Geometric mean
Great tit, across year	wav2vec 2.0 Base	0.425 (0.225 to 0.575)	0.603 (0.379 to 0.861)	0.100 (0.000 to 0.250)	0.075 (0.000 to 0.525)	0.100 (0.000 to 0.250)	0.925 (0.475 to 1.000)	0.304 (0.000 to 0.454)
Great tit, across year	wav2vec 2.0 Large (robust)	0.400 (0.200 to 0.700)	0.625 (0.409 to 0.828)	0.075 (0.000 to 0.225)	0.087 (0.000 to 0.250)	0.075 (0.000 to 0.225)	0.913 (0.750 to 1.000)	0.260 (0.000 to 0.462)
Great tit, across year	XLS-R 300M	0.388 (0.225 to 0.675)	0.578 (0.401 to 0.863)	0.100 (0.000 to 0.225)	0.087 (0.000 to 0.225)	0.100 (0.000 to 0.225)	0.913 (0.775 to 1.000)	0.300 (0.000 to 0.450)
Great tit, across year	wav2vec 2.0 Conformer Large	0.450 (0.250 to 0.775)	0.660 (0.482 to 0.873)	0.125 (0.000 to 0.300)	0.087 (0.000 to 0.475)	0.125 (0.000 to 0.300)	0.913 (0.525 to 1.000)	0.344 (0.000 to 0.518)
Great tit, across year	MMS 300M	0.388 (0.275 to 0.750)	0.606 (0.448 to 0.865)	0.125 (0.000 to 0.200)	0.100 (0.000 to 0.200)	0.125 (0.000 to 0.200)	0.900 (0.800 to 1.000)	0.321 (0.000 to 0.430)
Great tit, across year	HuBERT Base	0.450 (0.250 to 0.650)	0.658 (0.356 to 0.908)	0.100 (0.000 to 0.350)	0.087 (0.000 to 0.550)	0.100 (0.000 to 0.350)	0.913 (0.450 to 1.000)	0.277 (0.000 to 0.569)
Great tit, across year	HuBERT Large	0.500 (0.325 to 0.825)	0.664 (0.422 to 0.860)	0.125 (0.050 to 0.275)	0.087 (0.000 to 0.425)	0.125 (0.050 to 0.275)	0.913 (0.575 to 1.000)	0.342 (0.221 to 0.504)
Great tit, across year	data2vec Base (100 h)	0.463 (0.275 to 0.850)	0.657 (0.437 to 0.770)	0.113 (0.000 to 0.350)	0.100 (0.000 to 0.500)	0.113 (0.000 to 0.350)	0.900 (0.500 to 1.000)	0.311 (0.000 to 0.495)
Great tit, across year	data2vec Base (960 h)	0.475 (0.300 to 0.825)	0.637 (0.436 to 0.772)	0.113 (0.000 to 0.400)	0.087 (0.000 to 0.500)	0.113 (0.000 to 0.400)	0.913 (0.500 to 1.000)	0.316 (0.000 to 0.520)
Great tit, across year	WavLM Base+	0.450 (0.250 to 0.600)	0.650 (0.372 to 0.879)	0.138 (0.000 to 0.350)	0.087 (0.000 to 0.575)	0.138 (0.000 to 0.350)	0.913 (0.425 to 1.000)	0.314 (0.000 to 0.561)
Great tit, across year	WavLM Large	0.475 (0.250 to 0.775)	0.638 (0.458 to 0.885)	0.125 (0.025 to 0.375)	0.087 (0.000 to 0.375)	0.125 (0.025 to 0.375)	0.913 (0.625 to 1.000)	0.345 (0.152 to 0.556)
Great tit, across year	UniSpeech-SAT Base+	0.425 (0.275 to 0.575)	0.659 (0.359 to 0.888)	0.138 (0.000 to 0.400)	0.100 (0.000 to 0.550)	0.138 (0.000 to 0.400)	0.900 (0.450 to 1.000)	0.322 (0.000 to 0.616)
Great tit, across year	XEUS	0.425 (0.200 to 0.700)	0.638 (0.472 to 0.843)	0.100 (0.000 to 0.350)	0.087 (0.000 to 0.450)	0.100 (0.000 to 0.350)	0.912 (0.550 to 1.000)	0.310 (0.000 to 0.545)
Great tit, 50 birds, across year	BirdNET v2.4	0.348 (0.275 to 0.450)	0.545 (0.414 to 0.740)	0.092 (0.015 to 0.177)	0.092 (0.025 to 0.275)	0.092 (0.015 to 0.177)	0.908 (0.725 to 0.975)	0.288 (0.116 to 0.395)
Great tit, 50 birds, across year	Perch 2.0	0.436 (0.223 to 0.508)	0.549 (0.430 to 0.718)	0.112 (0.015 to 0.250)	0.103 (0.025 to 0.277)	0.112 (0.015 to 0.250)	0.897 (0.723 to 0.975)	0.317 (0.121 to 0.462)
Great tit, 50 birds, across year	Perch	0.204 (0.131 to 0.285)	0.525 (0.391 to 0.616)	0.023 (0.008 to 0.046)	0.092 (0.042 to 0.208)	0.023 (0.008 to 0.046)	0.908 (0.792 to 0.958)	0.147 (0.084 to 0.207)
Great tit, 50 birds, across year	SurfPerch	0.308 (0.177 to 0.450)	0.552 (0.425 to 0.641)	0.080 (0.023 to 0.192)	0.092 (0.017 to 0.262)	0.080 (0.023 to 0.192)	0.908 (0.738 to 0.983)	0.272 (0.140 to 0.392)
Great tit, 50 birds, across year	BirdNET v3 (preview)	0.144 (0.092 to 0.223)	0.509 (0.418 to 0.596)	0.004 (0.000 to 0.031)	0.096 (0.015 to 0.217)	0.004 (0.000 to 0.031)	0.904 (0.783 to 0.985)	0.042 (0.000 to 0.160)
Great tit, 50 birds, across year	AVEX sl-BEATs	0.192 (0.100 to 0.333)	0.506 (0.387 to 0.629)	0.016 (0.000 to 0.054)	0.088 (0.000 to 0.315)	0.016 (0.000 to 0.054)	0.912 (0.685 to 1.000)	0.120 (0.000 to 0.208)
Great tit, 50 birds, across year	AVEX EfficientNet-B0	0.332 (0.177 to 0.483)	0.555 (0.438 to 0.638)	0.068 (0.025 to 0.183)	0.096 (0.050 to 0.169)	0.068 (0.025 to 0.183)	0.904 (0.831 to 0.950)	0.250 (0.154 to 0.393)
Great tit, 50 birds, across year	AvesEcho	0.168 (0.100 to 0.275)	0.510 (0.440 to 0.637)	0.023 (0.008 to 0.092)	0.104 (0.000 to 0.154)	0.023 (0.008 to 0.092)	0.896 (0.846 to 1.000)	0.143 (0.081 to 0.280)
Great tit, 50 birds, across year	AudioProtoPNet	0.272 (0.200 to 0.358)	0.504 (0.362 to 0.665)	0.033 (0.008 to 0.075)	0.096 (0.017 to 0.192)	0.033 (0.008 to 0.075)	0.904 (0.808 to 0.983)	0.176 (0.084 to 0.258)
Great tit, 50 birds, across year	ConvNeXt (BirdSet)	0.260 (0.200 to 0.325)	0.498 (0.410 to 0.612)	0.028 (0.008 to 0.067)	0.096 (0.067 to 0.177)	0.028 (0.008 to 0.067)	0.904 (0.823 to 0.933)	0.155 (0.084 to 0.242)
Great tit, 50 birds, across year	Bird-MAE	0.154 (0.092 to 0.258)	0.518 (0.411 to 0.595)	0.015 (0.000 to 0.046)	0.096 (0.008 to 0.231)	0.015 (0.000 to 0.046)	0.904 (0.769 to 0.992)	0.114 (0.000 to 0.189)

Dataset	Embedding	Names correctly	Separation	Accepted and named, calibrated	Strangers accepted, calibrated	Balanced accuracy, known	Balanced accuracy, strangers	Geometric mean
Great tit, 50 birds, across year	ProtoCLR	0.162 (0.069 to 0.238)	0.507 (0.370 to 0.619)	0.012 (0.000 to 0.046)	0.096 (0.033 to 0.338)	0.012 (0.000 to 0.046)	0.904 (0.662 to 0.967)	0.100 (0.000 to 0.194)
Great tit, 50 birds, across year	RCL_FS_BSED	0.254 (0.158 to 0.392)	0.520 (0.449 to 0.670)	0.044 (0.008 to 0.125)	0.096 (0.025 to 0.262)	0.044 (0.008 to 0.125)	0.904 (0.738 to 0.975)	0.194 (0.085 to 0.329)
Great tit, 50 birds, across year	BirdAVES	0.240 (0.162 to 0.325)	0.559 (0.467 to 0.662)	0.076 (0.008 to 0.125)	0.096 (0.042 to 0.215)	0.076 (0.008 to 0.125)	0.904 (0.785 to 0.958)	0.263 (0.089 to 0.321)
Great tit, 50 birds, across year	AVES	0.212 (0.138 to 0.250)	0.531 (0.419 to 0.676)	0.050 (0.015 to 0.085)	0.096 (0.025 to 0.215)	0.050 (0.015 to 0.085)	0.904 (0.785 to 0.975)	0.214 (0.120 to 0.267)
Great tit, 50 birds, across year	NatureBEATs	0.196 (0.123 to 0.292)	0.504 (0.379 to 0.632)	0.015 (0.008 to 0.023)	0.096 (0.000 to 0.300)	0.015 (0.008 to 0.023)	0.904 (0.700 to 1.000)	0.110 (0.080 to 0.135)
Great tit, 50 birds, across year	BioLingual	0.128 (0.069 to 0.208)	0.471 (0.418 to 0.603)	0.008 (0.000 to 0.038)	0.096 (0.000 to 0.385)	0.008 (0.000 to 0.038)	0.904 (0.615 to 1.000)	0.086 (0.000 to 0.176)
Great tit, 50 birds, across year	BEATs	0.216 (0.131 to 0.325)	0.527 (0.456 to 0.690)	0.040 (0.000 to 0.108)	0.097 (0.017 to 0.331)	0.040 (0.000 to 0.108)	0.903 (0.669 to 0.983)	0.189 (0.000 to 0.303)
Great tit, 50 birds, across year	AudioMAE	0.124 (0.100 to 0.233)	0.519 (0.394 to 0.594)	0.016 (0.000 to 0.038)	0.096 (0.017 to 0.215)	0.016 (0.000 to 0.038)	0.904 (0.785 to 0.983)	0.122 (0.000 to 0.177)
Great tit, 50 birds, across year	VGGish	0.168 (0.115 to 0.342)	0.533 (0.434 to 0.662)	0.032 (0.000 to 0.069)	0.090 (0.031 to 0.238)	0.032 (0.000 to 0.069)	0.910 (0.762 to 0.969)	0.162 (0.000 to 0.237)
Great tit, 50 birds, across year	ECAPA-TDNN (speaker)	0.242 (0.123 to 0.323)	0.513 (0.388 to 0.631)	0.062 (0.000 to 0.092)	0.088 (0.017 to 0.223)	0.062 (0.000 to 0.092)	0.912 (0.777 to 0.983)	0.236 (0.000 to 0.283)
Great tit, 50 birds, across year	ResNet (speaker)	0.232 (0.167 to 0.358)	0.510 (0.422 to 0.657)	0.015 (0.000 to 0.067)	0.096 (0.000 to 0.223)	0.015 (0.000 to 0.067)	0.904 (0.777 to 1.000)	0.121 (0.000 to 0.236)
Great tit, 50 birds, across year	x-vector (speaker)	0.232 (0.146 to 0.375)	0.525 (0.402 to 0.716)	0.052 (0.008 to 0.138)	0.092 (0.031 to 0.225)	0.052 (0.008 to 0.138)	0.908 (0.775 to 0.969)	0.222 (0.082 to 0.340)
Great tit, 50 birds, across year	wav2vec 2.0 Base	0.212 (0.175 to 0.342)	0.531 (0.424 to 0.621)	0.031 (0.008 to 0.083)	0.099 (0.017 to 0.292)	0.031 (0.008 to 0.083)	0.901 (0.708 to 0.983)	0.160 (0.086 to 0.272)
Great tit, 50 birds, across year	wav2vec 2.0 Large (robust)	0.224 (0.146 to 0.275)	0.555 (0.390 to 0.685)	0.024 (0.000 to 0.083)	0.093 (0.025 to 0.223)	0.024 (0.000 to 0.083)	0.907 (0.777 to 0.975)	0.143 (0.000 to 0.262)
Great tit, 50 birds, across year	XLS-R 300M	0.208 (0.167 to 0.317)	0.542 (0.416 to 0.694)	0.031 (0.008 to 0.077)	0.097 (0.015 to 0.267)	0.031 (0.008 to 0.077)	0.903 (0.733 to 0.985)	0.165 (0.087 to 0.267)
Great tit, 50 birds, across year	wav2vec 2.0 Conformer Large	0.220 (0.158 to 0.338)	0.553 (0.425 to 0.692)	0.035 (0.008 to 0.123)	0.092 (0.017 to 0.200)	0.035 (0.008 to 0.123)	0.908 (0.800 to 0.983)	0.174 (0.089 to 0.333)
Great tit, 50 birds, across year	MMS 300M	0.225 (0.169 to 0.315)	0.556 (0.408 to 0.689)	0.044 (0.015 to 0.131)	0.096 (0.008 to 0.283)	0.044 (0.015 to 0.131)	0.904 (0.717 to 0.992)	0.196 (0.111 to 0.307)
Great tit, 50 birds, across year	HuBERT Base	0.232 (0.125 to 0.292)	0.535 (0.417 to 0.672)	0.028 (0.000 to 0.083)	0.092 (0.008 to 0.285)	0.028 (0.000 to 0.083)	0.908 (0.715 to 0.992)	0.158 (0.000 to 0.254)
Great tit, 50 birds, across year	HuBERT Large	0.227 (0.133 to 0.300)	0.560 (0.425 to 0.689)	0.044 (0.000 to 0.115)	0.096 (0.017 to 0.262)	0.044 (0.000 to 0.115)	0.904 (0.738 to 0.983)	0.195 (0.000 to 0.299)
Great tit, 50 birds, across year	data2vec Base (100 h)	0.250 (0.183 to 0.300)	0.564 (0.428 to 0.662)	0.058 (0.023 to 0.115)	0.096 (0.008 to 0.300)	0.058 (0.023 to 0.115)	0.904 (0.700 to 0.992)	0.225 (0.143 to 0.284)
Great tit, 50 birds, across year	data2vec Base (960 h)	0.258 (0.175 to 0.308)	0.572 (0.429 to 0.671)	0.062 (0.008 to 0.115)	0.092 (0.008 to 0.308)	0.062 (0.008 to 0.115)	0.908 (0.692 to 0.992)	0.230 (0.085 to 0.300)
Great tit, 50 birds, across year	WavLM Base+	0.208 (0.133 to 0.315)	0.536 (0.383 to 0.668)	0.036 (0.000 to 0.083)	0.099 (0.000 to 0.246)	0.036 (0.000 to 0.083)	0.901 (0.754 to 1.000)	0.179 (0.000 to 0.272)
Great tit, 50 birds, across year	WavLM Large	0.216 (0.150 to 0.300)	0.545 (0.386 to 0.691)	0.036 (0.000 to 0.077)	0.096 (0.017 to 0.215)	0.036 (0.000 to 0.077)	0.904 (0.785 to 0.983)	0.183 (0.000 to 0.246)
Great tit, 50 birds, across year	UniSpeech-SAT Base+	0.204 (0.108 to 0.285)	0.531 (0.412 to 0.668)	0.025 (0.000 to 0.067)	0.099 (0.008 to 0.262)	0.025 (0.000 to 0.067)	0.901 (0.738 to 0.992)	0.146 (0.000 to 0.243)

Dataset	Embedding	Names correctly	Separation	Accepted and named, calibrated	Strangers accepted, calibrated	Balanced accuracy, known	Balanced accuracy, strangers	Geometric mean
Great tit, 50 birds, across year	XEUS	0.192 (0.117 to 0.331)	0.543 (0.431 to 0.719)	0.040 (0.000 to 0.083)	0.100 (0.025 to 0.254)	0.040 (0.000 to 0.083)	0.900 (0.746 to 0.975)	0.183 (0.000 to 0.257)
Little owl, across year	BirdNET v2.4	0.659 (0.480 to 0.892)	0.691 (0.436 to 0.881)	0.200 (0.000 to 0.520)	0.084 (0.000 to 0.697)	0.213 (0.000 to 0.538)	0.894 (0.294 to 1.000)	0.388 (0.000 to 0.525)
Little owl, across year	Perch 2.0	0.751 (0.500 to 0.957)	0.743 (0.468 to 0.880)	0.374 (0.000 to 0.500)	0.100 (0.000 to 0.587)	0.367 (0.000 to 0.533)	0.918 (0.422 to 1.000)	0.481 (0.000 to 0.614)
Little owl, across year	Perch	0.742 (0.500 to 0.968)	0.759 (0.532 to 0.890)	0.322 (0.000 to 0.649)	0.091 (0.000 to 0.688)	0.340 (0.000 to 0.615)	0.910 (0.317 to 1.000)	0.497 (0.000 to 0.607)
Little owl, across year	SurfPerch	0.598 (0.337 to 0.902)	0.619 (0.501 to 0.837)	0.197 (0.000 to 0.431)	0.080 (0.000 to 0.447)	0.201 (0.000 to 0.502)	0.923 (0.555 to 1.000)	0.412 (0.000 to 0.553)
Little owl, across year	BirdNET v3 (preview)	0.423 (0.275 to 0.771)	0.593 (0.279 to 0.827)	0.125 (0.000 to 0.276)	0.085 (0.000 to 0.660)	0.156 (0.000 to 0.324)	0.921 (0.319 to 1.000)	0.320 (0.000 to 0.489)
Little owl, across year	AVEX sl-BEATs	0.519 (0.354 to 0.800)	0.584 (0.358 to 0.817)	0.107 (0.000 to 0.360)	0.086 (0.000 to 0.670)	0.098 (0.000 to 0.388)	0.917 (0.317 to 1.000)	0.264 (0.000 to 0.441)
Little owl, across year	AVEX EfficientNet-B0	0.614 (0.429 to 0.849)	0.665 (0.408 to 0.816)	0.194 (0.000 to 0.459)	0.091 (0.000 to 0.523)	0.219 (0.000 to 0.436)	0.930 (0.480 to 1.000)	0.372 (0.000 to 0.581)
Little owl, across year	AvesEcho	0.449 (0.255 to 0.567)	0.514 (0.381 to 0.817)	0.071 (0.000 to 0.174)	0.096 (0.022 to 0.367)	0.072 (0.000 to 0.192)	0.898 (0.621 to 0.977)	0.245 (0.000 to 0.417)
Little owl, across year	AudioProtoPNet	0.618 (0.469 to 0.926)	0.698 (0.376 to 0.904)	0.328 (0.000 to 0.713)	0.090 (0.000 to 0.578)	0.321 (0.000 to 0.633)	0.914 (0.451 to 1.000)	0.463 (0.000 to 0.626)
Little owl, across year	ConvNeXt (BirdSet)	0.777 (0.520 to 0.925)	0.741 (0.412 to 0.925)	0.284 (0.000 to 0.585)	0.100 (0.000 to 0.664)	0.275 (0.000 to 0.568)	0.885 (0.293 to 1.000)	0.418 (0.000 to 0.601)
Little owl, across year	Bird-MAE	0.631 (0.245 to 0.850)	0.692 (0.447 to 0.816)	0.166 (0.000 to 0.542)	0.086 (0.000 to 0.679)	0.152 (0.000 to 0.557)	0.917 (0.304 to 1.000)	0.320 (0.000 to 0.596)
Little owl, across year	ProtoCLR	0.588 (0.357 to 0.950)	0.578 (0.419 to 0.810)	0.106 (0.010 to 0.330)	0.107 (0.000 to 0.431)	0.100 (0.010 to 0.327)	0.897 (0.538 to 1.000)	0.299 (0.100 to 0.457)
Little owl, across year	RCL_FS_BSED	0.404 (0.229 to 0.657)	0.545 (0.401 to 0.814)	0.052 (0.000 to 0.153)	0.104 (0.000 to 0.367)	0.056 (0.000 to 0.149)	0.887 (0.638 to 1.000)	0.231 (0.000 to 0.345)
Little owl, across year	BirdAVES	0.474 (0.284 to 0.608)	0.580 (0.215 to 0.949)	0.096 (0.000 to 0.341)	0.110 (0.000 to 0.670)	0.117 (0.000 to 0.362)	0.903 (0.324 to 1.000)	0.281 (0.000 to 0.510)
Little owl, across year	AVES	0.525 (0.202 to 0.703)	0.595 (0.267 to 0.813)	0.099 (0.000 to 0.383)	0.088 (0.000 to 0.553)	0.104 (0.000 to 0.398)	0.904 (0.453 to 1.000)	0.274 (0.000 to 0.492)
Little owl, across year	NatureBEATs	0.544 (0.357 to 0.706)	0.611 (0.396 to 0.864)	0.071 (0.000 to 0.470)	0.086 (0.000 to 0.714)	0.071 (0.000 to 0.523)	0.907 (0.316 to 1.000)	0.263 (0.000 to 0.472)
Little owl, across year	BioLingual	0.534 (0.286 to 0.840)	0.530 (0.392 to 0.757)	0.071 (0.000 to 0.360)	0.091 (0.000 to 0.358)	0.061 (0.000 to 0.392)	0.913 (0.631 to 1.000)	0.226 (0.000 to 0.524)
Little owl, across year	BEATs	0.481 (0.380 to 0.810)	0.597 (0.345 to 0.850)	0.097 (0.000 to 0.490)	0.096 (0.000 to 0.765)	0.088 (0.000 to 0.553)	0.914 (0.240 to 1.000)	0.249 (0.000 to 0.505)
Little owl, across year	AudioMAE	0.579 (0.276 to 0.775)	0.610 (0.362 to 0.795)	0.095 (0.000 to 0.250)	0.086 (0.000 to 0.495)	0.085 (0.000 to 0.291)	0.921 (0.494 to 1.000)	0.279 (0.000 to 0.427)
Little owl, across year	VGGish	0.375 (0.153 to 0.784)	0.554 (0.242 to 0.771)	0.070 (0.000 to 0.160)	0.083 (0.000 to 0.442)	0.071 (0.000 to 0.171)	0.932 (0.515 to 1.000)	0.233 (0.000 to 0.361)
Little owl, across year	ECAPA-TDNN (speaker)	0.559 (0.286 to 0.882)	0.660 (0.393 to 0.830)	0.210 (0.024 to 0.392)	0.091 (0.000 to 0.385)	0.184 (0.022 to 0.434)	0.904 (0.579 to 1.000)	0.389 (0.143 to 0.592)
Little owl, across year	ResNet (speaker)	0.599 (0.424 to 0.951)	0.646 (0.488 to 0.789)	0.126 (0.010 to 0.417)	0.093 (0.000 to 0.495)	0.124 (0.009 to 0.441)	0.903 (0.477 to 1.000)	0.347 (0.096 to 0.504)
Little owl, across year	x-vector (speaker)	0.569 (0.353 to 0.784)	0.575 (0.383 to 0.814)	0.109 (0.010 to 0.385)	0.105 (0.000 to 0.565)	0.100 (0.008 to 0.376)	0.900 (0.429 to 1.000)	0.304 (0.087 to 0.484)

Dataset	Embedding	Names correctly	Separation	Accepted and named, calibrated	Strangers accepted, calibrated	Balanced accuracy, known	Balanced accuracy, strangers	Geometric mean
Little owl, across year	wav2vec 2.0 Base	0.450 (0.240 to 0.863)	0.560 (0.158 to 0.833)	0.029 (0.000 to 0.324)	0.090 (0.000 to 0.638)	0.025 (0.000 to 0.316)	0.906 (0.304 to 1.000)	0.151 (0.000 to 0.481)
Little owl, across year	wav2vec 2.0 Large (robust)	0.515 (0.255 to 0.843)	0.578 (0.325 to 0.800)	0.020 (0.000 to 0.233)	0.102 (0.000 to 0.585)	0.021 (0.000 to 0.235)	0.879 (0.413 to 1.000)	0.134 (0.000 to 0.378)
Little owl, across year	XLS-R 300M	0.525 (0.276 to 0.814)	0.555 (0.307 to 0.769)	0.038 (0.000 to 0.402)	0.129 (0.000 to 0.691)	0.037 (0.000 to 0.389)	0.845 (0.286 to 1.000)	0.155 (0.000 to 0.373)
Little owl, across year	wav2vec 2.0 Conformer Large	0.523 (0.459 to 0.896)	0.607 (0.386 to 0.836)	0.061 (0.000 to 0.245)	0.086 (0.000 to 0.394)	0.059 (0.000 to 0.238)	0.898 (0.542 to 1.000)	0.226 (0.000 to 0.413)
Little owl, across year	MMS 300M	0.548 (0.235 to 0.863)	0.563 (0.332 to 0.810)	0.038 (0.000 to 0.324)	0.126 (0.000 to 0.617)	0.038 (0.000 to 0.295)	0.853 (0.344 to 1.000)	0.163 (0.000 to 0.386)
Little owl, across year	HuBERT Base	0.527 (0.237 to 0.900)	0.593 (0.348 to 0.835)	0.061 (0.000 to 0.239)	0.096 (0.000 to 0.426)	0.072 (0.000 to 0.240)	0.882 (0.464 to 1.000)	0.238 (0.000 to 0.448)
Little owl, across year	HuBERT Large	0.633 (0.220 to 0.858)	0.569 (0.418 to 0.811)	0.010 (0.000 to 0.533)	0.086 (0.000 to 0.643)	0.013 (0.000 to 0.537)	0.917 (0.354 to 1.000)	0.107 (0.000 to 0.500)
Little owl, across year	data2vec Base (100 h)	0.562 (0.431 to 0.906)	0.611 (0.362 to 0.810)	0.068 (0.000 to 0.458)	0.090 (0.000 to 0.659)	0.087 (0.000 to 0.452)	0.910 (0.348 to 1.000)	0.256 (0.000 to 0.479)
Little owl, across year	data2vec Base (960 h)	0.589 (0.479 to 0.917)	0.614 (0.385 to 0.803)	0.128 (0.000 to 0.450)	0.098 (0.000 to 0.565)	0.130 (0.000 to 0.451)	0.907 (0.438 to 1.000)	0.343 (0.000 to 0.451)
Little owl, across year	WavLM Base+	0.537 (0.247 to 0.730)	0.587 (0.308 to 0.786)	0.048 (0.000 to 0.206)	0.106 (0.000 to 0.532)	0.049 (0.000 to 0.226)	0.862 (0.398 to 1.000)	0.200 (0.000 to 0.402)
Little owl, across year	WavLM Large	0.514 (0.347 to 0.825)	0.558 (0.289 to 0.794)	0.047 (0.000 to 0.363)	0.086 (0.000 to 0.489)	0.047 (0.000 to 0.332)	0.903 (0.470 to 1.000)	0.185 (0.000 to 0.405)
Little owl, across year	UniSpeech-SAT Base+	0.531 (0.269 to 0.930)	0.576 (0.409 to 0.794)	0.061 (0.000 to 0.193)	0.091 (0.000 to 0.477)	0.062 (0.000 to 0.246)	0.891 (0.499 to 1.000)	0.241 (0.000 to 0.406)
Little owl, across year	XEUS	0.565 (0.172 to 0.833)	0.579 (0.290 to 0.804)	0.045 (0.000 to 0.353)	0.096 (0.000 to 0.755)	0.045 (0.000 to 0.333)	0.913 (0.221 to 1.000)	0.200 (0.000 to 0.428)
Rook, across year	BirdNET v2.4	0.611 (0.429 to 0.879)	0.523 (0.408 to 0.679)	0.080 (0.000 to 0.234)	0.100 (0.000 to 0.503)	0.064 (0.000 to 0.224)	0.867 (0.413 to 1.000)	0.216 (0.000 to 0.421)
Rook, across year	Perch 2.0	0.575 (0.401 to 0.842)	0.610 (0.417 to 0.904)	0.101 (0.000 to 0.291)	0.100 (0.000 to 0.403)	0.094 (0.000 to 0.373)	0.876 (0.475 to 1.000)	0.293 (0.000 to 0.550)
Rook, across year	Perch	0.523 (0.367 to 0.765)	0.570 (0.373 to 0.702)	0.036 (0.000 to 0.165)	0.102 (0.000 to 0.494)	0.023 (0.000 to 0.154)	0.880 (0.471 to 1.000)	0.130 (0.000 to 0.361)
Rook, across year	SurfPerch	0.500 (0.310 to 0.821)	0.557 (0.356 to 0.707)	0.049 (0.004 to 0.125)	0.098 (0.003 to 0.437)	0.054 (0.010 to 0.142)	0.897 (0.504 to 0.997)	0.216 (0.101 to 0.329)
Rook, across year	BirdNET v3 (preview)	0.504 (0.281 to 0.791)	0.578 (0.365 to 0.861)	0.071 (0.000 to 0.282)	0.100 (0.000 to 0.407)	0.063 (0.000 to 0.344)	0.877 (0.459 to 1.000)	0.239 (0.000 to 0.494)
Rook, across year	AVEX sl-BEATs	0.552 (0.361 to 0.835)	0.570 (0.332 to 0.737)	0.119 (0.001 to 0.451)	0.095 (0.004 to 0.270)	0.140 (0.000 to 0.395)	0.891 (0.695 to 0.996)	0.353 (0.022 to 0.583)
Rook, across year	AVEX EfficientNet-B0	0.540 (0.407 to 0.828)	0.582 (0.397 to 0.890)	0.064 (0.001 to 0.211)	0.100 (0.000 to 0.383)	0.043 (0.000 to 0.261)	0.881 (0.521 to 1.000)	0.188 (0.022 to 0.414)
Rook, across year	AvesEcho	0.571 (0.326 to 0.816)	0.560 (0.355 to 0.755)	0.060 (0.000 to 0.228)	0.093 (0.000 to 0.679)	0.045 (0.000 to 0.156)	0.888 (0.320 to 1.000)	0.192 (0.000 to 0.329)
Rook, across year	AudioProtoPNet	0.574 (0.457 to 0.805)	0.570 (0.382 to 0.849)	0.064 (0.000 to 0.222)	0.099 (0.000 to 0.508)	0.048 (0.000 to 0.213)	0.864 (0.448 to 1.000)	0.183 (0.000 to 0.420)
Rook, across year	ConvNeXt (BirdSet)	0.591 (0.444 to 0.889)	0.602 (0.412 to 0.879)	0.077 (0.000 to 0.279)	0.099 (0.000 to 0.352)	0.053 (0.000 to 0.341)	0.883 (0.532 to 1.000)	0.189 (0.000 to 0.487)
Rook, across year	Bird-MAE	0.469 (0.293 to 0.771)	0.582 (0.382 to 0.682)	0.073 (0.003 to 0.382)	0.100 (0.003 to 0.276)	0.113 (0.002 to 0.391)	0.921 (0.709 to 0.999)	0.322 (0.044 to 0.532)

Dataset	Embedding	Names correctly	Separation	Accepted and named, calibrated	Strangers accepted, calibrated	Balanced accuracy, known	Balanced accuracy, strangers	Geometric mean
Rook, across year	ProtoCLR	0.471 (0.256 to 0.881)	0.575 (0.348 to 0.739)	0.078 (0.000 to 0.440)	0.099 (0.000 to 0.255)	0.112 (0.000 to 0.410)	0.912 (0.723 to 1.000)	0.315 (0.000 to 0.544)
Rook, across year	RCL_FS_BSED	0.394 (0.225 to 0.859)	0.504 (0.341 to 0.727)	0.061 (0.001 to 0.522)	0.097 (0.001 to 0.280)	0.103 (0.000 to 0.573)	0.929 (0.695 to 1.000)	0.294 (0.021 to 0.631)
Rook, across year	BirdAVES	0.502 (0.156 to 0.764)	0.525 (0.383 to 0.786)	0.069 (0.018 to 0.428)	0.099 (0.016 to 0.293)	0.074 (0.012 to 0.411)	0.914 (0.730 to 0.992)	0.258 (0.107 to 0.623)
Rook, across year	AVES	0.494 (0.172 to 0.764)	0.534 (0.381 to 0.804)	0.105 (0.023 to 0.331)	0.097 (0.017 to 0.459)	0.074 (0.014 to 0.365)	0.906 (0.572 to 0.983)	0.239 (0.117 to 0.598)
Rook, across year	NatureBEATs	0.457 (0.299 to 0.867)	0.619 (0.312 to 0.778)	0.110 (0.016 to 0.491)	0.098 (0.007 to 0.382)	0.136 (0.012 to 0.416)	0.905 (0.609 to 0.996)	0.344 (0.105 to 0.607)
Rook, across year	BioLingual	0.585 (0.353 to 0.791)	0.550 (0.400 to 0.727)	0.060 (0.003 to 0.319)	0.105 (0.000 to 0.580)	0.048 (0.002 to 0.282)	0.883 (0.347 to 1.000)	0.209 (0.044 to 0.372)
Rook, across year	BEATs	0.486 (0.217 to 0.762)	0.592 (0.278 to 0.762)	0.091 (0.015 to 0.178)	0.096 (0.013 to 0.458)	0.069 (0.021 to 0.161)	0.894 (0.648 to 0.993)	0.244 (0.141 to 0.397)
Rook, across year	AudioMAE	0.442 (0.192 to 0.768)	0.602 (0.288 to 0.698)	0.090 (0.020 to 0.223)	0.105 (0.024 to 0.239)	0.082 (0.025 to 0.230)	0.896 (0.773 to 0.979)	0.268 (0.151 to 0.470)
Rook, across year	VGGish	0.407 (0.279 to 0.742)	0.547 (0.353 to 0.634)	0.026 (0.007 to 0.084)	0.099 (0.011 to 0.259)	0.038 (0.007 to 0.097)	0.918 (0.785 to 0.991)	0.190 (0.083 to 0.282)
Rook, across year	ECAPA-TDNN (speaker)	0.472 (0.227 to 0.794)	0.518 (0.311 to 0.679)	0.051 (0.000 to 0.129)	0.099 (0.001 to 0.513)	0.045 (0.000 to 0.093)	0.904 (0.583 to 0.999)	0.199 (0.000 to 0.284)
Rook, across year	ResNet (speaker)	0.496 (0.273 to 0.901)	0.537 (0.363 to 0.757)	0.081 (0.020 to 0.199)	0.097 (0.012 to 0.279)	0.076 (0.017 to 0.203)	0.918 (0.704 to 0.986)	0.264 (0.129 to 0.444)
Rook, across year	x-vector (speaker)	0.582 (0.147 to 0.847)	0.499 (0.371 to 0.735)	0.065 (0.002 to 0.329)	0.100 (0.007 to 0.338)	0.069 (0.001 to 0.297)	0.891 (0.639 to 0.989)	0.248 (0.030 to 0.528)
Rook, across year	wav2vec 2.0 Base	0.385 (0.281 to 0.780)	0.547 (0.329 to 0.695)	0.040 (0.000 to 0.136)	0.105 (0.005 to 0.303)	0.055 (0.000 to 0.142)	0.904 (0.750 to 0.998)	0.218 (0.000 to 0.327)
Rook, across year	wav2vec 2.0 Large (robust)	0.380 (0.205 to 0.776)	0.531 (0.400 to 0.584)	0.032 (0.012 to 0.104)	0.098 (0.031 to 0.223)	0.030 (0.015 to 0.089)	0.914 (0.821 to 0.970)	0.165 (0.118 to 0.285)
Rook, across year	XLS-R 300M	0.432 (0.222 to 0.751)	0.526 (0.315 to 0.665)	0.039 (0.018 to 0.088)	0.098 (0.032 to 0.197)	0.050 (0.015 to 0.098)	0.913 (0.848 to 0.975)	0.212 (0.122 to 0.308)
Rook, across year	wav2vec 2.0 Conformer Large	0.429 (0.261 to 0.716)	0.547 (0.363 to 0.665)	0.045 (0.003 to 0.164)	0.100 (0.003 to 0.343)	0.047 (0.007 to 0.161)	0.904 (0.639 to 0.998)	0.203 (0.080 to 0.343)
Rook, across year	MMS 300M	0.441 (0.228 to 0.772)	0.541 (0.424 to 0.633)	0.043 (0.007 to 0.133)	0.103 (0.020 to 0.209)	0.040 (0.008 to 0.102)	0.906 (0.818 to 0.967)	0.189 (0.088 to 0.302)
Rook, across year	HuBERT Base	0.371 (0.212 to 0.703)	0.552 (0.374 to 0.729)	0.088 (0.002 to 0.182)	0.099 (0.028 to 0.309)	0.058 (0.002 to 0.166)	0.913 (0.737 to 0.985)	0.228 (0.040 to 0.383)
Rook, across year	HuBERT Large	0.459 (0.221 to 0.796)	0.524 (0.336 to 0.682)	0.043 (0.000 to 0.280)	0.096 (0.000 to 0.504)	0.034 (0.000 to 0.181)	0.897 (0.610 to 1.000)	0.177 (0.000 to 0.332)
Rook, across year	data2vec Base (100 h)	0.354 (0.227 to 0.714)	0.532 (0.408 to 0.653)	0.053 (0.000 to 0.155)	0.097 (0.022 to 0.271)	0.052 (0.000 to 0.100)	0.896 (0.739 to 0.981)	0.210 (0.000 to 0.313)
Rook, across year	data2vec Base (960 h)	0.400 (0.236 to 0.739)	0.537 (0.349 to 0.668)	0.033 (0.011 to 0.100)	0.098 (0.001 to 0.333)	0.030 (0.014 to 0.117)	0.899 (0.651 to 0.998)	0.161 (0.113 to 0.308)
Rook, across year	WavLM Base+	0.405 (0.162 to 0.784)	0.495 (0.359 to 0.696)	0.033 (0.002 to 0.348)	0.105 (0.005 to 0.509)	0.044 (0.001 to 0.268)	0.917 (0.592 to 0.993)	0.193 (0.037 to 0.459)
Rook, across year	WavLM Large	0.464 (0.208 to 0.793)	0.531 (0.329 to 0.688)	0.042 (0.003 to 0.087)	0.104 (0.016 to 0.222)	0.043 (0.003 to 0.066)	0.905 (0.796 to 0.982)	0.194 (0.055 to 0.237)
Rook, across year	UniSpeech-SAT Base+	0.486 (0.321 to 0.797)	0.540 (0.397 to 0.634)	0.081 (0.012 to 0.246)	0.099 (0.046 to 0.176)	0.059 (0.015 to 0.315)	0.916 (0.836 to 0.972)	0.234 (0.119 to 0.513)

Dataset	Embedding	Names correctly	Separation	Accepted and named, calibrated	Strangers accepted, calibrated	Balanced accuracy, known	Balanced accuracy, strangers	Geometric mean
Rook, across year	XEUS	0.432 (0.279 to 0.736)	0.564 (0.344 to 0.669)	0.064 (0.015 to 0.145)	0.103 (0.000 to 0.475)	0.058 (0.009 to 0.129)	0.902 (0.632 to 1.000)	0.226 (0.092 to 0.343)
Red-tailed black cockatoo, random split	BirdNET v2.4	0.980 (0.899 to 1.000)	0.850 (0.632 to 0.997)	0.741 (0.029 to 0.921)	0.082 (0.000 to 0.584)	0.627 (0.013 to 0.940)	0.939 (0.530 to 1.000)	0.718 (0.115 to 0.939)
Red-tailed black cockatoo, random split	Perch 2.0	0.968 (0.879 to 1.000)	0.908 (0.642 to 1.000)	0.814 (0.029 to 0.941)	0.085 (0.000 to 0.632)	0.757 (0.013 to 0.961)	0.939 (0.497 to 1.000)	0.750 (0.115 to 0.921)
Red-tailed black cockatoo, random split	Perch	0.975 (0.874 to 1.000)	0.878 (0.718 to 0.998)	0.736 (0.318 to 0.891)	0.100 (0.000 to 0.579)	0.671 (0.288 to 0.920)	0.939 (0.675 to 1.000)	0.769 (0.494 to 0.899)
Red-tailed black cockatoo, random split	SurfPerch	0.975 (0.906 to 1.000)	0.899 (0.681 to 0.999)	0.644 (0.118 to 1.000)	0.089 (0.000 to 0.934)	0.626 (0.089 to 1.000)	0.945 (0.221 to 1.000)	0.697 (0.292 to 0.868)
Red-tailed black cockatoo, random split	BirdNET v3 (preview)	0.921 (0.752 to 1.000)	0.828 (0.616 to 0.983)	0.602 (0.088 to 0.880)	0.080 (0.000 to 0.624)	0.557 (0.099 to 0.810)	0.943 (0.431 to 1.000)	0.637 (0.314 to 0.851)
Red-tailed black cockatoo, random split	AVEX sl-BEATs	0.820 (0.582 to 0.896)	0.675 (0.452 to 0.960)	0.208 (0.058 to 0.710)	0.091 (0.000 to 0.441)	0.192 (0.051 to 0.531)	0.889 (0.692 to 1.000)	0.429 (0.207 to 0.667)
Red-tailed black cockatoo, random split	AVEX EfficientNet-B0	0.968 (0.854 to 0.990)	0.849 (0.602 to 0.983)	0.667 (0.029 to 0.812)	0.090 (0.000 to 0.548)	0.620 (0.013 to 0.802)	0.932 (0.676 to 1.000)	0.705 (0.115 to 0.854)
Red-tailed black cockatoo, random split	AvesEcho	0.964 (0.915 to 1.000)	0.928 (0.667 to 0.988)	0.862 (0.000 to 0.981)	0.107 (0.000 to 0.632)	0.826 (0.000 to 0.994)	0.906 (0.650 to 1.000)	0.836 (0.000 to 0.936)
Red-tailed black cockatoo, random split	AudioProtoPNet	0.968 (0.792 to 1.000)	0.846 (0.555 to 0.992)	0.670 (0.059 to 0.901)	0.090 (0.000 to 0.446)	0.600 (0.049 to 0.911)	0.949 (0.640 to 1.000)	0.706 (0.219 to 0.930)
Red-tailed black cockatoo, random split	ConvNeXt (BirdSet)	0.976 (0.819 to 1.000)	0.912 (0.596 to 1.000)	0.783 (0.059 to 0.970)	0.098 (0.000 to 0.678)	0.699 (0.049 to 0.974)	0.928 (0.534 to 1.000)	0.726 (0.220 to 0.960)
Red-tailed black cockatoo, random split	Bird-MAE	0.941 (0.836 to 1.000)	0.855 (0.670 to 0.996)	0.673 (0.136 to 0.940)	0.104 (0.000 to 0.702)	0.705 (0.111 to 0.905)	0.883 (0.377 to 1.000)	0.701 (0.333 to 0.880)
Red-tailed black cockatoo, random split	ProtoCLR	0.772 (0.617 to 0.907)	0.715 (0.576 to 0.850)	0.277 (0.059 to 0.443)	0.090 (0.000 to 0.439)	0.234 (0.049 to 0.463)	0.893 (0.655 to 1.000)	0.455 (0.220 to 0.636)
Red-tailed black cockatoo, random split	RCL_FS_BSED	0.889 (0.618 to 0.985)	0.668 (0.522 to 0.981)	0.352 (0.000 to 0.752)	0.094 (0.000 to 0.733)	0.500 (0.000 to 0.839)	0.760 (0.265 to 1.000)	0.515 (0.000 to 0.717)
Red-tailed black cockatoo, random split	BirdAVES	0.986 (0.881 to 1.000)	0.923 (0.733 to 0.998)	0.815 (0.136 to 0.980)	0.092 (0.000 to 0.947)	0.784 (0.138 to 0.986)	0.911 (0.106 to 1.000)	0.760 (0.320 to 0.898)
Red-tailed black cockatoo, random split	AVES	0.981 (0.901 to 1.000)	0.924 (0.778 to 1.000)	0.848 (0.106 to 0.990)	0.111 (0.000 to 0.719)	0.825 (0.069 to 0.991)	0.913 (0.482 to 1.000)	0.777 (0.262 to 0.913)
Red-tailed black cockatoo, random split	NatureBEATs	0.944 (0.860 to 1.000)	0.903 (0.712 to 0.993)	0.706 (0.212 to 0.947)	0.094 (0.000 to 0.667)	0.664 (0.148 to 0.965)	0.935 (0.510 to 1.000)	0.765 (0.385 to 0.885)
Red-tailed black cockatoo, random split	BioLingual	0.779 (0.585 to 0.955)	0.732 (0.418 to 0.974)	0.313 (0.088 to 0.653)	0.089 (0.000 to 0.801)	0.332 (0.088 to 0.615)	0.896 (0.246 to 1.000)	0.532 (0.285 to 0.660)
Red-tailed black cockatoo, random split	BEATs	0.868 (0.727 to 0.939)	0.835 (0.568 to 0.961)	0.501 (0.030 to 0.800)	0.104 (0.000 to 0.723)	0.537 (0.017 to 0.828)	0.862 (0.290 to 1.000)	0.600 (0.131 to 0.779)
Red-tailed black cockatoo, random split	AudioMAE	0.867 (0.655 to 0.916)	0.771 (0.476 to 0.954)	0.429 (0.076 to 0.782)	0.095 (0.000 to 0.485)	0.372 (0.073 to 0.735)	0.853 (0.519 to 1.000)	0.539 (0.270 to 0.662)
Red-tailed black cockatoo, random split	VGGish	0.849 (0.680 to 0.944)	0.731 (0.593 to 0.939)	0.403 (0.045 to 0.732)	0.090 (0.000 to 0.456)	0.489 (0.107 to 0.717)	0.940 (0.557 to 1.000)	0.654 (0.327 to 0.736)
Red-tailed black cockatoo, random split	ECAPA-TDNN (speaker)	0.866 (0.636 to 0.970)	0.634 (0.353 to 0.854)	0.224 (0.053 to 0.358)	0.090 (0.000 to 0.316)	0.172 (0.018 to 0.367)	0.946 (0.829 to 1.000)	0.400 (0.124 to 0.591)
Red-tailed black cockatoo, random split	ResNet (speaker)	0.773 (0.520 to 0.970)	0.584 (0.445 to 0.784)	0.144 (0.000 to 0.430)	0.105 (0.000 to 0.386)	0.124 (0.000 to 0.333)	0.927 (0.713 to 1.000)	0.343 (0.000 to 0.498)
Red-tailed black cockatoo, random split	x-vector (speaker)	0.933 (0.758 to 0.983)	0.752 (0.543 to 0.918)	0.373 (0.099 to 0.649)	0.079 (0.000 to 0.386)	0.398 (0.062 to 0.802)	0.944 (0.678 to 1.000)	0.624 (0.249 to 0.863)

Dataset	Embedding	Names correctly	Separation	Accepted and named, calibrated	Strangers accepted, calibrated	Balanced accuracy, known	Balanced accuracy, strangers	Geometric mean
Red-tailed black cockatoo, random split	wav2vec 2.0 Base	0.937 (0.868 to 0.990)	0.867 (0.618 to 0.987)	0.648 (0.015 to 0.888)	0.113 (0.000 to 0.635)	0.633 (0.009 to 0.815)	0.909 (0.494 to 1.000)	0.687 (0.093 to 0.874)
Red-tailed black cockatoo, random split	wav2vec 2.0 Large (robust)	0.930 (0.800 to 0.972)	0.839 (0.634 to 0.965)	0.727 (0.121 to 0.860)	0.085 (0.000 to 0.526)	0.623 (0.072 to 0.849)	0.935 (0.744 to 1.000)	0.750 (0.268 to 0.883)
Red-tailed black cockatoo, random split	XLS-R 300M	0.956 (0.836 to 0.985)	0.877 (0.686 to 0.983)	0.667 (0.182 to 0.879)	0.085 (0.000 to 0.614)	0.657 (0.131 to 0.836)	0.930 (0.653 to 1.000)	0.761 (0.360 to 0.875)
Red-tailed black cockatoo, random split	wav2vec 2.0 Conformer Large	0.962 (0.800 to 0.985)	0.874 (0.644 to 0.990)	0.695 (0.136 to 0.940)	0.091 (0.000 to 0.609)	0.670 (0.100 to 0.905)	0.954 (0.627 to 1.000)	0.760 (0.317 to 0.832)
Red-tailed black cockatoo, random split	MMS 300M	0.966 (0.800 to 0.985)	0.882 (0.685 to 0.990)	0.642 (0.182 to 0.880)	0.086 (0.000 to 0.614)	0.690 (0.131 to 0.839)	0.915 (0.683 to 1.000)	0.774 (0.358 to 0.855)
Red-tailed black cockatoo, random split	HuBERT Base	0.970 (0.891 to 1.000)	0.854 (0.555 to 0.992)	0.701 (0.076 to 0.912)	0.106 (0.000 to 0.743)	0.681 (0.043 to 0.893)	0.958 (0.268 to 1.000)	0.743 (0.208 to 0.888)
Red-tailed black cockatoo, random split	HuBERT Large	0.956 (0.818 to 0.985)	0.854 (0.648 to 0.984)	0.648 (0.106 to 0.877)	0.111 (0.000 to 0.609)	0.665 (0.083 to 0.864)	0.905 (0.539 to 1.000)	0.754 (0.288 to 0.845)
Red-tailed black cockatoo, random split	data2vec Base (100 h)	0.971 (0.891 to 1.000)	0.866 (0.577 to 0.994)	0.759 (0.136 to 0.912)	0.096 (0.000 to 0.456)	0.724 (0.100 to 0.893)	0.962 (0.552 to 1.000)	0.815 (0.317 to 0.878)
Red-tailed black cockatoo, random split	data2vec Base (960 h)	0.961 (0.862 to 1.000)	0.855 (0.578 to 0.991)	0.750 (0.136 to 0.912)	0.089 (0.000 to 0.525)	0.719 (0.100 to 0.887)	0.962 (0.373 to 1.000)	0.807 (0.317 to 0.887)
Red-tailed black cockatoo, random split	WavLM Base+	0.968 (0.883 to 1.000)	0.878 (0.595 to 0.995)	0.744 (0.045 to 0.930)	0.092 (0.000 to 0.559)	0.716 (0.026 to 0.923)	0.940 (0.445 to 1.000)	0.796 (0.161 to 0.906)
Red-tailed black cockatoo, random split	WavLM Large	0.961 (0.818 to 1.000)	0.852 (0.654 to 0.988)	0.671 (0.121 to 0.900)	0.124 (0.000 to 0.604)	0.686 (0.092 to 0.893)	0.929 (0.511 to 1.000)	0.774 (0.302 to 0.849)
Red-tailed black cockatoo, random split	UniSpeech-SAT Base+	0.971 (0.926 to 1.000)	0.860 (0.589 to 0.994)	0.731 (0.045 to 0.925)	0.083 (0.000 to 0.683)	0.710 (0.026 to 0.901)	0.957 (0.319 to 1.000)	0.768 (0.161 to 0.899)
Red-tailed black cockatoo, random split	XEUS	0.969 (0.873 to 1.000)	0.880 (0.663 to 0.997)	0.750 (0.121 to 0.947)	0.106 (0.000 to 0.644)	0.747 (0.069 to 0.965)	0.945 (0.391 to 1.000)	0.843 (0.263 to 0.884)

Notes: Calibrated: threshold set on other strangers to admit one in ten; strangers accepted: the proportion of test strangers it actually admitted. Balanced accuracy on known individuals: mean over individuals of the proportion of their calls accepted and correctly named; on strangers: proportion rejected. Geometric mean of the two, as in the AnimalCLEF 2025 benchmark (Adam et al. 2025). Duration and loudness controls are excluded because cosine normalisation removes their magnitude.

Table S16. BirdNET's internal layers, a learned combination of them, and learned pooling over time.

Dataset	Layer chosen by the Fisher rule	Its re-ID accuracy	Its adjusted p	Final embedding	Best layer chosen with the test labels
Zebra finch, one bird per recording	stage 4, mean	0.933	0.0012	0.967	embedding, mean (0.967)
Great tit, across year	stage 3, mean	0.338	0.0012	0.444	embedding, mean (0.444)
Little owl, across year	stage 2, mean	0.157	0.1932	0.501	post-convolution, mean and spread (0.516)
Chiffchaff, within year	stage 3, mean	0.704	0.0012	0.765	post-convolution, mean and spread (0.769)
Chiffchaff, across year	stage 3, mean	0.170	1.0000	0.160	stage 2, mean (0.195)
Tree pipit, within year	stage 1, mean	0.139	1.0000	0.376	embedding, mean (0.376)
Tree pipit, across year	stage 1, mean	0.070	1.0000	0.214	embedding, mean (0.214)
Zebra finch, group of four	stage 2, mean and spread	0.729	0.0024	0.807	embedding, mean (0.807)
Zebra finch, group of eight	stage 4, mean	0.410	0.0624	0.498	post-convolution, mean and spread (0.515)

Dataset	Embedding	Kernel ridge	Class mean	Nearest clip
Zebra finch, one bird per recording	final layer alone	0.967	0.933	0.956
Zebra finch, one bird per recording	final layer projected	0.978	0.989	0.989
Zebra finch, one bird per recording	layers concatenated	0.967	0.933	0.967
Zebra finch, one bird per recording	learned combination	0.978	0.967	0.967
Great tit, across year	final layer alone	0.444	0.375	0.425
Great tit, across year	final layer projected	0.469	0.487	0.487
Great tit, across year	layers concatenated	0.381	0.325	0.362
Great tit, across year	learned combination	0.481	0.481	0.475
Little owl, across year	final layer alone	0.501	0.393	0.548
Little owl, across year	final layer projected	0.472	0.494	0.494
Little owl, across year	layers concatenated	0.467	0.428	0.558
Little owl, across year	learned combination	0.553	0.577	0.577
Chiffchaff, within year	final layer alone	0.765	0.549	0.643
Chiffchaff, within year	final layer projected	0.767	0.763	0.737
Chiffchaff, within year	layers concatenated	0.763	0.476	0.631
Chiffchaff, within year	learned combination	0.784	0.803	0.778
Chiffchaff, across year	final layer alone	0.160	0.125	0.115
Chiffchaff, across year	final layer projected	0.200	0.195	0.195
Chiffchaff, across year	layers concatenated	0.165	0.145	0.135
Chiffchaff, across year	learned combination	0.205	0.205	0.205
Tree pipit, within year	final layer alone	0.376	0.327	0.363
Tree pipit, within year	final layer projected	0.300	0.333	0.333
Tree pipit, within year	layers concatenated	0.373	0.261	0.323
Tree pipit, within year	learned combination	0.314	0.356	0.360
Tree pipit, across year	final layer alone	0.214	0.173	0.176
Tree pipit, across year	final layer projected	0.208	0.198	0.198
Tree pipit, across year	layers concatenated	0.192	0.144	0.179
Tree pipit, across year	learned combination	0.204	0.192	0.195

Dataset	Layer	Individuals scored	Layer averaged over time	Untrained sequence average	Learned sequence pooling
Zebra finch, one bird per recording	post-activation, sequence	3	0.967	0.900	0.900
Zebra finch, one bird per recording	stage 1, sequence	3	0.867	0.700	0.833
Zebra finch, one bird per recording	stage 2, sequence	3	0.967	0.733	0.633
Zebra finch, one bird per recording	stage 3, sequence	3	0.900	0.933	0.800
Zebra finch, one bird per recording	stage 4, sequence	3	0.967	0.900	0.900
Great tit, across year	post-activation, sequence	7	0.429	0.243	0.314
Great tit, across year	stage 1, sequence	7	0.200	0.229	0.214
Great tit, across year	stage 2, sequence	7	0.200	0.229	0.214
Great tit, across year	stage 3, sequence	7	0.343	0.343	0.386
Great tit, across year	stage 4, sequence	7	0.500	0.243	0.314
Little owl, across year	post-activation, sequence	7	0.512	0.351	0.399
Little owl, across year	stage 1, sequence	7	0.262	0.286	0.375
Little owl, across year	stage 2, sequence	7	0.440	0.363	0.196
Little owl, across year	stage 3, sequence	7	0.458	0.363	0.250
Little owl, across year	stage 4, sequence	7	0.506	0.375	0.363
Chiffchaff, within year	post-activation, sequence	6	0.491	0.403	0.441
Chiffchaff, within year	stage 1, sequence	6	0.153	0.213	0.353
Chiffchaff, within year	stage 2, sequence	6	0.214	0.345	0.592
Chiffchaff, within year	stage 3, sequence	6	0.468	0.592	0.568
Chiffchaff, within year	stage 4, sequence	6	0.474	0.427	0.518
Egyptian fruit bat, across year	post-activation, sequence	3	0.368	0.361	0.265
Egyptian fruit bat, across year	stage 1, sequence	3	0.345	0.321	0.292
Egyptian fruit bat, across year	stage 2, sequence	3	0.311	0.302	0.247
Egyptian fruit bat, across year	stage 3, sequence	3	0.255	0.303	0.255
Egyptian fruit bat, across year	stage 4, sequence	3	0.373	0.332	0.302

Notes: Upper: the 14 candidates of Appendix S1.13, with the permutation p (9,999 shuffles) multiplied by the number of distinct candidates (12, since two of the 14 repeat others) and capped at one; the best layer chosen with the test labels cannot be chosen in practice. Middle: learned combination against the final embedding alone, the same projection fitted to the final embedding, and the seven layers joined; no interval was computed on the differences. Lower: learned pooling scored on individuals it never saw (3 to 7 per dataset), so its values are not comparable in scale with Table 2.

Table S17. The four pairings on North Atlantic right whale upcalls and tag noise from 11 whales (data and frequency shift of Tolkova et al. 2026), with BirdNET.

Frequency shift (Hz)	Fisher ratio on enrolment clips	C→C (accuracy / p)	B→B (accuracy / p)	C→B (accuracy / p)	B→C (accuracy / p)
0	0.660	0.833 / 0.0001	0.625 / 0.0001	0.250 / 0.0451	0.292 / 0.0105
1000	0.516	0.806 / 0.0001	0.750 / 0.0001	0.333 / 0.0094	0.306 / 0.0046
2000	0.559	0.694 / 0.0001	0.611 / 0.0001	0.181 / 0.2611	0.292 / 0.0048
3000	0.605	0.681 / 0.0001	0.625 / 0.0001	0.319 / 0.0175	0.417 / 0.0001
4000	0.585	0.764 / 0.0001	0.639 / 0.0001	0.389 / 0.0037	0.458 / 0.0001
5000	0.534	0.639 / 0.0001	0.611 / 0.0001	0.264 / 0.0531	0.306 / 0.0033
6000	0.715	0.708 / 0.0001	0.597 / 0.0001	0.264 / 0.0445	0.333 / 0.0023
7000	0.547	0.681 / 0.0001	0.611 / 0.0001	0.333 / 0.0074	0.347 / 0.0004
8000	0.691	0.708 / 0.0001	0.583 / 0.0001	0.222 / 0.1094	0.361 / 0.0003
9000	0.639	0.708 / 0.0001	0.569 / 0.0001	0.264 / 0.0372	0.264 / 0.0347
10000	0.726	0.583 / 0.0001	0.556 / 0.0001	0.250 / 0.0551	0.208 / 0.1054

Notes: Each whale carried one tag deployment, so enrolment and test clips come from one recording. The shift moves the 50 to 500 Hz upcall band upward before BirdNET reads it, identically for calls and noise. The shift for the result was specified in advance as the one with the highest Fisher ratio on the enrolment calls (10,000 Hz). p from 9,999 label shuffles. After Holm's correction over the 11 shifts, B→C exceeded chance at 9 shifts and C→B at 1, and neither at 10,000 Hz. Data: github.com/avokloti/narw-acoustic-identification, commit 3cad65d, which declares no licence; no clip is redistributed. No identity claim is made from this dataset.

Table S18. Beecher's information statistic (bits) for each neural network at its layer chosen by the Fisher rule.

Neural network	Zebra finch, group of four	Zebra finch, one bird per recording	Zebra finch, group of eight	Great tit, across year	Tree pipit, across year	Chiffchaff, across year	Tree pipit, within year	Little owl, across year	Little penguin, across night	Red-tailed black cockatoo, random split	Egyptian fruit bat, across year	Chiffchaff, within year	Rook, across year	Great tit, 50 birds, across year
BirdNET v2.4	1.12	7.92	2.10	2.80	2.08	3.45	2.34	7.58	4.01	5.82	0.81	4.37	1.19	8.19
Perch 2.0	1.37	7.33	2.16	2.67	1.71	2.20	2.15	7.91	3.94	5.86	0.50	3.18	1.28	7.69
Perch	1.12	7.12	2.11	2.15	1.72	1.71	1.84	8.32	3.62	5.79	0.26	2.38	1.04	5.09
SurfPerch	1.02	4.68	1.56	2.16	1.27	1.67	1.46	4.97	3.08	6.17	0.20	2.24	0.71	6.17
BirdNET v3 (preview)	1.24	6.50	1.52	1.05	1.16	1.17	1.29	4.98	2.25	3.91	0.33	1.96	0.81	2.91
AVEX sl-BEATs	0.27	5.18	0.88	2.49	1.00	1.89	1.35	3.74	2.42	2.83	0.59	3.16	0.73	7.00
AVEX EfficientNet-B0	0.99	5.04	1.62	1.98	1.11	1.52	1.42	5.41	2.65	4.19	0.55	2.03	0.89	5.45
AvesEcho	0.99	6.01	1.88	2.83	1.27	3.14	1.84	4.07	2.51	7.59	0.74	4.30	0.96	7.95
AudioProto PNet	0.97	7.58	2.07	2.24	1.68	1.97	2.17	7.56	3.55	5.34	0.68	2.86	1.00	5.12
ConvNeXt (BirdSet)	1.20	7.48	2.36	1.78	2.06	2.25	2.51	7.92	4.11	6.07	0.64	2.85	1.14	4.68
Bird-MAE	0.94	4.40	1.33	2.51	1.02	1.82	1.60	5.61	2.96	5.51	0.58	3.29	0.50	7.37
ProtoCLR	0.31	3.20	1.15	2.53	0.94	2.05	1.10	4.87	2.55	3.48	0.62	2.33	0.41	6.16
RCL_FS_B SED	0.82	3.50	1.21	2.98	1.73	2.63	2.14	3.91	3.84	4.57	0.93	3.01	0.47	8.07
BirdAVES	0.63	2.58	1.38	2.77	1.64	2.36	1.82	4.37	4.04	6.85	0.89	2.25	0.55	6.63
AVES	0.33	2.81	1.25	2.72	1.81	2.69	2.17	4.51	4.80	6.83	0.76	2.61	0.61	7.08
NatureBEATs	0.89	3.08	0.94	1.51	1.08	2.17	1.34	3.98	1.94	6.28	0.54	3.37	0.64	6.64
BioLingual	0.90	6.03	1.62	2.28	1.16	1.67	1.37	3.90	2.40	3.99	0.40	3.04	0.98	3.84
BEATs	0.65	3.35	1.35	2.43	1.45	2.05	1.54	3.82	2.51	4.28	0.58	2.44	0.77	7.02
AudioMAE	0.87	4.49	1.53	2.02	1.06	1.52	1.35	4.15	1.90	3.67	0.69	2.31	0.63	4.79
VGGish	0.66	2.68	1.15	2.03	1.36	1.72	1.23	2.88	2.48	4.03	0.45	2.09	0.40	4.89
ECAPA-TD NN (speaker)	1.17	5.75	1.73	1.52	1.25	1.68	1.47	4.90	3.13	3.73	0.14	1.49	0.77	3.95
ResNet (speaker)	1.01	6.35	1.73	1.73	1.17	1.90	1.47	4.22	2.87	3.11	0.34	1.61	0.68	4.41
x-vector (speaker)	1.02	4.88	1.39	2.02	1.69	2.26	2.13	5.15	3.54	4.66	0.38	1.97	0.85	4.79
wav2vec 2.0 Base	1.18	4.13	1.69	2.91	1.95	2.37	1.87	4.13	4.56	5.80	0.92	2.60	0.40	6.98
wav2vec 2.0 Large (robust)	1.24	3.35	1.81	3.09	1.70	2.63	1.66	5.52	5.30	6.60	0.93	2.60	0.47	7.68
XLS-R 300M	1.16	4.68	2.25	2.89	1.87	2.70	1.72	5.48	5.32	6.89	0.88	2.57	0.61	7.29

Neural network wav2vec 2.0 Conformer Large	Zebra finch, group of four 1.27	Zebra finch, one bird per recording 4.88	Zebra finch, group of eight 2.21	Great tit, across year 3.36	Tree pipit, across year 1.74	Chiffchaff, across year 2.71	Tree pipit, within year 1.75	Little owl, across year 6.02	Little penguin, across night 5.40	Red-tailed black cockatoo, random split 7.22	Egyptian fruit bat, across year 1.04	Chiffchaff, within year 2.92	Rook, across year 0.50	Great tit, 50 birds, across year 8.18
MMS 300M	1.16	4.47	2.32	2.81	1.89	2.70	1.66	5.51	5.36	6.95	0.93	2.59	0.58	7.45
HuBERT Base	0.81	4.46	1.80	2.95	1.85	2.30	2.01	5.38	5.09	6.77	0.96	2.51	0.52	7.09
HuBERT Large	0.99	4.22	1.06	3.29	1.74	2.66	1.87	5.59	5.09	7.04	0.98	2.72	0.56	7.76
data2vec Base (100 h)	1.20	2.82	0.99	3.26	2.20	2.96	1.97	5.98	5.06	7.44	0.91	2.96	0.34	8.18
data2vec Base (960 h)	1.17	2.64	1.14	3.28	2.14	2.91	1.98	6.05	4.85	7.06	0.96	2.68	0.51	8.09
WavLM Base+	1.20	5.06	1.96	2.89	1.81	2.39	1.97	5.67	5.18	6.60	0.96	2.69	0.30	7.28
WavLM Large	1.05	4.20	1.98	3.15	1.74	2.64	1.85	5.59	5.26	6.68	1.00	2.78	0.56	8.10
UniSpeech-SAT Base+	1.23	3.96	1.92	3.02	1.82	2.34	1.94	5.44	4.97	6.74	0.93	2.62	0.36	7.28
XEUS	1.39	4.85	2.10	3.07	2.13	2.70	2.16	5.76	5.54	7.68	0.82	2.96	0.63	7.82

Notes: Computed on all call clips of each dataset from principal components (Appendix S1.7). It measures how well clips group by label, whether the grouping comes from the animal or the recording.

Table S19. BirdNET, the 16 speech network checkpoints of the earlier run and two controls, run twice: candidate embeddings compared on each dataset.

Dataset	Candidates in the first run	Compared	Identical
Zebra finch, group of four	734	734	734
Zebra finch, one bird per recording	735	735	735
Zebra finch, group of eight	492	492	492
Great tit, across year	735	735	735
Tree pipit, across year	659	659	659
Chiffchaff, across year	710	710	708
Tree pipit, within year	659	659	659
Little owl, across year	735	735	735
Little penguin, across night	735	735	735
Red-tailed black cockatoo, random split	735	735	735
Egyptian fruit bat, across year	735	735	735
Chiffchaff, within year	633	633	633

Notes: Identical: the same re-ID accuracy and the same right or wrong answer on every test clip. The two candidates that differed (Appendix S1.3) were not chosen by either layer rule, and every reported embedding was identical.

Table S20. Datasets screened and not used, with the reason.

Dataset	Reported scale	Reason not used
Rorua (great spotted kiwi, <i>Apteryx maxima</i>), Bedoya & Molles (2021)	849 calls, 30 individuals	Call labels derive from the nest associated with each recorder and the published workflow divides calls at random, so nest, recorder, location and identity are not independently crossed.
Wild zebra finches, Chauhan et al. (2025)	2,915 clips, 173 individuals	Identity labels were assigned from distinctive song exemplars, with a median of eight clips per individual; too few to hold a group of strangers back.
House wren, Krieg & Wade (2023)	35 birds (17 male, 18 female)	No bird has ten calls on each of two recordings or dates, which the admission rule specified in advance requires.
Black-capped and Carolina chickadee hybrid zone, Palmer et al. (2025)	55 genotyped birds; song recordings from 10	Ten recorded birds are too few for a design across several days.
Ovenbird, Lapp et al. (2025)	Public code sample: 100 localised calls from 10 individuals; evaluation set described in the paper: 3,963 clips from 45 individuals	The public sample is too small, and the evaluation set named in the paper was not obtained.

Table S21. Within-recording permutation test for all 38 embeddings and controls on the datasets in which recordings hold several individuals.

Embedding	Zebra finch, group of four	Zebra finch, group of eight	Rook, across year	Zebra finch, one bird per recording
BirdNET v2.4	0.807; 0.587; 0.0001	0.498; 0.362; 0.0002	0.375; 0.224; 0.0001	0.967; 0.856; 0.0002
Perch 2.0	0.830; 0.651; 0.0001	0.559; 0.354; 0.0001	0.418; 0.234; 0.0001	0.956; 0.789; 0.0001
Perch	0.803; 0.587; 0.0001	0.528; 0.371; 0.0001	0.393; 0.212; 0.0001	0.878; 0.800; 0.0008
SurfPerch	0.780; 0.596; 0.0002	0.397; 0.297; 0.0014	0.306; 0.166; 0.0001	0.900; 0.689; 0.0001
BirdNET v3 (preview)	0.794; 0.633; 0.0013	0.515; 0.354; 0.0001	0.328; 0.193; 0.0001	0.844; 0.700; 0.0001
AVEX sl-BEATS	0.417; 0.472; 0.1958	0.201; 0.179; 0.0115	0.284; 0.238; 0.0005	0.733; 0.578; 0.0001
AVEX EfficientNet-B0	0.798; 0.578; 0.0001	0.485; 0.376; 0.0002	0.345; 0.192; 0.0001	0.878; 0.778; 0.0002
AvesEcho	0.541; 0.394; 0.0031	0.297; 0.240; 0.0066	0.343; 0.199; 0.0001	0.756; 0.633; 0.0001
AudioProtoPNet	0.835; 0.638; 0.0001	0.576; 0.419; 0.0002	0.386; 0.227; 0.0001	0.933; 0.822; 0.0005
ConvNeXt (BirdSet)	0.839; 0.615; 0.0001	0.502; 0.354; 0.0002	0.409; 0.225; 0.0001	0.978; 0.867; 0.0001
Bird-MAE	0.367; 0.422; 0.5322	0.188; 0.175; 0.0330	0.125; 0.221; 0.8013	0.567; 0.511; 0.0007
ProtoCLR	0.372; 0.394; 0.0938	0.118; 0.188; 0.5933	0.101; 0.233; 0.9833	0.378; 0.389; 0.0938
RCL FS BSSED	0.661; 0.454; 0.0001	0.210; 0.170; 0.0201	0.105; 0.219; 0.9063	0.456; 0.478; 0.3139
BirdAVES	0.638; 0.509; 0.0011	0.245; 0.245; 0.0530	0.151; 0.191; 0.3141	0.511; 0.433; 0.0001
AVES	0.445; 0.399; 0.0171	0.175; 0.183; 0.0942	0.179; 0.183; 0.0665	0.556; 0.489; 0.0011
NatureBEATs	0.734; 0.491; 0.0001	0.201; 0.166; 0.0069	0.166; 0.242; 0.7836	0.600; 0.456; 0.0001
BioLingual	0.417; 0.376; 0.0054	0.192; 0.201; 0.0741	0.281; 0.248; 0.0008	0.956; 0.811; 0.0001
BEATs	0.495; 0.500; 0.0881	0.197; 0.188; 0.0446	0.207; 0.193; 0.0167	0.500; 0.444; 0.0007
AudioMAE	0.509; 0.413; 0.0006	0.275; 0.179; 0.0005	0.185; 0.237; 0.4203	0.733; 0.656; 0.0015
VGGish	0.702; 0.477; 0.0001	0.445; 0.402; 0.0064	0.180; 0.152; 0.0009	0.533; 0.511; 0.0280
ECAPA-TDNN (speaker)	0.711; 0.537; 0.0001	0.454; 0.245; 0.0001	0.271; 0.188; 0.0001	0.844; 0.678; 0.0001
ResNet (speaker)	0.771; 0.532; 0.0001	0.467; 0.271; 0.0001	0.297; 0.227; 0.0001	0.878; 0.789; 0.0005
x-vector (speaker)	0.789; 0.518; 0.0001	0.454; 0.319; 0.0001	0.296; 0.186; 0.0001	0.767; 0.667; 0.0004
wav2vec 2.0 Base	0.826; 0.560; 0.0001	0.380; 0.240; 0.0010	0.153; 0.142; 0.0128	0.622; 0.622; 0.0530
wav2vec 2.0 Large (robust)	0.555; 0.399; 0.0048	0.310; 0.245; 0.0124	0.125; 0.217; 0.7978	0.256; 0.311; 0.3913
XLS-R 300M	0.789; 0.606; 0.0004	0.367; 0.227; 0.0025	0.187; 0.185; 0.0408	0.756; 0.700; 0.0023
wav2vec 2.0 Conformer Large	0.780; 0.555; 0.0001	0.406; 0.367; 0.0192	0.256; 0.179; 0.0001	0.700; 0.611; 0.0003
MMS 300M	0.798; 0.587; 0.0002	0.402; 0.271; 0.0015	0.163; 0.164; 0.0511	0.722; 0.667; 0.0013
HuBERT Base	0.638; 0.550; 0.0035	0.376; 0.314; 0.0083	0.235; 0.156; 0.0001	0.744; 0.644; 0.0001
HuBERT Large	0.716; 0.546; 0.0008	0.218; 0.197; 0.0175	0.197; 0.209; 0.2489	0.656; 0.600; 0.0011
data2vec Base (100 h)	0.743; 0.606; 0.0011	0.480; 0.371; 0.0001	0.169; 0.162; 0.0250	0.478; 0.433; 0.0005
data2vec Base (960 h)	0.743; 0.583; 0.0007	0.371; 0.419; 0.2285	0.236; 0.187; 0.0001	0.422; 0.378; 0.0020
WavLM Base+	0.817; 0.573; 0.0001	0.397; 0.310; 0.0022	0.105; 0.149; 0.8075	0.756; 0.689; 0.0010
WavLM Large	0.743; 0.500; 0.0001	0.314; 0.288; 0.0287	0.166; 0.192; 0.5178	0.711; 0.644; 0.0003
UniSpeech-SAT Base+	0.794; 0.573; 0.0001	0.410; 0.328; 0.0017	0.181; 0.162; 0.0067	0.789; 0.678; 0.0001
XEUS	0.817; 0.560; 0.0001	0.445; 0.354; 0.0019	0.247; 0.190; 0.0001	0.867; 0.767; 0.0002
Loudness control	0.381; 0.353; 0.0266	0.210; 0.275; 0.3593	0.126; 0.125; 0.0480	0.233; 0.278; 0.5674
Duration control	0.468; 0.468; 0.9617	0.284; 0.284; 0.1037	0.030; 0.180; 0.9615	0.189; 0.200; 0.7396

Notes: Each cell: re-ID accuracy; 95th percentile of the within-recording test; its p (9,999 shuffles of labels among clips of the same recording, or, for the zebra finch recorded singly, the same day). Entries with $p < 0.05$: Zebra finch, group of four 33; Zebra finch, group of eight 31; Rook, across year 25; Zebra finch, one bird per recording 32. Each embedding at its layer chosen by the Fisher rule.

Table S22. Exploratory analysis, specified after the fourth round of review: each neural network read at its own final output against BirdNET v2.4.*Zebra finch, group of four:* BirdNET v2.4 0.807 (0.669 to 0.963).

Neural network	Output read, re-ID accuracy	Difference from BirdNET	Holm p
Perch 2.0	embedding: 0.830 (0.679 to 0.923)	0.023 (−0.037 to 0.073)	1.0000
Perch	embedding: 0.803 (0.717 to 0.838)	−0.005 (−0.125 to 0.079)	1.0000
SurfPerch	embedding: 0.780 (0.682 to 0.860)	−0.028 (−0.092 to 0.057)	1.0000
BirdNET v3 (preview)	embedding: 0.794 (0.696 to 0.900)	−0.014 (−0.062 to 0.027)	1.0000
AVEX sl-BEATs	embedding: 0.716 (0.596 to 0.900)	−0.092 (−0.140 to 0.017)	0.9594
AVEX EfficientNet-B0	embedding: 0.798 (0.660 to 0.867)	−0.009 (−0.094 to 0.055)	1.0000
AvesEcho	embedding: 0.798 (0.702 to 0.942)	−0.009 (−0.036 to 0.075)	1.0000
AudioProtoPNet	embedding: 0.835 (0.679 to 0.950)	0.028 (−0.013 to 0.064)	1.0000
ConvNeXt (BirdSet)	embedding: 0.839 (0.698 to 0.950)	0.032 (−0.010 to 0.070)	1.0000
Bird-MAE	embedding: 0.532 (0.302 to 0.606)	−0.275 (−0.437 to −0.175)	< 0.0074
ProtoCLR	embedding: 0.610 (0.506 to 0.672)	−0.197 (−0.376 to −0.063)	< 0.0074
RCL FS_BSED	embedding: 0.661 (0.491 to 0.835)	−0.147 (−0.273 to −0.027)	0.6216
BirdAVES	embedding: 0.638 (0.396 to 0.774)	−0.170 (−0.319 to −0.123)	< 0.0074
AVES	embedding: 0.596 (0.385 to 0.742)	−0.211 (−0.287 to −0.152)	< 0.0074
NatureBEATs	embedding: 0.780 (0.660 to 0.908)	−0.028 (−0.054 to −0.009)	0.1716
BioLingual	embedding: 0.761 (0.623 to 0.938)	−0.046 (−0.072 to −0.012)	0.1716
BEATs	embedding: 0.725 (0.561 to 0.946)	−0.083 (−0.130 to −0.012)	0.1716
AudioMAE	embedding: 0.683 (0.482 to 0.958)	−0.124 (−0.251 to 0.023)	1.0000
VGGish	embedding: 0.702 (0.486 to 0.873)	−0.106 (−0.182 to −0.058)	< 0.0074
ECAPA-TDNN (speaker)	embedding: 0.702 (0.528 to 0.850)	−0.106 (−0.151 to −0.086)	< 0.0074
ResNet (speaker)	embedding: 0.794 (0.669 to 0.896)	−0.014 (−0.059 to 0.038)	1.0000
x-vector (speaker)	embedding: 0.729 (0.509 to 0.914)	−0.078 (−0.170 to −0.042)	< 0.0074
wav2vec 2.0 Base	layer 12, full speed: 0.390 (0.188 to 0.532)	−0.417 (−0.758 to −0.202)	< 0.0074
wav2vec 2.0 Large (robust)	layer 24, full speed: 0.298 (0.123 to 0.585)	−0.509 (−0.620 to −0.170)	< 0.0074
XLS-R 300M	layer 24, full speed: 0.683 (0.434 to 0.908)	−0.124 (−0.245 to −0.047)	< 0.0074
wav2vec 2.0 Conformer Large	layer 24, full speed: 0.651 (0.517 to 0.824)	−0.156 (−0.224 to −0.046)	0.1716
MMS 300M	layer 24, full speed: 0.665 (0.392 to 0.937)	−0.142 (−0.254 to −0.019)	0.1716
HuBERT Base	layer 12, full speed: 0.555 (0.424 to 0.648)	−0.252 (−0.483 to −0.088)	< 0.0074
HuBERT Large	layer 24, full speed: 0.683 (0.561 to 0.824)	−0.124 (−0.140 to −0.075)	< 0.0074
data2vec Base (100 h)	layer 12, full speed: 0.569 (0.447 to 0.647)	−0.239 (−0.459 to −0.098)	< 0.0074
data2vec Base (960 h)	layer 12, full speed: 0.390 (0.342 to 0.491)	−0.417 (−0.617 to −0.189)	0.1716
WavLM Base+	layer 12, full speed: 0.528 (0.423 to 0.623)	−0.280 (−0.538 to −0.088)	0.1716
WavLM Large	layer 24, full speed: 0.752 (0.604 to 0.892)	−0.055 (−0.108 to −0.008)	0.1716
UniSpeech-SAT Base+	layer 12, full speed: 0.670 (0.568 to 0.769)	−0.138 (−0.188 to −0.075)	< 0.0074
XEUS	layer 18, full speed: 0.495 (0.000 to 0.917)	−0.312 (−0.912 to 0.174)	1.0000
Loudness control	summary: 0.381 (0.018 to 0.846)	−0.427 (−0.693 to −0.118)	< 0.0074
Duration control	summary: 0.468 (0.000 to 0.850)	−0.339 (−0.906 to 0.107)	1.0000

Zebra finch, one bird per recording: BirdNET v2.4 0.967 (0.933 to 0.989).

Neural network	Output read, re-ID accuracy	Difference from BirdNET	Holm p
Perch 2.0	embedding: 0.956 (0.911 to 1.000)	−0.011 (−0.044 to 0.022)	1.0000

Neural network	Output read, re-ID accuracy	Difference from BirdNET	Holm p
Perch	embedding: 0.878 (0.722 to 0.978)	-0.089 (-0.222 to 0.000)	0.4338
SurfPerch	embedding: 0.900 (0.833 to 0.967)	-0.067 (-0.111 to -0.022)	0.0154
BirdNET v3 (preview)	embedding: 0.844 (0.744 to 0.933)	-0.122 (-0.200 to -0.044)	0.0240
AVEX sl-BEATs	embedding: 0.889 (0.811 to 0.967)	-0.078 (-0.156 to -0.022)	0.4338
AVEX EfficientNet-B0	embedding: 0.878 (0.767 to 0.978)	-0.089 (-0.189 to 0.000)	0.4416
AvesEcho	embedding: 0.889 (0.756 to 0.978)	-0.078 (-0.189 to 0.000)	0.4338
AudioProtoPNet	embedding: 0.933 (0.844 to 1.000)	-0.033 (-0.111 to 0.022)	1.0000
ConvNeXt (BirdSet)	embedding: 0.978 (0.933 to 1.000)	0.011 (-0.022 to 0.044)	1.0000
Bird-MAE	embedding: 0.689 (0.522 to 0.833)	-0.278 (-0.444 to -0.122)	< 0.0074
ProtoCLR	embedding: 0.389 (0.189 to 0.589)	-0.578 (-0.767 to -0.389)	< 0.0074
RCL FS BSSED	embedding: 0.456 (0.189 to 0.722)	-0.511 (-0.756 to -0.267)	< 0.0074
BirdAVES	embedding: 0.511 (0.333 to 0.678)	-0.456 (-0.611 to -0.300)	< 0.0074
AVES	embedding: 0.567 (0.411 to 0.722)	-0.400 (-0.533 to -0.267)	< 0.0074
NatureBEATs	embedding: 0.856 (0.767 to 0.944)	-0.111 (-0.178 to -0.044)	0.0052
BioLingual	embedding: 0.956 (0.911 to 0.989)	-0.011 (-0.044 to 0.022)	1.0000
BEATs	embedding: 0.700 (0.500 to 0.878)	-0.267 (-0.444 to -0.111)	< 0.0074
AudioMAE	embedding: 0.700 (0.522 to 0.845)	-0.267 (-0.467 to -0.100)	0.0052
VGGish	embedding: 0.533 (0.333 to 0.733)	-0.433 (-0.622 to -0.244)	< 0.0074
ECAPA-TDNN (speaker)	embedding: 0.722 (0.533 to 0.878)	-0.244 (-0.411 to -0.100)	< 0.0074
ResNet (speaker)	embedding: 0.878 (0.756 to 0.967)	-0.089 (-0.200 to 0.000)	0.4416
x-vector (speaker)	embedding: 0.767 (0.622 to 0.900)	-0.200 (-0.333 to -0.089)	0.0028
wav2vec 2.0 Base	layer 12, full speed: 0.311 (0.144 to 0.489)	-0.656 (-0.822 to -0.478)	< 0.0074
wav2vec 2.0 Large (robust)	layer 24, full speed: 0.256 (0.056 to 0.478)	-0.711 (-0.889 to -0.511)	< 0.0074
XLS-R 300M	layer 24, full speed: 0.356 (0.178 to 0.556)	-0.611 (-0.778 to -0.422)	< 0.0074
wav2vec 2.0 Conformer Large	layer 24, full speed: 0.478 (0.267 to 0.678)	-0.489 (-0.667 to -0.322)	< 0.0074
MMS 300M	layer 24, full speed: 0.578 (0.367 to 0.778)	-0.389 (-0.589 to -0.189)	< 0.0074
HuBERT Base	layer 12, full speed: 0.433 (0.267 to 0.589)	-0.533 (-0.689 to -0.378)	< 0.0074
HuBERT Large	layer 24, full speed: 0.422 (0.244 to 0.622)	-0.544 (-0.700 to -0.367)	< 0.0074
data2vec Base (100 h)	layer 12, full speed: 0.389 (0.167 to 0.622)	-0.578 (-0.778 to -0.367)	< 0.0074
data2vec Base (960 h)	layer 12, full speed: 0.367 (0.156 to 0.589)	-0.600 (-0.789 to -0.400)	< 0.0074
WavLM Base+	layer 12, full speed: 0.422 (0.233 to 0.622)	-0.544 (-0.722 to -0.356)	< 0.0074
WavLM Large	layer 24, full speed: 0.611 (0.411 to 0.800)	-0.356 (-0.556 to -0.167)	< 0.0074
UniSpeech-SAT Base+	layer 12, full speed: 0.433 (0.233 to 0.633)	-0.533 (-0.711 to -0.344)	< 0.0074
XEUS	layer 18, full speed: 0.278 (0.100 to 0.478)	-0.689 (-0.856 to -0.500)	< 0.0074
Loudness control	summary: 0.233 (0.100 to 0.389)	-0.733 (-0.878 to -0.578)	< 0.0074
Duration control	summary: 0.189 (0.000 to 0.456)	-0.778 (-0.967 to -0.533)	< 0.0074

Zebra finch, group of eight: BirdNET v2.4 0.498 (0.181 to 0.731).

Neural network	Output read, re-ID accuracy	Difference from BirdNET	Holm p
Perch 2.0	embedding: 0.559 (0.374 to 0.719)	0.061 (-0.034 to 0.209)	1.0000
Perch	embedding: 0.528 (0.365 to 0.725)	0.031 (-0.093 to 0.199)	1.0000
SurfPerch	embedding: 0.397 (0.171 to 0.689)	-0.100 (-0.213 to 0.016)	1.0000
BirdNET v3 (preview)	embedding: 0.515 (0.347 to 0.692)	0.017 (-0.099 to 0.179)	1.0000
AVEX sl-BEATs	embedding: 0.437 (0.232 to 0.664)	-0.061 (-0.163 to 0.085)	1.0000
AVEX EfficientNet-B0	embedding: 0.485 (0.234 to 0.705)	-0.013 (-0.070 to 0.067)	1.0000
AvesEcho	embedding: 0.568 (0.434 to 0.759)	0.070 (-0.151 to 0.341)	1.0000

Neural network	Output read, re-ID accuracy	Difference from BirdNET	Holm p
AudioProtoPNet	embedding: 0.576 (0.397 to 0.748)	0.079 (−0.037 to 0.228)	1.0000
ConvNeXt (BirdSet)	embedding: 0.502 (0.284 to 0.743)	0.004 (−0.095 to 0.128)	1.0000
Bird-MAE	embedding: 0.223 (0.030 to 0.611)	−0.275 (−0.530 to −0.026)	< 0.0074
ProtoCLR	embedding: 0.118 (0.000 to 0.450)	−0.380 (−0.640 to −0.047)	0.4898
RCL FS BSSED	embedding: 0.210 (0.013 to 0.627)	−0.288 (−0.539 to −0.004)	1.0000
BirdAVES	embedding: 0.245 (0.091 to 0.481)	−0.253 (−0.409 to −0.059)	< 0.0074
AVES	embedding: 0.175 (0.061 to 0.406)	−0.323 (−0.550 to −0.038)	0.4288
NatureBEATs	embedding: 0.472 (0.309 to 0.722)	−0.026 (−0.312 to 0.367)	1.0000
BioLingual	embedding: 0.397 (0.226 to 0.726)	−0.100 (−0.296 to 0.099)	1.0000
BEATs	embedding: 0.419 (0.131 to 0.670)	−0.079 (−0.179 to 0.011)	1.0000
AudioMAE	embedding: 0.376 (0.229 to 0.609)	−0.122 (−0.291 to 0.093)	1.0000
VGGish	embedding: 0.445 (0.114 to 0.681)	−0.052 (−0.182 to 0.010)	1.0000
ECAPA-TDNN (speaker)	embedding: 0.358 (0.222 to 0.535)	−0.140 (−0.291 to 0.076)	1.0000
ResNet (speaker)	embedding: 0.467 (0.314 to 0.720)	−0.031 (−0.180 to 0.155)	1.0000
x-vector (speaker)	embedding: 0.310 (0.148 to 0.607)	−0.188 (−0.408 to 0.100)	1.0000
wav2vec 2.0 Base	layer 12, full speed: 0.192 (0.115 to 0.310)	−0.306 (−0.462 to −0.054)	0.3894
wav2vec 2.0 Large (robust)	layer 24, full speed: 0.245 (0.000 to 0.639)	−0.253 (−0.706 to 0.451)	1.0000
XLS-R 300M	layer 24, full speed: 0.249 (0.064 to 0.573)	−0.249 (−0.532 to 0.100)	1.0000
wav2vec 2.0 Conformer Large	layer 24, full speed: 0.389 (0.201 to 0.552)	−0.109 (−0.328 to 0.174)	1.0000
MMS 300M	layer 24, full speed: 0.223 (0.037 to 0.607)	−0.275 (−0.528 to 0.013)	1.0000
HuBERT Base	layer 12, full speed: 0.310 (0.146 to 0.562)	−0.188 (−0.379 to 0.065)	1.0000
HuBERT Large	layer 24, full speed: 0.507 (0.260 to 0.684)	0.009 (−0.204 to 0.199)	1.0000
data2vec Base (100 h)	layer 12, full speed: 0.210 (0.087 to 0.429)	−0.288 (−0.495 to 0.000)	1.0000
data2vec Base (960 h)	layer 12, full speed: 0.205 (0.064 to 0.319)	−0.293 (−0.416 to −0.116)	< 0.0074
WavLM Base+	layer 12, full speed: 0.376 (0.147 to 0.552)	−0.122 (−0.311 to 0.047)	1.0000
WavLM Large	layer 24, full speed: 0.489 (0.257 to 0.692)	−0.009 (−0.161 to 0.143)	1.0000
UniSpeech-SAT Base+	layer 12, full speed: 0.397 (0.208 to 0.581)	−0.100 (−0.331 to 0.220)	1.0000
XEUS	layer 18, full speed: 0.236 (0.016 to 0.500)	−0.262 (−0.453 to −0.059)	0.3604
Loudness control	summary: 0.210 (0.014 to 0.608)	−0.288 (−0.565 to 0.079)	1.0000
Duration control	summary: 0.284 (0.000 to 0.572)	−0.214 (−0.661 to 0.000)	1.0000

Great tit, across year. BirdNET v2.4 0.444 (0.319 to 0.569).

Neural network	Output read, re-ID accuracy	Difference from BirdNET	Holm p
Perch 2.0	embedding: 0.450 (0.312 to 0.588)	0.006 (−0.094 to 0.100)	1.0000
Perch	embedding: 0.294 (0.194 to 0.400)	−0.150 (−0.256 to −0.056)	0.0088
SurfPerch	embedding: 0.375 (0.275 to 0.475)	−0.069 (−0.131 to −0.012)	0.1452
BirdNET v3 (preview)	embedding: 0.156 (0.106 to 0.206)	−0.287 (−0.437 to −0.144)	< 0.0074
AVEX sl-BEATs	embedding: 0.381 (0.287 to 0.475)	−0.062 (−0.131 to 0.006)	0.3304
AVEX EfficientNet-B0	embedding: 0.338 (0.225 to 0.456)	−0.106 (−0.231 to 0.000)	0.2970
AvesEcho	embedding: 0.438 (0.294 to 0.581)	−0.006 (−0.069 to 0.056)	1.0000
AudioProtoPNet	embedding: 0.294 (0.194 to 0.394)	−0.150 (−0.269 to −0.044)	0.0484
ConvNeXt (BirdSet)	embedding: 0.325 (0.200 to 0.463)	−0.119 (−0.225 to −0.025)	0.0960
Bird-MAE	embedding: 0.138 (0.044 to 0.244)	−0.306 (−0.469 to −0.131)	0.0168
ProtoCLR	embedding: 0.113 (0.031 to 0.212)	−0.331 (−0.500 to −0.150)	0.0128
RCL FS BSSED	embedding: 0.219 (0.094 to 0.362)	−0.225 (−0.338 to −0.106)	0.0088
BirdAVES	embedding: 0.362 (0.231 to 0.506)	−0.081 (−0.175 to 0.019)	0.3462

Neural network	Output read, re-ID accuracy	Difference from BirdNET	Holm p
AVES	embedding: 0.287 (0.169 to 0.425)	-0.156 (-0.281 to -0.037)	0.0960
NatureBEATs	embedding: 0.269 (0.131 to 0.412)	-0.175 (-0.294 to -0.056)	0.0522
BioLingual	embedding: 0.319 (0.206 to 0.444)	-0.125 (-0.206 to -0.050)	0.0150
BEATs	embedding: 0.250 (0.163 to 0.344)	-0.194 (-0.325 to -0.075)	0.0114
AudioMAE	embedding: 0.219 (0.131 to 0.306)	-0.225 (-0.356 to -0.112)	< 0.0074
VGGish	embedding: 0.169 (0.087 to 0.250)	-0.275 (-0.394 to -0.150)	< 0.0074
ECAPA-TDNN (speaker)	embedding: 0.256 (0.175 to 0.344)	-0.188 (-0.331 to -0.056)	0.0480
ResNet (speaker)	embedding: 0.237 (0.144 to 0.338)	-0.206 (-0.319 to -0.106)	< 0.0074
x-vector (speaker)	embedding: 0.212 (0.106 to 0.338)	-0.231 (-0.350 to -0.125)	< 0.0074
wav2vec 2.0 Base	layer 12, full speed: 0.219 (0.106 to 0.350)	-0.225 (-0.356 to -0.094)	0.0114
wav2vec 2.0 Large (robust)	layer 24, full speed: 0.119 (0.031 to 0.231)	-0.325 (-0.506 to -0.131)	0.0168
XLS-R 300M	layer 24, full speed: 0.119 (0.056 to 0.194)	-0.325 (-0.462 to -0.181)	< 0.0074
wav2vec 2.0 Conformer Large	layer 24, full speed: 0.181 (0.100 to 0.269)	-0.262 (-0.412 to -0.113)	0.0114
MMS 300M	layer 24, full speed: 0.138 (0.056 to 0.231)	-0.306 (-0.463 to -0.138)	0.0088
HuBERT Base	layer 12, full speed: 0.156 (0.100 to 0.212)	-0.287 (-0.425 to -0.150)	< 0.0074
HuBERT Large	layer 24, full speed: 0.094 (0.037 to 0.163)	-0.350 (-0.494 to -0.212)	< 0.0074
data2vec Base (100 h)	layer 12, full speed: 0.131 (0.056 to 0.219)	-0.312 (-0.462 to -0.163)	< 0.0074
data2vec Base (960 h)	layer 12, full speed: 0.056 (0.013 to 0.119)	-0.387 (-0.537 to -0.237)	< 0.0074
WavLM Base+	layer 12, full speed: 0.175 (0.113 to 0.244)	-0.269 (-0.375 to -0.169)	< 0.0074
WavLM Large	layer 24, full speed: 0.131 (0.069 to 0.200)	-0.312 (-0.450 to -0.181)	< 0.0074
UniSpeech-SAT Base+	layer 12, full speed: 0.194 (0.119 to 0.275)	-0.250 (-0.363 to -0.144)	< 0.0074
XEUS	layer 18, full speed: 0.237 (0.131 to 0.350)	-0.206 (-0.344 to -0.069)	0.0484
Loudness control	summary: 0.113 (0.031 to 0.206)	-0.331 (-0.494 to -0.169)	< 0.0074
Duration control	summary: 0.069 (0.000 to 0.175)	-0.375 (-0.537 to -0.200)	< 0.0074

Tree pipit, across year. BirdNET v2.4 0.214 (0.117 to 0.336).

Neural network	Output read, re-ID accuracy	Difference from BirdNET	Holm p
Perch 2.0	embedding: 0.447 (0.294 to 0.625)	0.233 (0.149 to 0.310)	< 0.0074
Perch	embedding: 0.351 (0.219 to 0.517)	0.137 (0.064 to 0.207)	0.0348
SurfPerch	embedding: 0.243 (0.153 to 0.358)	0.029 (-0.015 to 0.083)	1.0000
BirdNET v3 (preview)	embedding: 0.160 (0.096 to 0.228)	-0.054 (-0.173 to 0.035)	1.0000
AVEX sl-BEATs	embedding: 0.240 (0.121 to 0.379)	0.026 (-0.051 to 0.100)	1.0000
AVEX EfficientNet-B0	embedding: 0.211 (0.114 to 0.319)	-0.003 (-0.091 to 0.091)	1.0000
AvesEcho	embedding: 0.089 (0.019 to 0.190)	-0.125 (-0.192 to -0.055)	0.0448
AudioProtoPNet	embedding: 0.201 (0.102 to 0.317)	-0.013 (-0.070 to 0.036)	1.0000
ConvNeXt (BirdSet)	embedding: 0.288 (0.138 to 0.464)	0.073 (-0.023 to 0.156)	1.0000
Bird-MAE	embedding: 0.051 (0.018 to 0.095)	-0.163 (-0.294 to -0.052)	0.0300
ProtoCLR	embedding: 0.083 (0.030 to 0.156)	-0.131 (-0.261 to 0.000)	1.0000
RCL FS_BSED	embedding: 0.045 (0.017 to 0.071)	-0.169 (-0.283 to -0.080)	< 0.0074
BirdAVES	embedding: 0.214 (0.116 to 0.334)	0.000 (-0.048 to 0.039)	1.0000
AVES	embedding: 0.096 (0.033 to 0.196)	-0.118 (-0.202 to -0.057)	< 0.0074
NatureBEATs	embedding: 0.192 (0.087 to 0.301)	-0.022 (-0.106 to 0.064)	1.0000
BioLingual	embedding: 0.227 (0.101 to 0.375)	0.013 (-0.044 to 0.058)	1.0000
BEATs	embedding: 0.058 (0.037 to 0.083)	-0.157 (-0.275 to -0.065)	< 0.0074
AudioMAE	embedding: 0.169 (0.113 to 0.237)	-0.045 (-0.135 to 0.054)	1.0000
VGGish	embedding: 0.070 (0.034 to 0.109)	-0.144 (-0.249 to -0.062)	< 0.0074

Neural network	Output read, re-ID accuracy	Difference from BirdNET	Holm p
ECAPA-TDNN (speaker)	embedding: 0.131 (0.082 to 0.180)	-0.083 (-0.227 to 0.034)	1.0000
ResNet (speaker)	embedding: 0.150 (0.077 to 0.247)	-0.064 (-0.143 to 0.011)	1.0000
x-vector (speaker)	embedding: 0.182 (0.098 to 0.303)	-0.032 (-0.118 to 0.045)	1.0000
wav2vec 2.0 Base	layer 12, full speed: 0.086 (0.019 to 0.172)	-0.128 (-0.255 to 0.014)	1.0000
wav2vec 2.0 Large (robust)	layer 24, full speed: 0.064 (0.000 to 0.142)	-0.150 (-0.296 to -0.013)	0.7550
XLS-R 300M	layer 24, full speed: 0.083 (0.028 to 0.143)	-0.131 (-0.239 to -0.049)	< 0.0074
wav2vec 2.0 Conformer Large	layer 24, full speed: 0.195 (0.130 to 0.270)	-0.019 (-0.085 to 0.041)	1.0000
MMS 300M	layer 24, full speed: 0.115 (0.040 to 0.196)	-0.099 (-0.257 to 0.041)	1.0000
HuBERT Base	layer 12, full speed: 0.147 (0.053 to 0.300)	-0.067 (-0.153 to 0.020)	1.0000
HuBERT Large	layer 24, full speed: 0.147 (0.053 to 0.269)	-0.067 (-0.175 to 0.022)	1.0000
data2vec Base (100 h)	layer 12, full speed: 0.141 (0.044 to 0.266)	-0.073 (-0.191 to 0.056)	1.0000
data2vec Base (960 h)	layer 12, full speed: 0.134 (0.048 to 0.223)	-0.080 (-0.221 to 0.052)	1.0000
WavLM Base+	layer 12, full speed: 0.144 (0.064 to 0.263)	-0.070 (-0.162 to 0.000)	1.0000
WavLM Large	layer 24, full speed: 0.137 (0.053 to 0.273)	-0.077 (-0.164 to -0.009)	0.6396
UniSpeech-SAT Base+	layer 12, full speed: 0.144 (0.059 to 0.262)	-0.070 (-0.134 to -0.003)	1.0000
XEUS	layer 18, full speed: 0.070 (0.019 to 0.140)	-0.144 (-0.237 to -0.072)	< 0.0074
Loudness control	summary: 0.073 (0.018 to 0.138)	-0.141 (-0.261 to -0.033)	0.2484
Duration control	summary: 0.083 (0.000 to 0.234)	-0.131 (-0.307 to 0.079)	1.0000

Chiffchaff, across year: BirdNET v2.4 0.160 (0.035 to 0.330).

Neural network	Output read, re-ID accuracy	Difference from BirdNET	Holm p
Perch 2.0	embedding: 0.160 (0.034 to 0.289)	0.000 (-0.091 to 0.087)	1.0000
Perch	embedding: 0.150 (0.044 to 0.277)	-0.010 (-0.078 to 0.068)	1.0000
SurfPerch	embedding: 0.255 (0.103 to 0.421)	0.095 (0.015 to 0.192)	0.6808
BirdNET v3 (preview)	embedding: 0.085 (0.033 to 0.151)	-0.075 (-0.245 to 0.077)	1.0000
AVEX sl-BEATs	embedding: 0.150 (0.054 to 0.267)	-0.010 (-0.090 to 0.086)	1.0000
AVEX EfficientNet-B0	embedding: 0.120 (0.036 to 0.234)	-0.040 (-0.116 to 0.033)	1.0000
AvesEcho	embedding: 0.220 (0.018 to 0.432)	0.060 (-0.043 to 0.207)	1.0000
AudioProtoPNet	embedding: 0.115 (0.047 to 0.197)	-0.045 (-0.155 to 0.054)	1.0000
ConvNeXt (BirdSet)	embedding: 0.135 (0.034 to 0.262)	-0.025 (-0.098 to 0.051)	1.0000
Bird-MAE	embedding: 0.095 (0.025 to 0.175)	-0.065 (-0.227 to 0.083)	1.0000
ProtoCLR	embedding: 0.195 (0.041 to 0.379)	0.035 (-0.141 to 0.222)	1.0000
RCL FS_BSED	embedding: 0.150 (0.016 to 0.312)	-0.010 (-0.083 to 0.064)	1.0000
BirdAVES	embedding: 0.175 (0.061 to 0.310)	0.015 (-0.054 to 0.095)	1.0000
AVES	embedding: 0.140 (0.015 to 0.305)	-0.020 (-0.081 to 0.053)	1.0000
NatureBEATs	embedding: 0.170 (0.050 to 0.307)	0.010 (-0.041 to 0.056)	1.0000
BioLingual	embedding: 0.160 (0.031 to 0.289)	0.000 (-0.086 to 0.078)	1.0000
BEATs	embedding: 0.140 (0.049 to 0.234)	-0.020 (-0.123 to 0.058)	1.0000
AudioMAE	embedding: 0.150 (0.046 to 0.279)	-0.010 (-0.071 to 0.045)	1.0000
VGGish	embedding: 0.095 (0.000 to 0.275)	-0.065 (-0.126 to -0.005)	1.0000
ECAPA-TDNN (speaker)	embedding: 0.155 (0.044 to 0.295)	-0.005 (-0.087 to 0.092)	1.0000
ResNet (speaker)	embedding: 0.080 (0.028 to 0.140)	-0.080 (-0.203 to 0.027)	1.0000
x-vector (speaker)	embedding: 0.105 (0.042 to 0.181)	-0.055 (-0.157 to 0.026)	1.0000
wav2vec 2.0 Base	layer 12, full speed: 0.195 (0.017 to 0.390)	0.035 (-0.053 to 0.112)	1.0000
wav2vec 2.0 Large (robust)	layer 24, full speed: 0.140 (0.000 to 0.341)	-0.020 (-0.247 to 0.218)	1.0000
XLS-R 300M	layer 24, full speed: 0.165 (0.028 to 0.349)	0.005 (-0.083 to 0.109)	1.0000

Neural network	Output read, re-ID accuracy	Difference from BirdNET	Holm p
wav2vec 2.0 Conformer Large	layer 24, full speed: 0.195 (0.094 to 0.308)	0.035 (−0.054 to 0.116)	1.0000
MMS 300M	layer 24, full speed: 0.170 (0.000 to 0.404)	0.010 (−0.093 to 0.151)	1.0000
HuBERT Base	layer 12, full speed: 0.195 (0.016 to 0.438)	0.035 (−0.073 to 0.190)	1.0000
HuBERT Large	layer 24, full speed: 0.190 (0.049 to 0.354)	0.030 (−0.062 to 0.126)	1.0000
data2vec Base (100 h)	layer 12, full speed: 0.130 (0.028 to 0.262)	−0.030 (−0.107 to 0.049)	1.0000
data2vec Base (960 h)	layer 12, full speed: 0.170 (0.066 to 0.294)	0.010 (−0.071 to 0.093)	1.0000
WavLM Base+	layer 12, full speed: 0.135 (0.011 to 0.323)	−0.025 (−0.078 to 0.023)	1.0000
WavLM Large	layer 24, full speed: 0.195 (0.044 to 0.375)	0.035 (−0.074 to 0.170)	1.0000
UniSpeech-SAT Base+	layer 12, full speed: 0.165 (0.029 to 0.332)	0.005 (−0.057 to 0.065)	1.0000
XEUS	layer 18, full speed: 0.165 (0.000 to 0.379)	0.005 (−0.086 to 0.122)	1.0000
Loudness control	summary: 0.025 (0.000 to 0.058)	−0.135 (−0.308 to 0.000)	1.0000
Duration control	summary: 0.105 (0.000 to 0.298)	−0.055 (−0.271 to 0.156)	1.0000

Tree pipit, within year: BirdNET v2.4 0.376 (0.195 to 0.564).

Neural network	Output read, re-ID accuracy	Difference from BirdNET	Holm p
Perch 2.0	embedding: 0.548 (0.361 to 0.738)	0.172 (0.103 to 0.251)	< 0.0074
Perch	embedding: 0.426 (0.230 to 0.628)	0.050 (−0.054 to 0.171)	1.0000
SurfPerch	embedding: 0.455 (0.265 to 0.634)	0.079 (−0.010 to 0.171)	1.0000
BirdNET v3 (preview)	embedding: 0.257 (0.128 to 0.417)	−0.119 (−0.304 to 0.054)	1.0000
AVEX sl-BEATs	embedding: 0.380 (0.220 to 0.543)	0.003 (−0.125 to 0.125)	1.0000
AVEX EfficientNet-B0	embedding: 0.281 (0.111 to 0.475)	−0.096 (−0.244 to 0.051)	1.0000
AvesEcho	embedding: 0.208 (0.075 to 0.364)	−0.168 (−0.336 to 0.024)	1.0000
AudioProtoPNet	embedding: 0.442 (0.239 to 0.657)	0.066 (−0.087 to 0.237)	1.0000
ConvNeXt (BirdSet)	embedding: 0.502 (0.298 to 0.704)	0.125 (0.012 to 0.265)	0.8758
Bird-MAE	embedding: 0.191 (0.055 to 0.384)	−0.185 (−0.414 to 0.042)	1.0000
ProtoCLR	embedding: 0.152 (0.025 to 0.308)	−0.224 (−0.448 to −0.012)	1.0000
RCL FS BSED	embedding: 0.172 (0.054 to 0.337)	−0.205 (−0.441 to 0.051)	1.0000
BirdAVES	embedding: 0.419 (0.241 to 0.607)	0.043 (−0.101 to 0.194)	1.0000
AVES	embedding: 0.360 (0.190 to 0.554)	−0.017 (−0.217 to 0.212)	1.0000
NatureBEATs	embedding: 0.297 (0.148 to 0.452)	−0.079 (−0.200 to 0.061)	1.0000
BioLingual	embedding: 0.327 (0.179 to 0.475)	−0.050 (−0.141 to 0.053)	1.0000
BEATs	embedding: 0.195 (0.084 to 0.314)	−0.182 (−0.332 to −0.042)	0.2380
AudioMAE	embedding: 0.211 (0.137 to 0.281)	−0.165 (−0.319 to −0.020)	0.7320
VGGish	embedding: 0.116 (0.028 to 0.216)	−0.261 (−0.489 to −0.052)	0.3776
ECAPA-TDNN (speaker)	embedding: 0.244 (0.121 to 0.395)	−0.132 (−0.325 to 0.034)	1.0000
ResNet (speaker)	embedding: 0.211 (0.105 to 0.345)	−0.165 (−0.342 to 0.000)	1.0000
x-vector (speaker)	embedding: 0.300 (0.131 to 0.490)	−0.076 (−0.232 to 0.086)	1.0000
wav2vec 2.0 Base	layer 12, full speed: 0.165 (0.049 to 0.281)	−0.211 (−0.361 to −0.043)	0.4340
wav2vec 2.0 Large (robust)	layer 24, full speed: 0.162 (0.016 to 0.345)	−0.215 (−0.470 to 0.036)	1.0000
XLS-R 300M	layer 24, full speed: 0.145 (0.048 to 0.261)	−0.231 (−0.430 to −0.043)	0.3498
wav2vec 2.0 Conformer Large	layer 24, full speed: 0.277 (0.120 to 0.453)	−0.099 (−0.227 to 0.014)	1.0000
MMS 300M	layer 24, full speed: 0.178 (0.040 to 0.382)	−0.198 (−0.437 to 0.040)	1.0000
HuBERT Base	layer 12, full speed: 0.248 (0.049 to 0.474)	−0.129 (−0.315 to 0.055)	1.0000
HuBERT Large	layer 24, full speed: 0.241 (0.063 to 0.439)	−0.135 (−0.288 to −0.007)	0.9990
data2vec Base (100 h)	layer 12, full speed: 0.132 (0.047 to 0.234)	−0.244 (−0.405 to −0.101)	0.0070
data2vec Base (960 h)	layer 12, full speed: 0.112 (0.042 to 0.194)	−0.264 (−0.431 to −0.110)	< 0.0074

Neural network	Output read, re-ID accuracy	Difference from BirdNET	Holm p
WavLM Base+	layer 12, full speed: 0.274 (0.069 to 0.505)	-0.102 (-0.289 to 0.067)	1.0000
WavLM Large	layer 24, full speed: 0.234 (0.070 to 0.415)	-0.142 (-0.298 to -0.010)	0.9128
UniSpeech-SAT Base+	layer 12, full speed: 0.248 (0.086 to 0.428)	-0.129 (-0.293 to -0.003)	1.0000
XEUS	layer 18, full speed: 0.241 (0.078 to 0.425)	-0.135 (-0.314 to 0.027)	1.0000
Loudness control	summary: 0.228 (0.032 to 0.461)	-0.149 (-0.407 to 0.140)	1.0000
Duration control	summary: 0.145 (0.000 to 0.334)	-0.231 (-0.490 to 0.023)	1.0000

Little owl, across year: BirdNET v2.4 0.501 (0.372 to 0.631).

Neural network	Output read, re-ID accuracy	Difference from BirdNET	Holm p
Perch 2.0	embedding: 0.651 (0.494 to 0.786)	0.150 (0.044 to 0.244)	0.1840
Perch	embedding: 0.646 (0.501 to 0.775)	0.145 (0.057 to 0.229)	0.0432
SurfPerch	embedding: 0.442 (0.291 to 0.593)	-0.059 (-0.218 to 0.097)	1.0000
BirdNET v3 (preview)	embedding: 0.373 (0.237 to 0.523)	-0.128 (-0.258 to 0.003)	0.7392
AVEX sl-BEATS	embedding: 0.418 (0.238 to 0.611)	-0.084 (-0.238 to 0.067)	1.0000
AVEX EfficientNet-B0	embedding: 0.373 (0.237 to 0.522)	-0.128 (-0.253 to -0.003)	0.6104
AvesEcho	embedding: 0.464 (0.294 to 0.629)	-0.037 (-0.193 to 0.117)	1.0000
AudioProtoPNet	embedding: 0.631 (0.472 to 0.782)	0.130 (0.033 to 0.246)	0.0960
ConvNeXt (BirdSet)	embedding: 0.624 (0.479 to 0.754)	0.123 (0.006 to 0.229)	0.6104
Bird-MAE	embedding: 0.251 (0.103 to 0.412)	-0.251 (-0.453 to -0.036)	0.3424
ProtoCLR	embedding: 0.172 (0.067 to 0.305)	-0.329 (-0.467 to -0.182)	< 0.0074
RCL FS_BSED	embedding: 0.189 (0.071 to 0.331)	-0.312 (-0.495 to -0.111)	0.0936
BirdAVES	embedding: 0.285 (0.155 to 0.427)	-0.216 (-0.346 to -0.089)	0.0120
AVES	embedding: 0.351 (0.209 to 0.484)	-0.150 (-0.323 to 0.019)	0.7480
NatureBEATS	embedding: 0.364 (0.215 to 0.527)	-0.138 (-0.258 to -0.031)	0.2232
BioLingual	embedding: 0.408 (0.234 to 0.598)	-0.093 (-0.256 to 0.080)	1.0000
BEATS	embedding: 0.364 (0.203 to 0.539)	-0.138 (-0.295 to 0.017)	0.7480
AudioMAE	embedding: 0.437 (0.276 to 0.598)	-0.064 (-0.226 to 0.106)	1.0000
VGGish	embedding: 0.231 (0.098 to 0.379)	-0.270 (-0.398 to -0.157)	< 0.0074
ECAPA-TDNN (speaker)	embedding: 0.376 (0.256 to 0.492)	-0.125 (-0.236 to -0.027)	0.1862
ResNet (speaker)	embedding: 0.344 (0.231 to 0.456)	-0.157 (-0.308 to 0.009)	0.7392
x-vector (speaker)	embedding: 0.364 (0.220 to 0.515)	-0.138 (-0.321 to 0.068)	1.0000
wav2vec 2.0 Base	layer 12, full speed: 0.201 (0.095 to 0.325)	-0.300 (-0.442 to -0.128)	0.0174
wav2vec 2.0 Large (robust)	layer 24, full speed: 0.130 (0.028 to 0.262)	-0.371 (-0.526 to -0.215)	< 0.0074
XLS-R 300M	layer 24, full speed: 0.160 (0.074 to 0.250)	-0.342 (-0.483 to -0.206)	< 0.0074
wav2vec 2.0 Conformer Large	layer 24, full speed: 0.295 (0.166 to 0.433)	-0.206 (-0.371 to -0.024)	0.4590
MMS 300M	layer 24, full speed: 0.197 (0.102 to 0.298)	-0.305 (-0.479 to -0.125)	0.0224
HuBERT Base	layer 12, full speed: 0.275 (0.145 to 0.426)	-0.226 (-0.379 to -0.069)	0.1056
HuBERT Large	layer 24, full speed: 0.209 (0.097 to 0.334)	-0.292 (-0.428 to -0.155)	0.0062
data2vec Base (100 h)	layer 12, full speed: 0.258 (0.130 to 0.393)	-0.243 (-0.410 to -0.077)	0.0966
data2vec Base (960 h)	layer 12, full speed: 0.265 (0.132 to 0.414)	-0.236 (-0.406 to -0.068)	0.1176
WavLM Base+	layer 12, full speed: 0.310 (0.176 to 0.450)	-0.192 (-0.383 to 0.019)	0.7480
WavLM Large	layer 24, full speed: 0.327 (0.174 to 0.487)	-0.174 (-0.355 to 0.024)	0.7480
UniSpeech-SAT Base+	layer 12, full speed: 0.265 (0.127 to 0.424)	-0.236 (-0.381 to -0.080)	0.0936
XEUS	layer 18, full speed: 0.285 (0.155 to 0.424)	-0.216 (-0.389 to -0.036)	0.3366
Loudness control	summary: 0.069 (0.014 to 0.134)	-0.432 (-0.570 to -0.305)	< 0.0074
Duration control	summary: 0.115 (0.000 to 0.293)	-0.386 (-0.526 to -0.243)	< 0.0074

Little penguin, across night: BirdNET v2.4 0.693 (0.607 to 0.773).

Neural network	Output read, re-ID accuracy	Difference from BirdNET	Holm p
Perch 2.0	embedding: 0.705 (0.599 to 0.791)	0.011 (−0.051 to 0.069)	1.0000
Perch	embedding: 0.635 (0.528 to 0.727)	−0.058 (−0.134 to 0.013)	1.0000
SurfPerch	embedding: 0.646 (0.551 to 0.728)	−0.047 (−0.107 to 0.011)	1.0000
BirdNET v3 (preview)	embedding: 0.501 (0.371 to 0.611)	−0.192 (−0.268 to −0.124)	< 0.0074
AVEX sl-BEATs	embedding: 0.731 (0.673 to 0.795)	0.038 (−0.017 to 0.092)	1.0000
AVEX EfficientNet-B0	embedding: 0.569 (0.444 to 0.681)	−0.124 (−0.199 to −0.051)	0.0036
AvesEcho	embedding: 0.750 (0.650 to 0.834)	0.057 (−0.011 to 0.132)	1.0000
AudioProtoPNet	embedding: 0.674 (0.569 to 0.760)	−0.019 (−0.076 to 0.036)	1.0000
ConvNeXt (BirdSet)	embedding: 0.658 (0.538 to 0.755)	−0.035 (−0.130 to 0.052)	1.0000
Bird-MAE	embedding: 0.419 (0.271 to 0.541)	−0.274 (−0.388 to −0.171)	< 0.0074
ProtoCLR	embedding: 0.447 (0.310 to 0.560)	−0.246 (−0.366 to −0.132)	< 0.0074
RCL FS_BSED	embedding: 0.622 (0.532 to 0.711)	−0.071 (−0.117 to −0.022)	0.0728
BirdAVES	embedding: 0.634 (0.517 to 0.731)	−0.059 (−0.140 to 0.018)	1.0000
AVES	embedding: 0.670 (0.553 to 0.769)	−0.023 (−0.096 to 0.048)	1.0000
NatureBEATs	embedding: 0.759 (0.692 to 0.826)	0.066 (−0.001 to 0.138)	0.6336
BioLingual	embedding: 0.617 (0.500 to 0.724)	−0.076 (−0.145 to −0.006)	0.4392
BEATs	embedding: 0.569 (0.444 to 0.681)	−0.124 (−0.202 to −0.051)	0.0090
AudioMAE	embedding: 0.528 (0.398 to 0.636)	−0.165 (−0.246 to −0.099)	< 0.0074
VGGish	embedding: 0.463 (0.341 to 0.576)	−0.230 (−0.300 to −0.166)	< 0.0074
ECAPA-TDNN (speaker)	embedding: 0.442 (0.324 to 0.552)	−0.251 (−0.327 to −0.182)	< 0.0074
ResNet (speaker)	embedding: 0.423 (0.302 to 0.538)	−0.270 (−0.373 to −0.170)	< 0.0074
x-vector (speaker)	embedding: 0.441 (0.317 to 0.544)	−0.253 (−0.354 to −0.155)	< 0.0074
wav2vec 2.0 Base	layer 12, full speed: 0.475 (0.340 to 0.601)	−0.218 (−0.308 to −0.126)	0.0036
wav2vec 2.0 Large (robust)	layer 24, full speed: 0.237 (0.118 to 0.351)	−0.456 (−0.559 to −0.362)	< 0.0074
XLS-R 300M	layer 24, full speed: 0.510 (0.356 to 0.638)	−0.183 (−0.284 to −0.093)	< 0.0074
wav2vec 2.0 Conformer Large	layer 24, full speed: 0.432 (0.295 to 0.560)	−0.261 (−0.385 to −0.151)	< 0.0074
MMS 300M	layer 24, full speed: 0.577 (0.413 to 0.720)	−0.116 (−0.219 to −0.022)	0.1742
HuBERT Base	layer 12, full speed: 0.431 (0.305 to 0.557)	−0.263 (−0.356 to −0.165)	0.0036
HuBERT Large	layer 24, full speed: 0.434 (0.295 to 0.561)	−0.259 (−0.357 to −0.161)	< 0.0074
data2vec Base (100 h)	layer 12, full speed: 0.356 (0.209 to 0.484)	−0.337 (−0.440 to −0.250)	< 0.0074
data2vec Base (960 h)	layer 12, full speed: 0.289 (0.203 to 0.368)	−0.404 (−0.486 to −0.327)	< 0.0074
WavLM Base+	layer 12, full speed: 0.399 (0.281 to 0.516)	−0.294 (−0.391 to −0.192)	< 0.0074
WavLM Large	layer 24, full speed: 0.460 (0.322 to 0.585)	−0.234 (−0.327 to −0.142)	< 0.0074
UniSpeech-SAT Base+	layer 12, full speed: 0.391 (0.239 to 0.543)	−0.302 (−0.417 to −0.182)	< 0.0074
XEUS	layer 18, full speed: 0.601 (0.445 to 0.743)	−0.092 (−0.207 to 0.021)	1.0000
Loudness control	summary: 0.159 (0.048 to 0.267)	−0.534 (−0.659 to −0.411)	< 0.0074
Duration control	summary: 0.167 (0.000 to 0.394)	−0.527 (−0.738 to −0.287)	< 0.0074

Red-tailed black cockatoo, random split: BirdNET v2.4 0.961 (0.930 to 0.980).

Neural network	Output read, re-ID accuracy	Difference from BirdNET	Holm p
Perch 2.0	embedding: 0.961 (0.933 to 0.979)	0.000 (−0.015 to 0.023)	1.0000
Perch	embedding: 0.947 (0.920 to 0.973)	−0.014 (−0.043 to 0.029)	1.0000
SurfPerch	embedding: 0.958 (0.920 to 0.981)	−0.003 (−0.019 to 0.017)	1.0000

Neural network	Output read, re-ID accuracy	Difference from BirdNET	Holm p
BirdNET v3 (preview)	embedding: 0.854 (0.791 to 0.892)	-0.106 (-0.158 to -0.070)	< 0.0074
AVEX sl-BEATS	embedding: 0.936 (0.894 to 0.964)	-0.025 (-0.053 to 0.000)	0.5742
AVEX EfficientNet-B0	embedding: 0.891 (0.839 to 0.925)	-0.070 (-0.107 to -0.034)	0.0048
AvesEcho	embedding: 0.969 (0.926 to 0.993)	0.008 (-0.020 to 0.032)	1.0000
AudioProtoPNet	embedding: 0.941 (0.910 to 0.967)	-0.020 (-0.050 to 0.023)	1.0000
ConvNeXt (BirdSet)	embedding: 0.969 (0.936 to 0.991)	0.008 (-0.019 to 0.045)	1.0000
Bird-MAE	embedding: 0.527 (0.169 to 0.753)	-0.434 (-0.778 to -0.216)	< 0.0074
ProtoCLR	embedding: 0.515 (0.219 to 0.713)	-0.445 (-0.726 to -0.259)	< 0.0074
RCL FS_BSED	embedding: 0.709 (0.436 to 0.841)	-0.252 (-0.510 to -0.122)	< 0.0074
BirdAVES	embedding: 0.947 (0.893 to 0.980)	-0.014 (-0.050 to 0.014)	1.0000
AVES	embedding: 0.924 (0.848 to 0.963)	-0.036 (-0.092 to -0.010)	0.0154
NatureBEATS	embedding: 0.944 (0.873 to 0.978)	-0.017 (-0.070 to 0.013)	1.0000
BioLingual	embedding: 0.933 (0.886 to 0.960)	-0.028 (-0.057 to -0.004)	0.3060
BEATS	embedding: 0.908 (0.845 to 0.945)	-0.053 (-0.091 to -0.029)	< 0.0074
AudioMAE	embedding: 0.782 (0.615 to 0.871)	-0.179 (-0.327 to -0.101)	< 0.0074
VGGish	embedding: 0.812 (0.680 to 0.902)	-0.148 (-0.258 to -0.071)	< 0.0074
ECAPA-TDNN (speaker)	embedding: 0.725 (0.614 to 0.796)	-0.235 (-0.328 to -0.166)	< 0.0074
ResNet (speaker)	embedding: 0.812 (0.752 to 0.849)	-0.148 (-0.187 to -0.116)	< 0.0074
x-vector (speaker)	embedding: 0.796 (0.652 to 0.868)	-0.165 (-0.290 to -0.100)	< 0.0074
wav2vec 2.0 Base	layer 12, full speed: 0.768 (0.676 to 0.824)	-0.193 (-0.266 to -0.144)	< 0.0074
wav2vec 2.0 Large (robust)	layer 24, full speed: 0.499 (0.237 to 0.705)	-0.462 (-0.712 to -0.266)	< 0.0074
XLS-R 300M	layer 24, full speed: 0.737 (0.519 to 0.859)	-0.224 (-0.424 to -0.109)	< 0.0074
wav2vec 2.0 Conformer Large	layer 24, full speed: 0.739 (0.591 to 0.831)	-0.221 (-0.345 to -0.141)	< 0.0074
MMS 300M	layer 24, full speed: 0.742 (0.548 to 0.854)	-0.218 (-0.401 to -0.114)	< 0.0074
HuBERT Base	layer 12, full speed: 0.798 (0.675 to 0.872)	-0.162 (-0.261 to -0.103)	< 0.0074
HuBERT Large	layer 24, full speed: 0.641 (0.422 to 0.818)	-0.319 (-0.524 to -0.154)	< 0.0074
data2vec Base (100 h)	layer 12, full speed: 0.616 (0.384 to 0.770)	-0.345 (-0.560 to -0.205)	< 0.0074
data2vec Base (960 h)	layer 12, full speed: 0.655 (0.443 to 0.791)	-0.305 (-0.497 to -0.184)	< 0.0074
WavLM Base+	layer 12, full speed: 0.826 (0.733 to 0.887)	-0.134 (-0.204 to -0.087)	< 0.0074
WavLM Large	layer 24, full speed: 0.832 (0.693 to 0.913)	-0.129 (-0.243 to -0.063)	< 0.0074
UniSpeech-SAT Base+	layer 12, full speed: 0.812 (0.680 to 0.902)	-0.148 (-0.260 to -0.070)	< 0.0074
XEUS	layer 18, full speed: 0.863 (0.753 to 0.927)	-0.098 (-0.190 to -0.046)	< 0.0074
Loudness control	summary: 0.406 (0.153 to 0.587)	-0.555 (-0.793 to -0.375)	< 0.0074
Duration control	summary: 0.336 (0.000 to 0.666)	-0.625 (-0.958 to -0.304)	< 0.0074

Egyptian fruit bat, across year: BirdNET v2.4 0.381 (0.189 to 0.561).

Neural network	Output read, re-ID accuracy	Difference from BirdNET	Holm p
Perch 2.0	embedding: 0.350 (0.161 to 0.532)	-0.031 (-0.115 to 0.104)	1.0000
Perch	embedding: 0.268 (0.131 to 0.537)	-0.113 (-0.353 to 0.180)	1.0000
SurfPerch	embedding: 0.353 (0.202 to 0.496)	-0.028 (-0.136 to 0.104)	1.0000
BirdNET v3 (preview)	embedding: 0.293 (0.194 to 0.475)	-0.088 (-0.272 to 0.103)	1.0000
AVEX sl-BEATS	embedding: 0.371 (0.233 to 0.490)	-0.011 (-0.110 to 0.083)	1.0000
AVEX EfficientNet-B0	embedding: 0.307 (0.258 to 0.359)	-0.074 (-0.237 to 0.088)	1.0000
AvesEcho	embedding: 0.223 (0.160 to 0.292)	-0.158 (-0.299 to -0.027)	0.0504
AudioProtoPNet	embedding: 0.305 (0.131 to 0.512)	-0.077 (-0.189 to 0.100)	1.0000
ConvNeXt (BirdSet)	embedding: 0.333 (0.127 to 0.530)	-0.048 (-0.121 to 0.061)	1.0000

Neural network	Output read, re-ID accuracy	Difference from BirdNET	Holm p
Bird-MAE	embedding: 0.121 (0.054 to 0.218)	-0.261 (-0.467 to -0.056)	< 0.0074
ProtoCLR	embedding: 0.133 (0.052 to 0.233)	-0.249 (-0.494 to -0.001)	1.0000
RCL FS_BSED	embedding: 0.191 (0.077 to 0.327)	-0.191 (-0.467 to 0.056)	1.0000
BirdAVES	embedding: 0.347 (0.142 to 0.540)	-0.034 (-0.124 to 0.019)	1.0000
AVES	embedding: 0.287 (0.055 to 0.517)	-0.095 (-0.275 to 0.011)	1.0000
NatureBEATs	embedding: 0.337 (0.107 to 0.579)	-0.044 (-0.122 to 0.027)	1.0000
BioLingual	embedding: 0.209 (0.122 to 0.319)	-0.172 (-0.417 to 0.054)	1.0000
BEATs	embedding: 0.248 (0.173 to 0.361)	-0.134 (-0.358 to 0.052)	1.0000
AudioMAE	embedding: 0.182 (0.111 to 0.300)	-0.199 (-0.430 to 0.027)	1.0000
VGGish	embedding: 0.166 (0.081 to 0.216)	-0.216 (-0.370 to -0.033)	0.5250
ECAPA-TDNN (speaker)	embedding: 0.258 (0.183 to 0.354)	-0.124 (-0.333 to 0.080)	1.0000
ResNet (speaker)	embedding: 0.319 (0.204 to 0.412)	-0.062 (-0.244 to 0.144)	1.0000
x-vector (speaker)	embedding: 0.331 (0.201 to 0.437)	-0.051 (-0.151 to 0.061)	1.0000
wav2vec 2.0 Base	layer 12, full speed: 0.237 (0.015 to 0.376)	-0.144 (-0.367 to 0.109)	1.0000
wav2vec 2.0 Large (robust)	layer 24, full speed: 0.210 (0.046 to 0.395)	-0.171 (-0.462 to 0.167)	1.0000
XLS-R 300M	layer 24, full speed: 0.248 (0.037 to 0.406)	-0.134 (-0.328 to 0.161)	1.0000
wav2vec 2.0 Conformer Large	layer 24, full speed: 0.287 (0.082 to 0.422)	-0.095 (-0.289 to 0.072)	1.0000
MMS 300M	layer 24, full speed: 0.204 (0.091 to 0.292)	-0.178 (-0.340 to 0.019)	1.0000
HuBERT Base	layer 12, full speed: 0.309 (0.054 to 0.495)	-0.072 (-0.279 to 0.086)	1.0000
HuBERT Large	layer 24, full speed: 0.232 (0.091 to 0.393)	-0.150 (-0.405 to 0.176)	1.0000
data2vec Base (100 h)	layer 12, full speed: 0.219 (0.086 to 0.378)	-0.163 (-0.441 to 0.157)	1.0000
data2vec Base (960 h)	layer 12, full speed: 0.225 (0.097 to 0.365)	-0.156 (-0.396 to 0.117)	1.0000
WavLM Base+	layer 12, full speed: 0.268 (0.100 to 0.405)	-0.113 (-0.319 to 0.156)	1.0000
WavLM Large	layer 24, full speed: 0.309 (0.138 to 0.405)	-0.072 (-0.223 to 0.131)	1.0000
UniSpeech-SAT Base+	layer 12, full speed: 0.323 (0.146 to 0.430)	-0.058 (-0.231 to 0.164)	1.0000
XEUS	layer 18, full speed: 0.287 (0.052 to 0.458)	-0.095 (-0.251 to 0.044)	1.0000
Loudness control	summary: 0.248 (0.013 to 0.620)	-0.134 (-0.531 to 0.395)	1.0000
Duration control	summary: 0.208 (0.000 to 0.462)	-0.173 (-0.537 to 0.249)	1.0000

Chiffchaff, within year: BirdNET v2.4 0.765 (0.680 to 0.841).

Neural network	Output read, re-ID accuracy	Difference from BirdNET	Holm p
Perch 2.0	embedding: 0.831 (0.756 to 0.884)	0.066 (-0.008 to 0.122)	0.5782
Perch	embedding: 0.778 (0.656 to 0.858)	0.013 (-0.105 to 0.094)	1.0000
SurfPerch	embedding: 0.729 (0.632 to 0.808)	-0.036 (-0.115 to 0.013)	0.8892
BirdNET v3 (preview)	embedding: 0.624 (0.370 to 0.770)	-0.141 (-0.381 to -0.003)	0.3504
AVEX sl-BEATs	embedding: 0.655 (0.572 to 0.748)	-0.110 (-0.161 to -0.058)	< 0.0074
AVEX EfficientNet-B0	embedding: 0.691 (0.577 to 0.760)	-0.074 (-0.173 to -0.021)	0.0440
AvesEcho	embedding: 0.691 (0.518 to 0.772)	-0.073 (-0.217 to -0.008)	0.2400
AudioProtoPNet	embedding: 0.806 (0.646 to 0.890)	0.042 (-0.096 to 0.114)	1.0000
ConvNeXt (BirdSet)	embedding: 0.805 (0.687 to 0.866)	0.041 (-0.066 to 0.089)	1.0000
Bird-MAE	embedding: 0.341 (0.145 to 0.470)	-0.424 (-0.603 to -0.323)	< 0.0074
ProtoCLR	embedding: 0.325 (0.211 to 0.477)	-0.439 (-0.548 to -0.305)	< 0.0074
RCL FS_BSED	embedding: 0.256 (0.121 to 0.494)	-0.508 (-0.643 to -0.289)	< 0.0074
BirdAVES	embedding: 0.749 (0.683 to 0.812)	-0.016 (-0.083 to 0.050)	1.0000
AVES	embedding: 0.515 (0.396 to 0.645)	-0.250 (-0.329 to -0.168)	< 0.0074
NatureBEATs	embedding: 0.739 (0.558 to 0.824)	-0.026 (-0.164 to 0.042)	1.0000

Neural network	Output read, re-ID accuracy	Difference from BirdNET	Holm p
BioLingual	embedding: 0.659 (0.452 to 0.769)	-0.106 (-0.283 to -0.008)	0.2610
BEATs	embedding: 0.514 (0.310 to 0.624)	-0.251 (-0.432 to -0.163)	< 0.0074
AudioMAE	embedding: 0.570 (0.494 to 0.694)	-0.195 (-0.271 to -0.083)	0.0072
VGGish	embedding: 0.458 (0.256 to 0.561)	-0.307 (-0.501 to -0.223)	< 0.0074
ECAPA-TDNN (speaker)	embedding: 0.437 (0.309 to 0.531)	-0.328 (-0.441 to -0.263)	< 0.0074
ResNet (speaker)	embedding: 0.392 (0.273 to 0.534)	-0.373 (-0.470 to -0.271)	< 0.0074
x-vector (speaker)	embedding: 0.385 (0.258 to 0.523)	-0.379 (-0.470 to -0.290)	< 0.0074
wav2vec 2.0 Base	layer 12, full speed: 0.347 (0.251 to 0.480)	-0.418 (-0.502 to -0.314)	< 0.0074
wav2vec 2.0 Large (robust)	layer 24, full speed: 0.274 (0.070 to 0.397)	-0.491 (-0.698 to -0.376)	< 0.0074
XLS-R 300M	layer 24, full speed: 0.384 (0.193 to 0.526)	-0.381 (-0.553 to -0.272)	< 0.0074
wav2vec 2.0 Conformer Large	layer 24, full speed: 0.501 (0.351 to 0.592)	-0.263 (-0.412 to -0.185)	< 0.0074
MMS 300M	layer 24, full speed: 0.349 (0.187 to 0.499)	-0.416 (-0.559 to -0.301)	< 0.0074
HuBERT Base	layer 12, full speed: 0.492 (0.329 to 0.584)	-0.273 (-0.415 to -0.201)	< 0.0074
HuBERT Large	layer 24, full speed: 0.469 (0.313 to 0.575)	-0.296 (-0.455 to -0.203)	< 0.0074
data2vec Base (100 h)	layer 12, full speed: 0.316 (0.176 to 0.451)	-0.449 (-0.573 to -0.348)	< 0.0074
data2vec Base (960 h)	layer 12, full speed: 0.310 (0.208 to 0.424)	-0.454 (-0.568 to -0.349)	< 0.0074
WavLM Base+	layer 12, full speed: 0.485 (0.328 to 0.575)	-0.280 (-0.425 to -0.211)	< 0.0074
WavLM Large	layer 24, full speed: 0.585 (0.426 to 0.682)	-0.179 (-0.307 to -0.113)	< 0.0074
UniSpeech-SAT Base+	layer 12, full speed: 0.498 (0.346 to 0.594)	-0.267 (-0.399 to -0.201)	< 0.0074
XEUS	layer 18, full speed: 0.440 (0.290 to 0.576)	-0.324 (-0.444 to -0.235)	< 0.0074
Loudness control	summary: 0.094 (0.010 to 0.267)	-0.671 (-0.747 to -0.522)	< 0.0074
Duration control	summary: 0.040 (0.000 to 0.147)	-0.725 (-0.794 to -0.607)	< 0.0074

Rook, across year: BirdNET v2.4 0.375 (0.261 to 0.506).

Neural network	Output read, re-ID accuracy	Difference from BirdNET	Holm p
Perch 2.0	embedding: 0.418 (0.306 to 0.535)	0.043 (0.017 to 0.065)	0.0240
Perch	embedding: 0.393 (0.282 to 0.516)	0.018 (-0.027 to 0.062)	0.9816
SurfPerch	embedding: 0.306 (0.200 to 0.425)	-0.069 (-0.106 to -0.038)	0.0038
BirdNET v3 (preview)	embedding: 0.328 (0.212 to 0.439)	-0.048 (-0.098 to -0.008)	0.1632
AVEX sl-BEATs	embedding: 0.332 (0.191 to 0.444)	-0.043 (-0.119 to 0.011)	0.6640
AVEX EfficientNet-B0	embedding: 0.345 (0.247 to 0.442)	-0.031 (-0.073 to -0.001)	0.3010
AvesEcho	embedding: 0.343 (0.223 to 0.437)	-0.033 (-0.125 to 0.024)	0.9816
AudioProtoPNet	embedding: 0.386 (0.269 to 0.506)	0.011 (-0.008 to 0.024)	0.9248
ConvNeXt (BirdSet)	embedding: 0.409 (0.260 to 0.536)	0.034 (-0.044 to 0.083)	0.9816
Bird-MAE	embedding: 0.194 (0.065 to 0.365)	-0.182 (-0.255 to -0.107)	< 0.0074
ProtoCLR	embedding: 0.084 (0.018 to 0.219)	-0.292 (-0.374 to -0.165)	< 0.0074
RCL FS_BSED	embedding: 0.105 (0.032 to 0.215)	-0.270 (-0.350 to -0.178)	< 0.0074
BirdAVES	embedding: 0.174 (0.053 to 0.384)	-0.202 (-0.305 to -0.073)	0.0120
AVES	embedding: 0.176 (0.058 to 0.379)	-0.199 (-0.284 to -0.087)	0.0096
NatureBEATs	embedding: 0.314 (0.154 to 0.475)	-0.061 (-0.136 to -0.0004)	0.3010
BioLingual	embedding: 0.303 (0.186 to 0.404)	-0.073 (-0.139 to -0.025)	0.0120
BEATs	embedding: 0.315 (0.192 to 0.428)	-0.060 (-0.107 to -0.034)	< 0.0074
AudioMAE	embedding: 0.236 (0.119 to 0.411)	-0.140 (-0.224 to -0.047)	0.0198
VGGish	embedding: 0.180 (0.086 to 0.329)	-0.195 (-0.266 to -0.117)	< 0.0074
ECAPA-TDNN (speaker)	embedding: 0.254 (0.145 to 0.402)	-0.121 (-0.161 to -0.076)	< 0.0074
ResNet (speaker)	embedding: 0.239 (0.125 to 0.435)	-0.137 (-0.221 to -0.035)	0.0414

Neural network	Output read, re-ID accuracy	Difference from BirdNET	Holm p
x-vector (speaker)	embedding: 0.262 (0.136 to 0.417)	-0.114 (-0.163 to -0.064)	< 0.0074
wav2vec 2.0 Base	layer 12, full speed: 0.157 (0.095 to 0.261)	-0.218 (-0.301 to -0.132)	< 0.0074
wav2vec 2.0 Large (robust)	layer 24, full speed: 0.125 (0.012 to 0.267)	-0.250 (-0.409 to -0.089)	0.0120
XLS-R 300M	layer 24, full speed: 0.177 (0.060 to 0.373)	-0.198 (-0.297 to -0.089)	< 0.0074
wav2vec 2.0 Conformer Large	layer 24, full speed: 0.196 (0.105 to 0.302)	-0.179 (-0.234 to -0.138)	< 0.0074
MMS 300M	layer 24, full speed: 0.184 (0.039 to 0.375)	-0.191 (-0.312 to -0.076)	< 0.0074
HuBERT Base	layer 12, full speed: 0.194 (0.077 to 0.378)	-0.181 (-0.268 to -0.084)	0.0038
HuBERT Large	layer 24, full speed: 0.190 (0.086 to 0.335)	-0.185 (-0.269 to -0.114)	< 0.0074
data2vec Base (100 h)	layer 12, full speed: 0.158 (0.061 to 0.300)	-0.218 (-0.293 to -0.147)	< 0.0074
data2vec Base (960 h)	layer 12, full speed: 0.147 (0.045 to 0.318)	-0.229 (-0.330 to -0.126)	< 0.0074
WavLM Base+	layer 12, full speed: 0.121 (0.038 to 0.298)	-0.254 (-0.351 to -0.139)	< 0.0074
WavLM Large	layer 24, full speed: 0.163 (0.041 to 0.372)	-0.212 (-0.309 to -0.090)	0.0120
UniSpeech-SAT Base+	layer 12, full speed: 0.164 (0.063 to 0.356)	-0.211 (-0.309 to -0.097)	< 0.0074
XEUS	layer 18, full speed: 0.218 (0.112 to 0.387)	-0.157 (-0.241 to -0.073)	< 0.0074
Loudness control	summary: 0.126 (0.019 to 0.342)	-0.249 (-0.363 to -0.108)	< 0.0074
Duration control	summary: 0.030 (0.000 to 0.113)	-0.345 (-0.467 to -0.210)	0.0038

Notes: Own final output: the output embedding for networks trained on animal or general sound, the last transformer layer at the recording's own speed for speech networks, and the output at the recording's own speed for speaker networks; networks with one embedding are as in Table S13. Difference and Holm p as in Table S13. Against the layer chosen by the Fisher rule, the final output was higher in 90 of 130 network-dataset pairs for the ten animal networks with several layers (median change +0.069) and in 53 of 169 for the 13 speech networks with several layers (median -0.041).

References cited only in the supporting information

Babu, A., Wang, C., Tjandra, A., Lakhotia, K., Xu, Q., Goyal, N., Singh, K., von Platen, P., Saraf, Y., Pino, J., Baevski, A., Conneau, A. & Auli, M.

(2022) XLS-R: self-supervised cross-lingual speech representation learning at scale. Proceedings of Interspeech 2022, pp. 2278-2282.

<https://doi.org/10.21437/Interspeech.2022-143>

Baevski, A., Zhou, Y., Mohamed, A. & Auli, M. (2020) wav2vec 2.0: a framework for self-supervised learning of speech representations. Advances in Neural Information Processing Systems, 33, 12449-12460.

Baevski, A., Hsu, W.-N., Xu, Q., Babu, A., Gu, J. & Auli, M. (2022) data2vec: a general framework for self-supervised learning in speech, vision and language. Proceedings of the 39th International Conference on Machine Learning, Proceedings of Machine Learning Research, 162, 1298-1312.

Chen, S., Wang, C., Chen, Z., Wu, Y., Liu, S., Chen, Z., Li, J., Kanda, N., Yoshioka, T., Xiao, X., Wu, J., Zhou, L., Ren, S., Qian, Y., Qian, Y., Wu, J., Zeng, M., Yu, X. & Wei, F. (2022a) WavLM: large-scale self-supervised pre-training for full stack speech processing. *IEEE Journal of Selected Topics in Signal Processing*, 16, 1505-1518. <https://doi.org/10.1109/JSTSP.2022.3188113>

Chen, S., Wu, Y., Wang, C., Chen, Z., Chen, Z., Liu, S., Wu, J., Qian, Y., Wei, F., Li, J. & Yu, X. (2022b) UniSpeech-SAT: universal speech representation learning with speaker aware pre-training. *Proceedings of ICASSP 2022*, pp. 6152-6156. <https://doi.org/10.1109/ICASSP43922.2022.9747077>

Chen, S., Wu, Y., Wang, C., Liu, S., Tompkins, D., Chen, Z., Che, W., Yu, X. & Wei, F. (2023) BEATs: audio pre-training with acoustic tokenizers. *Proceedings of the 40th International Conference on Machine Learning, Proceedings of Machine Learning Research*, 202, 5178-5193. <https://proceedings.mlr.press/v202/chen23ag.html>

Desplanques, B., Thienpondt, J. & Demuynck, K. (2020) ECAPA-TDNN: emphasized channel attention, propagation and aggregation in TDNN based speaker verification. *Proceedings of Interspeech 2020*, pp. 3830-3834. <https://doi.org/10.21437/Interspeech.2020-2650>

Ghani, B., Kalkman, V.J., Planqué, B., Vellinga, W.-P., Gill, L. & Stowell, D. (2025) Impact of transfer learning methods and dataset characteristics on generalization in birdsong classification. *Scientific Reports*, 15, 16273. <https://doi.org/10.1038/s41598-025-00996-2>

Gulati, A., Qin, J., Chiu, C.-C., Parmar, N., Zhang, Y., Yu, J., Han, W., Wang, S., Zhang, Z., Wu, Y. & Pang, R. (2020) Conformer: Convolution-augmented Transformer for Speech Recognition. *Proceedings of Interspeech 2020*, pp. 5036-5040. <https://doi.org/10.21437/Interspeech.2020-3015>

Hatch, A.O., Kajarekar, S. & Stolcke, A. (2006) Within-class covariance normalization for SVM-based speaker recognition. *Proceedings of Interspeech 2006*, paper 1874. <https://doi.org/10.21437/Interspeech.2006-183>

- Heinrich, R., Rauch, L., Sick, B. & Scholz, C. (2025) AudioProtoPNet: an interpretable deep learning model for bird sound classification. *Ecological Informatics*, 87, 103081. <https://doi.org/10.1016/j.ecoinf.2025.103081>
- Hershey, S., Chaudhuri, S., Ellis, D.P.W., Gemmeke, J.F., Jansen, A., Moore, R.C., Plakal, M., Platt, D., Saurous, R.A., Seybold, B., Slaney, M., Weiss, R.J. & Wilson, K. (2017) CNN architectures for large-scale audio classification. In 2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 131-135. <https://doi.org/10.1109/ICASSP.2017.7952132>
- Hsu, W.-N., Bolte, B., Tsai, Y.-H.H., Lakhota, K., Salakhutdinov, R. & Mohamed, A. (2021a) HuBERT: self-supervised speech representation learning by masked prediction of hidden units. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 29, 3451-3460. <https://doi.org/10.1109/TASLP.2021.3122291>
- Hsu, W.-N., Sriram, A., Baevski, A., Likhomanenko, T., Xu, Q., Pratap, V., Kahn, J., Lee, A., Collobert, R., Synnaeve, G. & Auli, M. (2021b) Robust wav2vec 2.0: analyzing domain shift in self-supervised pre-training. *Proceedings of Interspeech 2021*, pp. 721-725. <https://doi.org/10.21437/Interspeech.2021-236>
- Huang, P.-Y., Xu, H., Li, J., Baevski, A., Auli, M., Galuba, W., Metze, F. & Feichtenhofer, C. (2022) Masked autoencoders that listen. *Advances in Neural Information Processing Systems*, 35, 28708-28720. <https://doi.org/10.52202/068431-2081>
- Kather, V.S., Hauptert, S., Ghani, B. & Stowell, D. (2026) Bacpipe: a Python package to make bioacoustic deep learning models accessible. *Methods in Ecology and Evolution*. <https://doi.org/10.1111/2041-210X.70406>
- Matějka, P., Novotný, O., Plchot, O., Burget, L., Díez Sánchez, M. & Černocký, J. (2017) Analysis of score normalization in multilingual speaker recognition. In *Interspeech 2017*, pp. 1567-1571. <https://doi.org/10.21437/Interspeech.2017-803>

- Moummad, I., Farrugia, N. & Serizel, R. (2024) Regularized contrastive pre-training for few-shot bioacoustic sound detection. In ICASSP 2024, IEEE International Conference on Acoustics, Speech and Signal Processing, pp. 1436-1440. <https://doi.org/10.1109/ICASSP48485.2024.10446409>
- Moummad, I., Serizel, R., Benetos, E. & Farrugia, N. (2026) Domain-invariant representation learning of bird sounds. In ICASSP 2026, IEEE International Conference on Acoustics, Speech and Signal Processing, pp. 15237-15241. <https://doi.org/10.1109/ICASSP55912.2026.11463533>
- Pratap, V., Tjandra, A., Shi, B., Tomasello, P., Babu, A., Kundu, S., Elkahky, A., Ni, Z., Vyas, A., Fazel-Zarandi, M., Baevski, A., Adi, Y., Zhang, X., Hsu, W.-N., Conneau, A. & Auli, M. (2024) Scaling speech technology to 1,000+ languages. *Journal of Machine Learning Research*, 25(97), 1-52.
- Rauch, L., Heinrich, R., Moummad, I., Joly, A., Sick, B. & Scholz, C. (2025a) Can masked autoencoders also listen to birds? *Transactions on Machine Learning Research*. <https://openreview.net/forum?id=GIBWR0Xo2J>
- Rauch, L., Schwinger, R., Wirth, M., Heinrich, R., Huseljic, D., Herde, M., Lange, J., Kahl, S., Sick, B., Tomforde, S. & Scholz, C. (2025b) BirdSet: a large-scale dataset for audio classification in avian bioacoustics. In *International Conference on Learning Representations (ICLR 2025)*, pp. 29482-29520.
- Ravanelli, M., Parcollet, T., Plantinga, P., Rouhe, A., Cornell, S., Lugosch, L., Subakan, C., Dawalatabad, N., Heba, A., Zhong, J., Chou, J.-C., Yeh, S.-L., Fu, S.-W., Liao, C.-F., Rastorgueva, E., Grondin, F., Aris, W., Na, H., Gao, Y., De Mori, R. & Bengio, Y. (2021) SpeechBrain: a general-purpose speech toolkit. *arXiv:2106.04624*. <https://doi.org/10.48550/arXiv.2106.04624>
- Robinson, D., Robinson, A. & Akrapongpisak, L. (2024) Transferable models for bioacoustics with human language supervision. In ICASSP 2024, IEEE International Conference on Acoustics, Speech and Signal Processing, pp. 1316-1320. <https://doi.org/10.1109/ICASSP48485.2024.10447250>

Robinson, D., Miron, M., Hagiwara, M. & Pietquin, O. (2025) NatureLM-audio: an audio-language foundation model for bioacoustics. In International Conference on Learning Representations (ICLR 2025), pp. 21378-21398.

Snyder, D., Garcia-Romero, D., Sell, G., Povey, D. & Khudanpur, S. (2018) X-vectors: robust DNN embeddings for speaker recognition. Proceedings of ICASSP 2018, pp. 5329-5333. <https://doi.org/10.1109/ICASSP.2018.8461375>

Villalba, J., Chen, N., Snyder, D., Garcia-Romero, D., McCree, A., Sell, G., Borgstrom, J., García-Perera, L.P., Richardson, F., Dehak, R., Torres-Carrasquillo, P.A. & Dehak, N. (2020) State-of-the-art speaker recognition with neural network embeddings in NIST SRE18 and Speakers in the Wild evaluations. *Computer Speech & Language*, 60, 101026. <https://doi.org/10.1016/j.csl.2019.101026>

Wang, C., Tang, Y., Ma, X., Wu, A., Okhonko, D. & Pino, J. (2020) Fairseq S2T: fast speech-to-text modeling with fairseq. Proceedings of the 1st Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 10th International Joint Conference on Natural Language Processing: System Demonstrations, pp. 33-39. <https://doi.org/10.18653/v1/2020.aacl-demo.6>

Williams, B., van Merriënboer, B., Dumoulin, V., Hamer, J., Fleishman, A.B., McKown, M., Munger, J., Rice, A.N., Lillis, A., White, C., Hobbs, C., Razak, T., Curnick, D., Jones, K.E. & Denton, T. (2025) Using tropical reef, bird and unrelated sounds for superior transfer learning in marine bioacoustics. *Philosophical Transactions of the Royal Society B*, 380(1928), 20240280. <https://doi.org/10.1098/rstb.2024.0280>